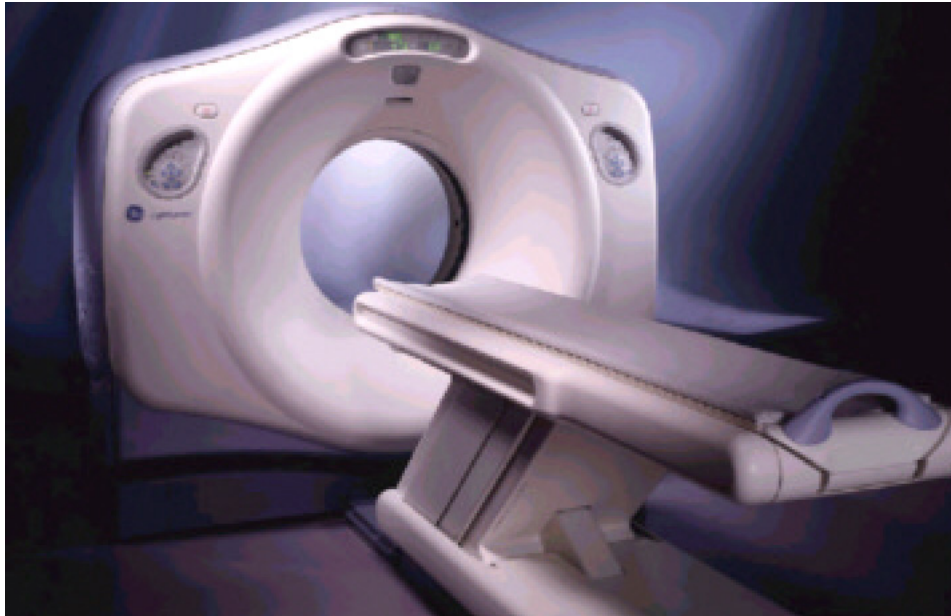




GE Lightspeed Series (1X-4X)



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Document Specifications

EPST Name: GE Lightspeed Series (1X-4X)

This EPST covers the 1X - 4X systems with Global Operator Console (GOC) GOC1 through GOC3. For

detailed console information and troubleshooting, refer to the documentation for your specific console.

Revision History

Rev	Author Initials	Sections Changed	Date	Brief Description of Change
0	n/a		n/a	n/a
1	Lynn Plowman	All	July 2006	Updates to all sections and new pages added



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Slip Ring Adjustments

Warning

Exposed Slip Rings pose an Electric Shock Hazard! Contact may cause serious injury or death! Voltages could be present on the slip ring. Power on the slip ring includes 120 VAC and 750 VDC.

Slip Ring Antenna Adjustment

Under normal circumstances, the position of the HSDCD antenna should not need adjustment. Adjust it only if there are indications of interference.

Under no circumstances should the HSDCD antenna be allowed to contact the rotating components.

Accurate placement of the HSDCD antenna above the emitter traces on the rotating slip ring is important to achieve the **optimal data transfer** performance and to **avoid contact** between the ring and the stationary antenna.

The **optimal position** of the **HSDCD receiver/antenna** above the emitter traces is 1.41 mm above (0.060 in.) and centered. Refer to **System Communication and Control** (Slip Ring) Troubleshooting.

Note Observe all anti-static precautions. Use an anti-static kit that has a mat, wrist strap and ground (GND).

1. Disable gantry power and apply Lock-Out Tag-Out.

2. Remove the Mylar window.
3. Remove gantry **side**, **top** and rear covers.
4. Make sure the HSDCD has no power applied; there should be no LEDs lit.
5. Loosen the axial **adjustment screws** and the radial adjustment screw for the HSDCD antenna.
6. The HSDCD receiver/antenna should now be loose above the emitter. Insert the HSDCD adjustment tool between the HSDCD antenna and the emitter traces on the ring; it should be a snug fit. For best results, space the HSDCD antenna so it is 1.41 mm (about 1/16 inch) above the emitter.
7. Tighten the radial screws. Press slightly on the top of the HSDCD antenna as the adjustment screws are tightened.
8. Carefully slide the adjustment tool from between the HSDCD antenna and the emitter.
9. Using the alignment sights at each end of the HSDCD antenna, center it above the center trace on the emitter PCB.
10. While holding the position of the HSDCD antenna, tighten the axial alignment screws.

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Potential for Ring Damage

The stationary and rotating components must never touch, even with the gantry tilted.

Caution

Do not rotate the gantry with the adjustment tool installed. Damage to the delicate PCB traces will result. There is no method to repair ring boards in the field; a ring swap is necessary.

Inspect the air-gap between the HSDCD antenna and the ring as the gantry rotates. It may be necessary to disable the gantry brake to rotate the gantry more easily. Look for clearances between the emitter and the HSDCD antenna. While rotating the ring, ensure the emitter trace is aligned with the HSDCD. During rotation, no parts of the HSDCD antenna should contact the emitter surface. The run-out of the platter slip ring traces should not exceed 0.83 mm axially and 0.81 mm radially. Especially check clearances near the emitter solder and PCB connections.

The **HSDCD receiver** has two LEDs. One LED indicates power is applied to the HSDCD and the other indicates the HSDCD is able to transmit a signal.

Scan Verification

1. Restore power to the system.
2. Observe diagnostic “DIP Stats” information. Before starting, log raw ring error count, FEC correct-able counts, and the date/time of the last file update.
3. Run 5 stationary and 50 rotational scans with X-ray. Technique is not important. It is important to exit the exam, because this triggers the “DIP Stats” update.
4. Observe diagnostic “DIP Stats” information. There should be no additional raw HSDCD ring errors or FEC correctable events.
5. Replace the gantry covers and secure. Re-install the Mylar window.
6. Locate the HSDCD Transmitter Power supply inside the Fuse Box Assembly. Measure the DC voltage on the wires leading to the Transmitter for +15 V (± 2.0). Use **indicator LEDs** to ensure proper operation.
7. Locate the HSDCD Receiver Power supply on the rear of the right fan assembly. Measure the DC voltage on the wires leading to the HSDCD Receiver. The specification is +15 V (± 2.0). Use indicator LEDs to ensure proper operation.



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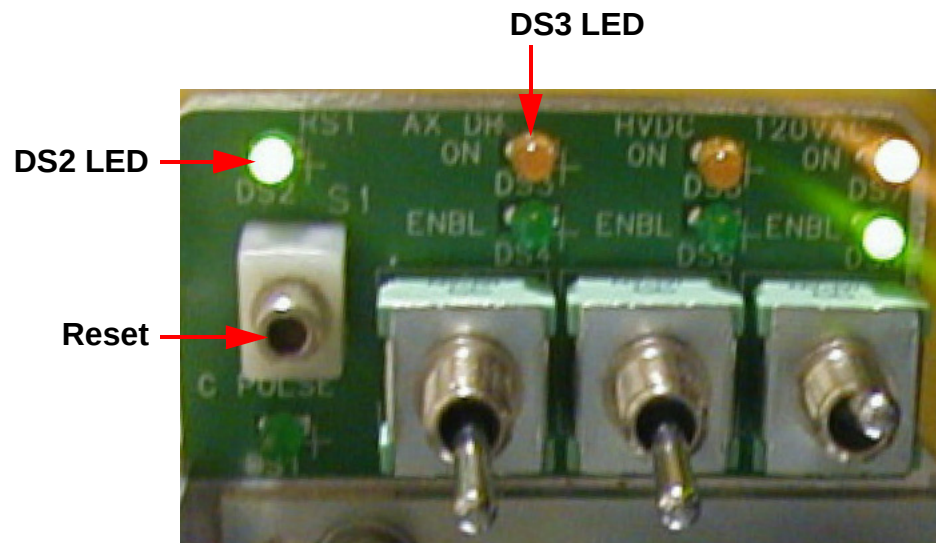


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How to Reset Axial Drive C Pulse (Home Flag Adjustment)

The **Axial Control Board** receives the "C" pulse from the **encoder** at J2D25 (AX_ENC_CHC) and J2C25 (AX_ENC_CHC_RTN). This signal is opto-isolated and becomes CHC. A green led DS2 (CHC) lights when a "C" pulse is received. CHC is processed and becomes CH_C*. This CH_C* signal provides the return path for REMOTE_C_L and REMOTE_C_H is the 5 VDC source used to illuminate the "C" pulse LED on the STC backplane. The "C" pulse LED located on the left fan assembly is also driven by this circuit.



Gantrol Control on STC Backplane

Procedure

1. Turn **OFF** the Axial Drive ENABLE and HVDC ENABLE switches on the **STC backplane**.
2. Rotate the gantry until the middle of the **home flag** is in the center of the opto-sensor window. Verify that the **DS3 LED** on the Axial Board lights up. Notice that when the gantry is rotated, LEDs DS7 and DS6 flash. The DS3 LED should only light when the home plate is in the opto-sensor window.
3. Once the DS3 LED is ON, rotate the gantry ± 1 degree. Verify that the DS2 LED lights up. Notice that the **DS2 LED** might flash rather than stay on. The key is to stop rotating the gantry as soon as the DS2 LED flashes. If the DS2 LED flashes or lights up, you have coincided with the C-pulse.

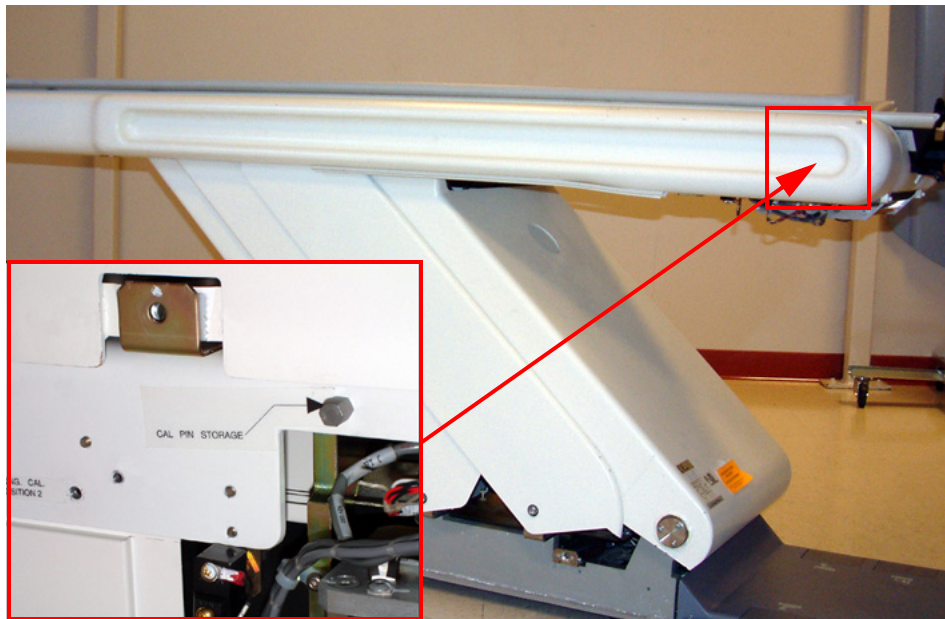
If the DS2 LED does not flash or light up, verify that the home plate is in the center of the opto-sensor and lift as well as rotate the encoder gear assembly until the DS2 lights up.





Cradle Characterization

Table Side View



Store Pin Location with Table Side Cover Removed

Procedure

1. From the operator console, select **Service Desktop > System Integration > Characterization > Cradle Characterization**.
2. Follow the Cradle Characterization instructions displayed on the screen.
3. In the Alignment Light procedure, the third and fourth screens, use the BLACK DOT on the cradle right side bumper.
4. Elevate the table to at least 40 mm from isocenter, so the black dot reaches the **external** alignment light before the cradle reaches its home position.
5. In the end-of-cradle position procedure, the fifth screen, use the same BLACK DOT on the cradle right-side bumper.



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Table Elevation Characterization

Important! Take care to measure the elevation distances exactly. Incorrect measurements will cause the Elevation Characterization to fail.

1. Remove upper and lower **right** side table covers.
2. Find and remove the small L-shaped measurement block stored on top of the bigger measurement block.
3. Install it in its characterization position at the **right rear** of the table.
4. Move the **measurement plate** on the right side of table to the "out" position.
5. Tighten the mounting screws for both plates.
6. Select **Service Desktop > Calibration > Mechanical Characterization > Table Elevation**.
7. Follow the Elevation Characterization instructions displayed on the screen.
8. Measure the elevation distance between the bottom surface of the each measurement block and the center of the rear leg upper pivot pin as vertically as possible.
9. Return the measurement plates to the storage position.





Video Monitor Adjustments

Note Allow the monitor to warm for at least 20 minutes, to reach normal operating temperature, before you adjust it so results will be stable.

Refer to [Image Quality](#).

Refer to the **Monitor Installation Manual** for details on how to adjust the Brightness and Contrast.

The light output from all color monitors is lower than the output from black and white monitors. For this reason, you need to be very careful when setting up the monitor brightness and contrast. Initially, the systems are set to factory defaults, but these can be adjusted.

The technologist may perceive the image on the monitor as "softer" than the image on the film, (i.e. they like the film, but they would like the image on the monitor to look like their film in terms of contrast and brightness). You may need to make the adjustment for Brightness and Contrast so that the technologist can see anatomical structure (window width) at the right amount of brightness (window level).

Adjust Brightness and Contrast

1. Display test image.
2. Darken the room.
3. Use the knobs on the back to adjust brightness and contrast.

Sony Adjustment

1. Press the MENU button (that looks like a monitor).
2. Press the UP or DOWN arrow buttons until the menu item you want to adjust is highlighted on the screen.
3. Press the MENU button to confirm this choice.
4. Press the UP or DOWN arrow buttons to select the item to adjust, and then the < or > buttons until the item's value that you want is selected on the screen. Select the COLOR menu to adjust Color Temperature (6500K is the desired setting).
5. Press the MENU button twice to store this value and exit.



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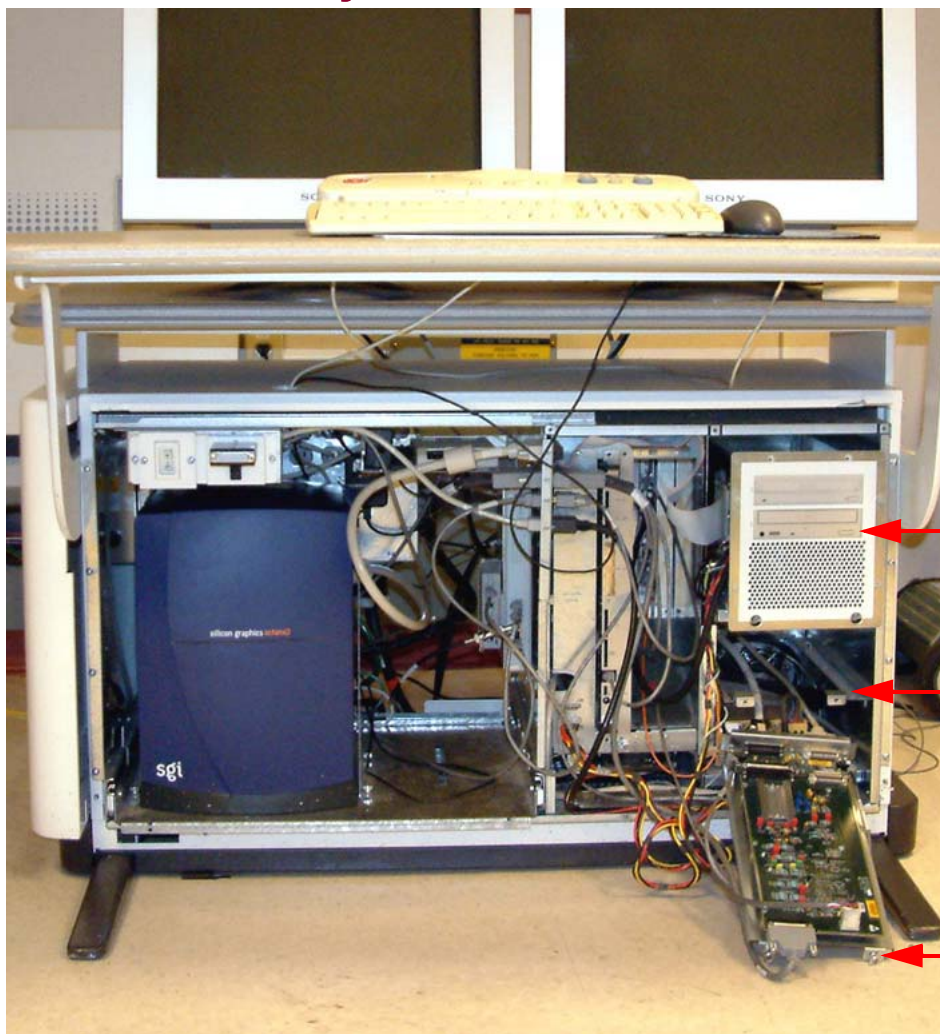
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Intercom Adjustments



Accessing Intercom Board with Console Front Cover Removed

How to Access The Intercom Board

1. Remove the front panel to the console.
The Intercom board is located underneath the MOD/CD-ROM drives assembly.
2. Loosen the two screws that secure the board to the chassis.
3. Carefully slide out the Intercom board, keeping any attached cables from snagging on the back side of the console or chassis frames.

MOD/CD-ROM
Drives Assembly

Intercom Board Chassis

Intercom Board Screw (2)

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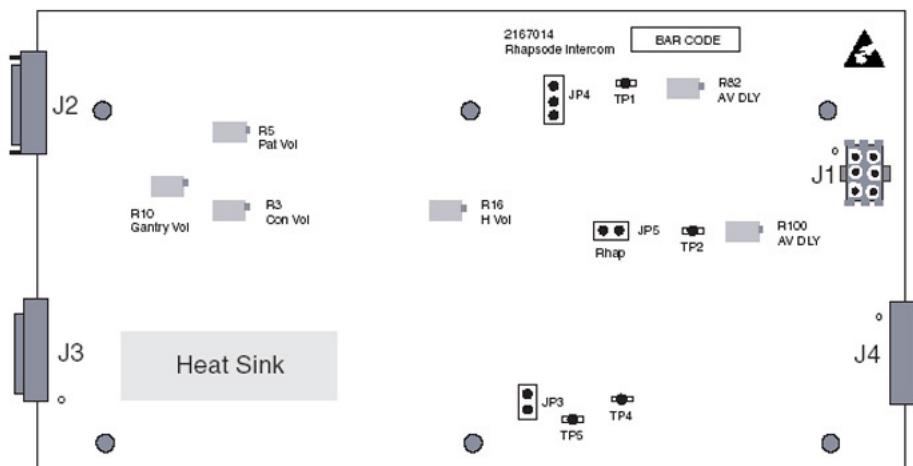
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Intercom Adjustments, *continued*



How to Adjust the Intercom Board (2167014)

Verify jumpers at:

- JP4, over pin 1 and 2 (pin 1 is the left most pin)
Connectors located on the left side of the board.
- JP5 jumper set.
- JP1, JP2, JP3, JP6, JP7, JP8 do NOT have jumpers.

Verify POT settings at:

- R3 - 0 turns CW (console speaker) more turns CW equals more volume.
- R5 - 4 turns CW (table speaker) more turns CW equals more volume.
- R10 - 3 turns CCW (min. gantry volume - user preference) fewer turns CCW equals less minimum volume.
- R100 - 10.25 turns CW (AutoVoice threshold).
- R82 and R16 are for non LightSpeed functions - no adjustment required.

How to Re-install the Intercom Board

1. Reverse the procedures in "How to Access the Intercom Board" on the previous page.
2. Power up and test intercom and auto voice functions.
3. Verify sound levels of intercom at console and gantry.
4. Verify auto voice level.

Refer to [Intercom Functional Checks](#).





Laser Lights

DANGER

When the covers are off, verify there are no objects in the gantry area before you press the gantry keypad button for the Laser Lights. The gantry rotates to the azimuth called 180 degrees. If someone's head is in the way, the sudden rotation could kill them.

Warning

Never stare at the laser beams because this can cause permanent eye damage.

Required Tools

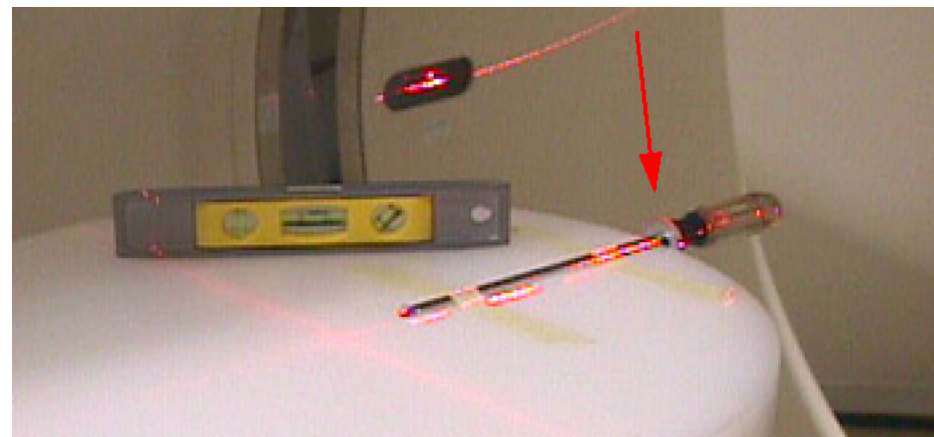
- Long Phillips #2 screwdriver
- 1 large washer 0.5 ID, 1.5 OD
- Masking Tape
- Torpedo Level
- Large 48 cm Phantom
- 5 mm Hex key

Procedure Preparation

1. Remove all gantry covers: **side**, **top**, **front** and rear.
2. Remove the cradle pad and associated accessories.
3. Place the large Phantom on the cradle end so it extends two inches beyond the cradle end.
4. Verify Phantom is level front to back and side to side.

Laser Alignment Jig Setup

1. Thread the large washer onto screwdriver and secure both to the Phantom as shown.



2. Turn on laser lights using the gantry control panel.
An **OBC backplane switch** also controls the alignment lights. LED DS1 "Lights Laser" indicates status.
3. Adjust jig position so that the internal lasers shine on the washer's edge center, and Sagittal and Coronal lasers shine on the center of the screwdriver shaft.
4. Choose either the left or right side of the jig as a **reference** laser for this procedure.

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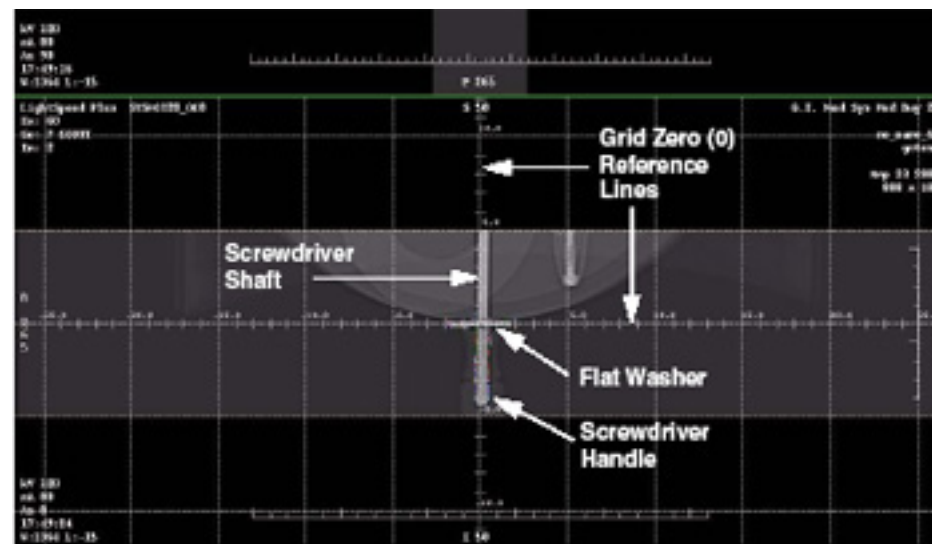
Laser Alignment Jig Setup, *continued*

5. Select Laser Align protocol: New Patient, Baby, 20.1 Service Generic Scan, Create New Series, Scout.

Scout	Scan Type	Start	End	kV	mA	Plane
1	Scout	S 50	I 50	120	80	90
2	Scout	S 50	I 50	120	80	0

6. Confirm and scan.
7. Use: IMAGE WORKS, Browser Select Exam, Series, Image. Select FORMAT one over one (lower left).
8. Once both scouts have displayed, adjust Window and Level so you can see the screwdriver handle.
9. Click in a viewport to activate it, and select **Grid**.
10. Record the error delta from the Zero Grid lines to the center of the screwdriver shaft and washer edge. Use Measure Distance if desired.
11. Use the gantry controls to position the washer and screwdriver shaft until they are centered on the zero X and Y grid lines. **DO NOT MOVE THE PHANTOM.**

Changing Elevation will post an Error Window. Ignore this and proceed.



12. Select **Repeat Series** and scan.
13. Repeat steps 5 through 11 until jig reference points are under the grid zero lines.

Important! Once the jig is aligned to ISO Center and the Z-Axis, **do not disturb it**. If it is disturbed, you will need to start over.

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Laser Lights, *continued*

DANGER

Never service the gantry with the Axial Drive enabled. Unexpected gantry rotation can result in serious injury or death.

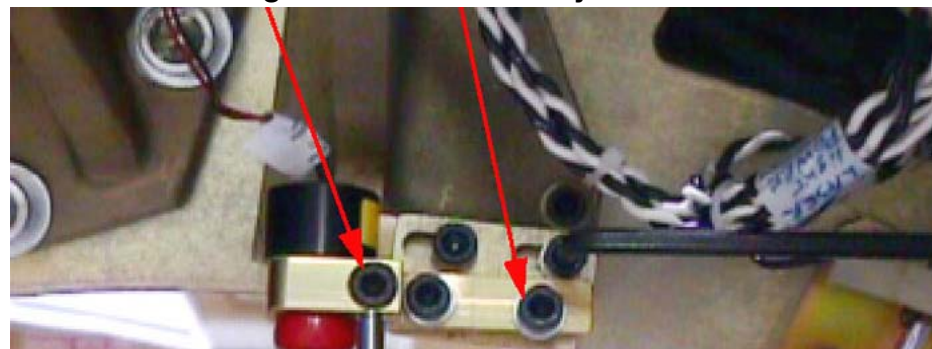
Adjusting Reference Lights

1. Press the **INTERNAL LANDMARK button** to make the cradle location the zero scan location.
2. After you verify that no one could be struck by gantry rotation, press the ALIGNMENT LIGHTS button.
3. **Turn OFF** the Axial Drive ENABLE switch on the **STC backplane**.
4. Adjust the previously chosen **reference Internal laser** to shine on the washer's edge center. Adjust the Sagittal and Coronal lasers so they shine on the screwdriver shaft at ISO Center. Set the Coronal lasers as level as possible and the Sagittal laser as parallel to the cradle as possible. Tracking adjustments will be done later.

Properly adjusted lasers will bisect the output port of the other two Internal lasers.

5. Move the cradle out of the gantry to the 240 mm Scan Location using the gantry control panel.

Laser Angular and Position Adjustment Screws



6. Adjust the Reference External Laser to shine on the washer's edge center.

Properly adjusted lasers will bisect the output port of the other External laser.

7. After Reference Lasers have been adjusted, raise and lower the table and verify both the External and Internal lasers track the washer's edge center.
8. If vertical angle adjustment is necessary, make sure you center the jig at ISO Center before you adjust the **internal** or **external** reference lasers.
9. Go back to step 4 if the laser is not on the edge. Otherwise, continue to step 10.
10. Remove the screwdriver jig without disturbing the Phantom.

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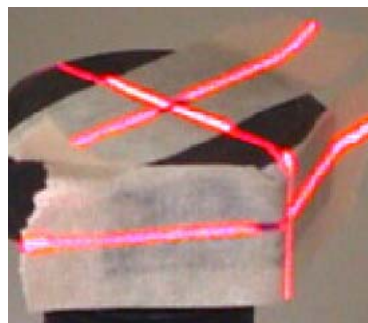
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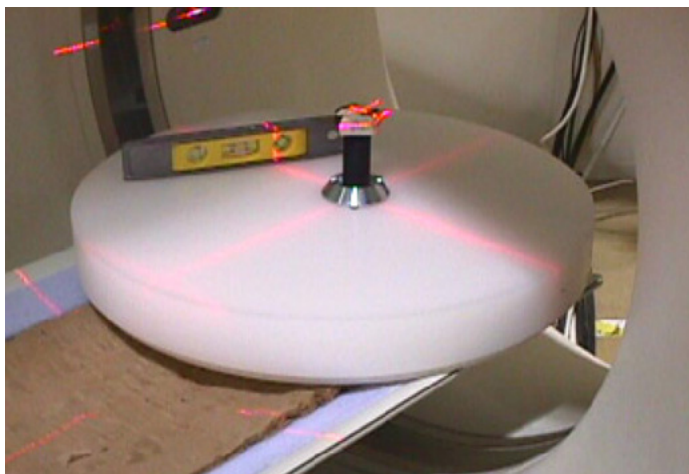
Laser Lights, *continued*

Adjusting Reference Lights, *continued*

11. Now that the **reference lasers have been set**, use a piece of notebook paper to adjust the other internal and external lasers to coincide with the reference lasers. Refer to [Laser Lights](#) Functional Checks.
12. Place another object such as a full contrast bottle or unopened soda can on the Phantom. Attach masking tape and use the Laser Lights to position the object at ISO Center.



Sagittal and Coronal Laser Alignment Jig Setup



13. Using a pen, carefully mark the laser intersection points on the masking tape. Once set, **DO NOT DISTURB THIS SECOND JIG**.
14. Using the gantry control panel, move the cradle out of the gantry to the 240 mm Scan Location. Verify the Sagittal and Coronal laser lights track your pen marks.
15. If Sagittal or Coronal angle adjustment is necessary, make these adjustments one at a time and verify the laser tracks from Internal to External landmarks before making the next adjustment. If the jig is disturbed, re-position the jig at ISO Center to reestablish your reference. Verify both Sagittal and Coronal lasers shine on the pen marks at ISO Center before proceeding. The loss of reference will require repeating both jig alignments. This would be necessary to ensure accuracy.
16. The last adjustment is the un-referenced Coronal laser to the Reference Coronal laser chosen earlier. Refer to [Laser Lights](#) Functional Checks.
17. Replace the cradle pad and associated accessories.
18. Replace all removed gantry covers.



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Plane of Rotation (POR) Alignment

Plane of Rotation puts the X-ray tube in the correct physical relationship to the detector. Typically this is only needed when the tube has been replaced or moved.

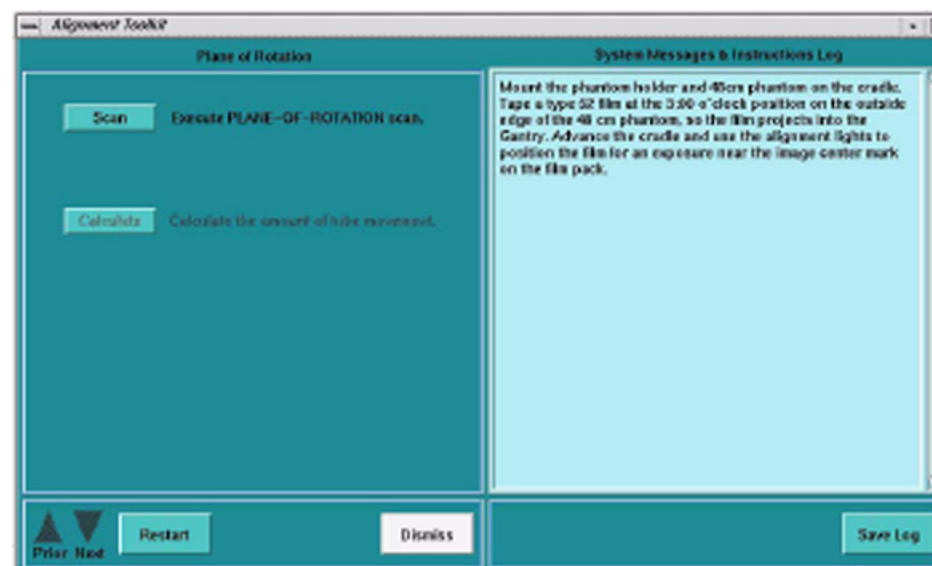
Required Tools

- Vernier Caliper and Dial Gauge Indicator
- Calibrated Vernier Caliper (millimeters or inches).
- Dial gauge Indicator (millimeters or inches)
- Type 52 Polaroid Film and Developer

Note The tube must be cool to do POR measurements. If you recently acquired any low technique scans, wait at least five minutes before you start the POR procedure. If you recently acquired image scans, wait at least 90 minutes. Wait 15 minutes between POR scans.

Procedure

1. Select **Service Desktop > Calibration > POR Alignment** -or- **Service Desktop > Replacement > Tube Change > POR Alignment** to access the POR Alignment screen.



2. Attach the dial gauge indicator onto the dial mount.
3. Zero the dial indicator when it's securely in place.

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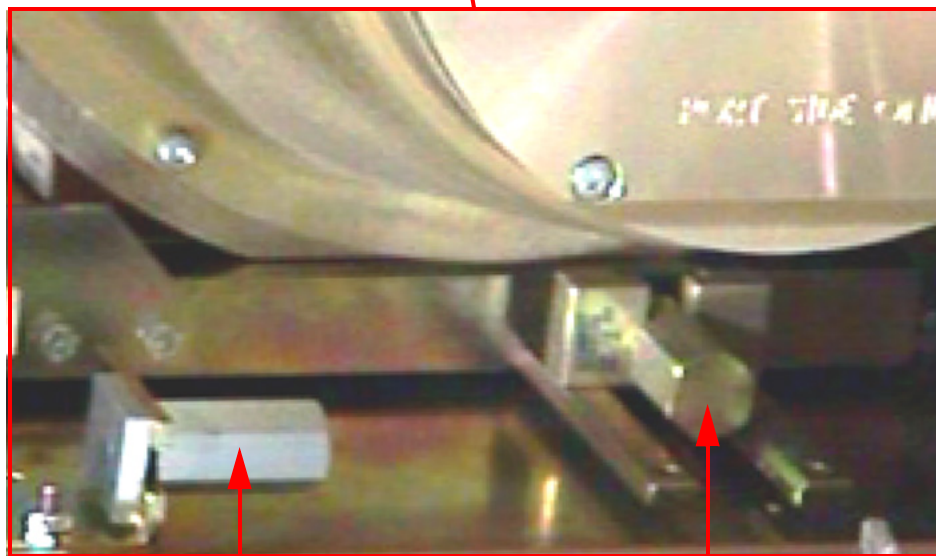
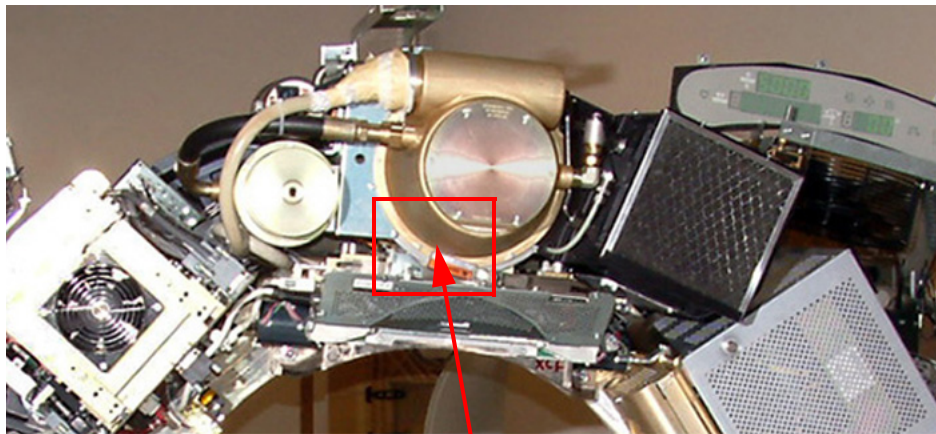


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Plane of Rotation (POR) Alignment, *continued*

Procedure (*continued*)



Dial Mount

POR Adjuster

4. Install the system's Phantom holder and place the 48 cm Phantom on it.
5. Attach one Polaroid Type 52 film on the outside edge of the 48 cm Phantom at the 3 o'clock position. Only the film should project into the gantry bore. The Phantom is used only to position the film.
6. Orient the side of the film side marked "This side toward lens" towards Isocenter.
7. Advance the cradle using the alignment lights to position the film's center marking on the alignment light marks; rotate the Phantom if necessary.
8. Take a scan.
9. Mark the outside of the film to show the table side. Use a pen and print the letter "T" on the side nearest the table. After the film is developed, transfer the marking to the photo.



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Procedure (*continued*)

10. Verify the film's narrow white beam lies within (between) the wider gray X-ray slits and the edges in both Z directions (in and out) are equally well defined by the collimator port. The edges of the narrow beam should be much sharper than the wide beam. If a difference in edge definition exists, check for gross Z misalignment; the tube's collimator adapter is a common cause of fuzzy film edges.
11. Use a Vernier calipers to measure the width of the two wider (dark gray) slits. They're the dark gray slits that extend past the edges of the narrower (off-white) slit to the blackest part of the film.
12. Take three measurements to obtain an average value for XF (front). Xf is the side of the film closest to the table. Using the same film measurement tool, take (1) Xf measurement at the top of the film, another near the middle and another near the bottom. Add these (3 or n) Xf distances together and divide this sum by 3 or n.
13. Now make three measurements to obtain an average value for XR (rear).
14. Select **Calculate** on the POR Alignment screen. Enter the values for XF and XR obtained in the steps above. The software will do the distance calculation.
The spec is 0.059 - 0.082 inches or 1.50 - 2.083 mm.
15. If the tube is out of spec, move it as indicated by the program. Loosen the four M12 bolts that secure the tube one-half turn out, no more. Clockwise rotation (in) of POR adjustment bolt moves the tube towards the table.
16. **Repeat POR procedure if you move the tube.**
Torque tube bolts to 49 ft-lbs (66.4 Nm).
17. Go to the **Beam on Window (BOW) Alignment** procedure.





Beam on Window (BOW) Alignment

BOW alignment puts the detector in the correct position so the X-ray beam hits the detector well, assuming the tube is positioned correctly through POR alignment.

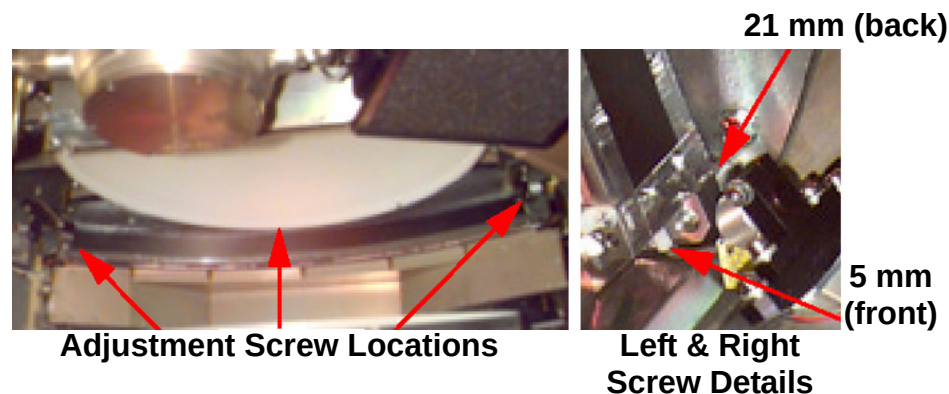
Important! Wait at least 15-30 minutes between scans to prevent unnecessary adjustments.

1. Select **Service Desktop > Calibration > BOW Alignment** -or- **Service Desktop > Replacement > Tube Change > BOW Alignment**.

2. Select **Move** to send the tube to 12 o'clock position.
3. Acquire a Beam on Window scan.
4. Select **Calculate** and make adjustments as indicated by the program.

Remove the rear gantry cover if necessary.

5. Loosen the M-6 caphead screw located at both ends of the detector.
6. Loosen the middle (jam) nut with a 10 mm wrench. Make indicated adjustments.
7. Acquire another BOW scan, then **Calculate**.
8. If your adjustments passed, proceed to step 9; otherwise, return to step 3.



Adjustment Screw	Affected Channel
Left	High
Middle	Medium
Right	Low

9. Tighten all three screws. Torque to 10 ft-lbs (13.5 Nm). Go to the **CBF/SAG Alignment** procedure.

The BOW specs are checked by the software. If an adjustment is needed for the first BOW scan, make adjustments per procedure. In the verification scan, the spec is different because the tube is warm. The spec is -1.5 mm (± 0.35).

Clockwise (CW) turns move the detector toward the mounting plate. Counter-clockwise (CCW) turns move the detector away from the mounting plate.



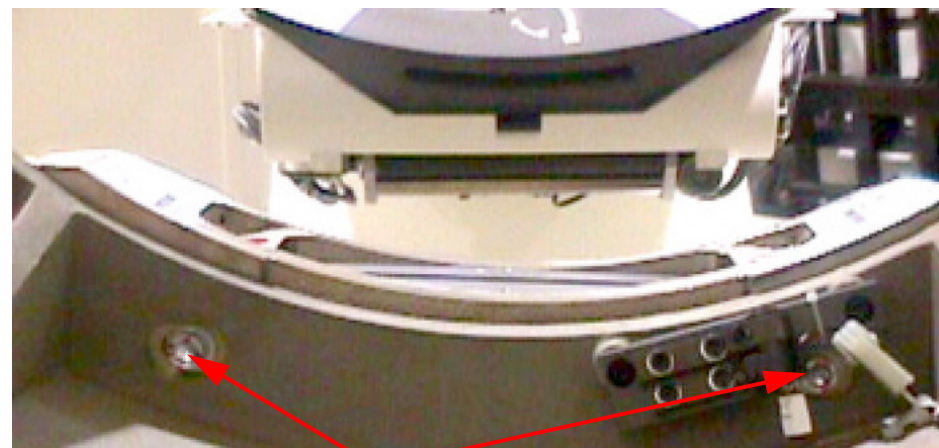


CBF/SAG Alignment

CBF/SAG Alignment ensures the focal spot is accurate, the bow-tie filter is centered and center of rotation is in a straight line.

Important! Wait at least 15-30 minutes between scans to prevent unnecessary adjustments.

1. Select **Service Desktop > Calibration > CBF and SAG Alignment** -or- **Service Desktop > Replacement > Tube Change > CBF and SAG Alignment**.
2. Click on the SCAN button to execute air filter scan.
3. Place the 1/8 inch screwdriver on the Phantom holder; it should be pointing into the Z direction.
4. Scan the pin.
5. Now scan air with the bow-tie filter.
6. Click on the CALCULATE button.
7. Install the dial indicator on the HEMRC and zero it.
8. Loosen the six M12 **collimator** cap screws. Four cap screws on the front of the collimator. One cap screw is behind the cable connections. Use a swivel adapter for the ratchet wrench. Two cap screws are on the rear.
9. **Adjust** the collimator as indicated by the results of the CBF calculation. Ignore the negative sign.
10. Tighten the collimator.
11. Rescan; CALCULATE.
12. Proceed to the next step if the adjustment is within limit, otherwise go back to step 8.
13. Torque all six M12 cap screws to 49 ft-lbs (66.4 Nm).
14. Go to the **Cold ISO Alignment** procedure.



Cap Screw Locations on Rear of Collimator



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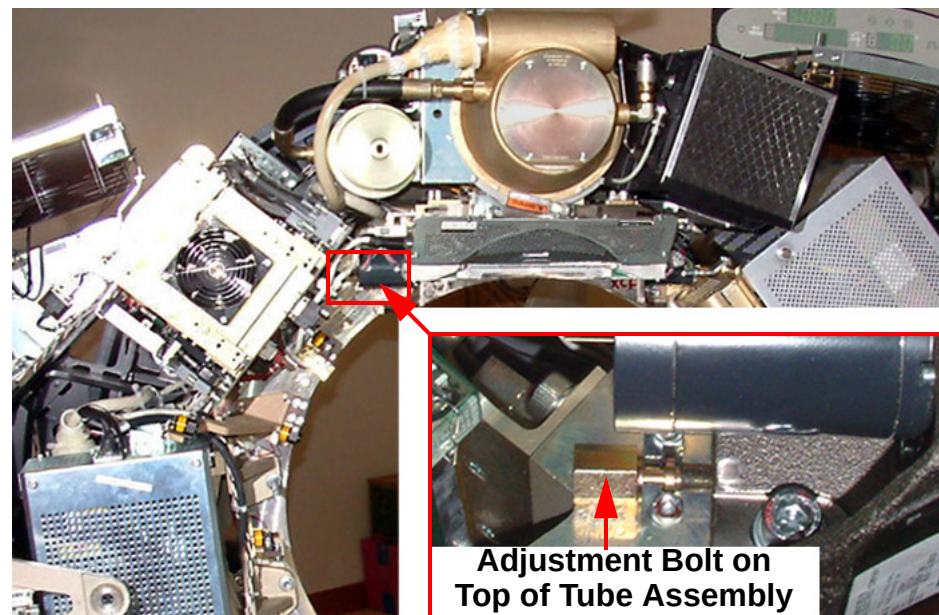


Cold ISO Alignment

Cold ISO alignment is done before the tube is heated by Generator calibrations. Hot ISO is done later.

Important! Wait at least 15-30 minutes between scans to prevent unnecessary adjustments.

1. Select **Service Desktop > Calibration > ISO Alignment** -or- **Service Desktop > Replacement > Tube Change > ISO Alignment**.
2. Execute Air scan (small spot).
3. Execute Air scan (large spot).
4. Place the 1/8 inch screwdriver on the Phantom holder (should be pointing into the Z direction).
5. Scan the pin (small spot).
6. Scan the pin (large spot).
7. CALCULATE cold ISO.
8. Mount a dial gauge on tube assembly and zero it.
9. Loosen the four M-12 bolts on the tube assembly.
10. Use the adjustment bolt on the top of the tube to adjust it UP/DOWN as indicated by the software.
11. Torque the four M12 bolts and verify the dial gauge still reads the correct adjustment value.
12. If the system indicates you are not within spec, repeat steps 2 through 7.
13. If the adjustments are within limits, proceed to the next step, otherwise go to step 11.
14. Torque the four M12 bolts to 49 ft-lbs (66.4 Nm).
15. Go to the **Generator Characterization** procedure.

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Generator Characterization

Generator characterization updates the small and large spot characterization file. This iterative process makes several scans at a different kV/mA/spot size. It calculates corrections, repeats the scan until the results fall within tolerance, then updates the characterization file.

1. Select **Service Desktop > Calibration > Generator Characterization** -or- **Service Desktop > Replacement -> HV Subsystem Change > Generator Characterization**.
2. Select **Read Metering** to check internal **kV metering circuits**.
3. Select **Run** to start the test. During the test, the firmware reads the metering circuits in the OFF state, then ON state, and reports the readings.
4. Compare the data in the "Delta" column on the Read Meter screen to the data in the "Limit" column.
"Delta" = DVM - A/D

Circuit "OFF"	Circuit "ON"
Anode kV = 0 (± 0.5)	Anode kV = 50 (± 7.5)
Cathode kV = 0 (± 0.5)	Cathode kV = 50 (± 7.5)
Total kV = 0 (± 0.5)	Total kV = 100 (± 15.0)

5. **Turn OFF** the Axial Drive ENABLE and HVDC ENABLE switches on the **STC backplane**.
6. Rotate the Anode tank to the 2 o'clock position.
7. Engage the gantry **rotational lock**.
8. Use a DVM as an mA meter; connect it to the mA board on the anode side:
 - the black lead to TP8 (ACAL1) on the mA board.
 - the red lead to TP11 (ACAL2) on the mA board.
9. Select **Read Metering** to check internal **mA metering circuits**.
10. Enter a delay in seconds that gives you enough time to walk from the console to the DVM and record the reading. The test will not begin until this time delay expires. Once it begins, the test enables the meter circuit for only four seconds.
11. Select the ACCEPT button.
12. Record the displayed and measured, Anode mA values for "Circuit OFF" and "Circuit ON."

The system has a test wire to TP5 included in the harness; the Cathode side should read approximately 19 mA during "Circuit On." Disconnect the test equipment from the Anode side, if used.

13. Go to the **High Voltage (HV) Calibration** procedure.





High Voltage (HV) Calibration

High voltage (HV) calibration for a Lightspeed Series system requires you to do the following:

- **Install the HV Divider (Bleeder)**
- **Calibrate the Cathode**
- **Calibrate the Anode**
- **Measure Total kV**
- **Verify kV Meter**
- **Exposure Time Accuracy**
- **Exposure Time Accuracy**
- **Use INSTALL NEW TUBE if Tube Is New**
- **Use AUTO MA CAL if Tube Is Not New**
- **Measure kV Rise and Fall Times**
- **Verify Scan Duration**
- **Scan Duration Spec**

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Install the HV Divider (Bleeder)

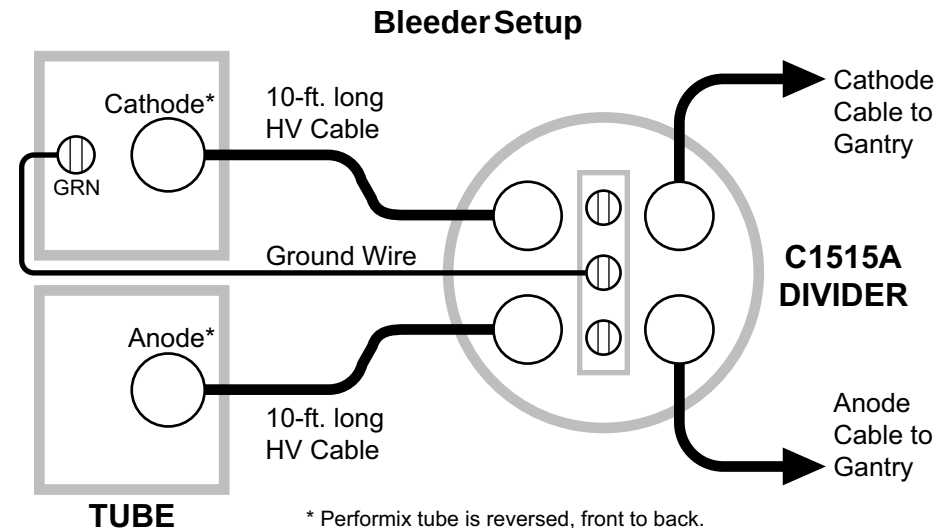
1. Turn **OFF** the Axial Drive ENABLE and HVDC ENABLE switches on the **STC backplane**.
2. Rotate the tube to the 3 o'clock position.
3. Engage the gantry **rotational lock**.
4. Turn **OFF** the 120 VAC ENABLE switch on the **STC backplane**.
5. Install the HV Divider between the tube and tanks. Place it on a table or use tube hoist, so the cables reach the tube. Add a ground wire (minimum size of AWG 12) from Tube ground to bleeder ground.

Caution The Performix tube unit **MUST** be grounded to the gantry during testing in order not to destroy the insert.

Caution Incorrect installation of anode and cathode HV cables can destroy the Performix tube.

6. Turn **ON** all three switches on the **STC backplane**. Press the **RESET DRIVES** button on any of the gantry keypads.

7. Connect a scope with 10X probes to the gantry service outlet. This will reduce waveform noise. To minimize bleeder-induced ripple on the kV waveform, connect a 30-foot Belden shielded twisted pair cable between the probes and bleeder.



8. Connect channel **one** to the **anode** output of the divider. Connect the scope ground to bleeder ground. Connect channel **two** to the **cathode** output of the divider. Trigger channel one, positive, DC couple, trigger mode normal. Channel one and two, 10 V/div, time base 200 ms. Invert Channel two.

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Calibrate the Cathode

1. Select **Service Desktop > Replacement > Diagnostic Data Collection (DDC) > Protocol Name > Bleeder Setup and Load** -or- **Service Desktop > Replacement > HV Subsystem Change > Bleeder Setup (DDC)**.

2. Verify/set-up the following DDC parameters:

STATIC X-RAY ON	Interscan Delay 2.00
1 SECOND	DAS Gain 31
1 SCAN	Gantry Tilt 0.0
FOCAL SPOT LARGE	Trigger Rate 984
100 kV	Filter BLOCKED
50 mA	Calibration Vector NONE
MONITOR ENABLE	

3. Select **Accept RX**. The Reading spec for the Cathode and Anode kV equals 50 kV (± 0.5).
4. If you use scope cursors to window the trace, position the left vertical cursor to the right of the rising edge of the waveform. Position the right vertical cursor to the left of the falling edge of the waveform.

5. **Adjust the kV board** until the scope reading for the Cathode kV and the displayed reading for the Cathode kV in the message log fall within ± 0.5 kV of each other. Use the POT labeled **CAKV, R316**.

Note Counterclockwise (CCW) turns decrease the scope kV and clockwise (CW) turns increase it. One-half (1/2) turn equals approximately 0.5 kV.

6. Record the results on FORM 4879.

Calibrate the Anode

1. Select **Bleeder Setup**.
2. Use the same DDC parameters as for cathode.
3. Select **Accept RX**.
4. **Adjust the kV board** until the scope reading for the Anode kV and the displayed reading for the Anode kV in the message log fall within ± 0.5 kV of each other. Use the POT labeled **ANKV, R318**.

Note Counterclockwise (CCW) turns decrease the scope kV and clockwise (CW) turns increase it. One-half (1/2) turn equals approximately 0.5 kV.

5. Record the results on FORM 4879.

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Measure Total kV

1. Select **Service Desktop > Replacement > Diagnostic Data Collection (DDC) > Protocol Name > Bleeder Setup and Load** -or- **Service Desktop > Replacement > HV Subsystem Change > Bleeder Setup (DDC)**.
2. Verify/set-up the following DDC parameters:

STATIC X-RAY ON	Interscan Delay 2.00
1 SECOND	DAS Gain 31
1 SCAN	Gantry Tilt 0.0
FOCAL SPOT LARGE	Trigger Rate 984
100 kV	Filter BLOCKED
50 mA	Calibration Vector NONE
MONITOR ENABLE	

3. Change the oscilloscope to add Ch.1 And Ch.2 to read total kV from the HV divider.

Channel one and two, 20 V/div, time base 200 ms, channel one, positive.
4. Select **Accept RX**.

5. Record the scope reading, and the Average kV displayed in the message log, in FORM 4879.
6. Display the Generator Characterization menu.
7. Toggle the softkey Monitor ENABLE to OFF so the message log no longer displays kV and mA readings.

Verify kV Meter

Verify the calibration of the internal kV metering circuits.

1. Select **Service Desktop > Calibration > Generator Characterization** -or- **Service Desktop > Replacement > HV Subsystem Change > Generator Characterization**.
2. Select **Read Metering**.
3. Select **Accept**. The test begins after the time delay expires. Once the test begins, the software enables the meter circuit for 4 seconds.
4. Record the displayed Anode kV, Cathode kV and Total kV values in FORM 4879 "Circuit OFF" and "Circuit ON" tables.
5. Select **Backup**.

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Exposure Time Accuracy

1. Select **Service Desktop > Diagnostics > Diagnostic Data Collection (DDC)**.
2. Select **Monitor Enable** to display scan times in the Message Log.
3. Change to the EXAM RX Desktop.
4. Select **New Patient** and prescribe the following **scout scans**. Take six scout scans of the following lengths: 20, 25, 30, 40, 150, 300, 480 mm. Use 120 kV and 40 mA.
5. Record the scan time displayed in the Message Log, on the HHS data sheet.
6. Select **New Patient** and prescribe the following **axial scans**. Use 120 kV and 40 mA. Use these scan times and Fields of View (FOV):

Select Time	FOV
0.6 sec.	Small
0.6 sec.	Large
1.0 sec.	Large
2.0 sec.	Large
3.0 sec.	Large

Select Time	FOV
4.0 sec.	Large

7. Select **New Patient** and prescribe the following **helical scans**. Use 120 kV, 40 mA and 10 mm slice. Use the following scan times and locations:

Scan Time	Scan Location
15.0 sec.	S70-170
28.0 sec.	S133-1135
30.0 sec.	S145-1145

8. When you finish the scan time tests, switch back to the SERVICE DESKTOP to display the Diagnostic Data Collection screen and toggle the Monitor ENABLE to OFF. Otherwise, the Message Log fills with kV, mA and scan statistics.
9. Go to the **N Number Adjustment** procedure.





High Voltage (HV) Calibration, *continued*

Remove the HV Divider

1. Press the **RESET DRIVES** button on any of the gantry keypads to **turn OFF** the DRIVES power.
2. **Turn OFF** all three switches on the **STC backplane**.
3. Remove the HV Divider.
4. Reconnect the HV cables for normal operation.

Caution

Incorrect installation of anode and cathode HV cables can destroy the Performix tube.

5. Re-apply paper toweling around the tube locking ring to absorb excess oil.
6. Disengage the gantry **rotational lock**.
7. **Turn ON** all three switches on the **STC backplane**.
8. Press the **RESET DRIVES** button on any of the gantry keypads to **turn ON** the DRIVES power.

Use INSTALL NEW TUBE if Tube Is New

1. Use the INSTALL NEW TUBE program to do mA Cal on a new tube. Run this program only on a new tube. Select **Service Desktop > Calibration > Generator Characterization** -or- **Service Desktop > Replacement > HV Subsystem Change > Generator Characterization > Install New Tube**.

The system automatically warms up the tube. Then it prompts you with the tube type.

2. Verify the number corresponds to your tube type; answer Y or N.

Software Token	Housing Number	Insert Number
12-MX_135CT	46-274800G1	46-274600G1
13-MX_165CT	46-309500G2	46-309300G1
14-MX_165CT_1	46-309500G2	46-309300G2
15-MX_200CT	2137130-2	2120785

3. Press START SCAN when it flashes to automatically run the program.

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Use AUTO MA CAL if Tube Is Not New

1. Use the AUTO MA CAL program to do mA Cal on an installed tube. Select **Service Desktop > Calibration > Generator Characterization** -or- **Service Desktop > Replacement > HV Subsystem Change > Generator Characterization > AUTO MA CAL.**

The system automatically warms up the tube.

2. Press START SCAN when it flashes to automatically run the program.

Measure kV Rise and Fall Times

1. At the OBC, connect a scope to the kV board.

Channel 1: Exposure Command EXCM, TP5. Scope ground to LGND, TP3, 2 V/div

Channel 2: Total kV KVTB, TP11. (At this test point kV = 20 kV per volt.)

Scope ground to AGND SGND, TP12, 1 V/div

Set the Scope Time base to 200 usec.

Positive or Negative trigger as required.

2. Select **Diagnostic Data Collection (DDC) > Protocol Name > Rise/fall and Load** -or- Select **Service Desktop -> Replacement > HV Subsystem Change > Rise and Fall Time (DDC).**

3. Verify/set-up the following DDC parameters:

STATIC X-RAY ON	Inter Scan Delay 2.00
1 SECOND	DAS Gain 31
1 SCAN	Gantry Tilt 0.0
FOCAL SPOT LARGE	Trigger Rate 984
80 kV	Filter BLOCKED
400 mA	Calibration Vector NONE

4. Measure **rise time** only on the 80 kV/400 mA scan.
5. Select **Accept RX.**
6. Select **Pause** after the start of the scan to prevent the scope from displaying the fall time.
7. After you record the rise time, change the technique to 140 kV and 40 mA, and select RESUME to start the **fall time** scan.

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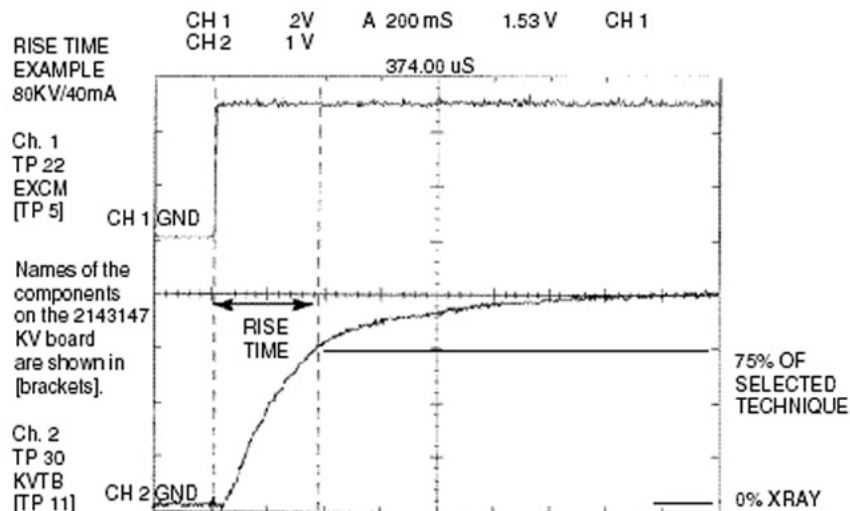
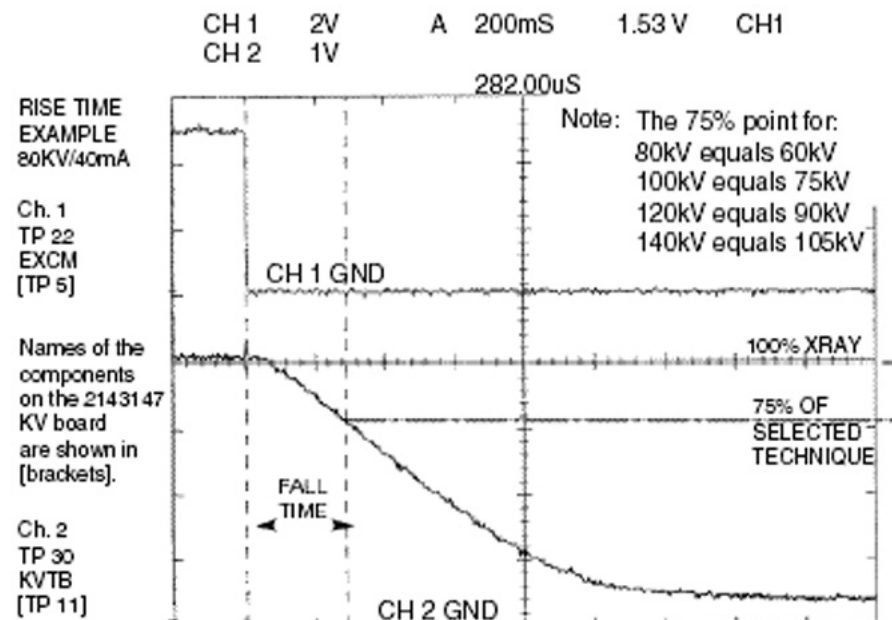
High Voltage (HV) Calibration, *continued*

Measure kV Rise and Fall Times, *continued*

8. Record the delay between the rise of the EXCM signal and the 75% threshold crossing of the selected kV on FORM 4879. Do not include the waveform overshoot. The 75% point for 80 kV equals 60 kV.

kVRise and Fall Times

kV	mA	Rise		Fall	
		Delay ms	Limit	Delay ms	Limit
80	400		0 + 1.9 ms	N/R	N/A
140	40	N/R	N/A		0 + 0.5 ms

Rise Waveform**Fall Waveform***Continued on next page*

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Verify Scan Duration

1. Select **Diagnostic Data Collection (DDC)** > Select **Protocol Name** > **Scan Timer and Load** -or -
Select **Service Desktop** > **Replacement** > **HV Subsystem Change** > **Scan Timer and Load**.
2. Toggle the softkey Monitor ENABLE to ON to display the scan time in the Message Log.
3. Verify/set the following DDC parameters:

STATIC X-RAY ON	Interscan Delay 2.00
1 SECOND	DAS Gain 31
1 SCAN	Gantry Tilt 0.0
FOCAL SPOT LARGE	Trigger Rate 984
100 kV	Filter BLOCKED
40 mA	Calibration Vector NONE

4. At the OBC, connect a scope to the kV board:
Channel 1, **Exposure Command** (EXCM, TP22).
Scope ground to TP3, 2 V/div
Channel 2, **Total kV**, TP11.
Scope ground to SIG, TP12, 1 V/div
Set the Time base to 200 msec, positive trigger.
5. Record the measured scan time from the oscilloscope and the displayed scan time from the message log.

Scan Duration Spec

Scope Exposure Duration = 0.96 to 1.04 s.

Displayed Exposure Duration = 0.99 to 1.02 s.

6. Toggle the softkey Monitor ENABLE to OFF to stop the Scan Time display in the message log. If you don't, the system log will continue to record exposure data.
7. Disconnect the scope from the kV board.
8. Replace the OBC cover.
9. Go to the **HOT ISO Alignment** procedure.



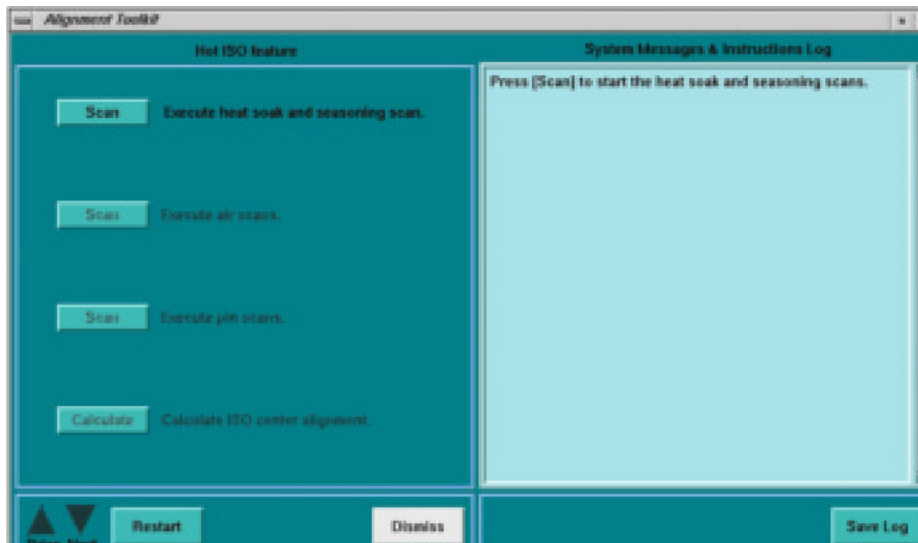


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HOT ISO Alignment

Hot ISO is a software correction calculated from two scan sets after a heat run.

1. Select **Service Desktop** > **Calibration** > **Hot ISO Alignment** -or-
Select **Service Desktop** > **Replacement** > **Hot ISO Alignment**.



5. Place the 1/8 inch screw driver on the Phantom holder and put it into the scan field.
6. Execute the Pin Scan (small spot).
7. Execute the Pin Scan (large spot)
8. Select **Calculate**.
9. Go to the **Data Acquisition System (DAS) Gain Calibration** procedure.

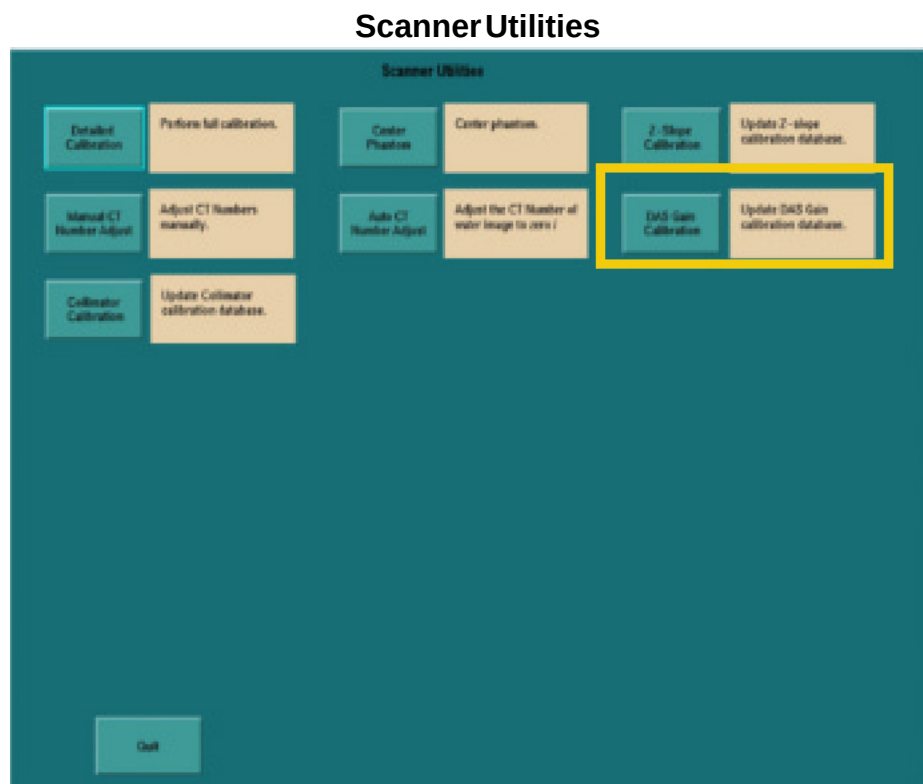
2. Execute the Heat Soak and Seasoning scan.
3. Execute the Air Scan (small spot).
4. Execute the Air Scan (large spot).





Data Acquisition System (DAS) Gain Calibration

1. Select **Scanner Utilities** on left monitor, then **DAS Gain Calibration**.



2. Ensure that there is nothing in the X-ray beam.
3. Select **Continue**.

The system will perform a Mylar window check and provide appropriate messages if the window needs cleaning. Then it will take 31 scans and save the results in the Calibration Data Base. It provides appropriate messages if the DAS Gain calibration should fail.

4. Go to the **Collimator Calibration** procedure.

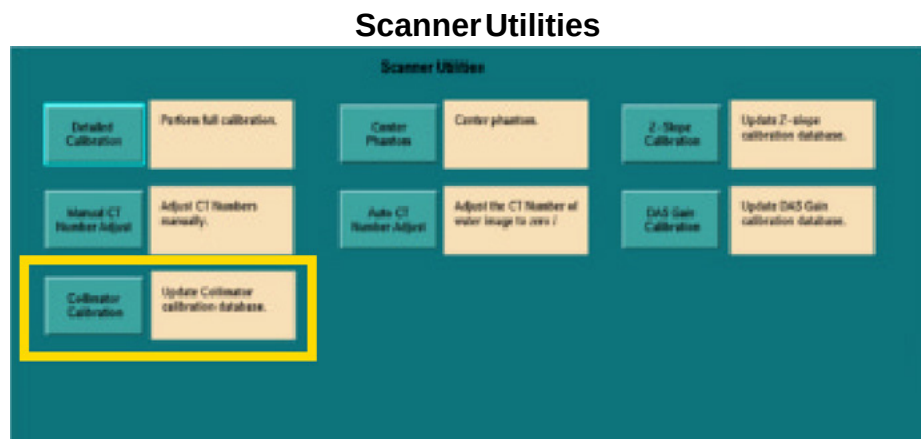




Collimator Calibration

This scanner tracks focal spot motion so the collimator opening can be reduced to restrict dosage. With this tracking, the collimator position is no longer a fixed function of aperture and focal spot size. The two cams, which operate independently, form the sides of the collimator and must move with focal spot motion. Information regarding the focal spot position is sensed through special channels called the Z channels. The Z information is translated into the position of the beam on the detector at the ISO channel. The translation process depends on calibration polynomials and operating points which are determined by the Collimator Calibration.

1. On the left monitor, select **Scanner Utilities > Collimator Calibration**.



2. Ensure there is nothing in the X-ray beam.
3. Select **Continue**.

The system will perform a Mylar window check and provide appropriate messages if the window needs cleaning. The system then takes eight scans and saves the results in the Calibration Data Base. It will provide appropriate messages if the Collimator calibration should fail.

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N Number Adjustment

Note If it has been more than 30 minutes since calibration scans, rerun FASTCAL (it will take approximately 15 minutes). N number adjustment needs the tube to be warm or CT numbers will change.

Run the small and large phantom CT N number adjustment before you acquire the image series, or any time you doubt the N# values. The calibration procedure modifies the pcal (phantom cal) vector to assign water a CT number equal to zero. After a successful N number adjustment, any water phantom scanned against a small or large CAL has a CT number near zero.

1. Place the QA Phantom in the holder.
2. Center the Phantom.
3. From the application desktop, select **Scanner Utilities > Auto CT Number Adjustment**.
4. Follow directions on the screen.
 - a. When centering the Phantom:
 - Centering may be easier to perform from the gantry right side.

- Center on the water portion of the Phantom, not over the internal block.
- b. The program will automatically make adjustments. If the program is unable to make adjustments, it will stop at the technique it is unable to adjust. Troubleshoot with a manual CT # Adjust before continuing.

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X-ray Tube Replacement

For X-ray tube replacement procedures, refer to
Directive 2394243-100 - LightSpeed 1.X-4.X Quick Tube
Change Procedure.



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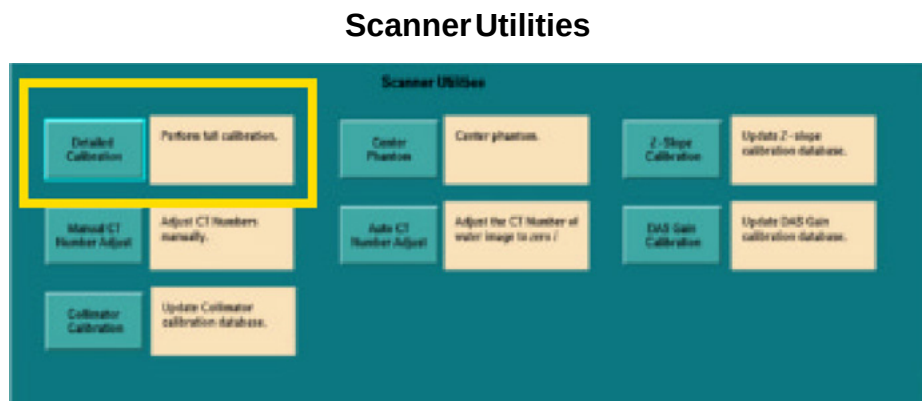
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Detailed Calibration

Calibrating the X-ray tube is a lengthy process. The software defaults select all techniques and aperture settings, and it tells you what to do. The customer has the option to select specific kV's that they use most often. It's preferred to calibrate all kV stations. An aperture choice that is not calibrated cannot be used for scanning.

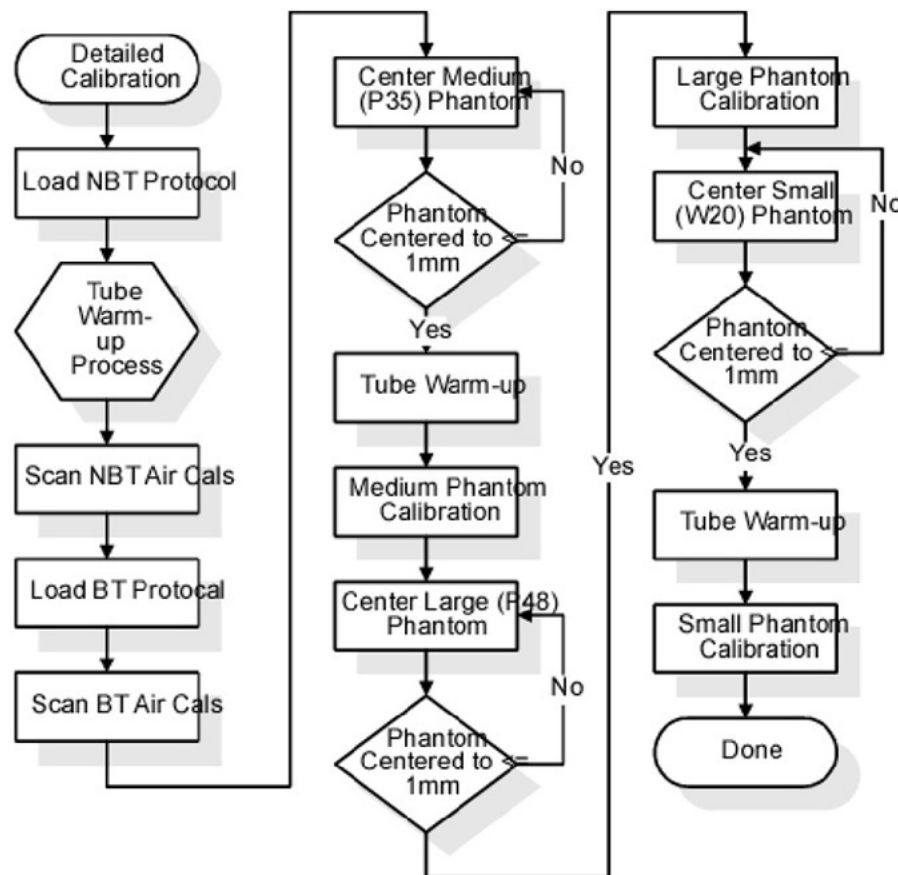
1. On the left monitor, select **Scanner Utilities** > **Detailed Calibration**.



2. Place the correct size Phantom on the Phantom holder of the gantry.
3. Align the Phantom manually using the alignment lights as a guide.

On the small water Phantom, make sure the alignment lights are centered on the water section.

The black markers on the Phantom are centered on the resolution section. The center of the water section is 60 mm (2 3/8 inches) in front of the markers.



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Detailed Calibration, *continued*

4. Select the CONFIRM button to calculate the accuracy of your alignment.
5. If either the X or Y coordinate is off by more than 1 mm, repeat steps 3 and 4 until both values are less than or equal to 1 mm.
6. Once the Phantom is centered, select **Continue**.

Tube temperature affects the results so it is tightly controlled by software. If the Phantom centering process takes more than ten minutes, Detailed Calibrations is aborted. If the tube temperature is not warm enough, software will perform warm up scans.

7. Go to the **N Number Adjustment** procedure.



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- Gantry Components
- Gantry Interior (Front View)
- Gantry Interior (Rear View)
- Gantry Interior - On Board Computer (OBC)
- Gantry Safety
- Gantry Tilt Components
- Emergency Stops (E-Stops)
- Slip Ring Access
- Slip Ring Components
- Stationary Controller (STC) Assembly
- Axial Drive Belt and Tensioner
- Axial Motor Drive (AMD)
- Axial (Rotation) Control Components
- Collision Sensors for Table and Gantry

Table

- Table - Enhanced Table Control (ETC)
- Table Overview (Left Side)
- Table Overview (Right Side)
- Collision Sensors for Table and Gantry

Power Distribution

- Power Distribution Unit (PDU)

Console

- Console
- Scan Control Intercom Module (SCIM)



Alignment &
Calibration



Component
Locations



Error
Codes



Functional
Checks



Logic & Block
Diagrams



Planned
Maintenance



Software &
Diagnostics



Theory



Trouble-
shooting



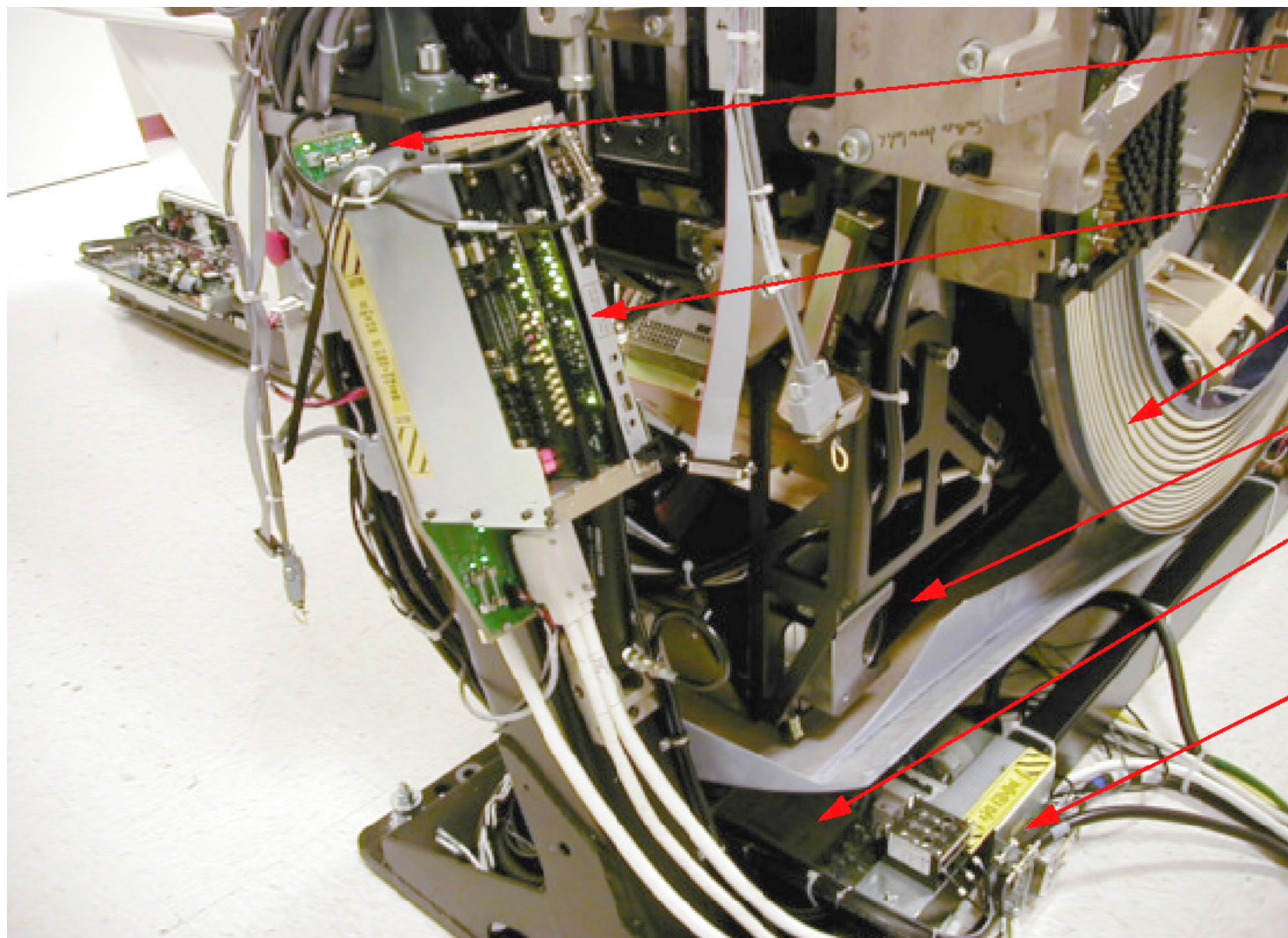
Safety



Notes



Gantry Components



Gantry Power Switch

STC

Slip Rings

Axial Drive Belt

Axial Motor (**AMD** is in
the front)

DAS Data Fiber

Alignment &
CalibrationComponent
LocationsError
CodesFunctional
ChecksLogic & Block
DiagramsPlanned
MaintenanceSoftware &
Diagnostics

Theory

Trouble-
shooting

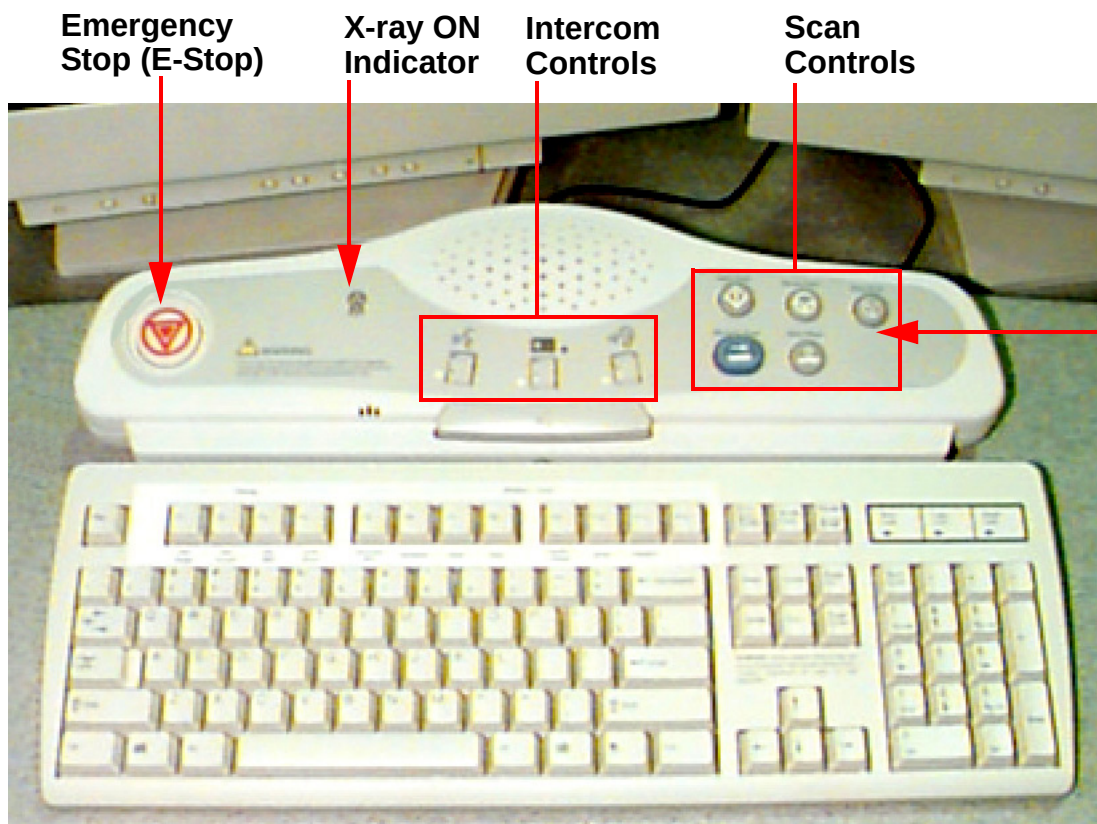
Safety



Notes



Scan Control Intercom Module (SCIM)



Note Some keyboards will have a sixth button, Prescribed Tilt, as shown above. However, this button may be disabled and non-functional.

The new SCIM offers enhanced operator controls from the console.

Refer to the Functional Checks section for How to Use the SCIM Scan Controls.



Alignment & Calibration



Component Locations



Error Codes



Functional Checks



Logic & Block Diagrams



Planned Maintenance



Software & Diagnostics



Theory



Troubleshooting



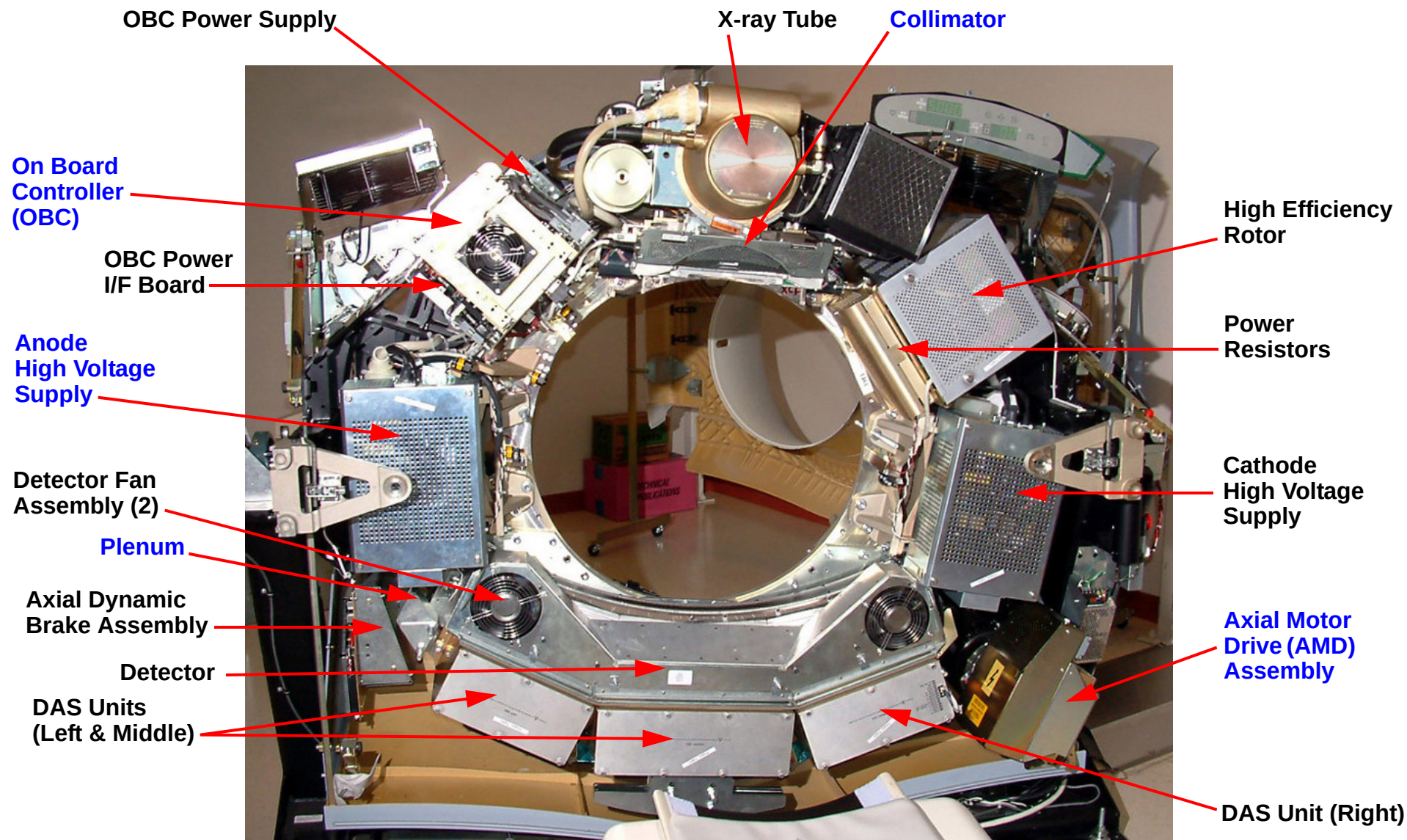
Safety



Notes



Gantry Interior (Front View)

Alignment &
CalibrationComponent
LocationsError
CodesFunctional
ChecksLogic & Block
DiagramsPlanned
MaintenanceSoftware &
Diagnostics

Theory

Trouble-
shooting

Safety

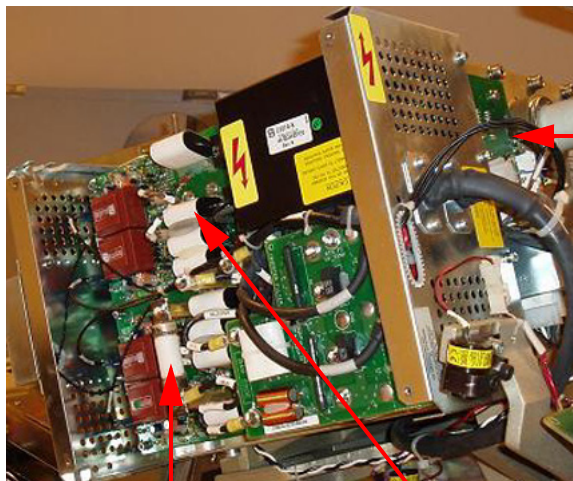


Notes



Gantry Interior (Front View), *continued*

Anode HV Supplyw/Cover Removed

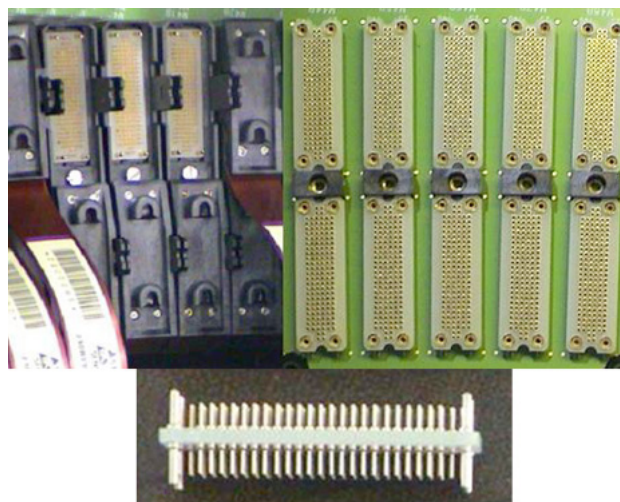


Fuse

Inverter

Measurement Board

DAS

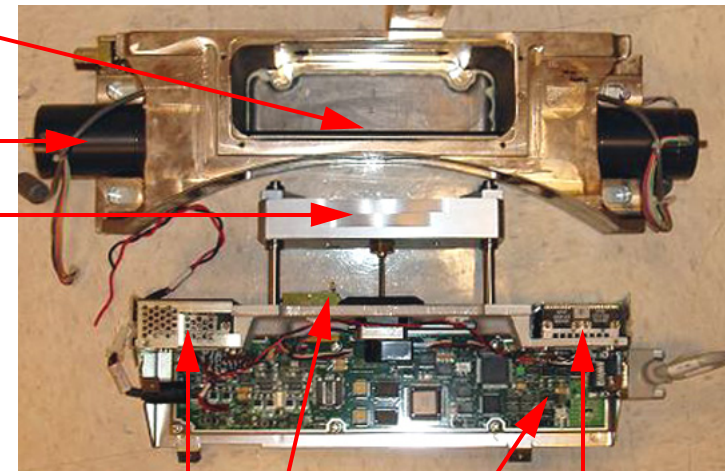


CAMs

CAM Motor (2)

Filter (2)

Collimator



Driver

Home Switch

CCB

Driver



Alignment &
Calibration



Component
Locations



Error
Codes



Functional
Checks



Logic & Block
Diagrams



Planned
Maintenance



Software &
Diagnostics



Theory



Trouble-
shooting



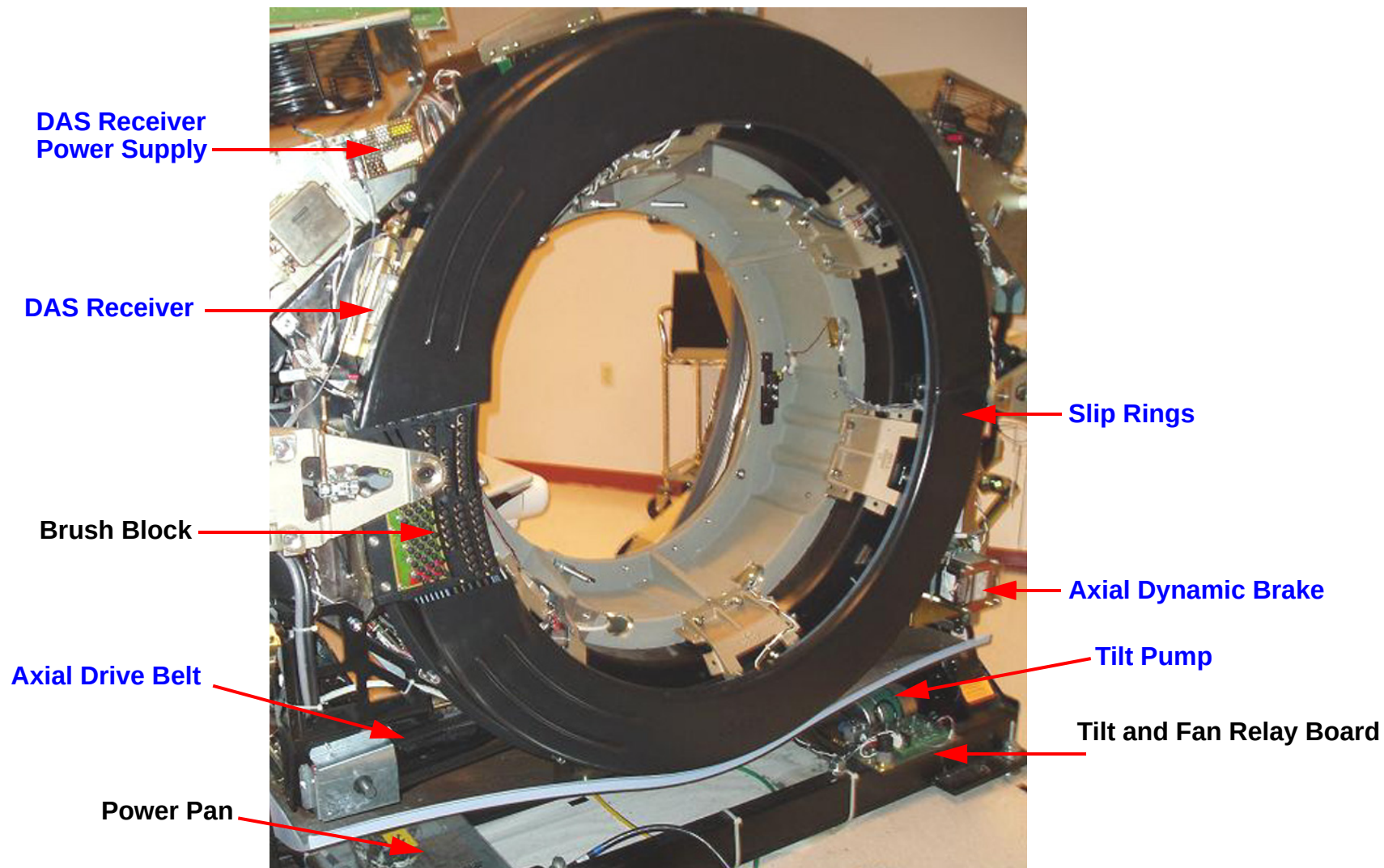
Safety



Notes



Gantry Interior (Rear View)



Alignment &
Calibration



Component
Locations



Error
Codes



Functional
Checks



Logic & Block
Diagrams



Planned
Maintenance



Software &
Diagnostics



Theory



Trouble-
shooting



Safety

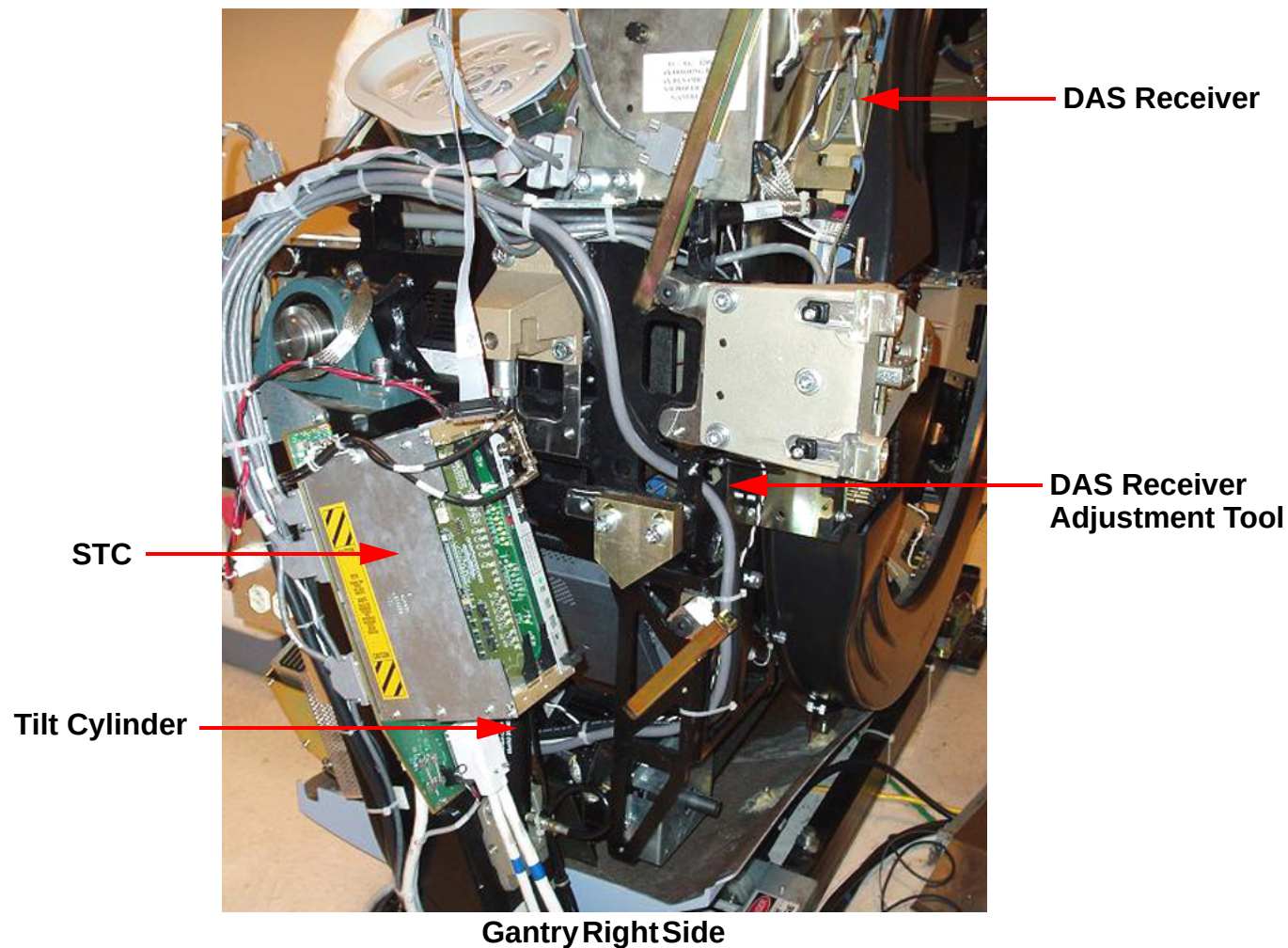


Notes



Component Locations

Gantry Interior (Rear View), *continued*



Alignment &
Calibration



Component
Locations



Error
Codes



Functional
Checks



Logic & Block
Diagrams



Planned
Maintenance



Software &
Diagnostics



Theory



Trouble-
shooting



Safety

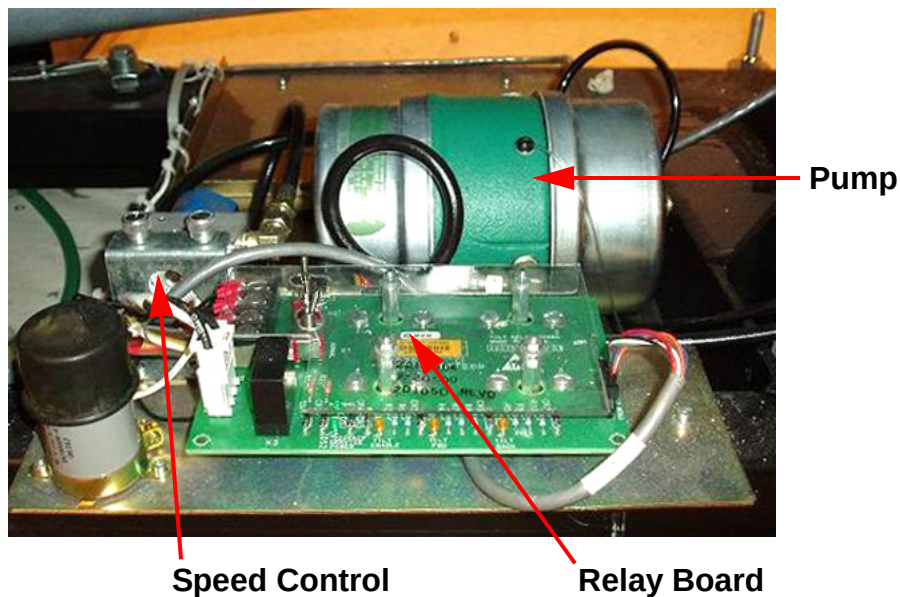


Notes

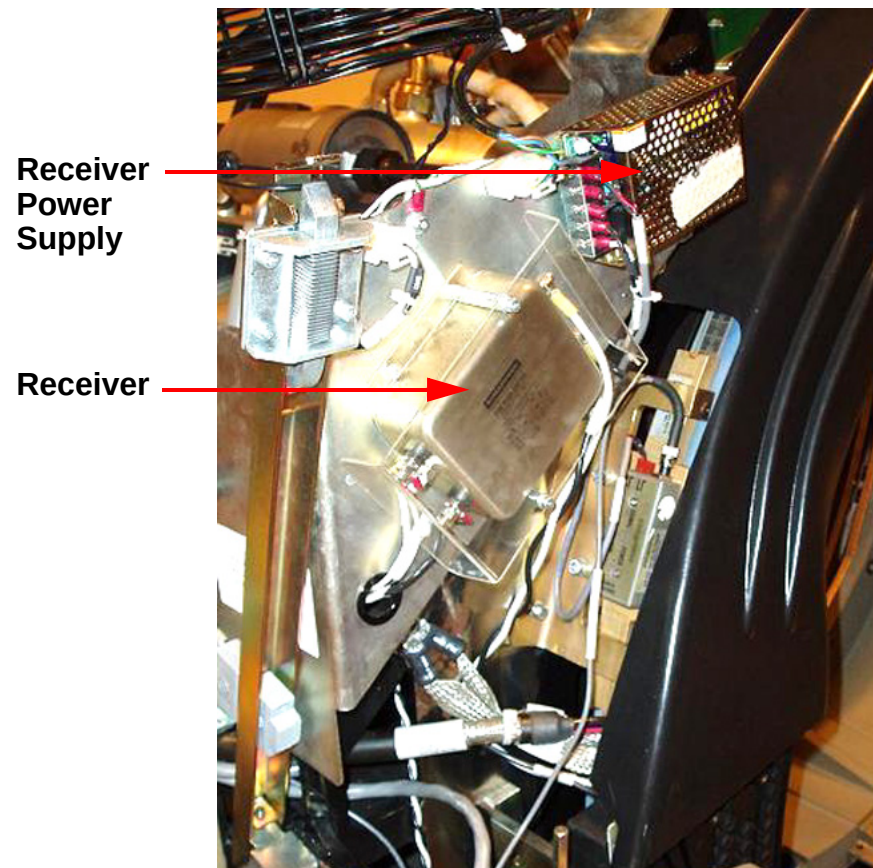


Gantry Interior (Rear View), *continued*

Tilt Pump Components



DAS Receiver Components



Alignment & Calibration



Component Locations



Error Codes



Functional Checks



Logic & Block Diagrams



Planned Maintenance



Software & Diagnostics



Theory



Troubleshooting



Safety



Notes



Emergency Stops (E-Stops)



On Console **SCIM**

Above Gantry Keypad



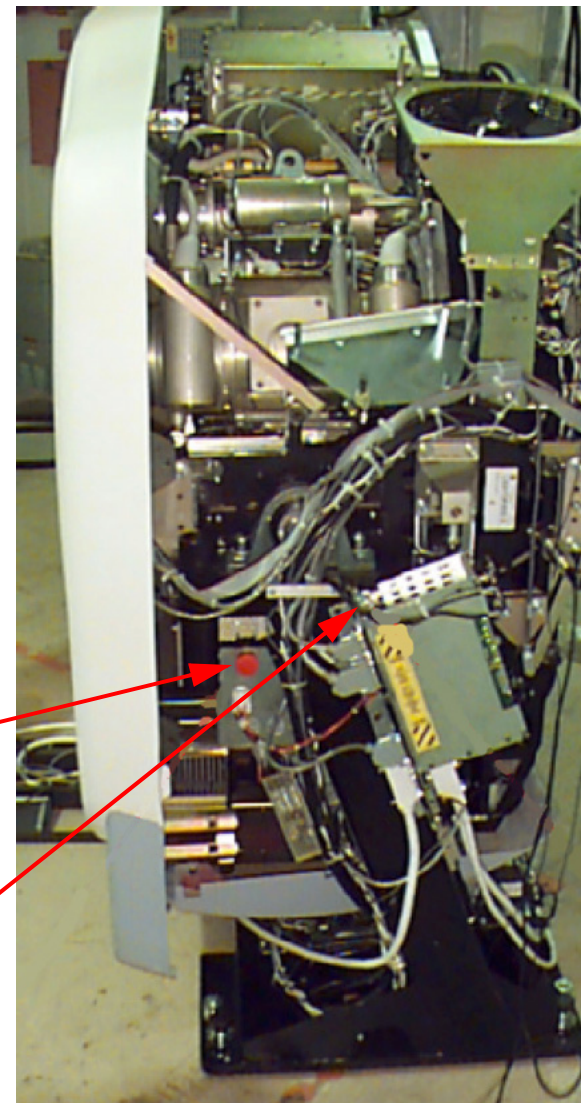
RESET
Button

Reset
LED

Gantry keypads have LEDs to indicate active control. They also have Gantry Start/Stop Scan controls that require informed usage.

On Gantry
Right Side

Gantry Power Control
Switches on STC Backplane



Alignment &
Calibration



Component
Locations



Error
Codes



Functional
Checks



Logic & Block
Diagrams



Planned
Maintenance



Software &
Diagnostics



Theory



Trouble-
shooting



Safety

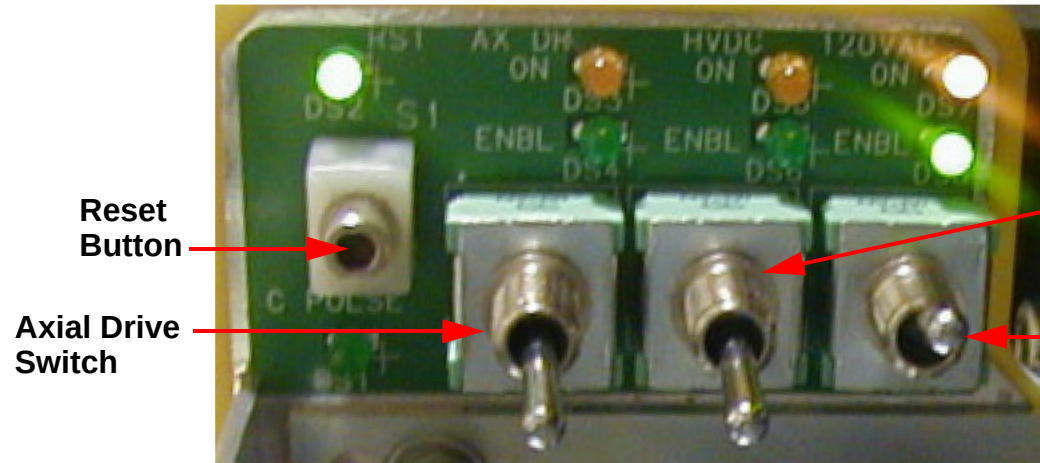
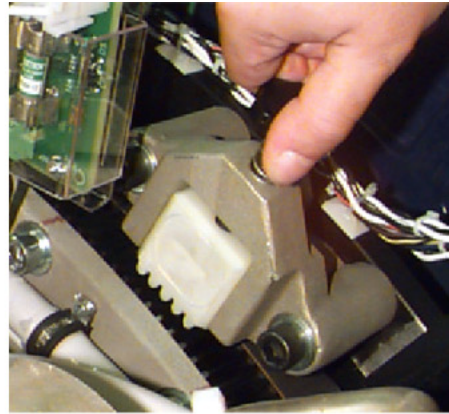


Notes



Gantry Safety

Turn the **Rotational Lock** handle clockwise until the teeth on the lock fully engage the teeth on the rotating assembly. You can rock the rotating assembly slightly if it's necessary to align the teeth. Hand tighten until snug. Do not over tighten. Visually verify that the teeth are engaged.



Rotational
Lock
Location

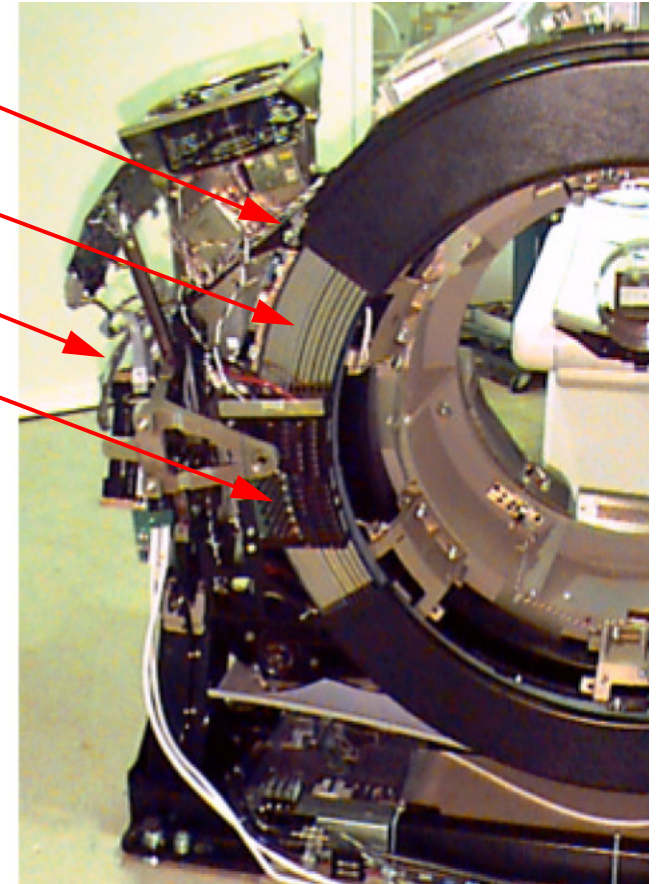
Slip Rings

STC
Switches
Location

Slip Ring
Brush
Assembly

HVDC
Switch

120 VAC
Switch



Rear View of Gantry with Covers Removed

STC Switches control **Gantry Power**. The LEDs indicate Axial Drive, Tube HV DC Power and Rotating Gantry AC Status.

You can also check the **C Pulse** here.



Alignment &
Calibration



Component
Locations



Error
Codes



Functional
Checks



Logic & Block
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Trouble-
shooting



Safety

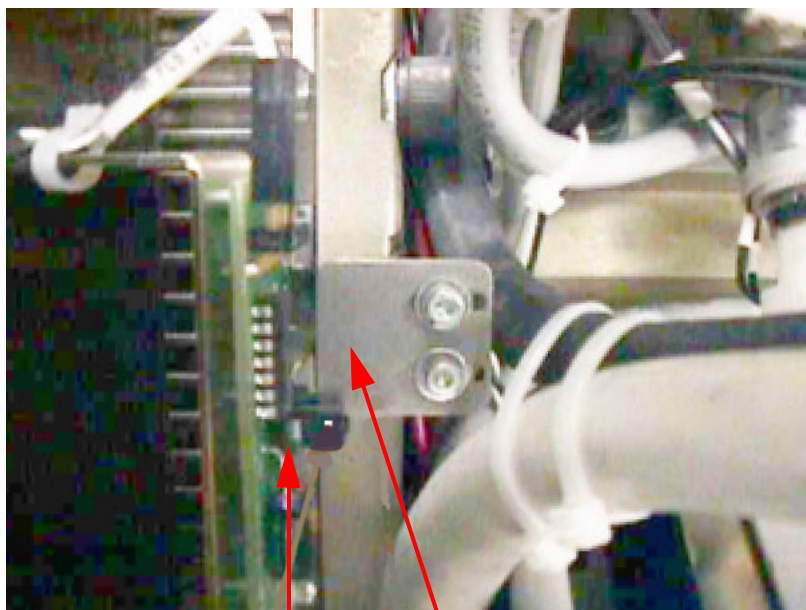
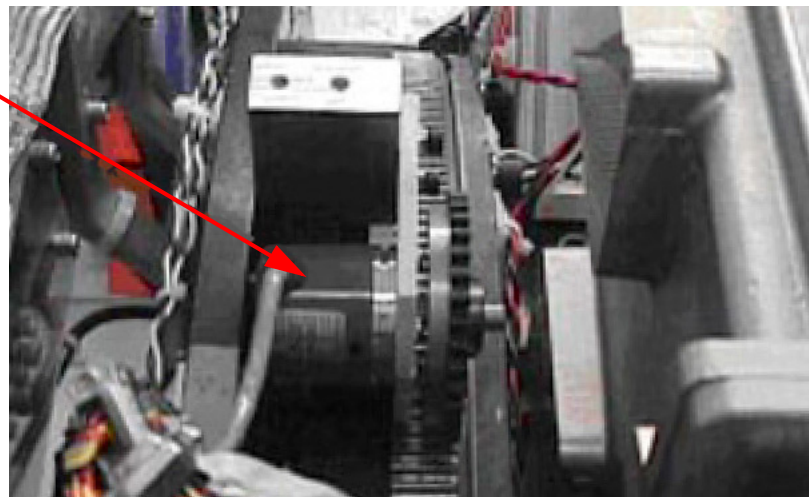


Notes



Axial (Rotation) Control Components

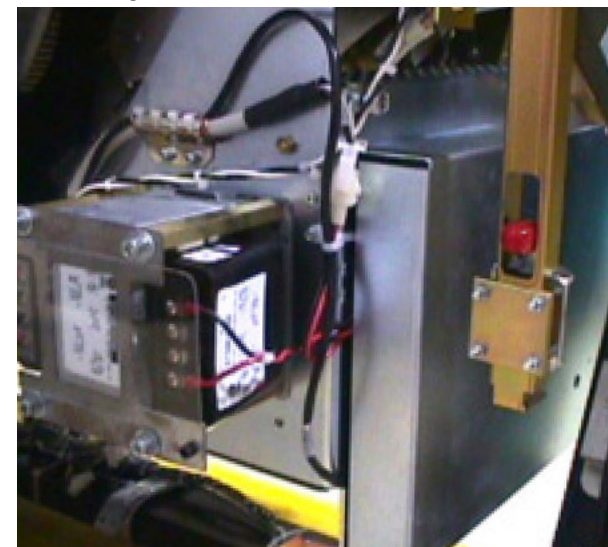
Axial Encoder is on the top, left side



Opto-Sensor

Home Flag

Dynamic Axial Brake Module



Alignment &
Calibration



Component
Locations



Error
Codes



Functional
Checks



Logic & Block
Diagrams



Planned
Maintenance



Software &
Diagnostics



Theory



Trouble-
shooting



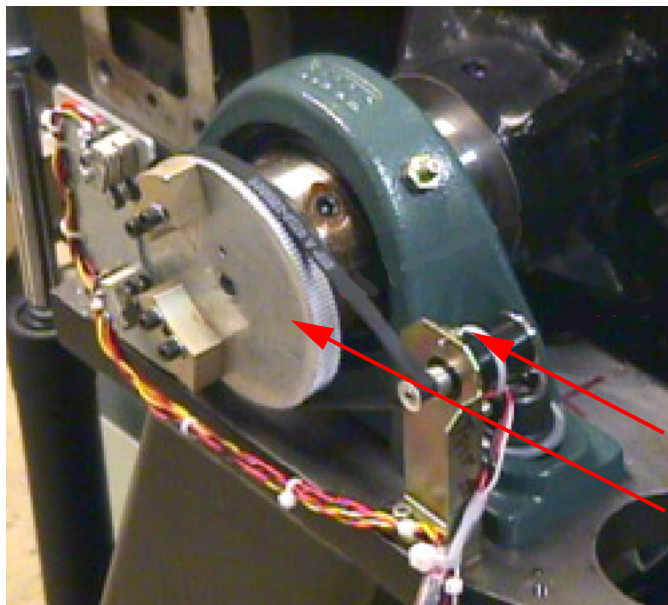
Safety



Notes



Gantry Tilt Components

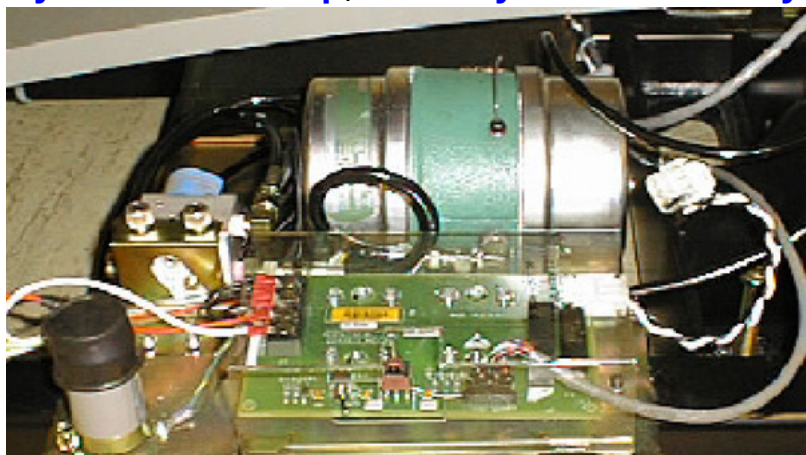
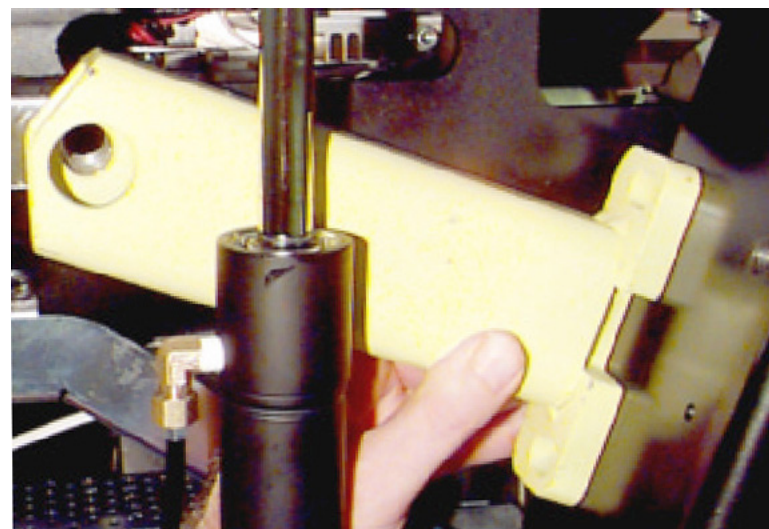
**Caution**

Do not attempt to adjust the 150-tooth pulley. This is factory set. Disturbing this pulley invalidates the Tilt Label values and cause inaccurate application tilt angles. Care should also be used to prevent damaging the legibility of the Tilt Label.

Tilt Pot

150-tooth pulley

Hydraulic Tilt Pump, Tilt Relay Board on Gantry

**Two Tilt Lock Brackets,**
often stored in the top of the PDU**Caution**

Always check and fill the hydraulic tilt assembly with the gantry tilted full backwards. Failure to do this will result in hydraulic fluid spills via the overflow tube when the system is pressurized.

Alignment &
CalibrationComponent
LocationsError
CodesFunctional
ChecksLogic & Block
DiagramsPlanned
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Diagnostics

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Trouble-
shooting

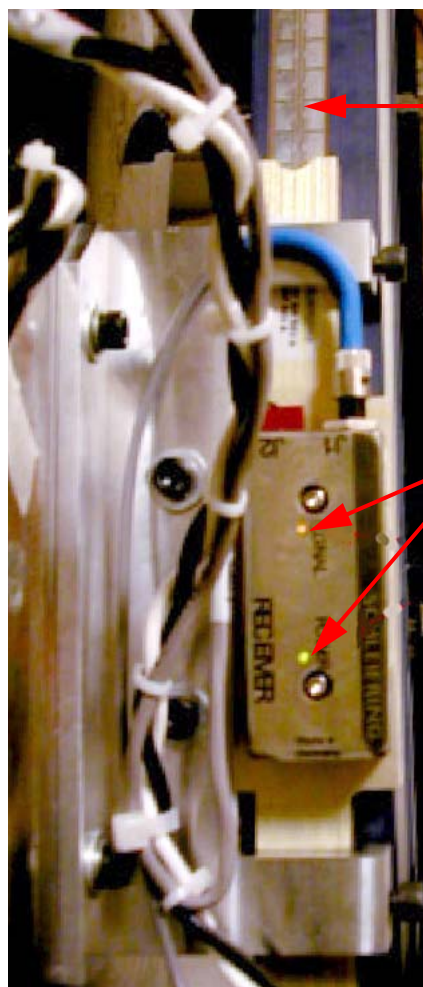
Safety



Notes



Slip Ring Access



View From Right Side

HSDCD Ring

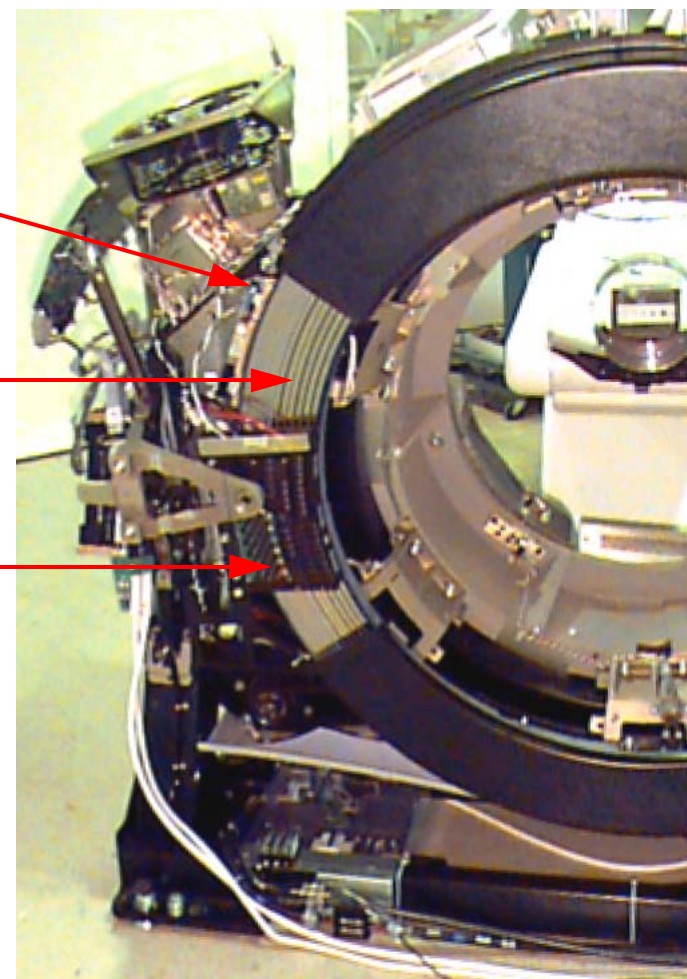
LEDs
on HSDCD Receiver**Caution**

Exposed Slip Rings pose an Electric Shock Hazard! Contact may cause serious injury or death unless High Voltages present on the slip ring are disabled or discharged. Power on the inner slip rings includes 120 VAC and 550 VDC.

Rotational Lock

Slip Rings

Slip Ring Brushes



View From Rear of Gantry

Refer to Slip Ring Communication **Checks**.Alignment &
CalibrationComponent
LocationsError
CodesFunctional
ChecksLogic & Block
DiagramsPlanned
MaintenanceSoftware &
Diagnostics

Theory

Trouble-
shooting

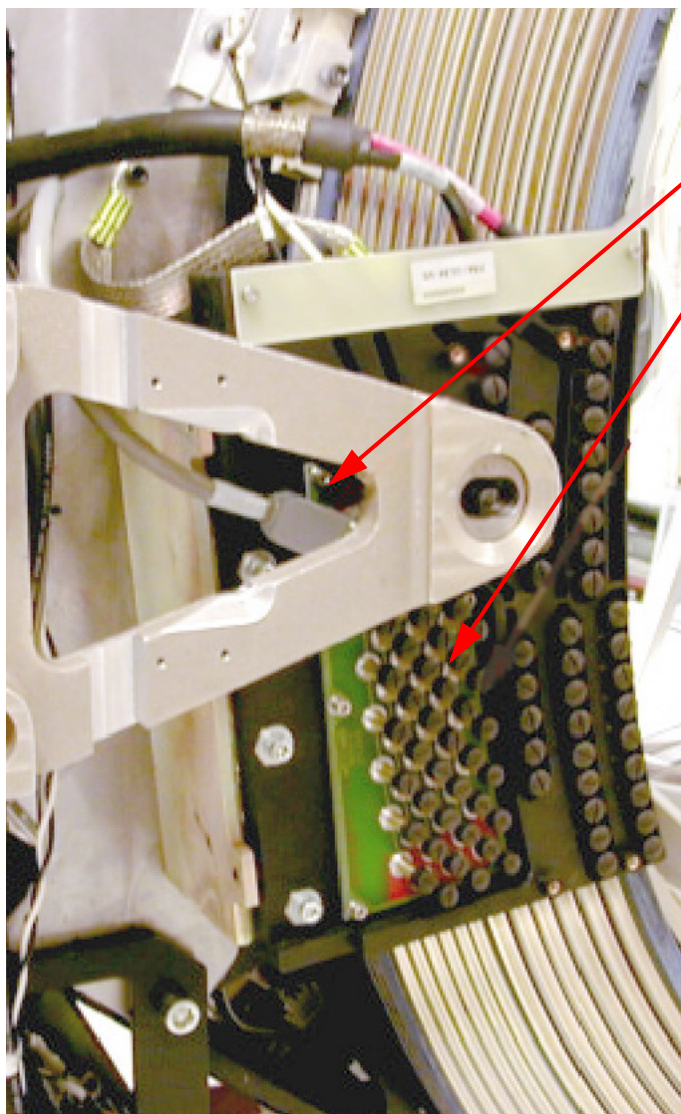
Safety



Notes



Slip Ring Components

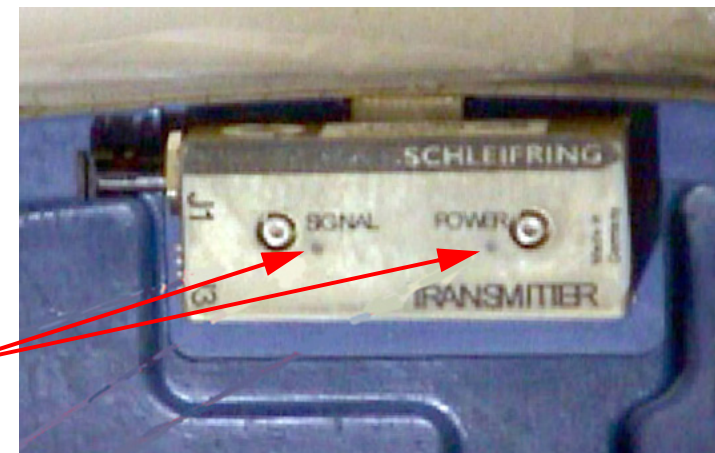


Connect Receiver
Data J1 to DCB J1 to
bypass DAS slip ring

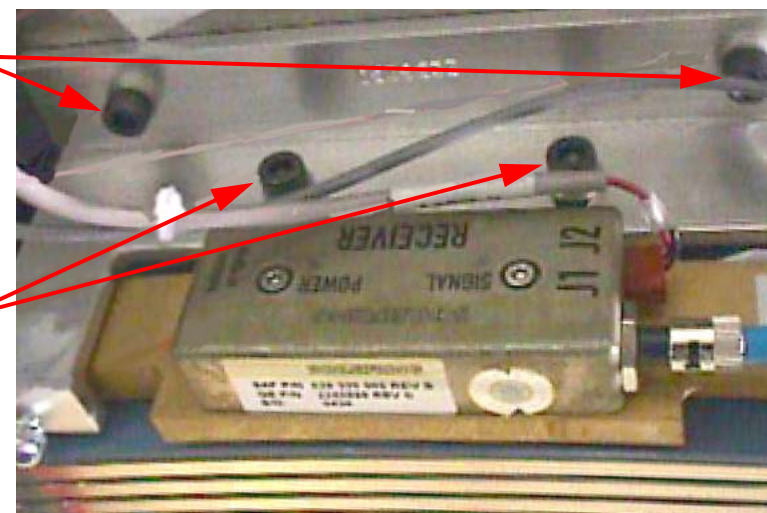
Slip Ring Brushes

LEDs indicate DAS Data
Transfer alive and Power
present

HSDCD Transmitter



HSDCD Receiver



Radial Height Adjust

Axial Center Adjust



Alignment &
Calibration



Component
Locations



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Codes



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Checks



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Maintenance



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Trouble-
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Safety



Notes



Collision Sensors for Table and Gantry

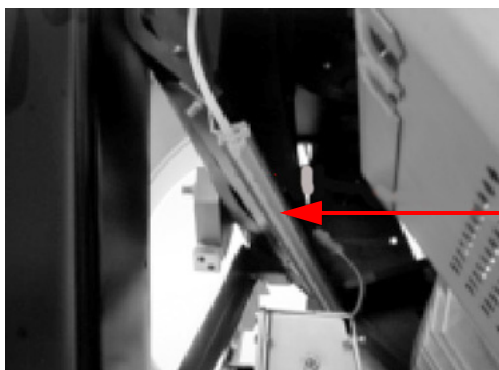
Table



Table Collision Tape Switch

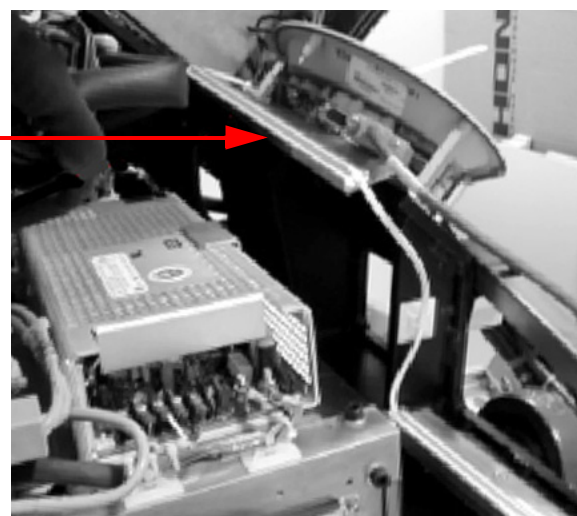
Patient Touch
Sensor of Front
Cover
(there is another on
the Rear Cone)

Gantry



Internal Gantry
Collision on Right
Side (High Side)

Internal Gantry
Collision on Left
Side (Low Side)



Alignment &
Calibration



Component
Locations



Error
Codes



Functional
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Notes



Axial Drive Belt and Tensioner



Refer to [Gantry Components](#) to see a larger, overall location of the belt and tensioner.



Alignment &
Calibration



Component
Locations



Error
Codes



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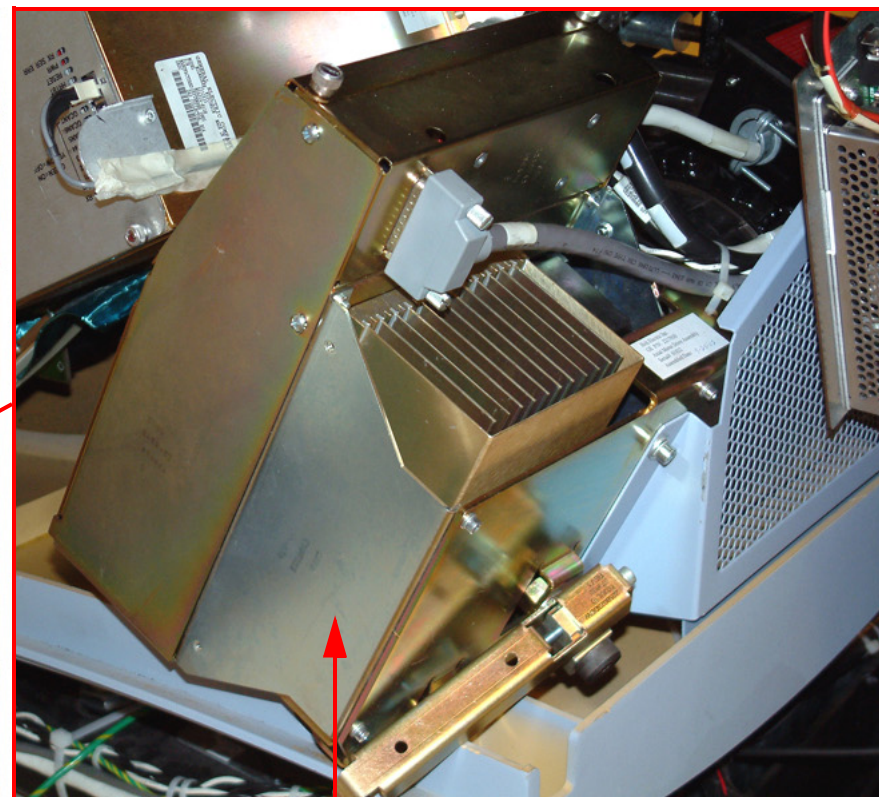
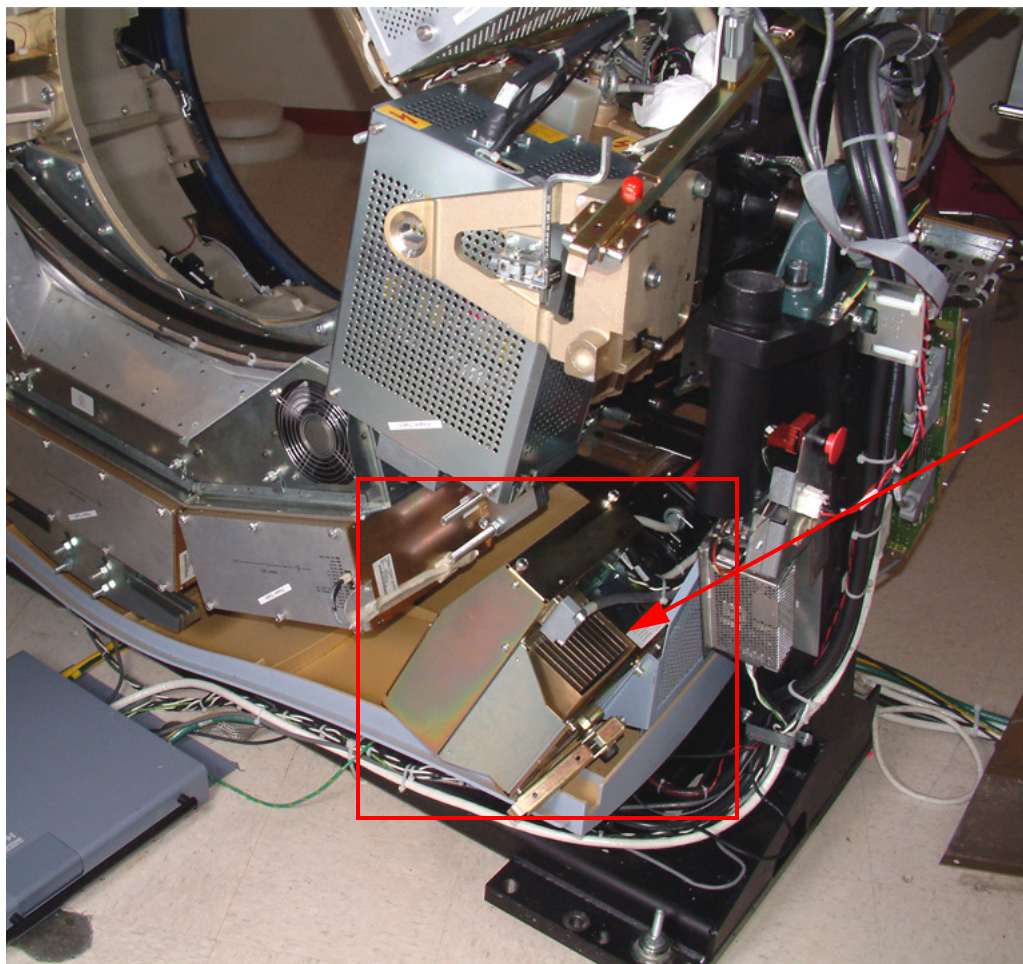
Notes



Axial Motor Drive (AMD)

The Axial Motor Drive (AMD) provides controlled power to the motor behind it. The AMD monitors axial speed

and receives axial encoder feedback via the Axial Control Board (ACB) for fine speed control.



Axial Control Board
(440 VDC Inside)



Alignment &
Calibration



Component
Locations



Error
Codes



Functional
Checks



Logic & Block
Diagrams



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Maintenance



Software &
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Trouble-
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Safety



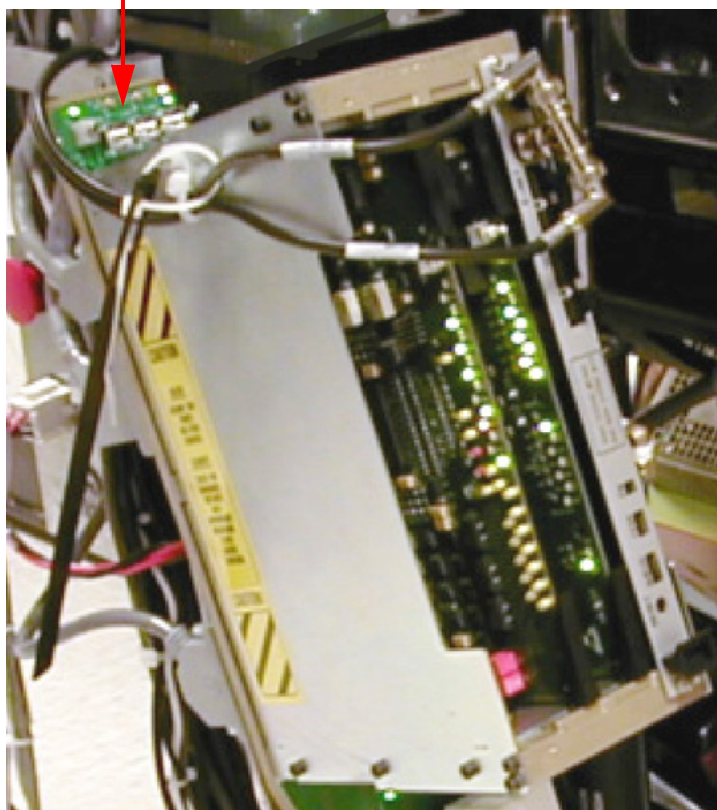
Notes



Stationary Controller (STC) Assembly

Enable Switches

Here is where Gantry Power and Axial Drive are controlled.

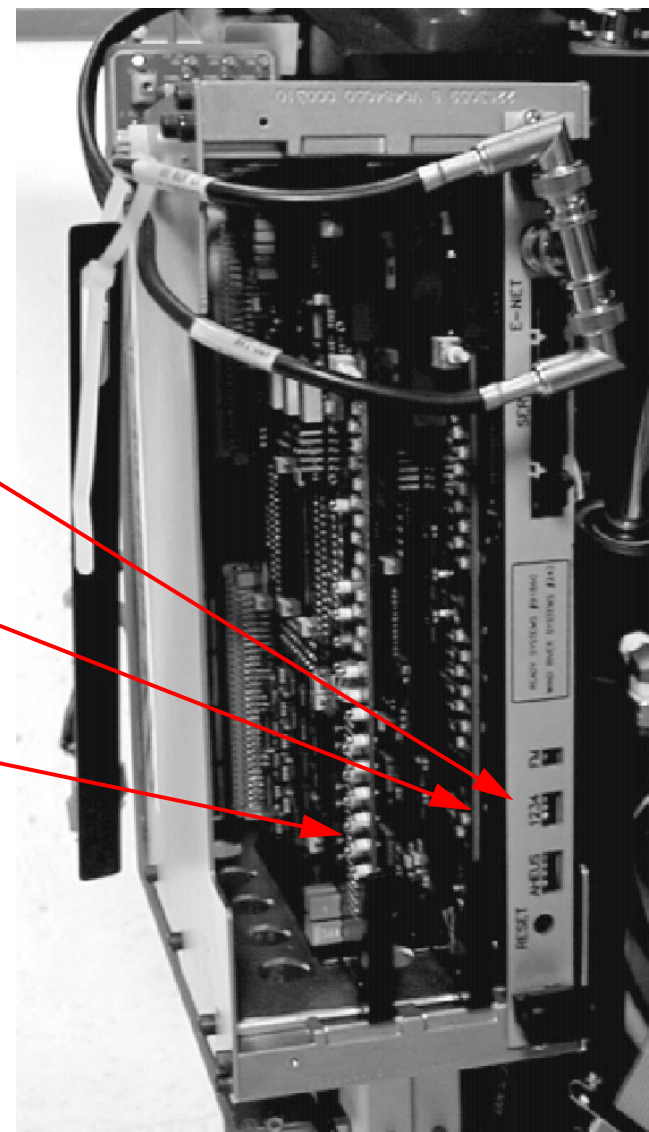


Refer to **STC Power** Checks.

Refer to **STC CPU LEDs** Checks.

Refer to **LSCOM LEDs** Checks.

Refer to **Axial Control Board** Theory.



Alignment &
Calibration



Component
Locations



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Trouble-
shooting



Safety



Notes



Table Overview (Right Side)

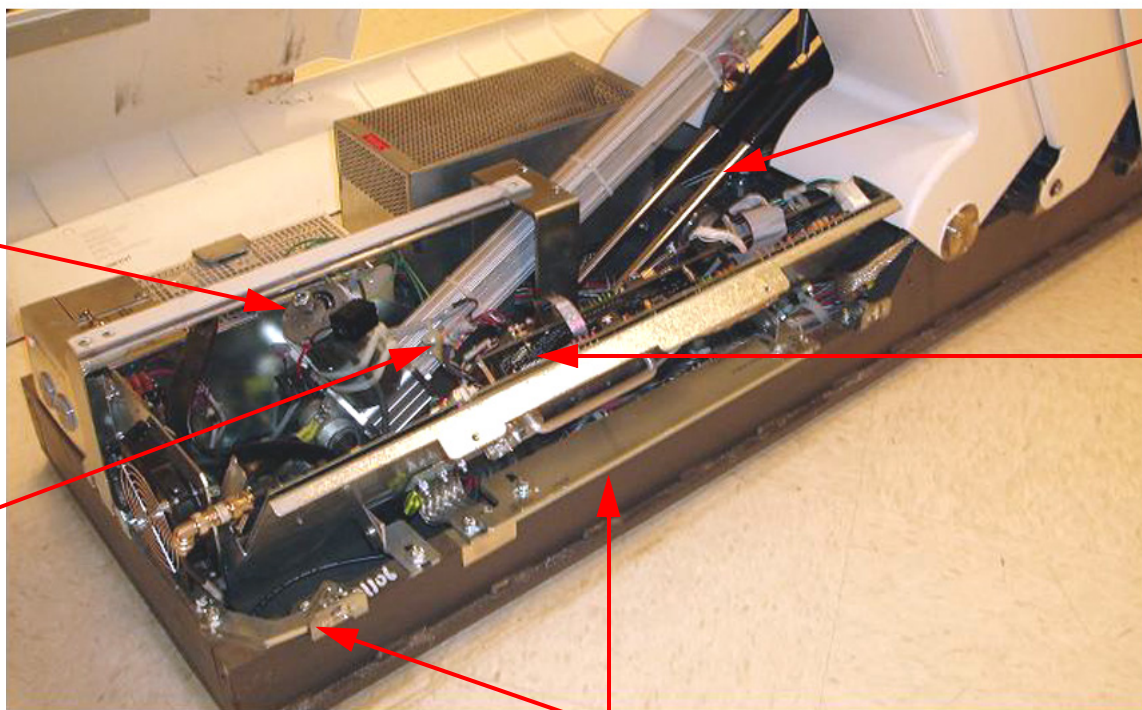
Home Latch
& Switch



Cradle Motor

Jumper Plug

Elevation
Motor



Air Shocks

Enhanced
Table Control
(ETC) with
cover removed

Elevation
Limit Switches

Measurement Plates



Alignment &
Calibration



Component
Locations



Error
Codes



Functional
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Logic & Block
Diagrams



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Trouble-
shooting



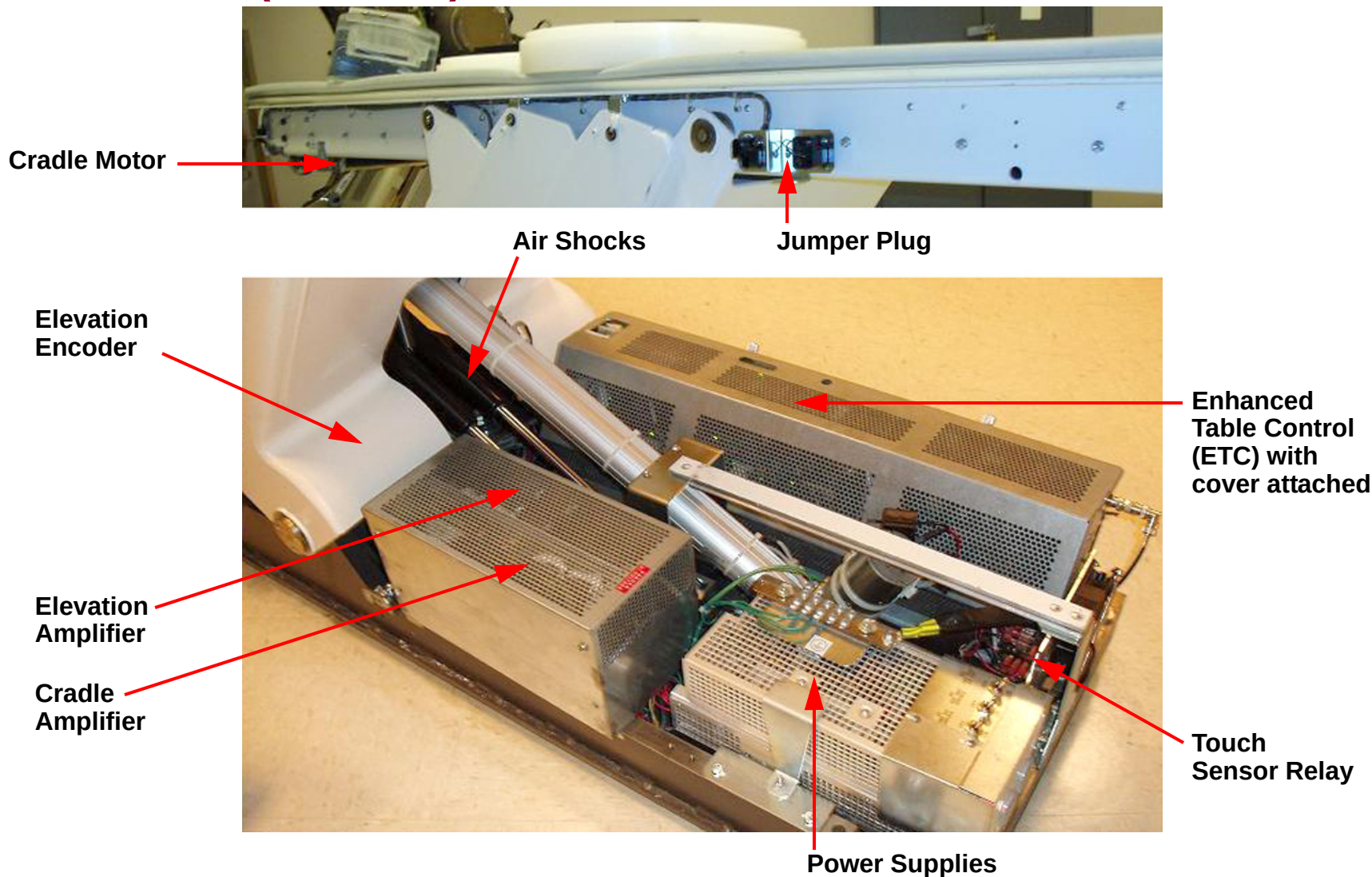
Safety



Notes



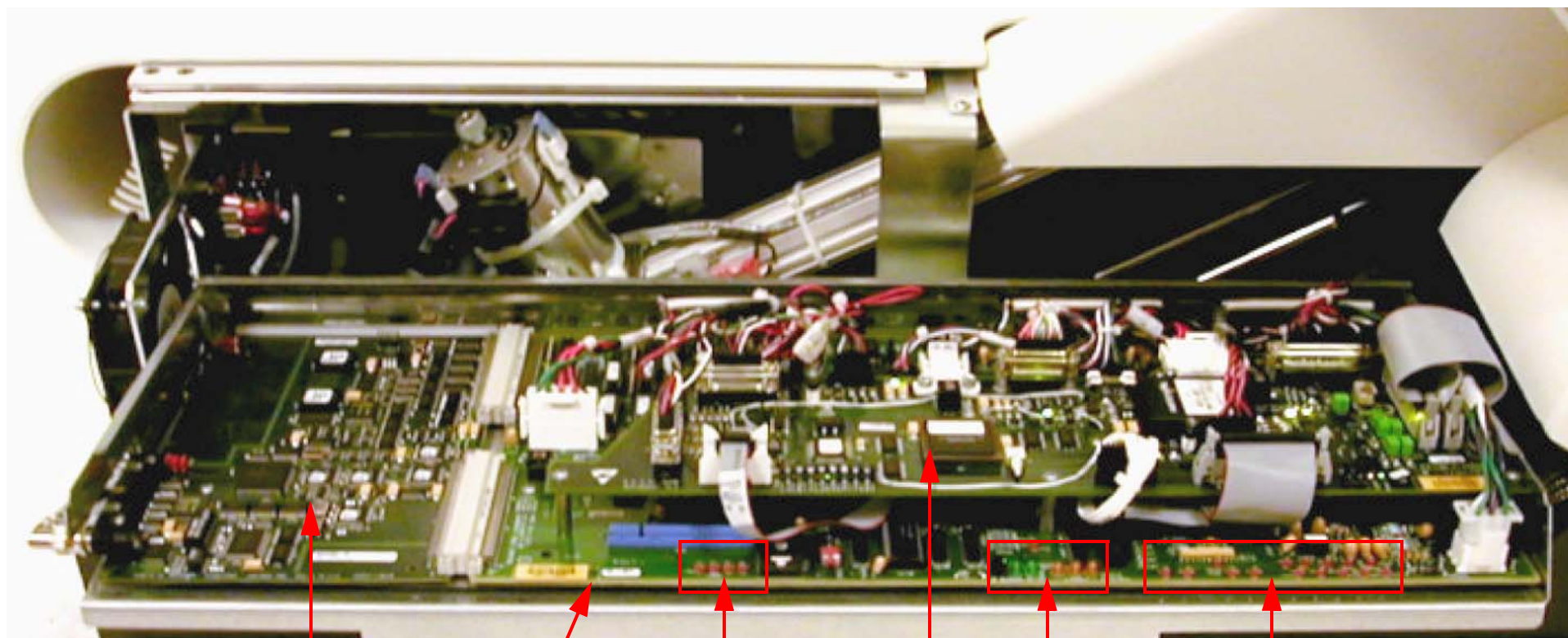
Table Overview (Left Side)





Component Locations

Table - Enhanced Table Control (ETC)



ETC-Artesyn

ETC Board

Test Points 1-5

ETC I/F Board

LEDs

Test Points 9-25
Others test points
are along back edge.Alignment &
CalibrationComponent
LocationsError
CodesFunctional
ChecksLogic & Block
DiagramsPlanned
MaintenanceSoftware &
Diagnostics

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Trouble-
shooting

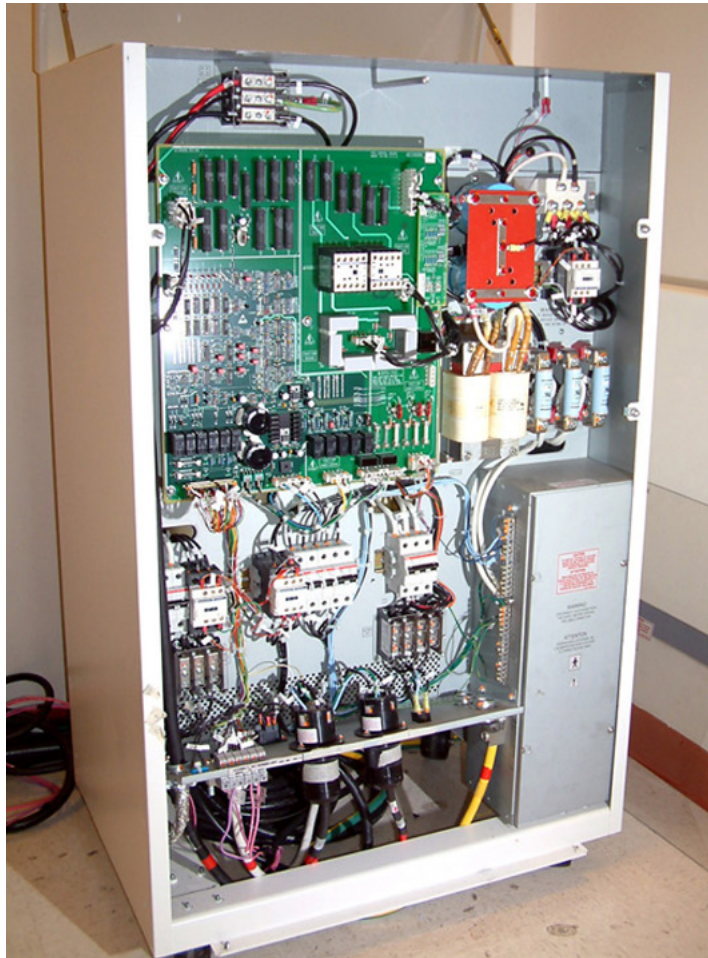
Safety



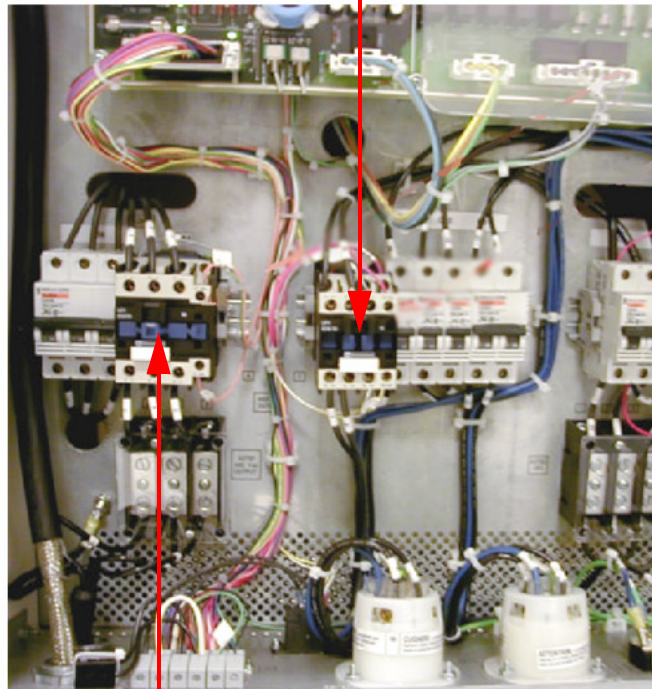
Notes



Power Distribution Unit (PDU)



Gantry/Table 24-hour
Power Relay



Axial Drive
Power Contactor

Power ON Lamp
(when you lift PDU lid)



Switch on Back of PDU
for Gantry 120 VAC



Alignment &
Calibration



Component
Locations



Error
Codes



Functional
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Planned
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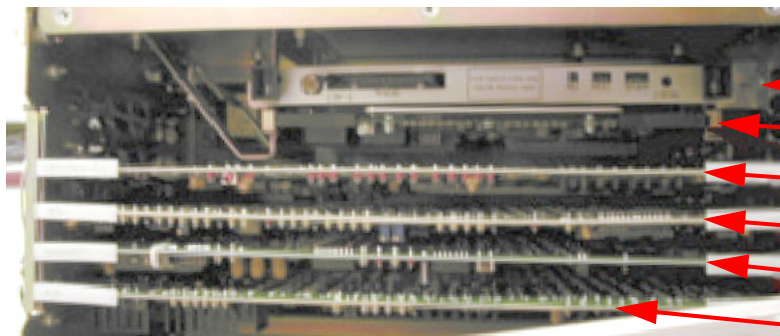


Notes

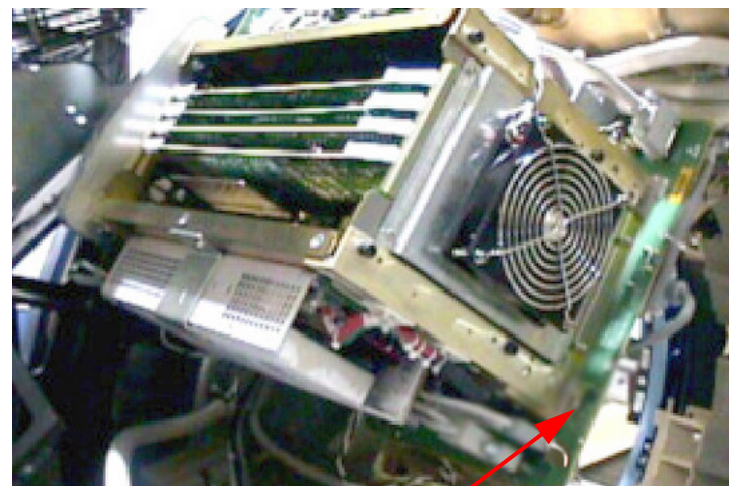
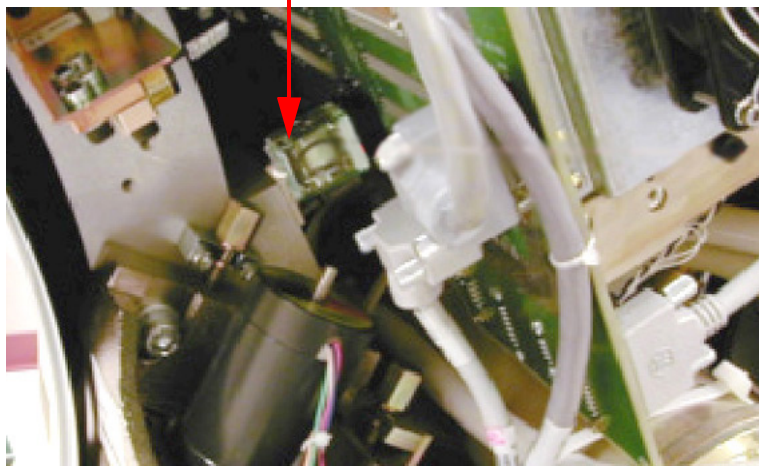


Component Locations

Gantry Interior - On Board Computer (OBC)

On Board Computer (OBC)
(as seen from gantry right side)The **OBC** on the rotating gantry contains:

- Artesyn CPU (half size)
- OBC LS Comm (half size)
- Gantry I/O
- kV Control Board
- HEMRC Control Board
- mA Control Board

OBC Backplane **K78****Laser Switch**
(OBC as seen from gantry front side)Alignment &
CalibrationComponent
LocationsError
CodesFunctional
ChecksLogic & Block
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Trouble-
shooting

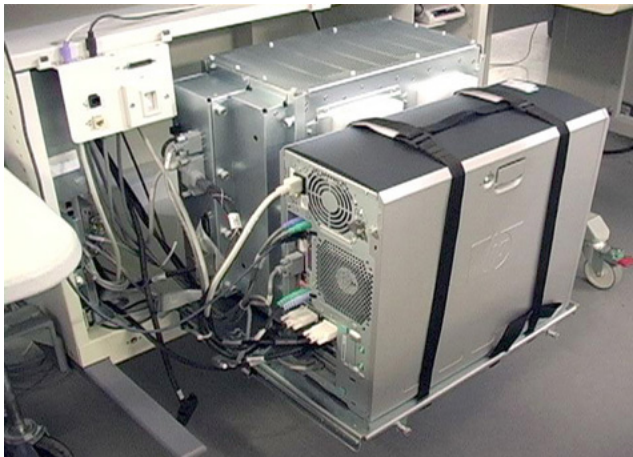
Safety



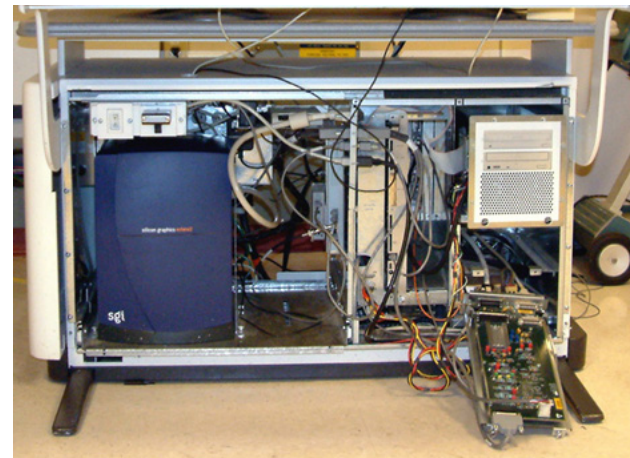
Notes



Console



GOC2



Octane with ICE Box on right



Alignment &
Calibration



Component
Locations



Error
Codes



Functional
Checks



Logic & Block
Diagrams



Planned
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Software &
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Trouble-
shooting



Safety



Notes



ERROR CODES

Overview of Error Codes and Logs

System Logs

Accessing Calibration Reports

X-ray Generation Errors

- Tube Current (mA Loop)

- Tube Filament

- X-ray Collimation and Filtration

- Rotor Control and DC BUS

Testing System Communication

- LAN Error Detection

- DIP Statistics

Gantry Rotation Errors



Alignment &
Calibration



Component
Locations



Error
Codes



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Logic & Block
Diagrams



Planned
Maintenance



Software &
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Trouble-
shooting



Safety



Notes



Overview of Error Codes and Logs

The Lightspeed system provides different types of error logs and codes that are useful for troubleshooting. The primary errors/logs that are useful include:

- [System Logs](#)
- [Calibration Reports](#)
- [System Communication](#)
- [X-ray Generation Errors](#)
- [Gantry Rotation Errors](#)



Alignment &
Calibration



Component
Locations



Error
Codes



Functional
Checks



Logic & Block
Diagrams



Planned
Maintenance



Software &
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Trouble-
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Notes



System Logs

The System Browser allows you to view a majority of error logs for the Lightspeed system. These include:

- GE System Log
- SYSLOG
- IOS Logs
- Tube Usage
- Run Time Statistics
- OC Information
- Configuration Files
- Software Healthpage
- Scan Usage
- VxWorks Logs
- Service Notepad
- Session Log

GE System Log

The GE System Log presents errors generated by failing application processes. These take about a minute to appear on the screen.

Interpreting Messages

Error messages allow you to:

- Interpret system errors
- Determine which tests to perform when troubleshooting a subsystem problem
- Identify and access the appropriate documentation and functional tests that are available for each major function.

There are three steps in interpreting each GE System Log entry:

1. Identify the function and use
2. Access the service desktop to view it
3. Assess its function (major or minor) to better understand its error log

Continued on next page



Alignment &
Calibration



Component
Locations



Error
Codes



Functional
Checks



Logic & Block
Diagrams



Planned
Maintenance



Software &
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Trouble-
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Notes



System Logs, *continued*

GE System Log, *continued*

Interpreting Messages, *continued*

Example: What would happen if the gantry stopped rotating during a helical scan?

The following errors would typically be reported:

- The **STC** reports the gantry stopped rotating
- The **ETC** reports the table stopped moving prematurely
- The **OBC** reports X-ray was turned off prematurely
- The **ICE/DARC** reports no incoming DAS data because the scan was aborted
- The **OC** reports no image was received for the image database and display.

The Lightspeed unit is equipped with error handling software that presorts errors and assigns a severity level to each log entry.



Alignment &
Calibration



Component
Locations



Error
Codes



Functional
Checks



Logic & Block
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Maintenance



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Trouble-
shooting



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Notes

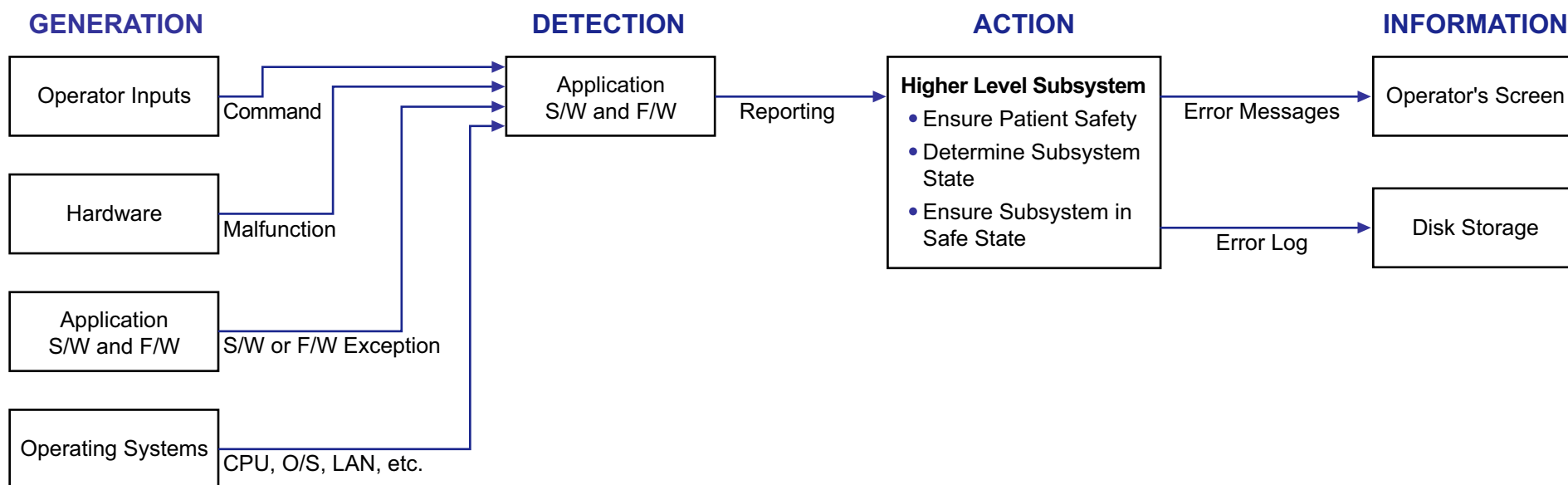


System Logs, *continued*

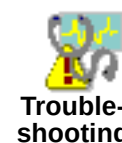
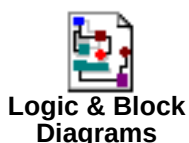
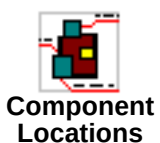
GE System Log, *continued*

Interpreting Messages, *continued*

This illustration shows how the system is designed to report errors and exceptions.



Continued on next page





System Logs, continued

GE System Log, continued

Message Log Layout

Date **Time**

Suite: Suite name **Host:** Host name **Proc:** Process name **Error:** Error number

File: File name **Method:** Method name **Line#:** Location ID (this row N/A for service view level)

Function: Major : Minor function

Scan/Image Type: Scan/image type **Scan:** Scan specifics **Image:** Image specifics

Exception Level: Level flag **Ticks:** ticks (In firmware only) **Log Series:** #####

Services lines: Specific and detailed information concerning the error including (but not limited to) additional problem description and cause information. Information indicating what was supposed to happen, what did happen, how it was detected (example: expected value vs. actual value, name of file attempted to open, etc.) including any information of abnormal conditions related to the error. Network input/output information. Company que information.

Error Log Field descriptions are summarized below:

- **Date** and **Time** – Date and time that the error message entered the error message log, **not the time the error occurred**.
- **Suite** – Suite name.
- **Host** – Host processor name (OC, STC, ETC, OBC) of the firmware that detected the error.
- **Proc** – Process name that detected the error.

- **Error** – Unique error identification. Number can be used for sorting, searching, and counting specific errors in error log. Can also be used as a pointer in documentation.
- **File, Method, Line #** – Does not appear when using the service view level.
- **Function** – Hierarchical breakdown of system functions uses major and minor functions.
- **Scan/Image Type** – Scan type, that is, axial, scout, helical, cine, diagnostic.
- **Scan** – Exam, series, scan # for raw file.
- **Image** – Exam, series, image # for image file.

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GE System Log, *continued*

Message Log Layout, *continued*

- **Exception Level:**
 - **Primary:** first error reported in a group of errors.
 - **Secondary:** rest of the errors in that group.
 - **Most Severe:** message that has the most severe scanner exception handling consequences in the group of errors.
- **Soft:** errors that have no effect on exception handling, but indicate system parameters that are beginning to approach the boundaries of satisfactory performance
- **Ticks** – Ticks will indicate real error occurrence order (clocks are synchronized).
- **Log Series** – relates all errors recorded as part of a verification series.
- **Message Text Body** - Specific and detailed information about the error including (but not limited to):
 - Additional problem description and cause information
 - Information indicating what was supposed to happen, what did happen, how was it detected (examples: expected value vs. actual value, name of file attempted to open, etc.) including any information concerning abnormal conditions related to the error
 - Network input/output information
 - Compute que information

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SYSLOG

Displays error messages stored in **/var/adm/SYSLOG**. These messages consist of the OC host bootup output and error messages that indicate bootup problems including HOST hardware failures and warnings.

IOS Logs

IOS Logfiles can be found under the following directories:

For UNIX – **/usr/g/ctuser/logfiles**

For LINUX – **/var/log/messages**

You can also access logfiles directly through the System Browser menu from the Common Service Desktop. Note, that each logfile size may be different from those of your workstation.

Use the **more** command to read the entire file content page-by-page or the **tail** command to see the most recent messages logged (or **tail -n**, with *n* being the number of lines you want to read).

Example:

more netslog [Return] OR
tail -n browserlog [Return] (last n lines)

Logfiles are periodically "cleaned", which means they are compressed when they exceed a certain size.

To read a file, uncompress it first and then read it using the **more** or **tail** commands. The "More" command allows the entire file to be viewed. The "Tail" command allows the last couple of lines of a file to be viewed.

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IOS Logs, continued

Application software logs for:

- Image Browser (browserlog)
- Image Database Read Server (dbrlog)
- Image Database Write Server (dbwlog)
- Image Server (imslog)
- DICOM Server (dsclog)
- Networking Server (netlog)
- Film Composer Log (fclog)
- Printer Server (prslog)
- Archive (arslog)
- Display
- Filming

```
## process: arsllog
## date of creation: Fri Sep 15 08:32:39 2005
## hospital name: BUC_ENG
## hostname: SdC
## operating system: Linux
## operating system release: 2.4.18-10smp
## operating system version: #1 SMP <Date>
## machine hardware: i686
#####
##date of process start: <Date>
#####
```

Example:

The first lines with ## signs describe the system from which the logfile is extracted, and the date and time of process start.

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IOS Logs, *continued*

Note The most commonly used log files are denoted with a "*".

arslog*

Information about the archive server log.

browserlog

Information about the browser, such as selecting a patient and opening the view, or receiving images and also returns of requests to servers.

dbrlog: Database Read server

Information about DATABASE in read mode.

dbwlog: Database Write server

Information about DATABASE in write mode.

dcsllog: Dicom Server log*

Information about the Dicom Server.

imslog: Image Server Log*

Information related to the IMAGE Server (image file management on disks).

netlog: Network activity Log*

Information related to the Network Server.

ppslog

Information about printer parameters.

prslog: Print Server Log

Information related to the Print Server.

terra log

Information about Image Database. Valid only for GOCIII and up consoles.



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Tube Usage

Shows the X-ray tube's model and serial numbers, its meter reading, and install date.

Run Time Statistics

These statistics allow you to view data such as DAS Interface Processor (**DIP**), **CAN** (HCAN, GCAN), **gantry temperature**, tube spits, rotor uptime, DIP statistics, gantry revolutions, and RCOM_SCOM.

OC Info

System Software Revisions (**showprods**)

Disk Usage (**df**)

OC Network Sockets (**netstat -ian**)

OC Route Table (**netstat -r**)

OC Network Configuration (**ifconfig**)

OC Current Processes (**ps -aef**)

OC Hardware Inventory (**hinv**)

ICE Box Log

Config Files

OC Host Configuration File, RIP Host Configuration File,
OC Scan Hardware Configuration File, RIP Scan
Hardware Configuration File.

Software Healthpage or Session Log

Shows when calibration, image analysis, diagnostic tools, collimator tests, and DAS tools were used. It helps you track what's been done. You can access this log the following ways:

- From the Common Service Desktop (CSD) select **Errorlogs** and then **SoftwareHistory log**.
- From the Common Service Desktop select **PM**, **Session log**, then **History**.

Scan Usage

Reports the parameters of each scan.

VxWorks Logs (UNIX only)

Logs errors found by **Common Service Desktop**, **continued** for the Motorola Power PC during bootup or diagnostics.

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Service Notepad

Use this to create a message for GE InSite and other Field Engineers. It will appear on the Health page.

Session Log

Enables you to enter or read FE comments, read InSite comments, tool and diagnostic start messages, and what images InSite has reviewed. The intent of Session Log is for Service Engineers to communicate their activities. It also records whenever the system is re-booted or reconfigured.



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Accessing Calibration Reports

There are two ways to access the Calibration (CAL) Report:

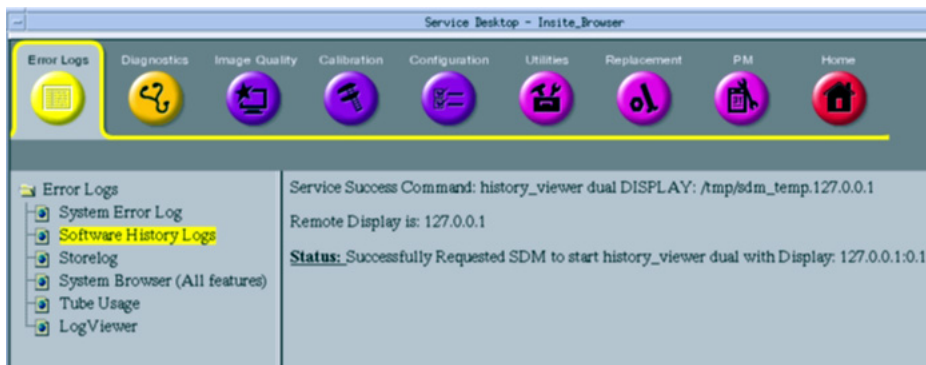
- UNIX shell (also used with remote connection)
- Common Service Desktop

When using a UNIX shell, do the following:

1. From prompt, type **/usr/g/service/state** and then press the **Enter** key.
2. At the next prompt, type **more ssw.calreport.hist** and then press the **Enter** key.

When using the Common Service Desktop, do the following:

1. Select **Error Logs** and then **Software History Logs**.



2. A history log list menu screen will appear. Select **calreport** and press the **LOAD** button.



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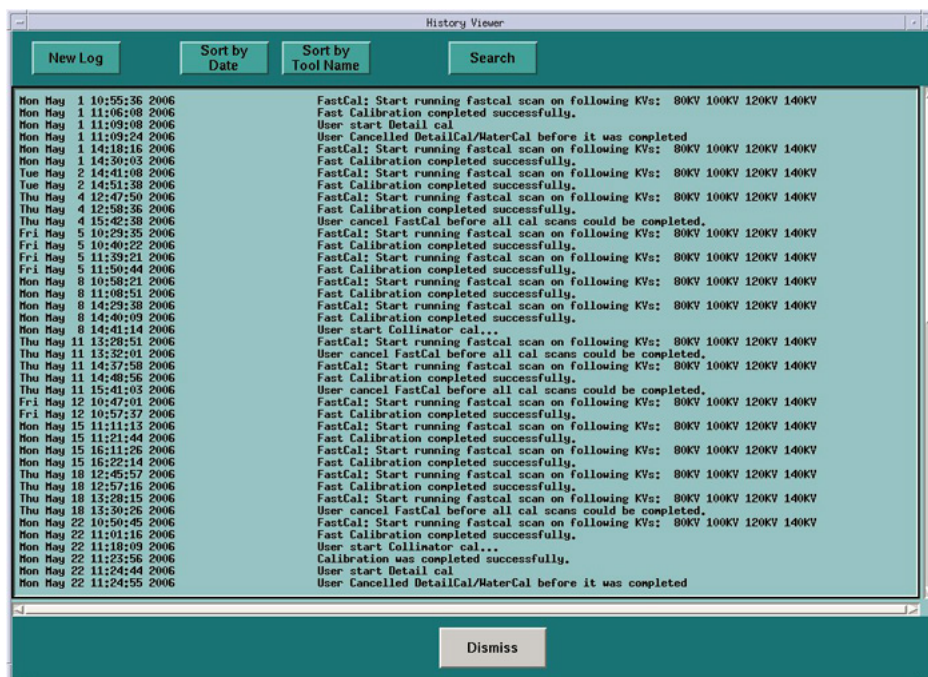
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Accessing Calibration Reports, *continued*

3. A CAL report history log screen will appear. Use the buttons above the list for search options. When you are finished, press the DISMISS button to return to the History Log List screen.



4. Press the DISMISS button on the History Log list screen to return to the Service Desktop directory.



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X-ray Generation Errors

When you exit **Generator Characterization**, this test may generate this false error message:

kV board tube spit counter = x error message

Tube Current (mA Loop)

Most times mA out of tolerance is a secondary error. Tube spits, kV overcurrents, kV shoot-throughs, large kV errors, or Open_Filament errors will cause mA out of tolerance errors.

Tube Filament

Determine whether the error is with the small or large filament only or with both. Large or small filament signal split after the OBC back panel.

Every time someone removes the cathode HV cable from the tube well, a **Filament Open Error** gets logged. This is not a problem.

X-ray Collimation and Filtration

The Filter failed to move to the requested position.

This is not a primary error. Look in the System Log for other Application or CAM errors.X

Rotor Control and DC BUS

Verify error in log is the primary error; note all pertinent details. Troubleshoot HVDC errors.



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Testing System Communication

LAN Error Detection

For Models with a Single Board Computer (SBC):

Before CTi

Enter the following commands at the SBC:

rsh sbc

become root

ping -s obcr 16 1000

Where:

ping = command

-s = statistics

obcr = host you are trying to ping

16 = packet size (ALWAYS use the number 16)

1000 = the number of times the command will run before stopping.

To stop the command, type **<CTRL> C**

Since CTi

Enter the following command at the SBC:

ping <host>

The command will run continuously until you type

<CTRL> C

Note The HSA system times will be longer than on a CTi, the CTi will be longer than Lightspeed and Octane (UNIX) will be longer than Linux (HP).

Testing Time Results

- Packet loss is not allowed: repair.
- If maximum time exceeds average time x 4 + 1ms: not looking good.
- If maximum time exceeds average time x 6 + 1ms: repair.

Tests can be run while scanning and reconstructing. Ignore any times during scan hardware reset; they are not valid times.

Troubleshooting Results

- Always test stationary first. If it passes, then electronics is fine.
- After testing stationary, next test using patient scan mode or via Diagnostic Data Collection (DDC). Scan 20 axials with 20 second ISD for 400 seconds of total test time. If it only fails rotate, then clean the slip rings and test OBCR after the cleaning.

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LAN Error Detection, *continued*

Ping Results

- If all hosts ping fine, the cable from OC to STC or Gantry is fine.
- If ICE pings fine but all gantry room pings are poor or bad, suspect the LAN switch, LAN power supply and/or transceiver.
- If all hosts ping 20ms, perform a Load from Cold (LFC) software reload.
- If you suspect dirty slip rings, always test the OBCR **before and after** the rings are cleaned.

Ping Result Examples

Note, that if you <ping ice> on a Global Recon Engine (GRE), it will ping the Data Acquisition and Recon Computer (DARC).

From a 04MW13.7.H2_P_M16_G:

--darc ping statistics---

472 packets transmitted, 472 received, 0% loss, time 470999ms
rtt min/avg/max/mdev = 0.048/0.192/0.649/0.080 ms
[root@lsct1]#

--ETC ping statistics---

494 packets transmitted, 494 received, 0% loss, time 493981ms

rtt min/avg/max/mdev = 2.484/3.893/9.057/1.551 ms
[root@lsct1]#

--OBCR ping statistics---

497 packets transmitted, 497 received, 0% loss, time 497022ms
rtt min/avg/max/mdev = 7.145/8.352/14.159/0.747 ms
[root@lsct1]#

From a 10.5_2.8.2I_H1.3M4 QXI:

--ct01_ice PING statistics---

223 packets transmitted, 223 packets received, 0%packet loss
round-trip min/avg/max = 0.810/1.143/1.460 ms
ct01_sbc 3#

--ETC PING statistics---

468 packets transmitted, 468 packets received, 0%packet loss
round-trip min/avg/max = 2.842/3.760/6.583 ms
ct01_sbc 2#

--OBCR PING statistics---

443 packets transmitted, 443 packets received, 0%packet loss
round-trip min/avg/max = 7.140/8.667/19.905 ms
ct01_sbc 2#

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Testing System Communication, *continued*

DIP Statistics

The **DIP Statistics** selection under the Service Menu reports **DAS data communication factors** such as how many: offset bytes were transmitted, taxi violations occurred, successful Forward Error Corrections (FEC), and scan aborts (unsuccessful FEC) since file creation (or reset).

Refer to **DIP Vendor Diagnostics**.



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Gantry Rotation Errors

Axial Drive error messages are posted as **260006500** in the "syslog". In the body of the error log entry, the fault code will be posted in the format of FXX.

Fault	Description
F01	Undefined Fault. This fault code is undefined and may indicate a defective Drive. Retry operation. If problems persist contact the factory for instructions.
F02	Auxiliary Fault. The interlock between the Chopper Control circuit on the Axial I/F Board and the Drive is open. Possible Chopper Control fault or connections to the I/F board. Also check the fuse on the chopper resistor pan.
F03	Power Loss Fault. The Drive internal DC bus remained low for >500 ms. Possible low voltage condition on 480 VAC in gantry or power interruption. Also may indicate excessive run or braking power required due to sluggish Motor.
F04	Undervoltage Fault. The Drive internal DC bus voltage dropped below 325 V. Possible low voltage condition on 480 VAC in gantry or power interruption. Also, may indicate excessive run or braking power required due to sluggish Motor.
F05	Overvoltage Fault. The Drive internal DC bus voltage has exceeded 810 V. Possible failure of axial I/F Board Chopper Control or excessive motor regeneration from AXMotor during braking.

Fault	Description
F06	Motor Stall Fault. The Drive output current has exceeded 12.6 A for >4 seconds. Possible Motor frozen bearing or shorted Motor winding. Defective Motor and/or cabling.
F07	Overload Fault. The Drive output current has exceeded 9.7 A for an extended time. Possible Motor sticky bearing or shorted Motor winding. Also, possible defective Motor and/or cabling.
F08	Overtemp Fault. The Drive heatsink temperature has exceeded 90°C (195°F). Check for blocked or dirty heat sink fins. Also check if the gantry ambient temperature has exceeded 40°C (104°F).
F09	Open Pot Fault. Potentiometer speed control is not used in this system. This fault code indicates a possible corrupted configuration parameter, defective Drive or Control Board. Retry operation.
F10	Serial Fault. This fault code indicates a possible corrupted configuration parameter, defective Drive or Control Board. Retry Operation.

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Fault	Description
F11	Op Error Fault. This fault code indicates a possible corrupted configuration parameter, defective Drive or Control Board. Retry Operation.
F12	Overcurrent Fault. Check for a short circuit at the drive output or excessive load conditions at the motor.
F13	Ground Fault. Check the motor and external wiring to the drive output terminals for a grounded condition.
F14	Option Error. This fault code is undefined and may indicate a defective Drive. Retry operation. If problems persist contact the factory for instructions.
F15	Motor Thermistor. System detected and open or short in the Motor Thermistor. Retry. Check connections to thermistor and motor heat.
F16	Bipolar directional fault detected by the axial drive.
F17	C167 Watchdog. Internal Drive Fault. Perform Reset of the Drive. If Retry is frequent, the Drives main control board is suspect; replace the Drive.
F18	Hardware Trap. Internal Drive Fault. Perform Reset of the Drive. If Retry is frequent, the Drives main control board is suspect; replace the Drive.
F19	Precharge Fault. This fault code indicates a possible hardware failure in the Drive. Retry operation. If problems persist, replace the Drive.

Fault	Description
F20	The load loss detect is set to enabled and the drive output torque current was below the load loss level for a time period greater then the load loss time.
F21	Undefined Fault. This fault code is undefined and may indicate a defective Drive. Retry operation. If problems persist contact the factory for instructions.
F22	DSP Reset Fault. Power up has occurred with an open Stop_AX2 or closed Start_AX2* signal. Check Control Board and wiring between Drive and STC.
F23	Loop Overrun Fault. This fault code indicates a possible hardware failure in the Drive. Retry operation. If problems persist, replace the Drive.
F24	Motor Mode Fault. This fault code indicates a possible hardware failure in the Drive. Retry operation. If problems persist, replace the Drive.

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Fault	Description
F25	Undefined Fault. This fault code is undefined and may indicate a defective Drive. Retry operation. If problems persist, contact the factory for instructions.
F26	Power Mode Fault. This fault code indicates a possible hardware failure in the Drive. Retry operation. If problems persist, replace the Drive.
F27	DSP Comm Fault. Internal Drive Fault. Perform Reset of the Drive. If Retry is frequent, the Drives main control board is suspect; replace the Drive.
F28	DSP Timeout Fault. Internal Drive Fault. Perform Reset of the Drive. If Retry is frequent, the Drives main control board is suspect; replace the Drive.
F29	Hertz Error Fault. This fault code indicates an operating frequency parameter was out of range. Possible corrupted configuration parameter, defective Drive or Control Board. Retry Operation.
F30	Hertz Select Fault. This fault code indicates an operating frequency parameter was out of range. Possible corrupted configuration parameter, defective Drive or Control Board. Retry Operation.
F31	DSP Queue Fault. Internal Drive Fault. Perform Reset of the Drive. If Retry is frequent, the Drives main control board is suspect; replace the Drive.

Fault	Description
F32	EEprom Fault. This fault code indicates a possible hardware failure in the Drive. Retry operation. If problems persist, replace the Drive.
F33	Max Retries Fault. The Drive unsuccessfully tried to reset a fault. See message(s) above for original problem.
F34	Prm Access Flt. Verify that [Run Boost] parameter is less than or equal to [Start Boost] parameter.
F35	Negative Slope Fault. This fault code indicates a Volts/ Hertz programming error. Possible corrupted configuration, defective Drive or Control Board. Retry.
F36	Diag C Lim Fault. Check programming of [Cur Lim Trip En] parameter. Check for excess load, improper DC boost setting, DC brake volts set too high or other causes of excess current.
F37	P Jump Error Fault.

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Fault	Description
F38	Phase U Fault. A phase to ground short has been detected in the U phase. Check wiring between the drive and Motor. Check Motor for grounded primary winding.
F39	Phase V Fault. A phase to ground short has been detected in the V phase. Check wiring between the drive and Motor. Check Motor for grounded primary winding.
F40	Phase W Fault. A phase to ground short has been detected in the W phase. Check wiring between the drive and Motor. Check Motor for grounded primary winding.
F41	UV Short Fault. A phase to phase short has been detected between the U & V phases. Check wiring between drive and Motor. Check Motor for shorted primary.
F42	UW Short Fault. A phase to phase short has been detected between the U & W phases. Check the wiring between the drive and Motor. Check Motor for shorted primary.
F43	VW Short Fault. A phase to phase short has been detected between the V & W phases. Check the wiring between the drive and Motor. Check Motor for shorted primary.

Fault	Description
F44	Undefined Fault. This fault code is undefined and may indicate a defective Drive. Retry operation. If problems persist, contact the factory for instructions.
F45	Undefined Fault. This fault code is undefined and may indicate a defective Drive. Retry operation. If problems persist, contact the factory for instructions.
F46	Power Test Fault. This fault code indicates a possible hardware failure in the Drive. Check all connections to the Power/Driver Board. Retry operation. If problems persist, replace the Drive.
F47	Transistor Saturation Fault. This fault code indicates a possible hardware failure in the Drive. Retry operation. If problems persist, replace the Drive.
F48	Reprogram Fault. Reset STC or cycle drive power.
F49	Undefined Fault. This fault code is undefined and may indicate a defective Drive. Retry operation. If problems persist, contact the factory for instructions
F50	Poles Calc Fault
F51	Background 10ms Over. This fault code indicates a possible hardware failure in the Drive. Retry operation. If problems persist, replace the Drive.

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Fault	Description
F52	Foreground 10ms Over. This fault code indicates a possible hardware failure in the Drive. Retry operation. If problems persist, replace the Drive.
F53	EE Init Read. This fault code indicates a possible hardware failure in the Drive. Retry operation. If problems persist, replace the Drive.
F54	EE Init Value. This fault code indicates a possible hardware failure in the Drive. Retry operation. If problems persist, replace the Drive.
F55	Temp Sense Open. Indicates a possible hardware failure in the Drive. Retry operation. If problems persist, replace the Drive.
F56	Precharge Open. This fault code indicates a possible hardware failure in the Drive. Retry operation. If problems persist, replace the Drive.
F57	Ground Warning. Check the motor and external wiring to the drive output terminals for a grounded condition.
F58	Blown Fuse Fault. This fault code indicates a possible hardware failure in the Drive. Retry operation. If problems persist, replace the Drive.
F59	Undefined Fault. This fault code is undefined and may indicate a defective Drive. Retry operation. If problems persist contact the factory for instructions.

Fault	Description
F60	Encoder Loss. This indicates that the axial drive can not sense the axial encoder. Possible causes are: Encoder, cabling, unplugged Axial Control Board or improper axial drive encoder jumper setting.
F61	Mult Prog Input. Multiple functions selected may indicate a defective Drive. Retry More than one function has been programmed reset hardware.
F62	III Prog Input. Multiple functions selected may indicate a defective Drive. Retry Config file or Drive Eprom may be corrupt reset hardware.
F63	Shear Pin Fault. Output Amps exceeds program limits Check Axial hardware and belt. Retry Look for slippage in Axial drive hardware or Axial Drive Belt.
F64	Power Overload. This fault code is undefined and may indicate a defective Drive. Retry operation. If problems persist contact the factory for instructions.
F65	Adapter Frequency Error. This fault code indicates an operating frequency parameter was out of range. Possible corrupted configuration parameter, defective Drive or Control Board. Retry Operation.

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Fault	Description
F66	EEprom Checksum Fault. This fault code indicates a possible hardware failure in the Drive. Retry operation. If problems persist, replace the Drive.
F67	Sync Loss Fault. This fault code is undefined and may indicate a defective Drive. Retry operation. If problems persist, contact the factory for instructions.
F68	ROM or RAM Loss Fault. Internal power-up tests did not execute properly. Check Language Module. Retry operation. If problems persist, replace the Drive.
F69	Open Output Fault. An undercurrent condition has been detected in one or more of the Drive output wires. Check the wiring between the drive and the Motor. Check the Motor for an open primary winding.
F70	Phase Unbalance Fault. An imbalance between Drive output phase currents has been detected. Check the wiring between the drive and the Motor.

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Detector Heater Power Supply

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Functional Checks and Replacement Procedures

Gantry and Table Power

Gantry and Table Power Supplies

Use a scope to check the **OBC** Power Supply. It should have no more than 500 mV of ripple in the **mA Loop**.

1. **Turn OFF** the Axial Drive ENABLE and HVDC ENABLE switches on the **STC backplane**.

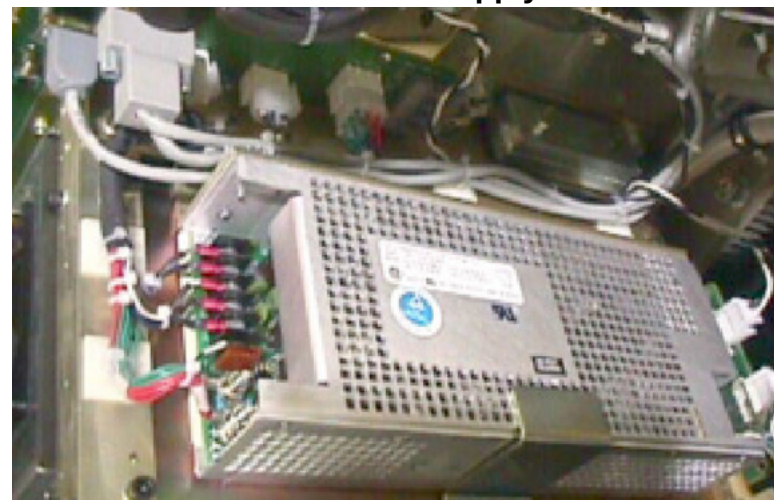


OBC and TABLE Power Supplies

Non-Adjustable Supplies	Adjustable Supplies
OBC +24	ETC, OBC, +5 and ± 15
Table (Display, E-Stop) +24	Table (Cradle Drive, Relays) +24
Detector Heater +26	Data Comm +12
Filament Supply +30	DAS +5, ± 5
Tilt/Elevation +170	Collimator +38.5

Note All figures shown above are for Volts DC (VDC).

OBC Power Supply



Tube Filament Selection

The mA Filament Control Board in the **OBC** monitors the **output voltage** of the Filament Inverter. An improper voltage is reported as an open filament. Both large and small filament use the same board circuitry and **relay K78 on the OBC backplane**. K78 is switched by filament circuitry on the mA Control board.

Every time someone removes the cathode HV cable from the tube well, a **Filament Open Error** gets logged.



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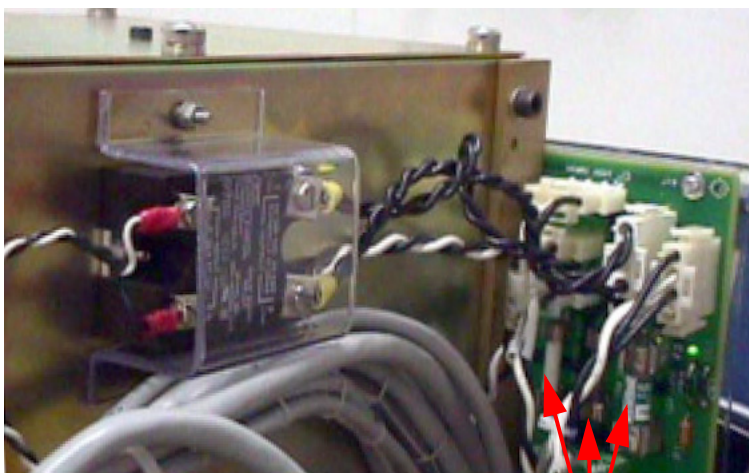


Functional Checks and Replacement Procedures

OBC PWR I/F

Caution

The **OBC PWR I/F** contains devices that can be damaged by ESD. This damage may not be immediately apparent; it may show up in the future as degraded operational performance. Therefore, use the proper ESD equipment.

OBC AC Power Fuses

Fuses F1 to F3

OBC Fuses

Fuse #	Rating	Description
F1	8A, MDA	Protects 120 VAC to the OBC and Inverters.
F2	15A, MDA	Protects 120 VAC to the HEMRC Assembly.
F3	12A, FNM	Protects 120 VAC to the X-ray Tube Assembly.

OBC LEDs

LED	Color	Label	Description
DS1	Green	DS1	Indicates 120 VAC is present.

Measurement Points for Line Power

Connector	Value	Comments
J8-1 & 2	120 VAC (Line)	Input externally protected to 30 A maximum.
J-8 & 4	0 VAC (Grounded Neutral)	n/a

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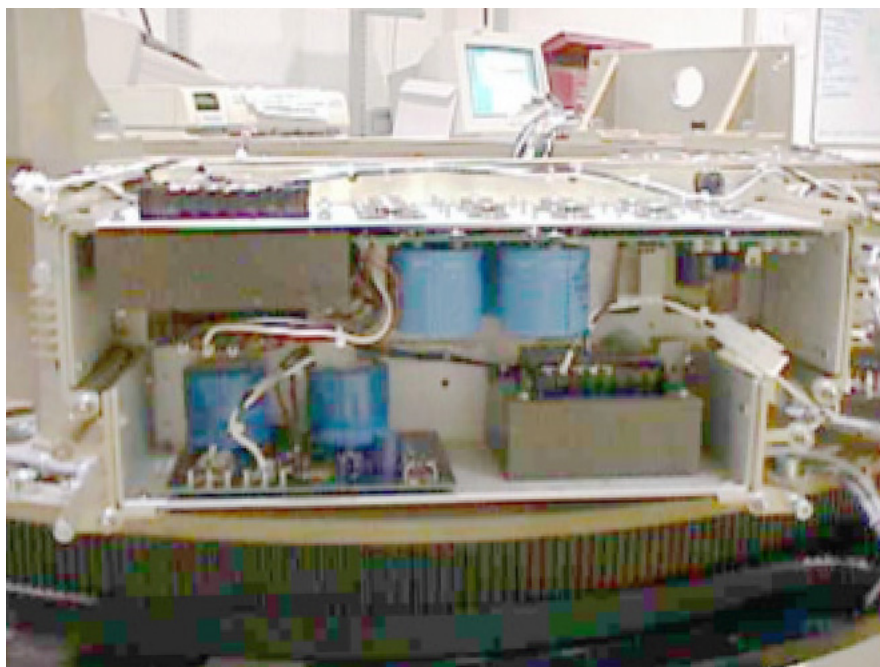
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DAS Power Supplies



Test points (TP) on the Center DAS Backplane allow you to measure Power Supply Voltages.

TP	Power Supply	Specification
1	+12 VDC Analog	+11.75 to +12.25
2	-12 VDC Analog	-11.75 to -12.25
3	+5.0 VDC Analog	+4.75 to +5.25
4	-5 VDC Analog	-4.75 to -5.25
5	+5.0 VDC Digital	+4.75 to +5.25
6	Analog Ground	n/a
7	Digital Ground	n/a



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Data Acquisition

Detector Heater Power Supply

The voltage reading across the output lugs of the detector heater power supply should be **24 VDC (± 1)**.

The DCB monitors detector temperature using the resistance of a thermistor inside the detector. The DCB will turn off the heater if it senses a temperature over spec.

The Detector Heater is part of the Detector FRU.

Elastomer

Important! Use ESD precautions: ESD wrist strap, finger cots, detector booties.

Elastomers make an electrical connection between the detector flexes and S-DAS backplane. In each DAS housing, there are eight elastomers. With the DAS at about 3 o'clock, elastomer number 1 is at the top so number 8 is at the bottom of that housing.

Since the output of the detector is a very low current signal, the connection must be very clean and well connected. Apply the correct torque to the flex housing cover and clamp. The torque spec is 9 in-lbs (1.02 Nm).

Related Checks

Refer to [DAS Power Supplies Checks](#).

Refer to [Slip Ring Communication Checks](#).

Refer to [System Communication and Control Troubleshooting](#).

Refer to [DAS Data \(Slip Ring\) Troubleshooting](#).



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STC Power

STC Power Switches

To verify proper power is supplied to the STC, do the following:

1. **Turn OFF** the Axial Drive ENABLE and HVDC ENABLE switches on the **STC backplane**.
2. **Turn ON** the 120 VAC ENABLE switch on the **STC backplane**.
3. Use a DVM to verify the STC power supply voltages at the following test points on the Axial Control Board:

Gantry Rotation Control Power

ACB Pwr Points	Specification	Source
TP12 (VCC) WRT TP9 (GND)	+5 VDC (± 0.25 V)	STC Power Supply
TP10 (3V3) WRT TP9 (GND)	+3.3 VDC (± 0.05 V)	Axial Control Board (No Adjustment)
TP3 (12 V) WRT TP1 (HGND)	+12 VDC (± 0.25 V)	Axial Motor Drive (No Adjustment)
TP2 (5 V CAN) WRT TP1 (GND)	+5 VDC (± 0.25 V)	Axial Control Board (No Adjustment)

WRT means "With Respect To"

Axial Encoder Power

Fuse	Value	Description
F1	1A	5 VDC to external Axial Encoder & Home Flag Assemblies.
F2	1A	5 VDC to external Cardiac Interface Assemblies.

STC Power Switches for Gantry

Switch	Description
S1	Momentary PB, resets gantry drives ENABLE circuit in PDU.
S2	Toggle switch enables Axial Drive .
S3	Toggle switch enables HVDC .
S4	Toggle switch enables Gantry 120 VAC .

Refer to **STC CPU LEDs Checks**.



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STC CPU LEDs

LED	Color	Label	Description
DS1	Green	C Pulse	C Pulse indicator from Axial Encoder.
DS2	Green	RST	Indicates status of the HVDC and gantry drives circuit in the PDU.
			On Steady = HVDC and Drives enabled.
			Slow Flash = E-Stop activated . HVDC and Drives Disabled.
			Fast Flash = Table Tape Switch activated . Cradle, Tilt and Elevation Disabled.
DS3	Yellow	AX DR ON	Indicates the Axial Drive Contactor in the PDU is energized.
DS4	Green	ENBL	Indicates the Axial Drive Contactor in the PDU is enabled.
DS5	Yellow	HVDC ON	Indicates the HVDC Contactor in the PDU is energized.
DS6	Green	ENBL	Indicates the HVDC Contactor in the PDU is enabled.
DS7	Yellow	120 VAC ON	Indicates the Gantry 120 VAC Contactor in the PDU is energized.
DS8	Green	ENBL	Indicates the Gantry 120 VAC Contactor in the PDU is enabled.
DS9	Green	ENC PWR	Indicates fuse F1, which feeds Axial Encoder, is OK.
DS10	Green	HHC PWR	Indicates fuse F2, which feeds the Cardiac Interface, is OK.
DS11	Green	ETC PWR	Indicates 5 VDC from the ETC, which feeds Hand-Held Control, is OK.

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Slip Ring Communication

Inspect

- Cables securely fastened.
- Terminators mounted correctly with correct screws.
- Fiber optic cables connected and locked.
- **Slip Ring Components**, rings, brushes not worn, dirty or damaged.
- Ground screws on **Slip Ring 6** and back of platter OK.

Slip Ring Status

At the host controller, type: **nbsClient obcr**

Then at nbsClient, enter: **s sr0**

At the host controller, type: **nbsClient stc**

Then at nbsClient, enter: **s sr0**

Ping Test

Ping test requires two people. Run an extended version of ping while someone manually rotates the gantry. The output will show where any rough spots exist. Clean those areas.

Use the following command: **ping -s obcr 16 100**

Refer to **Slip Ring Communication Maintenance**.

STC Test Points for Control Signals

STC	Color	Label	Description
TP1	YEL	TRX-SIN	Control signal inbound to STC on Slip Ring 10 .
TP2	BLK	TX-SREF	(Isolate) Control signal reference on Slip Ring 11.
TP3	YEL	TTX-SOUT	Control signals outbound from STC on Slip Ring 12.

HSDCD Transmitter or Receiver

The **DAS data slip ring transmitter** or **receiver** each has two LEDs. The green LED is ON for power, and the yellow LED is ON for data. The data LED is normally ON when the system is idle or collecting data.

LSCOM LEDs

Status of the **control signals** traveling the slip rings can be observed from the STC and OBC **LSCOM LEDs**.

DS6 RED Ring Disconnect

DS7 GRN TX Busy

DS8 GRN RX Busy



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Axial Drive Checks

Axial Home Flag Check

1. Turn the gantry by hand until the **Axial Home Flag** approaches the sensor, tube near the 7 o'clock.
2. Verify that the flag:
 - passes through sensor without making contact.
 - is centered in sensor window top to bottom.
3. With flag in the sensor window, verify the DS5 (Home) LED on the Axial Control Board lights.
4. Perform Axial Encoder Check.

Axial Encoder Check

1. Turn the gantry by hand until it passes through the Axial Home Flag; the tube will be near 7 o'clock.
2. Verify that:
 - Axial Control Board CHA and CHB LEDs toggle.
 - Axial Control Board CHC LED toggles on and off once while home LED is illuminated.
 - "C Pulse" LED on the **STC backplane** illuminates with the CHC LED.

Refer to [How to Reset Axial Drive C Pulse \(Home Flag Adjustment\)](#).

Axial Brake Check

Toggle the Axial Drive ENABLE switch on **STC backplane**. Listen to the brake; it should energize and de-energize.

The brake may not toggle if the system underwent a hardware reset since the last time you turned on gantry AC power. If the brake doesn't toggle, use the 120 VAC ENABLE switch on the **STC backplane** to turn OFF gantry power and then ON. Then toggle the Axial Drive ENABLE switch on the STC backplane. You should now hear the brake as it energizes and de-energizes.

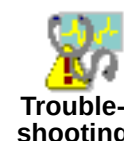
Turn OFF the Axial Drive ENABLE and verify that the switch pilot light turns OFF and the brake releases. Now you can rotate the gantry by hand.

Turn ON the Axial Drive ENABLE and verify that the switch pilot light turns ON and the brake energizes.

Turn ON all three switches on the **STC backplane**.

Axial Dynamic Brake Fuse

Fuse #	Value	Description
A4 F1	10A, 700 VDC	DCIN+ to Chopper Resistor Asm.





Functional Checks and Replacement Procedures

How to Replace the Axial Belt

Required Tools

- 10mm, 8mm 6mm, 5mm hex keys
- 12 inch extension for hex key

Note When ordering, be sure to order part #2217993.

Remove the Belt

1. Remove the **gantry covers** and lower slip ring cover.
2. **Turn OFF** all three switches on the **STC backplane**.
3. Remove system power at the Main Disconnect panel and use proper Lockout/Tagout procedures.
4. Remove the Home Flag assembly to prevent damage.
5. Remove the Axial Encoder to prevent damage to the encoder gear teeth.
6. Disconnect the power connector to OBC Power Interface board.
7. Remove the HEMRC cover.
8. Rotate the tube to the 3:00 position. Do not engage the rotational lock.
9. Loosen the two M12 screws with 10 mm hex key.
10. Use the 6 mm hex key and 12 inch extension to loosen the elongated hex screw and remove the drive belt from the drive gear.
11. Remove the belt from the drive gear. You will need every millimeter of length to clear the HEMRC. Work the belt toward the table on the rotating assembly. Keep all slack toward the tube side.
12. Work the belt around and behind the OBC to provide enough length to cross the tube. Work patiently to avoid damaging the Power Interface Board. Work the belt between the HEMRC and the tube radiator.
13. Once belt is over the hose, work belt around cathode end of the tube and radiator. You need to get in front of the tube to clear the corner of the HEMRC.
14. Once the belt is past the tube, carefully gather all the slack to clear the HEMRC corner.
15. When the HEMRC is cleared then carefully work the belt around the rest of the rotating gantry completing the removal process.

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**Functional Checks and Replacement Procedures**

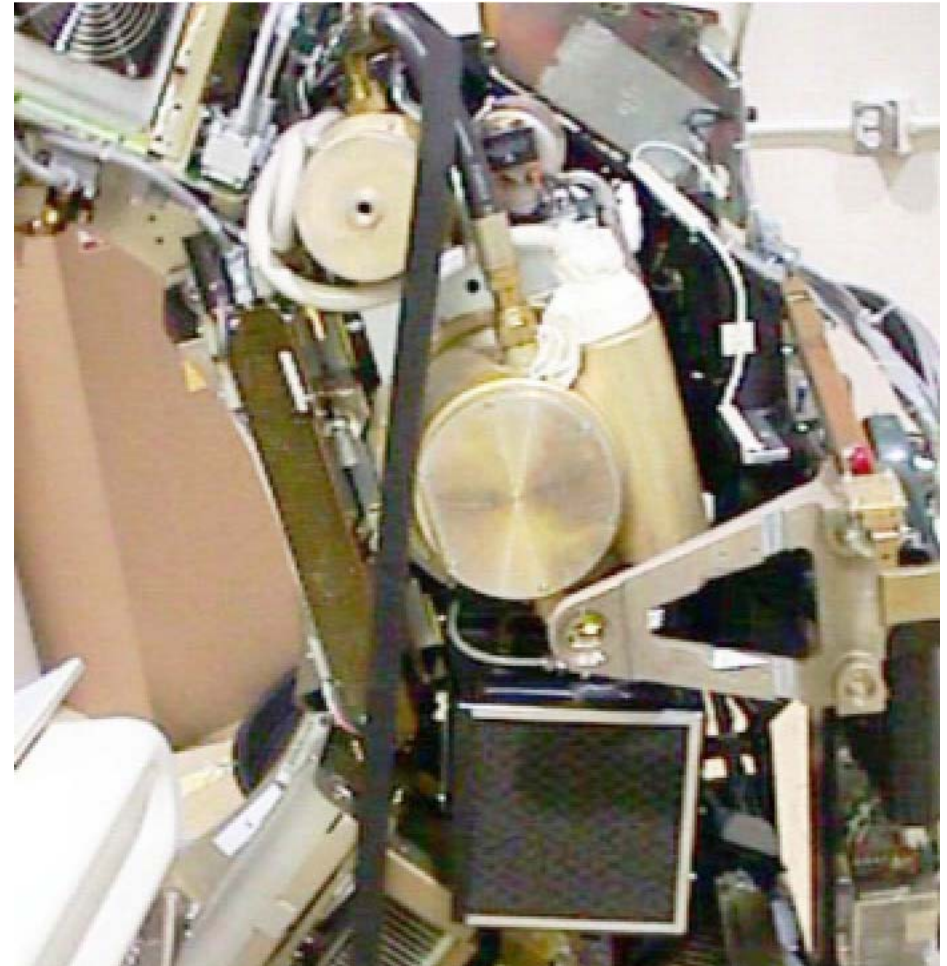
How to Replace the Axial Belt, *continued*

Install the Belt

1. Work the belt around the HEMRC, tube cathode and radiator, then the OBC.
2. Connect power plug to the OBC Power I/F board.
3. Install the home flag, axial encoder and HEMRC cover.
4. Slide the belt over the main drive gear and align it towards the back of the rotating assembly teeth. Check both top and bottom.
5. Work the belt through the pulley tensioner assembly and place on motor drive gear.
6. Tighten the elongated hex screw with a 6 mm hex key and a 12 inch extension. Apply enough tension so the washer can be rotated with your fingers.
7. Rotate gantry by hand several times and check tension. Make sure the belt does not slip off tensioning pulley and is tracking correctly toward the rear of the gantry. Correct tension is achieved when the washer can be turned with some difficulty with your fingers.
8. Tighten the two M12 screws to 30 ft-lbs (40.7 Nm). This locks the tensioner assembly.
9. Install the home flag assembly and then **check**.

10. Install the axial encoder and **align C pulse**.

Axial Drive Critical Path



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System Communication

Console LANVIEW LEDs

Refer to [Slip Ring Communication Checks](#).

The CTP100T adapter has five LANVIEW LEDs that monitor network activity (link, collision present, receive, and transmit) and power status. They serve as a useful tool to quickly diagnose physical layer problems.

LED	Description
LNK (green)	Link OK - ON when a twisted pair link has been established between the CTP100T and the device at the other end of the segment. It flashes when polarity is reversed on the twisted pair segment receive link. The CTP100T adapter still passes the data onto the twisted pair segment receive link. You should correct this to avoid future problems.
CLN (red)	Collision Present - ON when there is a jabber condition; flashes when a collision is detected.
RCV (yellow)	Receive - flashes when the CTP100T is receiving a data packet from the thinnet segment and forwarding it to the twisted pair segment.
XMT (green)	Transmit - flashes when a data packet is being transmitted on the thin-net segment.
PWR (green)	Power - ON when the transceiver is receiving power from the external wall transformer.





Enhanced Table Control (ETC)

Note In CT mode, ETC controls the table. In PET mode, the PET gantry controls the table.

ETC Board LEDs

DS263 CR-A: Cradle Encoder A-pulse active

DS264 CR-B: Cradle Encoder B-pulse active

DS265 CR-C: Cradle Encoder C-pulse active

DS266 EL-C: Elevation Encoder C-pulse active

DS310 CSTALL: Cradle stalled when on

DS311 ETFAULT: Elevation/Tilt Amp fault elevation or tilt drive shorted when on

DS312 ETSTALL: Elevation/Tilt Amp stall or tilt drive stalled when on

ETC Board Switch Settings

S162: Reset

ETC Test Point	Description
TP1	+5 V: (Digital +5 V) Power Supply
TP2	LGND: (Digital LGND) Power Supply Test Pt

ETC Test Point	Description
TP3	(Digital ± 12 V) RS232 Table XMIT
TP4	(Digital ± 12 V) RS232 Table REC
TP5	(Digital +5 V) Spare
TP6	(Digital 0 - 5 V) Long Encoder CH C
TP7	AGND: (Analog) AGND Power Supply Test Pt
TP8	(Analog ± 15 V) Cradle Tach (relatively noisy signal) Gain = 75 mm/sec = 6.3 V at TP
TP9	(Digital 0 - 5 V) Long Encoder CH B
TP10	(Digital 0 - 5 V) Long Encoder CH A
TP11	-15 V: (Analog -15 V) Power Supply
TP12	(Analog 0 - 10 V) Elev/Tilt Pulse Width Modulation (PWM) Control, (controls AR689 pulse width) Gain = 5 μ s pulse width/Volt command
TP13	(Analog +10 V) +10 Reference

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Enhanced Table Control (ETC), *continued*

ETC Board Switch Settings, *continued*

ETC Test Point	Description
TP14	(Analog 0 - 15 V) Cradle Current Integrator (averages out pulses from AR653) Wider pulses = more amp current = higher signal
TP15	-15V: (Analog 0 - 15 V) Elev/Tilt Current Integrator (averages out pulses from AR563) Wider pulses = more amp current = higher signal
TP16	(Analog 0 - 10 V) Cradle Pulse Width Modulation (PWM) Control (controls AR669 pulse width) Gain = 5 uSec pulse Width/volt command Analog Signal
TP17	(Digital 0 - 15 V) Cradle Pulse Width Modulation (PWM) sets output (v) of power amp. Volts out = Pulse width (uSec) * 24 / 56
TP18	AGND: (Analog Ground) Power Supply Test Point
TP19	(Analog ± 15 V) Elev/Tilt Motor Voltage Feedback (Filtered map pulse width modulation signal, fc = 39 cps, gain TP19/amp voltage = 0.05 v/v)
TP20	(Digital 0 - 15 V) Trigger SYNC Signal PWM trigger for power amps (17.72 KHz)

ETC Test Point	Description
TP21	+15V: (Analog +15 V) Power Supply Test Point
TP22	(Digital 0 - 5 V) Elev/Tilt Command Direction (+5V = Up/Forward)
TP23	(Digital 0 - 15 V) Elev/Tilt PWM (sets output (v) of power amp. volts out = pw (uSec) * 160 / 56
TP24	PGND: (Analog Physical Ground) Power Supply Test Point
TP25	+24V: (Analog +24 V) Power Supply Test Point
TP26	(Analog ± 15 V) Cradle motor voltage Feedback (filtered amp pwm signal), FC = 64 CPS Gain (TP26/amp voltage) = 0.41v/v



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Gantry Thermostat



How to Adjust

The gantry thermostat for fan control should be unlocked and properly set from the factory.

If you need to check or change its settings, use this procedure.

If the lock switch inside is set to the left, the settings are locked. Remove thermostat power before you remove the thermostat cover. Push the lock switch near the bottom to the right to enable adjustment. Reassemble, repower and adjust.

1. Use the **SET** key to jump through the categories; use the Up and Down arrow keys to change the value of a category.
2. Select or verify **C** for **Celsius**.
3. Select or verify **29** for **Setpoint**.
4. Select or verify **1** for **Differential**.
5. Select or verify **C1** for **Cooling**.

After 30 seconds of inactivity, the Set Mode accepts current settings and turns off.



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Gantry Display



The Gantry Display is controlled by a **CAN network** located on the Enhanced Table Controller (ETC) **Interface Board**.

Refer to **Gantry Display Theory**.

1. **Turn ON** the 120 VAC ENABLE switch on the **STC backplane**.
2. Make sure the gantry display goes through the power-up self-test. The display continues to cycle through its self-test until it completes the hardware reset and download of firmware parameters.
3. **Turn ON** "X-RAY DRIVES" power by pressing the RESET DRIVES button on the gantry keypad.



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Gantry Start/Stop Scan

Warning

Using the START and STOP buttons on the gantry keypad can expose you to X-ray. Know that the STOP SCAN button does NOT stop rotation. **Use the Gantry Interior (Front View), continued.** Knowing how these keys work is imperative.

The two gantry buttons, START SCAN and STOP SCAN, work as follows:

1. A New Patient scan prescription is accepted.
2. Technician enters scan room and attends patient.
3. Technician observes Delay time on Control Panel Timer Display and START SCAN LED flashing.
4. Technician presses START SCAN button.
5. Delay timer countdown begins.
6. Technician presses STOP SCAN button for any reason and Timer Display goes blank.
7. START SCAN LED becomes Solid ON (Resume Mode).
8. Technician now presses START SCAN button.
9. START SCAN LED begins flashing (Start Scan Mode). Timer Display is still blank.

10. If the START SCAN button is pressed a second time, X-ray will start. Technician should exit scan room and initiate exposure by pressing the flashing START SCAN on the console.

Gantry Keypad

Stop Scan LED & Button

Start Scan LED & Button



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Gantry Controls



- **(1) Breathing Light Demo** - Press to cycle through the four steps of **breath lights** demonstration:
 - Flash Breath Light.
 - Hold Breath On w/ 30 sec. showing.
 - 30 Second countdown on display.
 - Breath Light on.
- **(2) Tilt and Table Travel Limits** - Press and hold to cycle through the **allowable motion ranges** on the display.

- **(3) Internal Landmark** - Press to designate the anatomy under the **internal** lights as the 0.0 mm scan location. The alignment lights intersect at the three dimensional isocenter. Dim the scan room lights to improve alignment laser visibility.
- **(4) External Landmark** - Press to designate the anatomy under the **external** lights as the 0.0 mm scan location. The distance between the external lights and Scan Field of View (SFOV) equals about 31 cm (12.2 inches). After you prescribe the scan and initiate the scan sequence, the system prompts you to press the ADVANCE TO SCAN button to move the cradle into position for the first scan.
- **(5) Alignment Light** - Press to turn on the internal and external laser alignment lights. **This will cause gantry rotation to position lights through the covers.**

Never turn on alignment lights with gantry side covers removed unless you know no one is in the way.

Read the laser warning labels on the gantry. Never look directly into the laser beams.

Press (5) again to turn off the alignment lights.

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Gantry Controls, *continued*

- **(6) Cradle Release** - Press to latch or unlatch the cradle.
- **(7) Reset Drives** - Pressing when its LED is flashing will reset the gantry and table drives. When the LED is on, nothing will happen. When it is off, the circuit is disconnected.
- **(8) Home** - Press to restore the gantry and table to the Home position. The gantry returns to the 0_Tilt position while the cradle drives all the way out of the gantry. After the gantry and cradle reach their home positions, the table lower to the minimum height.
- **(9) Superior Tilt** - Press to tilt the top of the gantry away from the table.
- Press and hold both tilt buttons (9 and 15) at the same time, and a **tilt to Zero** is performed.
- A **tilt to Rx** is required within ScanRx, so one of the two tilt LEDs (next to 9 and 15) will flash to indicate which button to press. Holding the button down will move the tilt to the prescribed angle and the LED will turn off. If the tilt is moved from the correct angle, the corresponding LED will start flashing again.
- **(10) Table Down** - Press to lower the table.
- **(11) Cradle In** - Press to drive the cradle toward the gantry in slow speed.
- **(12) Table Up** - Press to raise the table.
- **(13) Center** - Press and hold at the same time as either (11) or (14) to increase cradle speed by a factor of 2.
- **(14) Cradle Out** - Press to drive the cradle away from the gantry in slow speed.
- **(15) Inferior Tilt** - Press to tilt the top of the gantry toward the table.

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Gantry Controls, *continued*

- **Timer Display:** This rectangle will display the Prep Countdown and ISD Countdown and IGD Countdown. It will also give feedback for errors by either *flashing* **ERR** (stuck button) or leaving **ERR** on *solid* (loss of communications) or OFF (panel is disconnected from Network and the ETC I/F board needs to be reset).
- Pressing the **START SCAN button** when its LED is flashing will start the prescribed scan sequence. If its LED is solid, the button functions as a RESUME.
- Pressing the **STOP SCAN button** at any time that a scan is prescribed, the button will result in stopping the scan. **When its LED is ON, X-rays are being emitted.**

Refer to [How to Use the SCIM Scan Controls](#).



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How to Use the SCIM Scan Controls

Scan Controls for Gantry at the Console



Note Some keyboards will have a sixth button, Prescribed Tilt, as shown above. However, this button may be disabled and non-functional.

Start Scan - Starts the X-ray exposure. Flashes green when scan is ready to be initiated. After a Pause Scan, it is illuminated. You can restart or resume the scan from the SCIM or user interface.

Pause Scan - Pauses the X-ray exposure after a scan is complete. Use it if you need to change parameters between scans.

Stop Scan - Immediately terminates the X-ray exposure. Use it if the patient is moving and you want to stop the exposure. The button will be lit solid green while an exposure is in progress.

Move to Scan - Moves the cradle to the prescribed start scan location. The button will flash green when movement is needed.

Stop Move - Stops cradle motion in progress. If a Scout or Helical scan is in progress, this button will also stop the exposure. Use it if there are health monitoring lines or blankets/sheets getting caught. The button will be lit solid green during cradle movement, during Scouts and Helical scans.

Refer to [Intercom](#).

Refer to [Gantry Controls](#).

Refer to [Gantry Display](#).



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**Functional Checks and Replacement Procedures**

Intercom

The intercom board is set at the factory and should not require re-adjustment. At install time, make sure the gantry cable has enough slack to pull the cable through to the front of cabinet. Recommended: 1 Meter (3 feet).

Intercom



Make sure the intercom's **Talk** and **Listen** functions work. If you hear feedback, or if the intercom does not seem loud enough with the console intercom controls set to maximum, then:

1. Close gantry covers.

2. To pull the intercom board from the console:

Shut down system software and turn off power to the computer/console.

Remove the console front cover.

If the intercom cable from the gantry does not have enough slack, remove the two screws that hold the intercom carrier assembly to the console, and remove the assembly from the console.

Refer to [Intercom Adjustments](#).

If there is not enough cable length to pull it out:

1. Disconnect keyboard cable and gantry cable from the interconnect board.
2. Disconnect Interface board power and Console I/F cable.
3. Remove intercom/interface board assembly.
4. Pull gantry interface cable through the unit (back to front).
5. Route the keyboard cable to the front of the console.
6. Place the assembly in front of console and reconnect the cables.



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Laser Lights

Warning

When you operate the alignment lights, do not stare at the laser beams because this can cause permanent eye damage.

Controls

- Press the **ALIGNMENT LIGHT button** on the gantry keypad to **turn ON** the lights and position them.
- Press the ALIGNMENT LIGHT button again to **turn OFF** the lights.
- Turn OFF** the Axial Drive ENABLE and HVDC ENABLE switches on the **STC backplane**.
- Use the switch on the OBC backplane to manually **turn ON** the alignment lights, table side upper left corner labeled Lights Laser. OBC LED DS1 next to the "Lights Laser" switch will light.

Internal Axial Lights

- Place a sheet of plain white paper over the output port of each light and verify that the two lines of laser light coincide with each other.

Do not try to adjust the internal axial lasers on the CT system to shine "down" on the collimator.

External Axial to Internal Axial Distance

- Raise the table to its highest elevation.
- Extend the cradle until both the internal and external laser lights shine on the cradle.
- Place a metric rule on the right edge of the cradle, and measure the distance from the internal axial laser line to the external axial line, generated by each laser.
- Verify this distance equals 240.0 mm (± 1.0). Place the rule on the left edge of the cradle and measure again.
- Extend the cradle until both internal and external laser lines shine on it.
- Lower the table to the interference limit. Verify the 240.0 mm (± 1.0) distance between the internal and external lights, on both edges of the cradle, as above.

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Laser Lights, *continued*

Coronal Lights

1. Place a sheet of plain white paper at the left side of the patient opening, in front of the coronal laser light.
2. Verify that the two coronal lines coincide with each other.
3. Check the right side in the same way.
4. Place the paper in the center of the gantry opening, and use a level to verify that the coronal lines are horizontal.
4. Repeat the procedure with the internal axial lights.
5. If you cannot see the external or internal axial lights under the conditions described above, obtain a DIGAPHOT model #3300 or 3303 light meter; measure the ambient light intensity at the cradle surface at the external and internal alignment light locations.

the external axial alignment lights on your hand. You only have to see the axial lights, not the sagittal or coronal lights.

Laser Light Visualization

H.H.S. requirements state that lights used to define the tomographic plane must be visible under ambient light conditions, up to 500 lux. To verify:

1. Turn the scan room lights on to their brightest normal level. Do not add localized spot lights to increase the brightness level.
2. Raise the table to its highest position, advance the cradle into the gantry, and turn on the alignment lights.
3. Center the back of your hand over the cradle, and hold it in the alignment light. Make sure you can see
6. If the ambient light reading(s) exceed 500 lux, reduce room lighting to the 500 lux level, and repeat steps 3 and 4. If the light meter readings equal 500 lux or less, replace the laser light(s) or their power supplies or both.

lux = Foot-candles x 10.76



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Interference Test

Note If the elevation or tilt readings fail to meet spec, DO NOT adjust the limit switches. Instead, recharacterize **elevation** or **tilt** or both.

Note Use the **FAST button** to get the elevation for each test.

1. Select **Hardware Reset** at the console to download the current characterization values to the ETC, OBC, and STC.
2. Lower and retract the cradle to the home position. Use the **HOME button**.
3. Verify the **elevation display** shows **500.0 mm (+3)**.
4. Raise the table to maximum height.
5. Verify the elevation shows **25.0 mm (+3)**.
6. Advance the cradle about a meter from home.
7. Tilt the gantry back to -23 degrees.
8. Lower the table until it stops.
9. Verify the elevation shows **115.5 mm (+3)** and the table clears the gantry by at least a cm or half inch.
10. Tilt the gantry to -20 degrees.
11. Lower the table until it stops.
12. Verify the elevation display reads **212.0 mm (+3)** and the table clears the gantry covers by at least a centimeter or half-inch.
13. Raise the table until the Elevation Display is between **115.5** and **212 mm**.
14. Tilt the gantry to from -20 to -10 degrees.
15. Tilt the gantry from -10 back again and make sure it stops at **-22.0 degrees (+0)**.
16. Tilt the gantry toward the table; make sure it stops at **+18.5 degrees (+0)**.
17. Return the cradle to Home position.
18. Make sure you can tilt the gantry to **30.0 degrees (+0)**.
19. Tilt gantry to **-30 degrees**.
20. Raise the table until you can advance the cradle about 30 cm or 12 inches from the Home position.

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Interference Test, *continued*

21. Now return the cradle to the Home position and the gantry to 0 degrees.
22. Make sure you cannot move the cradle at table heights below **210.0 mm (+8)**.
23. Verify the CRADLE UNLATCHED indicator on the gantry display is lit.

Cradle Unlatched Indicator



24. Select **File** > **Exit** from the pull-down menu.
25. Select **Dismiss** and then **Cleanup**.



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Emergency Stop (E-Stop) Check

1. Advance the cradle about 0.5 m (2 feet) from the home position.
2. Press one of the E-Stop buttons on the gantry.
3. Make sure the TABLE POWER shuts OFF, and the green RESET LED flashes.
4. Press one of the Table Elevation buttons to verify the Emergency Stop disabled table elevation.
5. Press one of the Cradle Drive buttons, to verify the Emergency Stop disabled the cradle drive.
6. Press one of the **RESET buttons** to turn ON X-RAY DRIVES POWER.
7. Press the other exterior E-Stop button on the gantry.
 - Verify the X-RAY DRIVES POWER shuts OFF.
 - Manually move the cradle to the home position to verify the cradle clutch released.
 - Verify the cradle latches securely once in the home position.
8. Press a RESET button to turn ON X-RAY DRIVES POWER.
9. Press one of the four **table tape switches** to make sure the X-RAY DRIVES POWER shuts OFF.
 - **Turn ON** TABLE POWER.
 - Repeat with the three remaining table tape switches.
10. Press the **console E-Stop** button (shown at right); verify the TABLE POWER shuts OFF.
11. **Turn ON** the 120 VAC ENABLE Switch on the **STC backplane**.
12. Exit.
13. Return to Service Desktop home page.

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X-ray Verification

Use this program to verify that X-ray is being detected properly. Besides being the last step of a tube replacement, it can be used to test the detector and DAS boards.

Pre-requisites for test to pass are:

- Tube calibration (kV and mA) is correct.
- Alignments (POR and BOW) are correct

Possible causes for failure are:

- Converter board
- Elastomer
- Detector Flex
- DAS Chassis backplane
- FET control signal

Procedure

1. Select **Service Desktop > Diagnostics > DAS Tools > X-ray Verification** -or- **Service Desktop > Replacement > DAS Board Change**.
2. Run tests and record results on Form 4879 when successfully completed.



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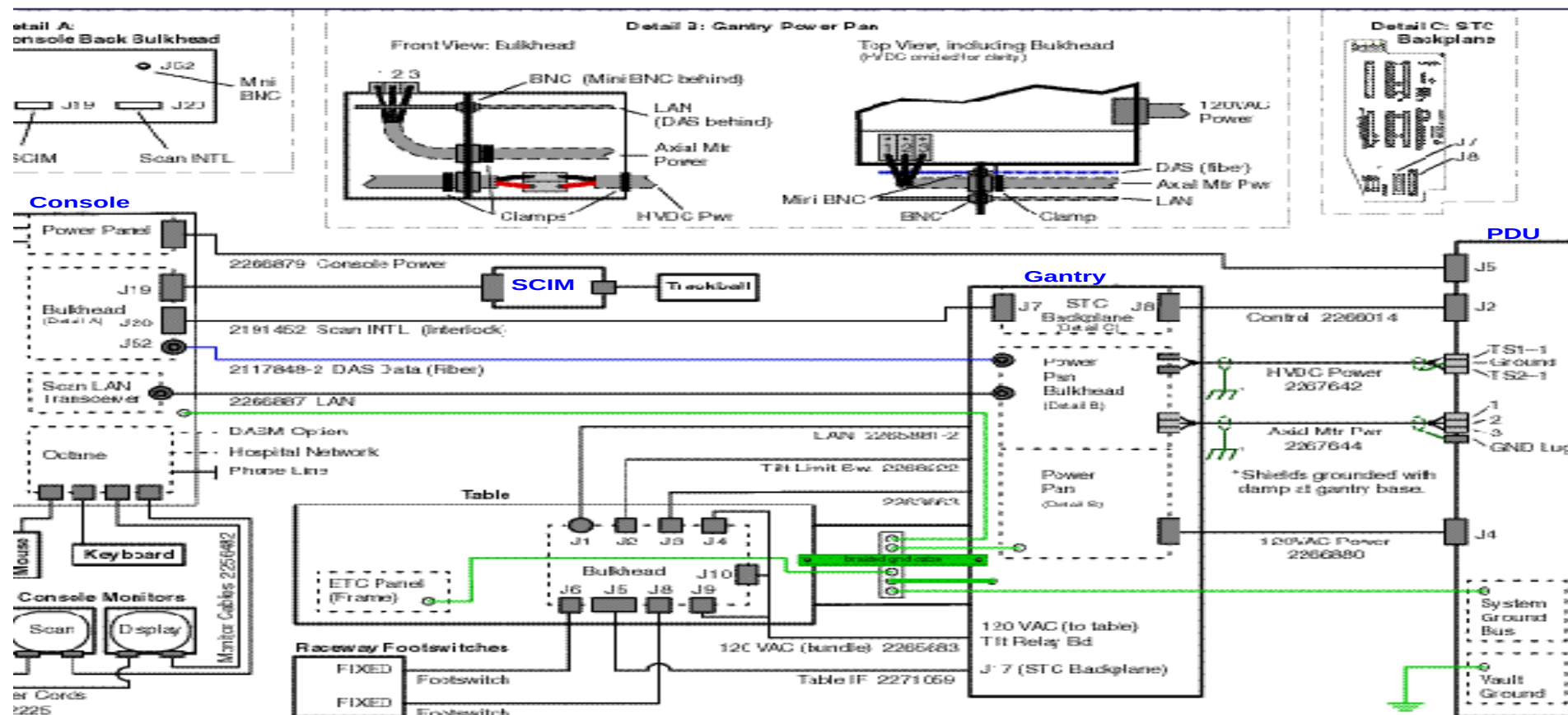
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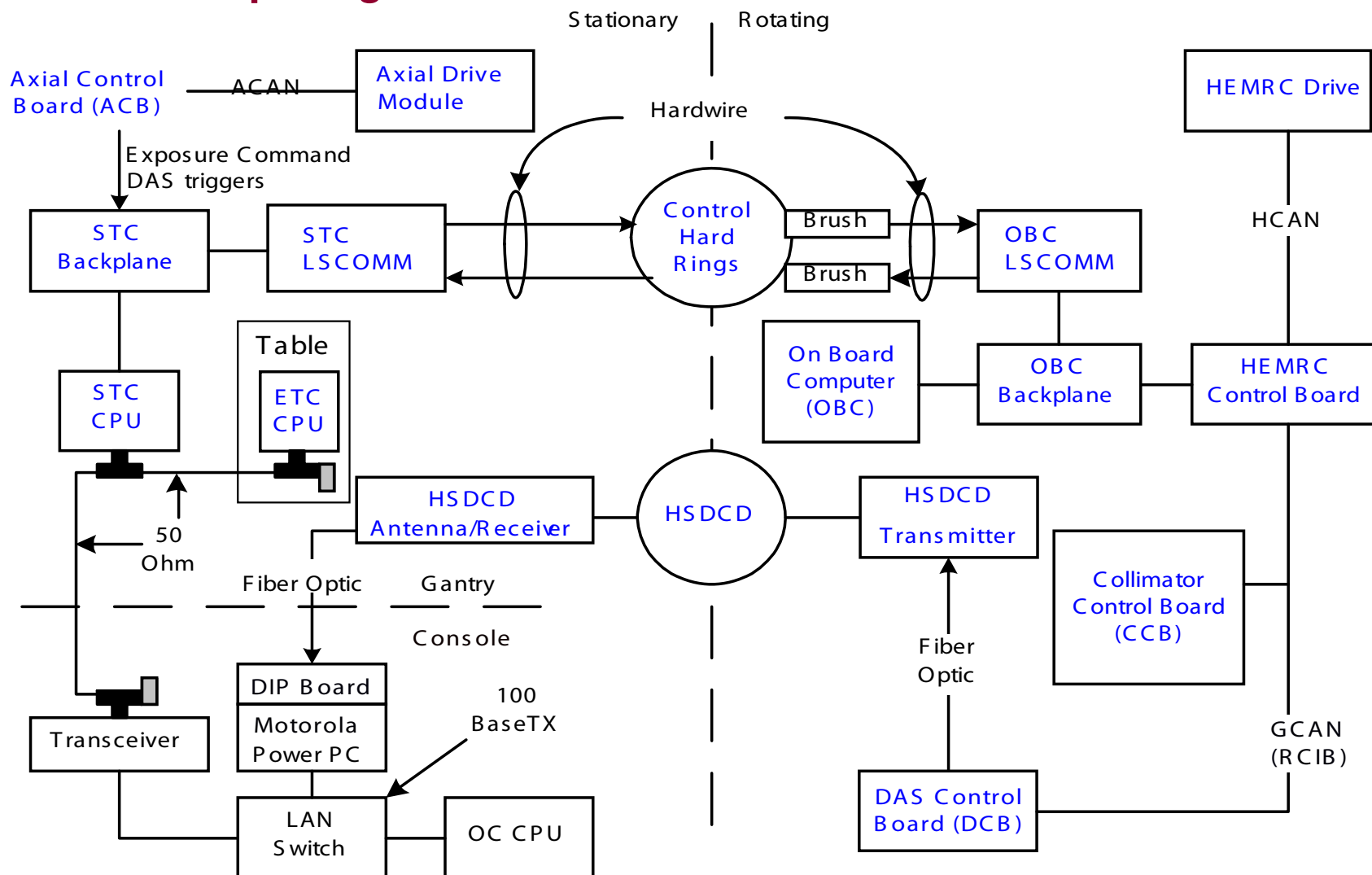
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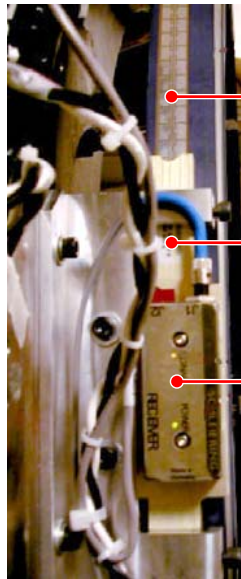
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Slip Ring Architecture



HSDCD

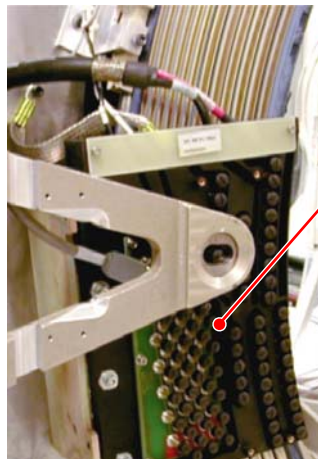
Emitter (Ring 13)

Antenna

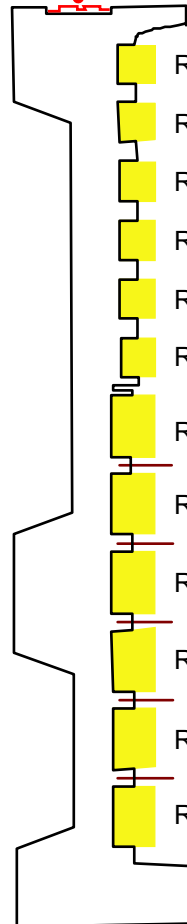
Receiver

Ring 12

Ring 1



Brush Block Assembly



Ring 12 TRX-IN (OBC) / TTX-OUT (SBC)

Ring 11 TRX-SIGNAL REF (Two-way Control Signals)

Ring 10 TRX-OUT (OBC) / TTX-IN (SBC)

Ring 9 SYSINTLK

Ring 8 CLOSEBC

Ring 7 PDU 24 A

Ring 6 GROUND

Ring 5 AC HOT

Ring 4 AC Neutral

Ring 3 (Unused)

Ring 2 -HVDC

Ring 1 +HVDC

WARNING

Exposed slip rings pose an electric shock hazard! Contact may cause serious injury or death unless high voltages present on the slip rings are disabled or discharged. Power on the slip ring includes 120 VAC and 550 VDC.

Rings 10 and 12 are both referenced to ring 11. Signal names depend on which host is "receiving" control data.

DAS Data is transmitted across the **HSDCD** from the transmitter to the antenna/receiver.

Rings 7, 8, and 9 comprise the safety interlock circuitry.

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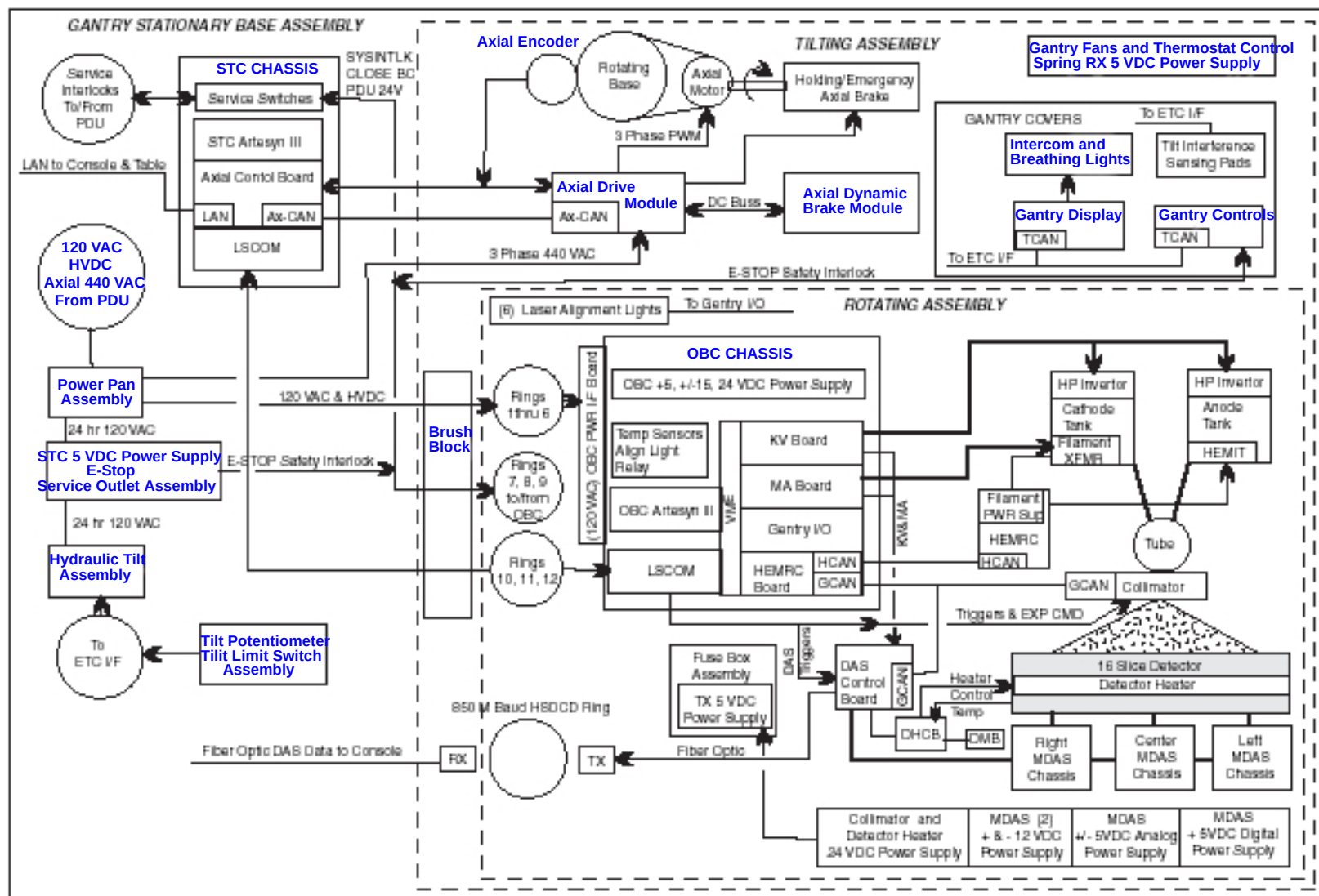
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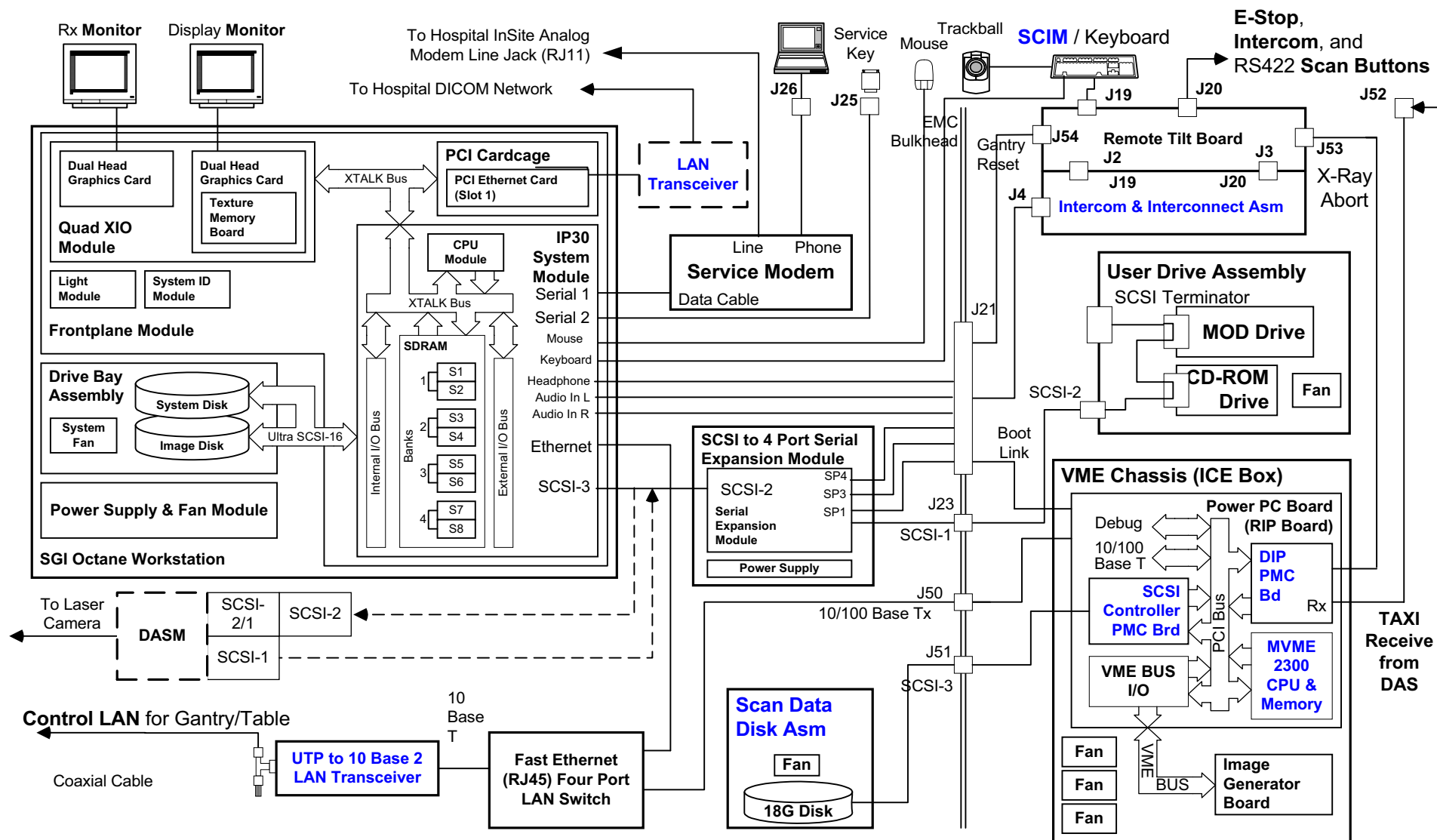
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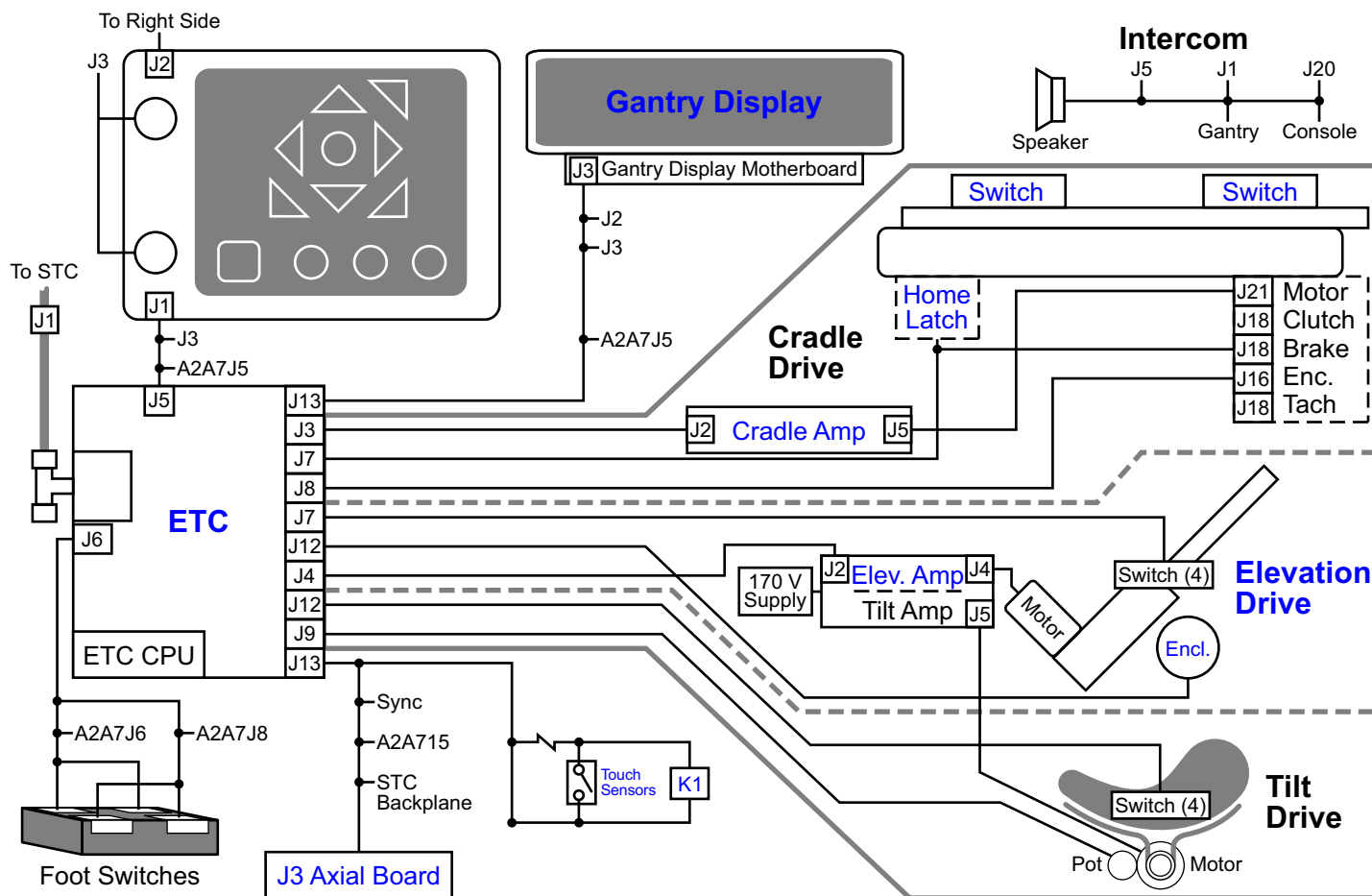
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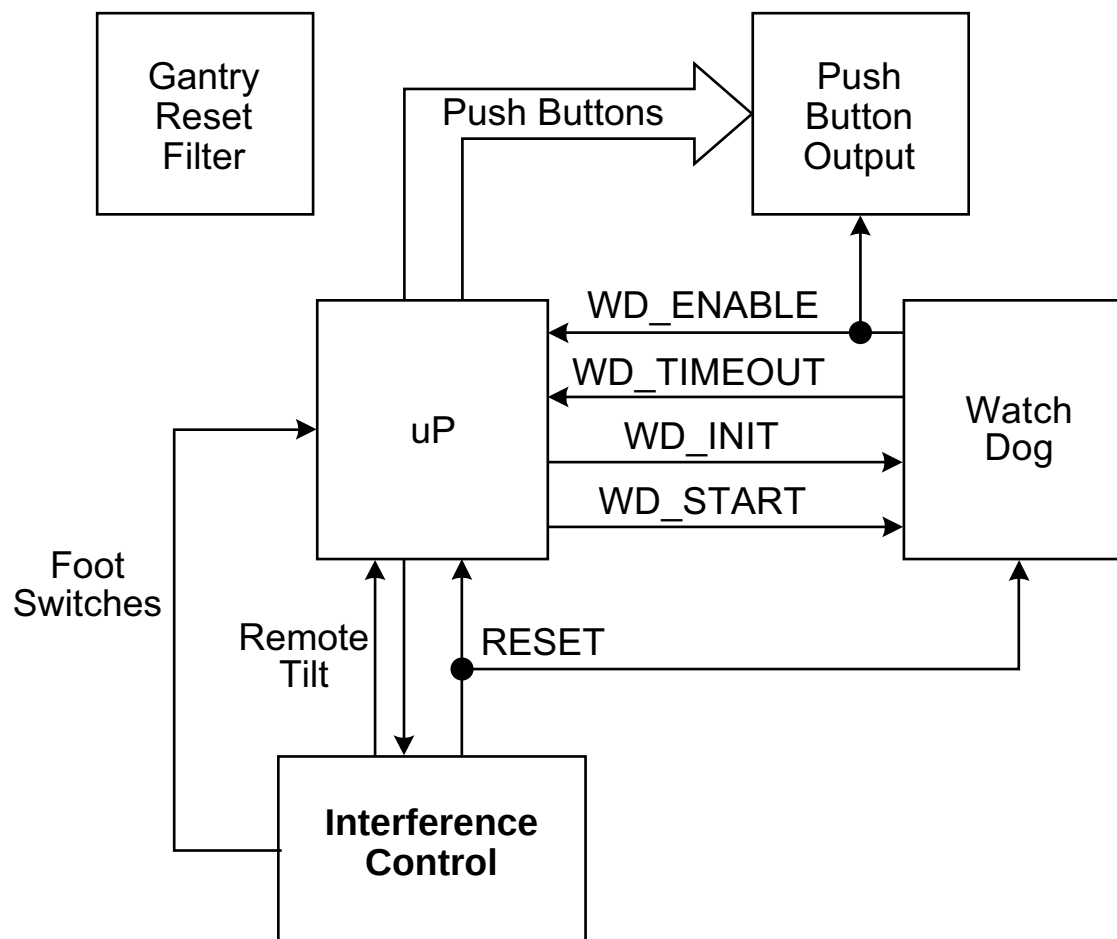
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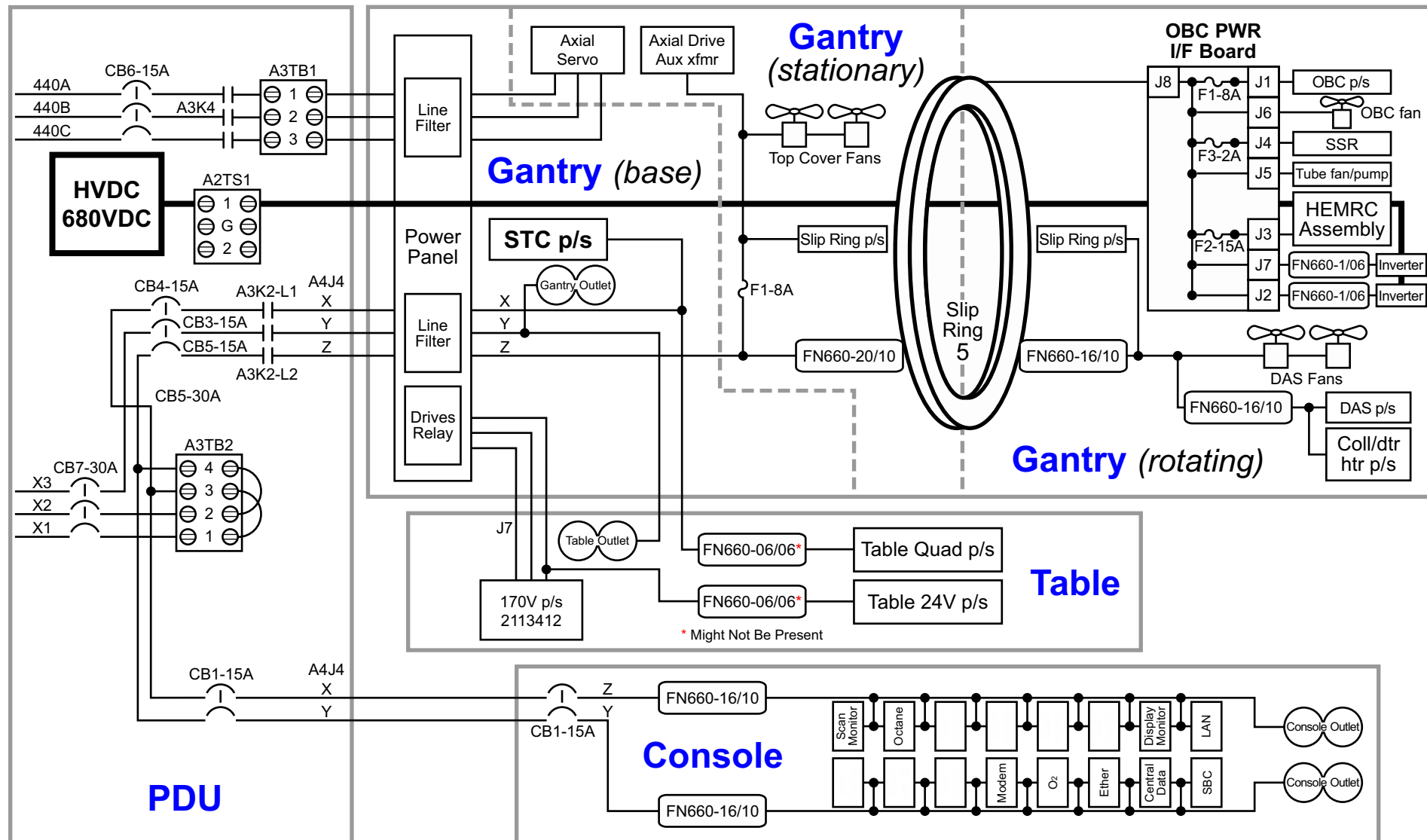
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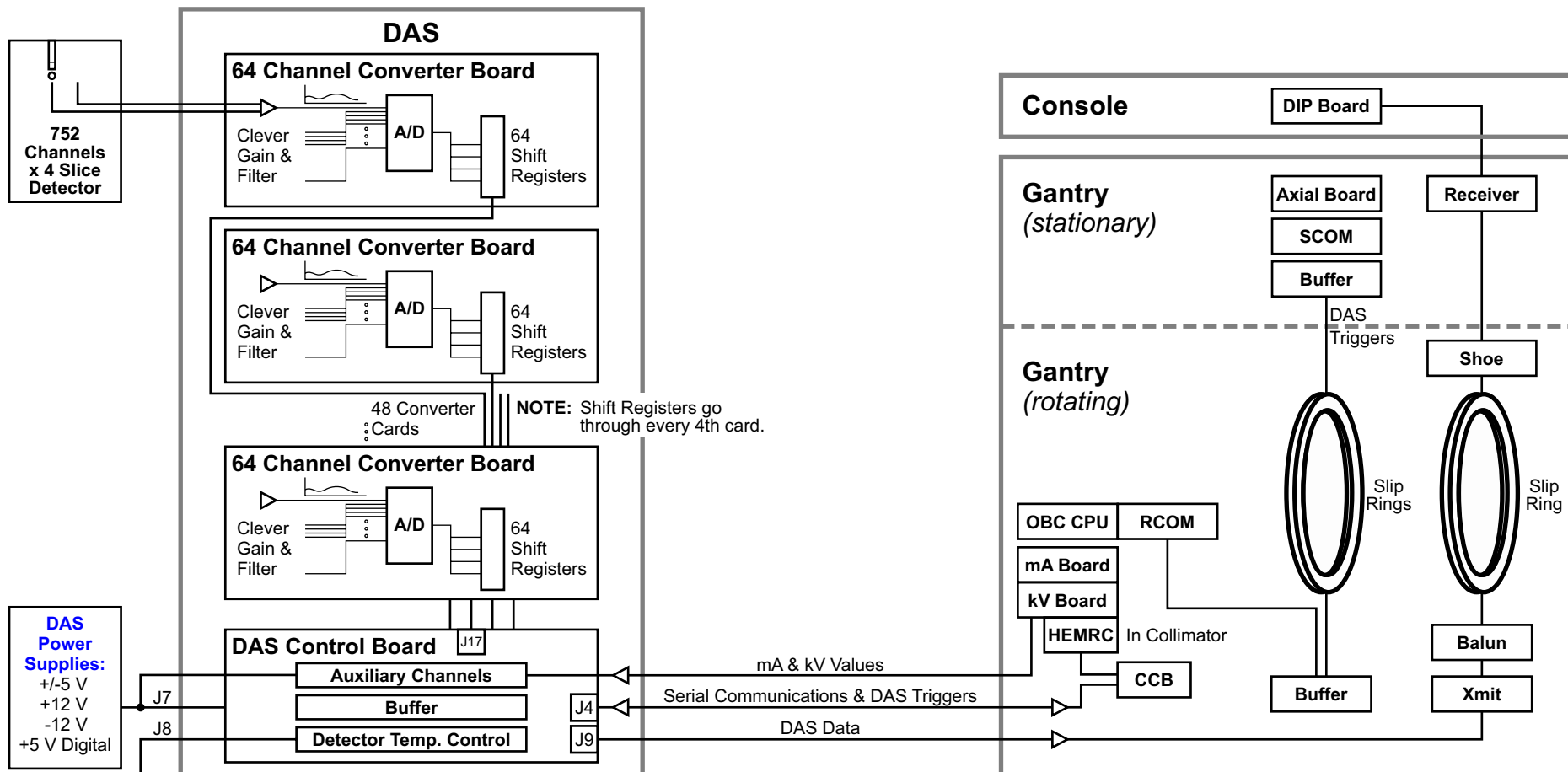
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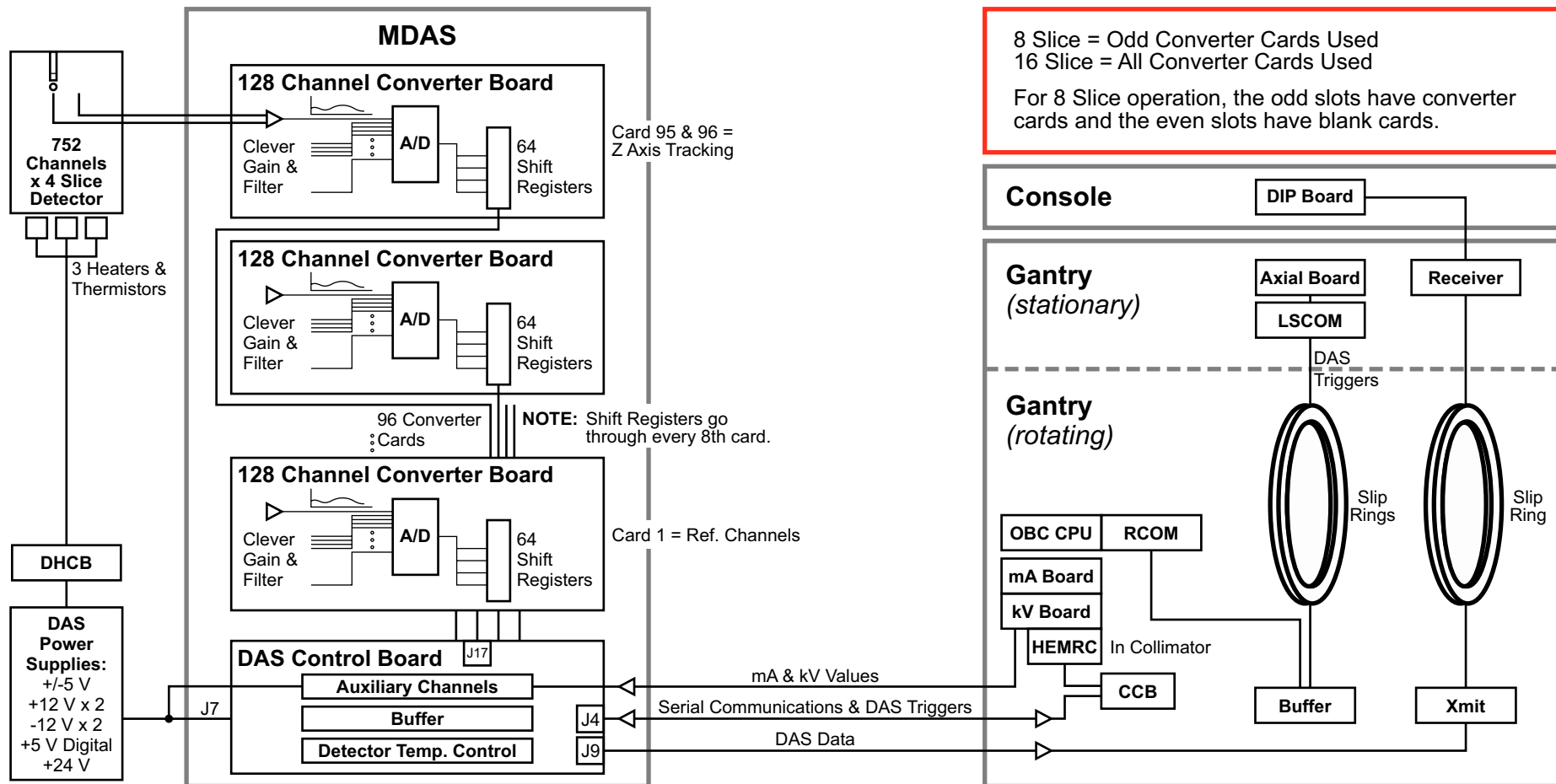
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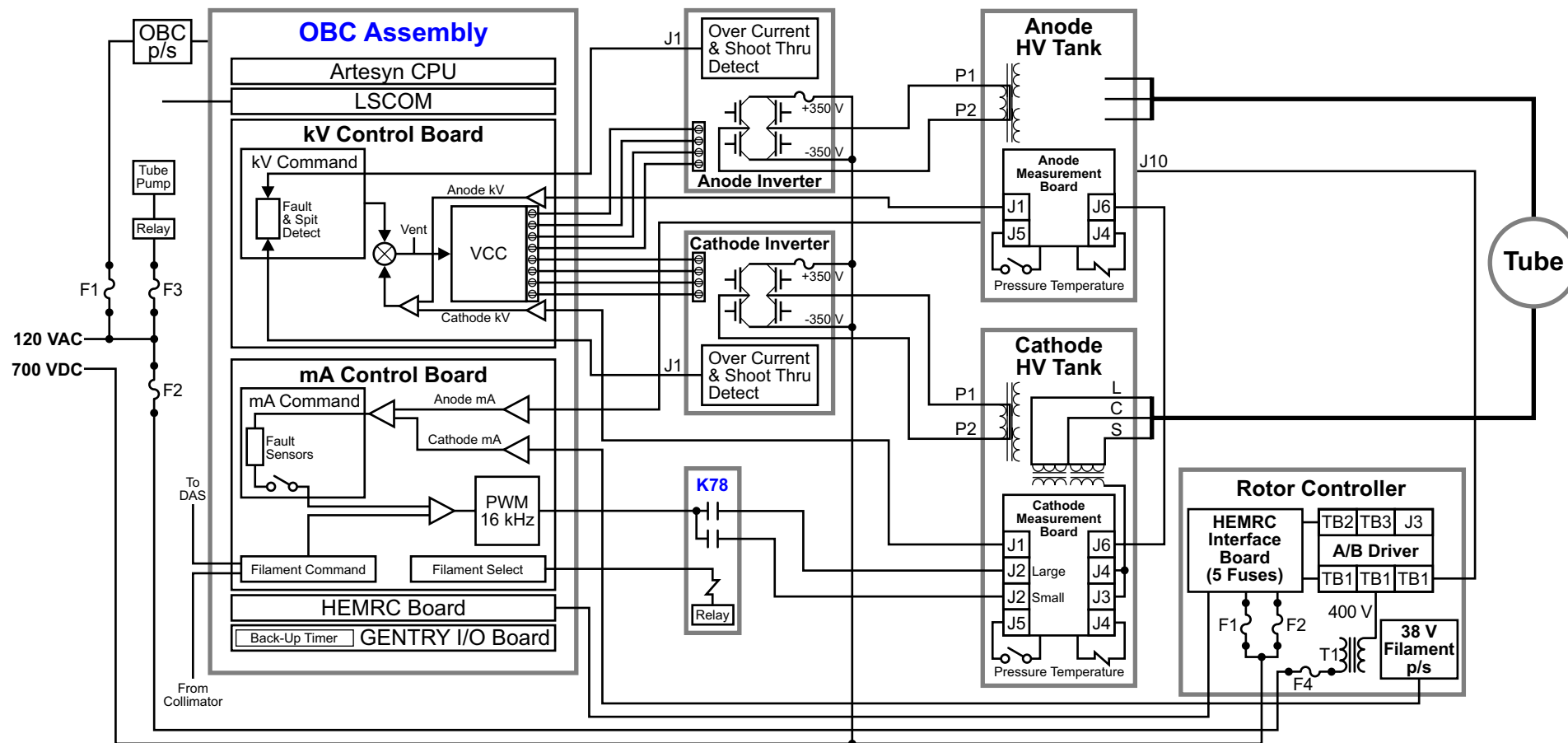


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Planned Maintenance (PM) Schedule

LightSpeed Series PM is based on two million gantry revolutions per year. Do PM four times a year, every 500,000 revolutions. Schedule is based on Rev Counter.

Job Card #	SubSystem	Description	Sch A xxx250	Sch B xxx500	Sch C xxx750	Sch D xxx000
CTPM-1308	Gantry	Grease Main Bearing				X
CTPM-1319	Gantry	Removal of Slip Ring Brush Debris	X	X	X	X
CTPM-1321	Gantry	Slip Ring Vacuuming	X	X	X	X
CTPM-1322	Gantry	Plenum Cleaning	X	X	X	X
CTPM-1323	Gantry	SCHLEIFRING Maintenance, Brush inspection	X	X	X	X
CTPM-1324	Gantry	Inspect Primary Copper Filter	X	X	X	X
CTPM-1421	High Voltage	mA Meter Verification			X	
CTPM-1423	High Voltage	HV Tank Feedback Resistors Calibration			X	
CTPM-1428	High Voltage	mA and kV Verification (HHS)			X	
CTPM-1429	High Voltage	Verify Rise and Fall Times (HHS)			X	
CTPM-1430	High Voltage	Verification of Internal Scan Timer (HHS)			X	
CTPM-1508	PDU	PM Checks		X		X

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Planned Maintenance (PM) Schedule

Job Card #	SubSystem	Description	Sch A xxx250	Sch B xxx500	Sch C xxx750	Sch D xxx000
CTPM-1601	System	Test Emergency Stops	X	X	X	X
CTPM-1603	System	Check Operation of Final Scan Abort Buttons	X	X	X	X
CTPM-1608	System	Update Site Log	X	X	X	X
Total time w/o camera (hours)			1.75	2.1	4.2	2.1

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Image Quality

Use the Image Quality menu to access the following tools to evaluate the site's system performance:

Image Quality Menu

Menu Item	What It Does
DD File Analysis	Analyze diagnostic data files
Image Analysis	Analyze system image quality
Install SMPTE image	Install SMPTE image Install the SMPTE (Society of Motion Picture and Television Engineers) test pattern into the ImageWorks Browser as Exam 1000. Use this test pattern to set up display monitors and filming cameras. This program also installs sample QA (Quality Assurance) images that show the typical image performance capability.
Scan Analysis	Analyze and plot scan data
Shell	Open a window that enables you to enter IRIX and UNIX commands, scripts that perform several commands or start programs. Press Alt+F12 to close the shell.



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Slip Ring Communication

A normal ring has a "patina" of brush material on the surface of the brass ring. This patina is about 3 mils thick and self renewing. It is the natural lubricant and completely normal. Do not attempt to remove this as future problems will arise and create a cycle of repeating failures on excess dust production.

Micro spots are acceptable provided they do not exceed the specifications detailed in the service manual.

1. Follow the slip ring vacuuming procedures outlined in the Service Manual. Vacuuming the slip rings will insure good communication.



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How to Safely Remove Covers

Warning Be aware that potential for injury exists when covers are removed and power is "ON."

Table Power

Removing the table side covers will activate the **safety interference** circuitry. A jumper plug is provided under the side covers to enable the drive circuits for service purposes. Once the jumper is in, use the gantry control RESET switch to re-enable table/gantry drives.

Gantry Warnings

DANGER The gantry rotates when you turn on the Alignment Lights, so don't do that with gantry side covers removed unless you first check that no one could be hurt.

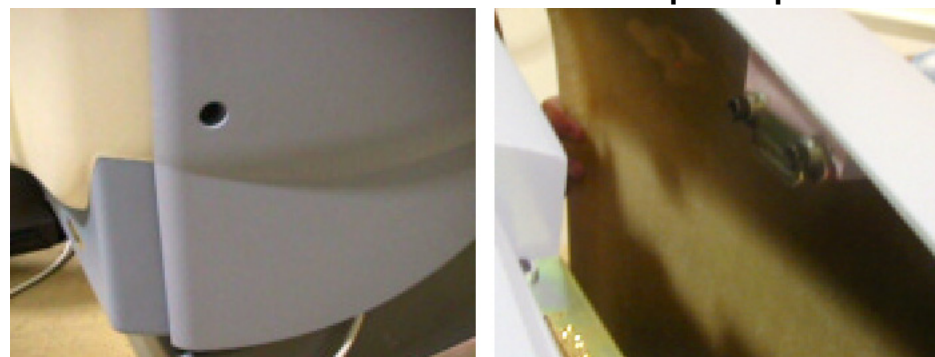
DANGER The most electrically dangerous area in the gantry is the exposed slip ring platter. Lock-Out and Tag-Out whenever you remove any of these gantry covers.

Caution When the gantry is rotating, objects could be ejected if not properly fastened. It's crucial that all hardware be fastened (torqued) to their proper specification.

Gantry Side Covers

1. Lower the table to the home (lowest) position.
2. The gantry cover on the right side **must be removed first** so you can turn OFF all power at the **STC backplane**.
3. Use an 8 mm hex wrench to unlatch the right side cover from the front cover.
4. Remove the right side cover by lifting it to release the two latches located on the top edge of the cover.

Side Latch on Front and Cover Clasp on Top



5. **Turn OFF** all three switches on the **STC backplane**.

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How to Safely Remove Covers, *continued*

Gantry Top Cover

1. Turn **OFF** all three switches at the **STC backplane**.
2. Remove the **side cover**.
3. Take hold of the top cover edge closest to the side cover and tilt upwards.
4. Slide the cover down 75 mm (2.95 inches). This allows the cover's tab to disengage from the mounting bracket.
5. Lift the cover clear.

Caution

Misuse can result in personal injury or damage to covers or other facility items.

The front cover dollies are a folding design for easy storage. Assembly can be done in 30 seconds.

Rotating arms on the stand are supposed to be stiff. If they fall freely, tighten the tensioning nuts. Loose rotating arms will reduce the stability of the dollies when supporting the front cover. Do not lubricate.

Front Cover Dolly Set-up

Front and rear cover removal and installation can be safely accomplished by one person using the dollies provided with the system.

Caution

Failure to use dollies will increase the likelihood of damaging the covers.

Do not lean covers against walls or lay them down.

Do not use dollies on uneven surfaces such as steps or elevator thresholds. The dollies are designed to be used on flat level floors within the scanning suite only.



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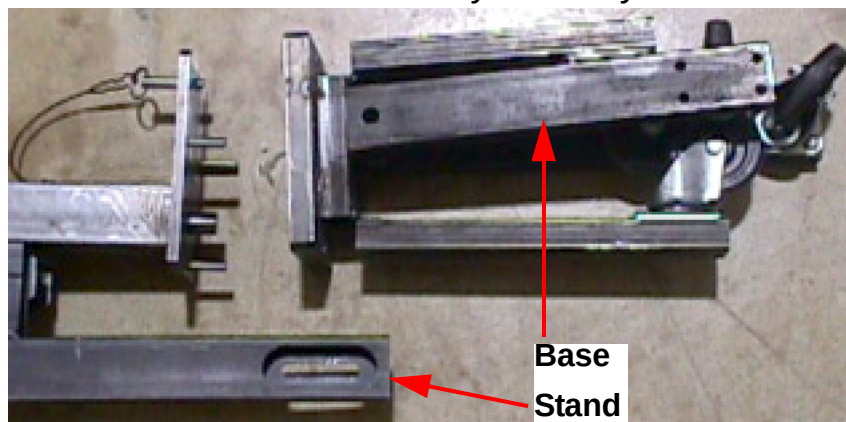
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How to Safely Remove Covers, *continued*

Front Cover Dolly Set-up, *continued*

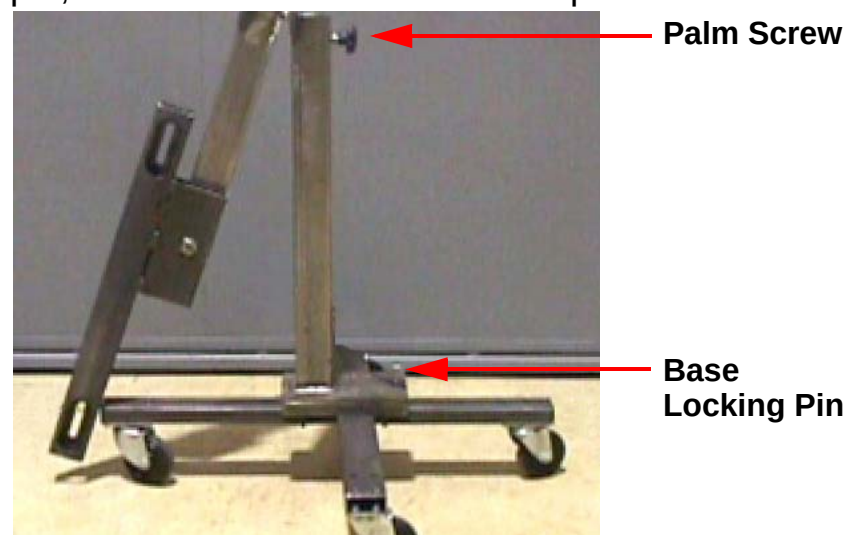
6. Arrange Dolly sections for assembly. The Base and Stand can be assembled only one way.



The Stand has a large stud and three smaller studs which engage the Base assembly.

- The large stud provides stability.
 - The three small studs engage and prevent the base arms from folding under if accidentally tipped.
 - The locking pin engages the fourth base arm and prevents accidental separation when assembled.
7. Unfold the Base so it looks like a cross and place it on flat surface.

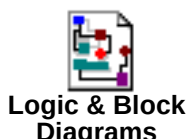
8. Install the Stand into the Base, insert base locking pin, unfold Stand and secure with palm screw.



9. Assemble the second dolly. The Stand portion of the dollies are side specific. The "Right" dolly has the locking pin for cover rotation. The "Left" dolly does not have a cover rotation locking pin.

Both dolly stands should have an "R" or an "L" indicating right or left. The base assemblies are interchangeable.

Continued on next page





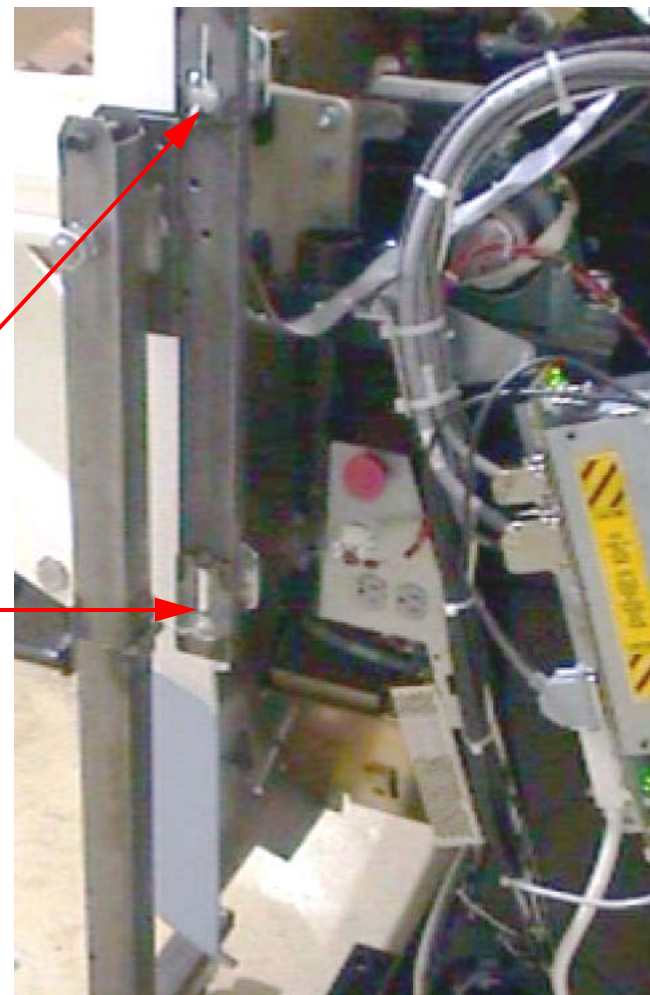
How to Safely Remove Covers, *continued*

Gantry Front Cover

1. Lower the table all the way.
2. Remove gantry **side** and **top** covers.
3. **Turn OFF** all three switches at the **STC backplane**.
4. Assemble both **front cover dollies**.
5. Position the Right dolly to the right side.
6. Tighten two shoulder bolts to the gantry securely. This will make it easier to re-install the cover.
7. Attach dolly to the shoulder bolts and secure assembly with two wing nuts.
8. Attach the Left dolly to the left side.
9. Detach front cover J3 and J2 and front cover BKHD J1 cables.

Shoulder Bolt
and Wing Nut

Shoulder Bolt
and Wing Nut



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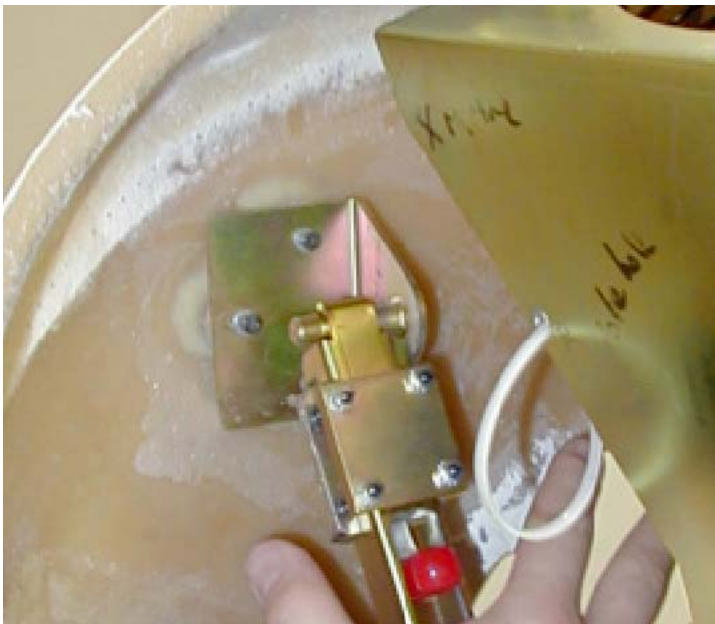


How to Safely Remove Covers, *continued*

Gantry Front Cover, *continued*

10. Disengage upper and lower cantrell brackets on both sides of the cover. Using steady but firm pressure, lift each of the lower cantrell brackets from their associated retainers.
11. Disengage the locking mechanism on the upper cantrell brackets by using your thumb to slide the trigger (red lever) back. This will release the locking
12. Disengage the rubber retaining straps on both sides. You may find it helpful to lift up on the cover to align the stud while attaching the rubber retaining straps.

Cantrell Brackets



11. Disengage the locking mechanism on the upper cantrell brackets by using your thumb to slide the trigger (red lever) back. This will release the locking

Rubber Retaining Straps



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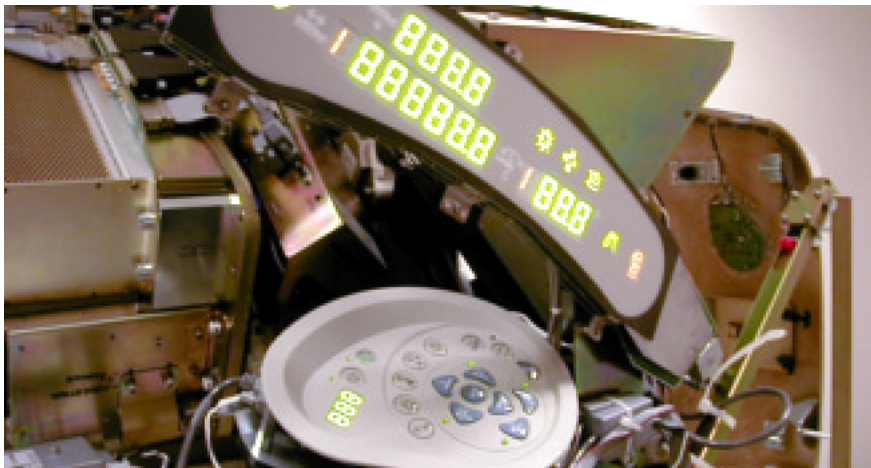
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How to Safely Remove Covers, *continued*

Gantry Front Cover, *continued*

13. Pull **locking pin** (red arrow) and rotate cover away from gantry.
14. Place **locking pin** in a dolly perforation.
15. Rotate the front cover more and roll it back a little.
16. Remove the **gantry display** and **control panel** and place them in their **service location**. Press on each ball stud until the panel is released. Keep one hand on the control panel to keep it from dropping.



17. Continue to roll the front cover over the table to a safe location where you could rotate it to be vertical and upside down.
18. Connect cable to terminator located on the cantrell arm. There are three cables. The ribbon cable is not used in the Service configuration. The other two will only fit in the terminator or the control panel, not both.

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Clean Plenum

Required Tools

HEPA Vacuum Cleaner, Part # 46-297933P1

DANGER PLEASE USE AND FOLLOW LOCK OUT/TAG OUT PROCEDURES.

Procedure

1. Move table to the lowest elevation.
2. Turn off facility power to PDU.
3. Remove gantry **side covers**.
4. Lift off gantry **top cover**.
5. Remove Mylar window.
6. Remove gantry **front cover**.
7. Rotate gantry until the DAS/Duct Fan assembly is within serviceable reach.
8. Engage **rotational lock**.
9. Disconnect fan harness.
10. Vacuum dust from fan "finger" guards.
11. Remove filters from plenum and vacuum dust from each filter. It's normal for the right chassis filter to be more dusty.
12. Inspect plenum for dust and remove and clean if necessary.
13. Replace filter. Note the DIRECTION OF AIRFLOW stamped on filter frame. Filter foam faces towards the fan.
14. Restore fan harness connection.
15. Reassemble gantry covers and Mylar window.
16. Restore power.



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SOFTWARE AND DIAGNOSTICS

Boot Up Sequence

Common Service Desktop(CSD)

Common Service Desktop

Overview

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Diagnostics Tab

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Data Acquisition

PMC DIP Board

DAS Data Processing VxWorks Diags

GE DIP Diags

VxWorks

Image Generation

Recon Interface Processor (RIP)

SCSI Control

Scan Data SCSI

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X-ray Generation: Diagnostic Data Collection

X-ray Functional Screen

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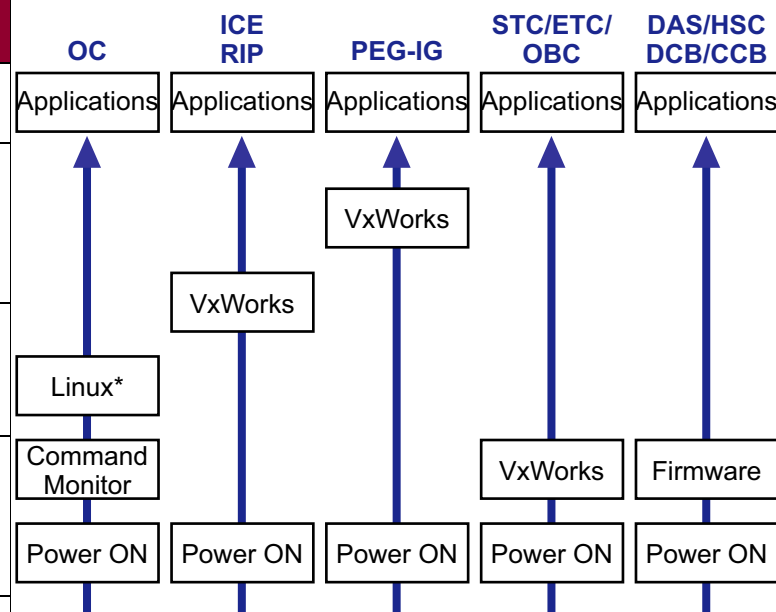


Notes



Boot Up Sequence

OC	ICE	PEG-IG	STC/ETC/OBC	DCB/CCB
1. Power up diagnostics.	1. Power up diagnostics.	1. Power up.	1. Power up diagnostics.	1. Power up diagnostics.
2. Boot Linux (or IRIX) from OC disk.	2. Wait for serial input activity.	2. Wait for ICE boot-up.	2. Attach ROM resident VxWorks. Wait for input on LAN.	2. Wait for input on CAN bus.
3. Start CT Applications software.	3. Load VxWorks and applications software via the LAN.			
	4. Find and initialize the Pegasus (PEG) IG board.	3. Boot VxWorks off of ICE.		
4. Start up Artesyn controllers via the Table/Gantry LAN.		4. Applications load through the ICE.	3. Applications firmware down-loaded and started.	3. OBC initializes via CAN bus.
Applications start-up complete.				



* Some systems may use SGI Octane2 computers with the IRIX Operating System



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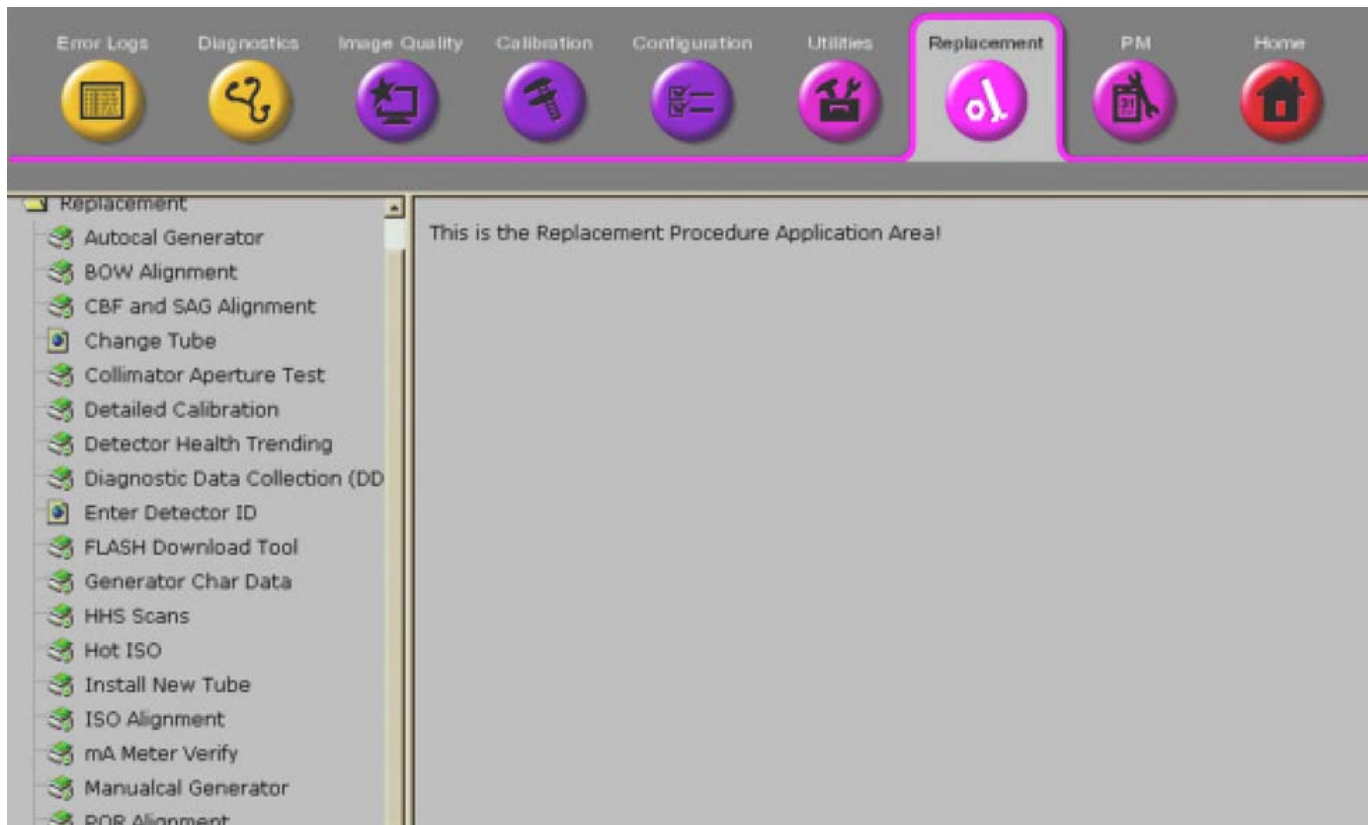


Common Service Desktop

Overview

The Common Service Desktop provides buttons, or **tabs**, that allow access to data compiled about the system. Use these tabs to access specific system information,

which may prove vital in properly diagnosing operating problems.



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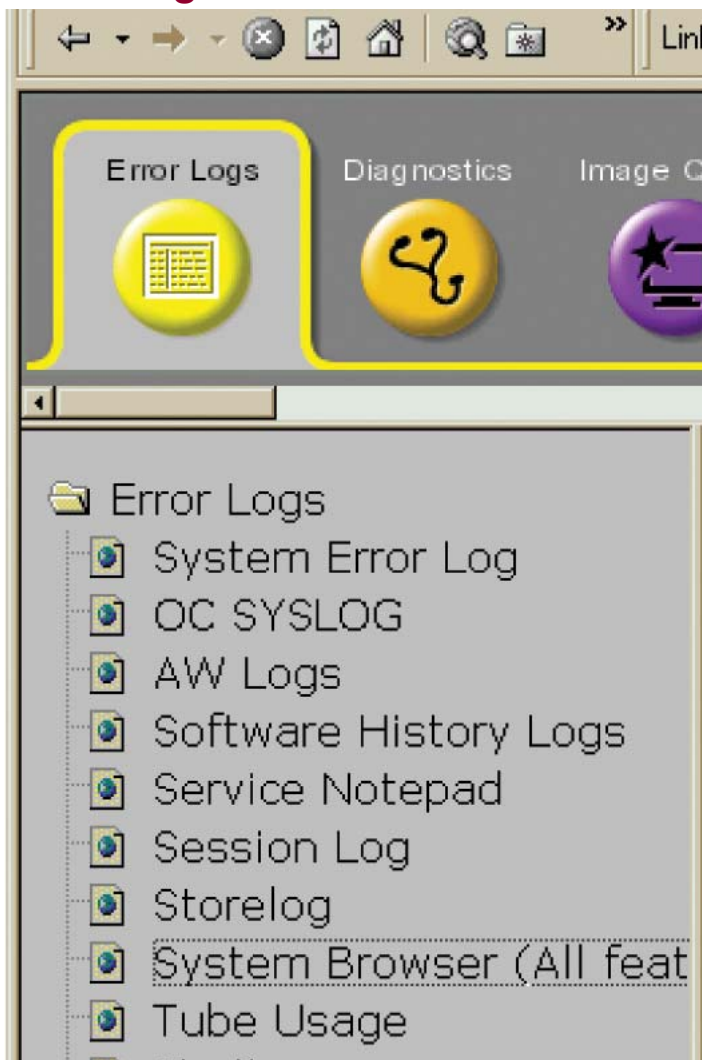


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Common Service Desktop, *continued*

Error Logs Tab



Select **System Browser** on the Error Logs tab to reach these **advanced log lists**:

- Searchable **System Error Log**
- SYSLOG OC
- IOS LOGS
- Tube Usage
- Run Time Statistics
- OC Info
- Config Files
- Software Health Page
- Scan Usage
- **VxWorks** Logs

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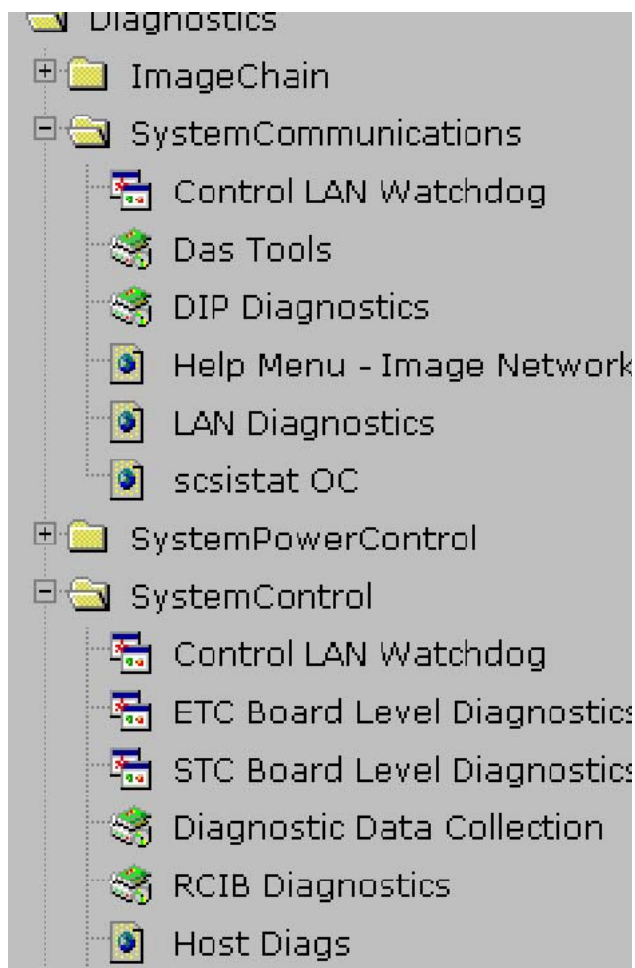
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Common Service Desktop, *continued*

Diagnostics Tab

System Communications and System Control



Diagnostic	What It Tests
Control LAN WD	OC, RIP, STC, OBC, ETC
DAS Tools	DAS and detector hardware
Diag Data Coll	Raw data acquisition control
DIP Diagnostics	DAS data receipt at the console
LAN	Scanner Network
ETC	Common Service Desktop, continued controls
Host Diags	Console CPU
RCIB	OBC, DCB, CCB fault comm
scsistat OC	Ability to access OC hard drives
STC	Axial Rotation Controls, Slip Ring Communication

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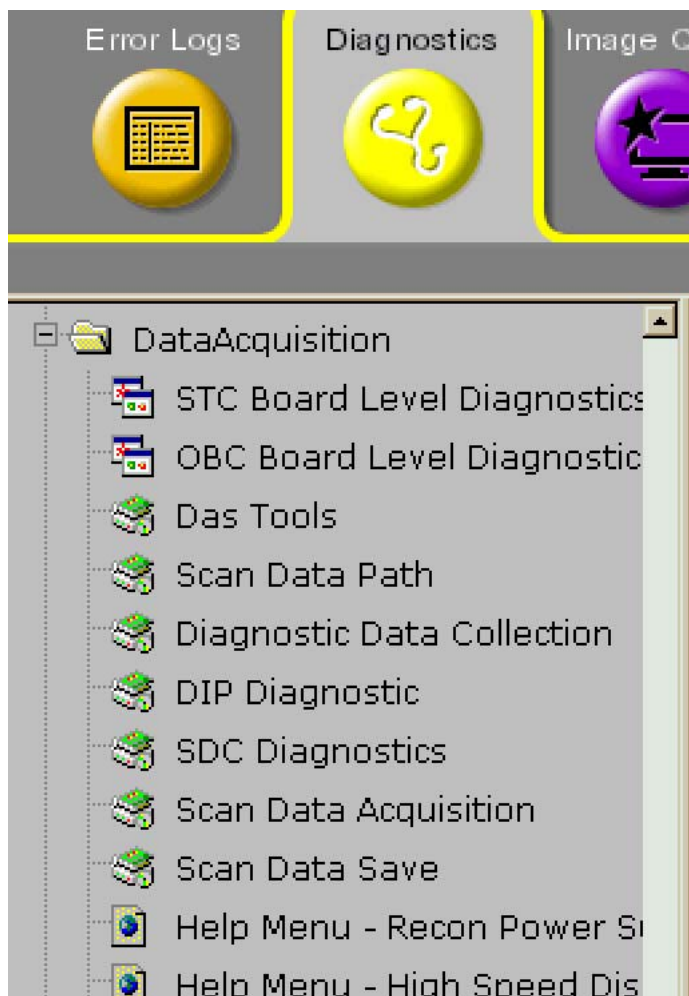
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Diagnostics Tab, *continued*

Data Acquisition



Note To see just the most recent DAS Tools error log as it updates, bring up a UNIX window and "tail" the log file on the left monitor. This makes looking at the errors more convenient.

1. Select **UNIX window** on Service Desktop
2. Enter: **cd/usr/g/system/state.**
3. Enter: **tail -f -n 50 ssw.dastools.hist.**
4. To exit the UNIX window, type: CTRL+C

Diagnostic	What It Tests
DAS Tools	DAS and detector hardware
Scan Data Path	DAS Digital HW to RIP scan data disk
Diag Data Coll	Raw data acquisition control
DIP	DAS data receipt at the console
SDC	Scan Data Correction Bd in ICE
STC	Slip Ring Comm and Axial Control
OBC	X-ray Generation Control

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Common Service Desktop, *continued*

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Data Acquisition, *continued*

PMC DIP Board

The PMC DAS Interface Processor (DIP) Board is attached (piggy-backed) to the RIP (Motorola PPC) Board in the VME Chassis. Because this board is intimately related to the RIP Board, it is advisable to run the tests identified for both in order to gain a full understanding of the nature of a problem.

Common Service Desktop, continued provides the interface for the PMC DIP Board and DAS data transfer from the gantry to the ICE BOX.

Shut down Applications before using VxWorks.

- First you should test whether the **DIP** Board is seen by VxWorks, use a VxWorks command.
- Once we know the DIP Board is seen by the low level software, use Service Desktop DIP diagnostics.

There are two types of tests (quick or extensive) and two sources of data (internal or external loopback). The external loop requires a fiber-optic cable to provide data. You have to provide the fiber-optic cable (2264931) to jumper the fiber-optic jacks (J2 and J3) on the DIP board.

DAS Data Processing VxWorks Diags

1. From the Service Desktop, select **Toolbox > Utilities > Application Shutdown**.
2. Open a UNIX Shell and enter the following to create a serial connection to the ICE BOX:
ice stop start connect.
3. Check whether the PCI Devices are seen, enter:
pciDeviceShow.

If the information identifies the DIP Board, this shows it is recognized by VxWorks.
4. To exit VxWorks and return to the Octane ctuser prompt, enter: **~**.

At the Octane ctuser prompt, restart Apps, enter: **st.**

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Common Service Desktop, *continued*

Diagnostics Tab, *continued*

Data Acquisition, *continued*

PMC DIP Board, *continued*

GE DIP Diags

Perform the following steps to launch the DIP diagnostic:

1. Click on the **Service Desktop > Diagnostics** (icon).
Select **Data Acquisition** (folder). Select **DIP Diagnostics**.

A diagnostic GUI will appear to enable you to select the DIP testing and loopback modes.

If you select *external* for the loopback mode, a pop-up reminding you of the need to install an external fiber-optic jumper will appear.



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VxWorks

Many vendors of UNIX controlled hardware use VxWorks software for developing software to operate or test their hardware. The computer boards supplied by Motorola, Mercury and Artesyn are examples.

Status and problems detected by VxWorks during **bootup** or diagnostic runs are logged. Refer to **System Logs**.

Testing these parts involved with data acquisition, image generation and data storage require Shutting Down Applications and using vendor provided tests. They also require restarting System Applications when done.

The Reconstruction Interface Processor (**RIP**) Board from Motorola has two PCI Mezzanine Cards (**PMC's**), piggybacked on the RIP board. These two PMC's are the SBS SCSI Controller Card and the GE DIP Card.

Outside the console, the ETC, STC and OBC have Artesyn computer boards that also use VxWorks.

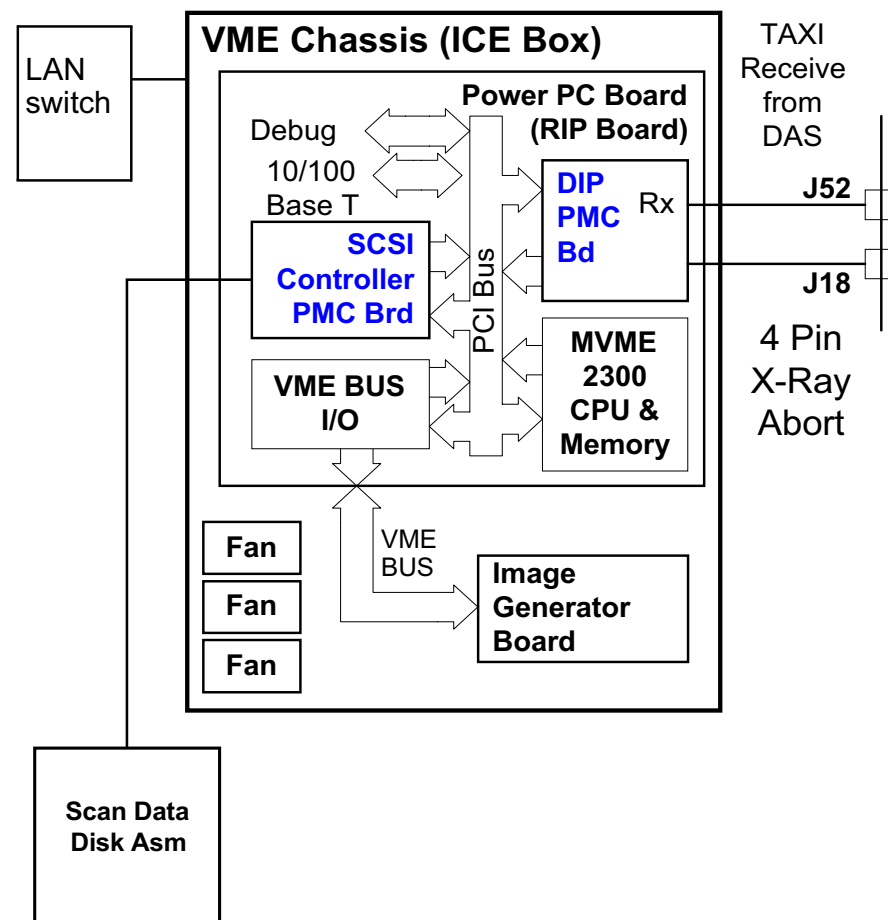
Refer to **RIP Diagnostics**

Refer to **Diagnostics, SCSI Control**.

Refer to **DIP Diagnostics**

Refer to **System Comm Drawing**

Refer to **Console Drawing**



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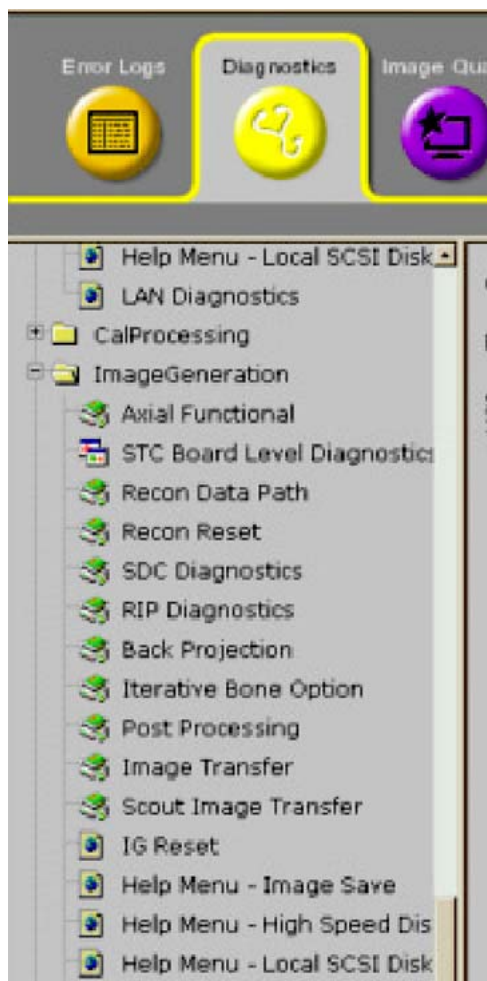
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Common Service Desktop, *continued*

Diagnostics Tab, *continued*

Image Generation



The RIP Board is a **Motorola Power PC**. It serves as the bus master. When rebooted, it reboots the IG and SDC. To check if the RIP is alive, open a UNIX window and enter: **ping ice**.

Diagnostic	What It Tests
Axial Functional	STC ACB DAS triggers depend on azimuth accuracy
Back Projection	How to Set Up InSite SDC to Image Generator (IG) in Image Chain Engine (ICE)
Iterative Bone Option	IG and RIP
LAN	Scanner control network
Post Processing	IG and RIP
RIP	Motorola Power PC
Image Transfer	OC ability to get image from ICE and put on OC Image Disk
Scout Image Transfer	IG not involved in transfer
IG Reset	Image Generation board

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Common Service Desktop, *continued*

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Recon Interface Processor (RIP)

The Reconstruction Interface Processor (RIP), or **Motorola Power PC board**, has built-in diagnostic routines that are run by either of two scripts: **diags** and **fulldiags**, from a UNIX Shell window.

The **diags** test is a quick board level diagnostic test that takes about one minute. The **fulldiags** test performs more extensive, lower level testing of the board's RAM and VME3 channels; it takes about six minutes.

These scripts are located in the **/usr/g/lice/bin** directory of the OC-Octane (O2-less consoles).

Starting the diagnostic will reboot the Motorola Board into a diagnostics mode. When the tests are completed, the script will automatically return to the Octane **ctuser** prompt at the **/usr/g/lice/bin** directory level.

Because the RIP Board is rebooted when these tests are launched, you must shut down Applications before running either of these tests.

Procedure

1. From the Service Desktop, select **Toolbox > Utilities > Application Shutdown**.

2. Open a UNIX Shell and enter: **rsh sbc**.
3. At the ctuser prompt, enter: **cd/usr/g/lice/bin**.
4. Launch either the short or long RIP diagnostic test, enter: **ice diags** -or- **ice fulldiags**.

Each test starts at the VME PPC1-Diag> prompt level. Verify that all tests listed have either a PASSED or BYPASS indication.
5. When the test routines complete, close the Remote Shell. At the ctuser prompt, enter: **exit**.

Restart Applications, enter: **st**.

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Diagnostics Tab, *continued*

SCSI Control

The PMC SCSI Controller Board is attached (piggyback) to the RIP (Motorola PPC) Board in the **VME Chassis**. Troubleshooting reconstruction, scan data control and storage problems involves testing several items.

VxWorks provides the interface for the PMC SCSI Controller Board and scan data transfer to the High Speed Disk. **Shut down applications before using VxWorks.**

- First you should test whether the **SCSI Controller** Board is seen by VxWorks, use a VxWorks command.
- Once we know the SCSI Board is seen, use VxWorks and the SCSI Controller to interrogate the **Scan Disk**.
- Finally, use a GE diagnostic script to exercise the High Speed Disk (writing and reading data) to ensure the **entire subsystem is working** correctly.

Scan Data SCSI

1. From the Service Desktop, select **Toolbox > Utilities > Application Shutdown**.
2. Open a UNIX Shell and enter the following to create a serial connection to the ICE BOX: **ice stop start connect**.
3. Check whether the PCI Devices are seen, enter: **pciDeviceShow**.

If the information identifies the SCSI Controller Board, this shows it is recognized by VxWorks.

4. Check whether the Scan Data Disk is seen, enter: **scsiShow**.

If the information identifies the SCSI Devices of the ICE BOX, they are seen by the Controller and recognized by VxWorks..

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SCSI Control, *continued*

Scan Data SCSI, *continued*

5. To test the RIP Board, SCSI Controller, and Scan Data Disk (HSD), to see if the SCSI subsystem is working reliably, use this script; enter:

test_high_speed.

If any errors are detected, they will be displayed on the terminal screen. If the test runs to completion without error, you can be assured that the High Speed Disk, SCSI Controller, and RIP Board are capable of handling scan data save and restore

6. To exit VxWorks and return to the Octane ctuser prompt, enter: **~.**
7. At the Octane ctuser prompt, restart Apps, enter: **st.**

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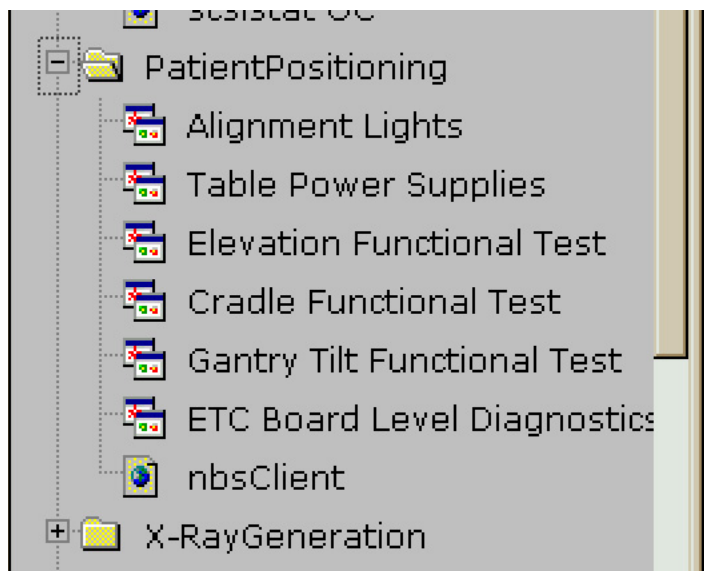
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Diagnostics Tab, *continued*

Patient Positioning



Diagnostic	What It Tests
Alignment Lights	OBC Gentry I/O
Elevation	ETC control and elevation hardware
Cradle	ETC control and cradle drive
Gantry Tilt	ETC control and tilt drive
ETC	Patient Positioning controls
nbsClient	STC, OBC, ETC control status

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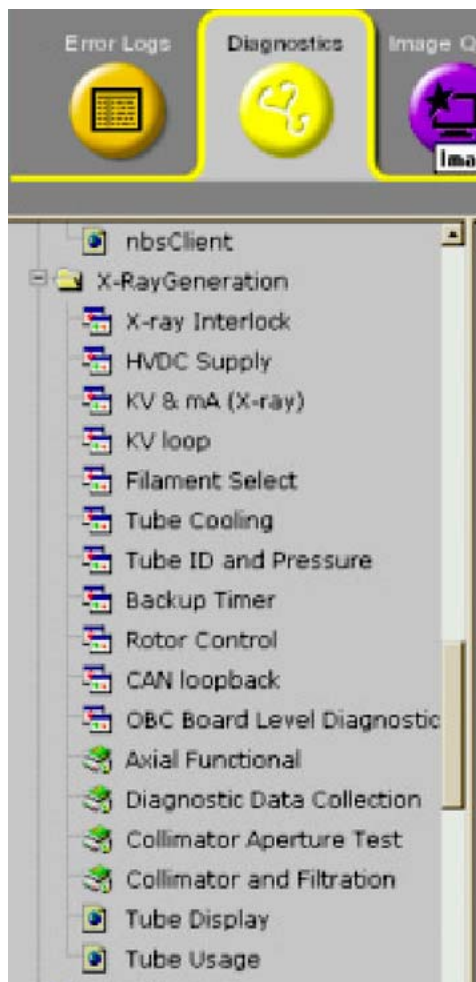
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Diagnostics Tab, *continued*

X-ray Generation



Diagnostic	What It Does
Collimation and Filtration	Verifies the collimator electro-mechanical elements can operate in a repeatable manner. This test commands the collimator electronics to move the cams and beam shaping filter to specific locations in their operating range checking the actual encoder counts to the expected position counts.
Collimator Aperture Test	Verifies the collimator control circuitry can correctly control the collimator cams independently of the detector z-channels.
Flash Download Tool	Updates firmware parameters after you upgrade software or replace the CCB , TCB , TNC , DCB , STC , ETC , and OBC . It uses files stored on the ICE disk. Run the tool if an ATTENTION window or error message indicates a file version mismatch.
kV Loop	Can be used to high pot system. The X-ray tube or HV cables can be replaced by a High Voltage Bleeder, thus eliminating (or confirming) the tube or HV cables as a cause.
mA & kV (X-ray)	Shows kV Board model.
Rotor Control	Checks the HCAN .

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X-ray Generation: Diagnostic Data Collection

Diagnostic Data Collection (DDC) can create scan files to help you troubleshoot the DAS and Detector.

Scan Analysis allows you to visualize and analyze raw data from DDC, patient scans, or Calibration. It calculates and graphs certain aspects of the data to give you insights into what kind of hardware problem may be causing poor system performance. Engineering uses these tools to also determine what software problems may exist in X-ray generation, beam shaping, detection, or data acquisition.

DDC Screen

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Diagnostics Tab, *continued*

X-ray Functional Screen

diagParentForm

X-ray Functional

X-ray Functional Test Selection

X-Ray Test Type:

kV Selection: Rotor Speed:

mA: Filter Type:

Min. ISD: Aperture:

Exposures: mA Loop:

Exp Time(sec): Focal Spot:

OBC Service Input:

Error Params

Max Errors:

Max Error Processing:

Log To File:

Min Results Update Rate:

Gantry Params

Gantry Enable:

Gantry Speed:

Tube Position:

X-ray Functional Results

High voltage status

No.	Device	Average Value	Selected Value	Last Sample
1.	Total KV:	119.9KV	120.0KV	120.0KV
2.	Cathode KV:	59.8KV	60.0KV	59.7KV
3.	Anode KV:	60.1KV	60.0KV	59.9KV
4.	Cathode mA:	201.4mA	200.0mA	201.6mA
5.	Anode mA:	185.3mA	200.0mA	185.3mA
6.	Cathode inverter current:	32.000A	---	31.925A
7.	Anode inverter current:	29.550A	---	29.550A
8.	Approx. KV inverter frequency (VCNT):	(2.32V)	---	24.6KHz
9.	Cathode inverter duty cycle:	87%	---	87%
10.	Anode inverter duty cycle:	75%	---	75%
11.	Filament current:	5.812A	5.737A	5.810A
12.	HVDC Bus voltage:	633V	---	633V
13.	Exposure duration:	---	1000mS	1007mS
14.	Exposure number:	---	100	11

Press the start scan button when ready.
Braking rotor...

Status

Testing has started.
Testing has completed with 3 errors.
Testing has started.

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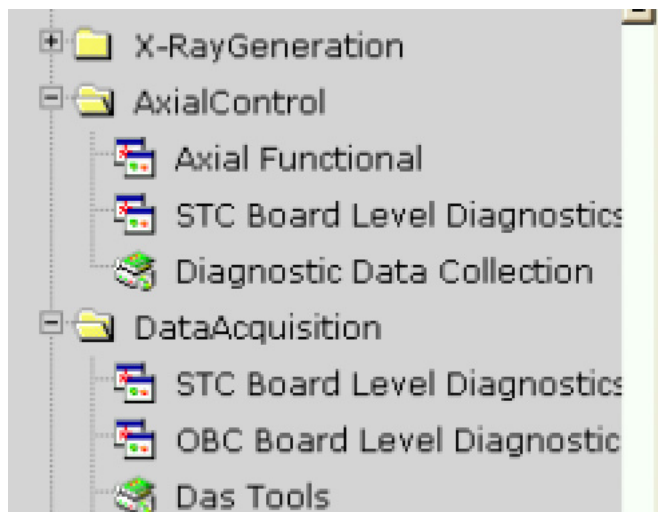
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Axial Control



Testing Gantry Rotation

Test gantry rotation with the **Axial Functional** diagnostic.

1. Select: **Service Desktop > Diagnostics > Axial Control > Axial Functional**.
2. Perform one pass each "Open Loop Rotation" at each scan speed. Verify no errors.
3. Perform one pass each "Closed Loop Rotation" at each scan speed. Verify no errors.
4. Perform several passes of "Go to Position" choosing different angles. Verify no errors.

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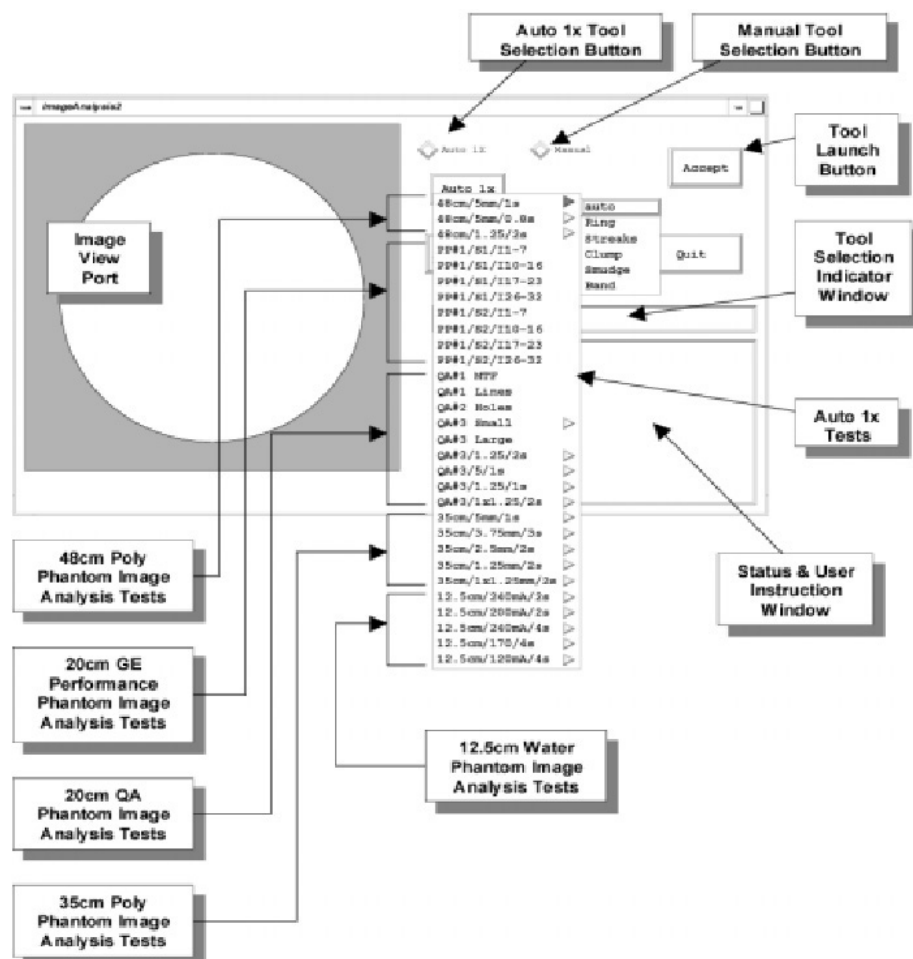


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Image Quality Tab

Image Analysis



The new Image Analysis Tool helps you easily evaluate **image quality**. The tool automatically sets and measures ROI's based on the tool or test and images you request. It can automatically compare the results against design specifications and indicate whether a standard Service Image Series passes or fails.

The tool has Manual and Auto 1x measurement sub-tools.

The **Manual** sub-tools set unique measurement ROI's based on one image or several within a series you select from the Service Browser Window. Results are displayed in a Report Window without comparison to specifications or an indication of Pass/Fail.

The **Auto** 1x sub-tests run appropriate combinations of the Manual sub-tools on a standard series of images acquired using the Service Image Series protocols. You must select the correct series from the Service Browser Window. Results of the Manual sub-tool tests are displayed in a Report Window with comparisons to specifications; these tests do indicate Pass or Fail.

The **security key** is required to select the Auto 1x Image Analysis Tool. Image Analysis is located on the Service Desktop under Image Quality..

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Common Service Desktop, *continued*

Image Quality Tab, *continued*

Image Analysis, *continued*

Available Manual Sub-tools Used

- **Clump**
- **Center Smudge**
- **Center Artifact**
- **Center Spot and Max Pixel**
- **Ring**
- **Band**
- **Streak**

Means and Standard Deviation averages four fixed outer ROI boxes and center ROI box of an image displayed in the tool image view-port. The size and position of the ROI boxes depends on the phantom type and SFOV.

Series Means: Reports the row averages of the ROI values for all images in a series you select from the Service Browser. It also reports the row average difference of the means of the outside and inside ROI boxes ($AvX_o - AvX_c$).

QA#1: computes the MTF and contrast scale for each image in a four image 20 cm QA Phantom series.

QA#2: designed for the 3rd eight-image series of the 20 cm QA Phantom series.

QA#3: designed for the 4th series of the 20 cm QA Series.

Manual ROI: places a 14cmx14cm ROI box at ISO on the image. Adjust the size and position of the ROI box using the left mouse button. Click the ACCEPT button to launch an analysis of the region's means and standard deviation.

Image Resolution: positions fixed ROI's on the image, reports the 50% and 10% MTF factors and calculates a four-image average of these values. The tool is designed for the two, 32 image, GE Performance Phantom series. The second series is a Edge Recon of the first.

GE Perf#1: positions two 17x17 pixel boxes, at (204,180) and other at (221, 102) and reports means, standard deviation and contrast scale.

GE Perf#2: positions two fixed boxes and performs the mean subtraction between these ROI's as contrast factor.

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Image Analysis, *continued*

Available Manual Sub-tools Used, *continued*

Visible Lines: designed for a four image series acquired using the scan parameters of the second Bone Recon of Series 1, 20 cm QA Phantom QA Image Series listed on the Miscellaneous Menu. Enter the **letter code** that represents the number of six line pair resolution patterns you see in the image.

Visible Holes: designed for the third 20 cm QA Phantom Image Series Scan Protocol under Miscellaneous Menu.

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Replacement Tab

The Replacement Tab contains the procedures for replacing components.



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Common Service Desktop, *continued*

Home Tab

The screen below shows the information typically displayed on the Home Tab of the Common Service

Desktop. Use this tab for an overall idea of system information and current system status.

System Information		
Item	Information	Status
Facility	G.E. Medical Systems	-
Suite Name	baya	-
System Type	LightSpeed Plus	-
System ID	0000CTBAYA	-
Service ID	CT_BAYA	-
IP Address	3.7.52.110	Interface: ef1
Access Level	Non-proprietary	Valid
Software Installation Date	10/10/2000	-
DASM Camera	"Laser Camera"	Installed
DICOM Network Cameras	9 Camera(s)	Installed
Installed Tube	15	-
Tube Install Date	Tue Oct 19 11:49:11 1999	Current mAs: 68411964.7
Data Acquisition System	SDAS	-
Power Distribution Unit	CT_Compact	-

System Health Information		
Item	Information	Status
Gantry Revolutions	1349424	Check PM Schedule
Spit Rate Exceeded Errors	N/A	N/A
Slipping Communication Bit Error Rate	1.09602e-11	Marginal
DIP Errors	2.50e-14	OK

Item	Past Five Days			Today
	Avg.	Min.	Max.	
Tube Spits / 10 Scan Seconds	0.03	0.00	0.08	Spits: 2 , Scan Seconds: 259
Slipping Brush Disconnects	0.00	0	0	0

Temperatures	Past Five Days			Today		
	Avg.	Min.	Max.	Avg.	Min.	Max.
Detector Temperatures	36.00	35.29	36.32	35.49	35.36	36.12
DAS Converter Temperatures	35.97	29.50	43.50	35.86	29.50	43.50
Gantry Ambient Temperatures	35.82	35.50	36.46	35.57	35.50	36.28
OBC Temperatures	28.21	27.05	31.42	27.54	27.07	31.42

Current System Status		
Item	Information	Status
System Date	Tue, Oct 24, 2000	-
System Time	13:30:28 CDT	-
Application Software	9.3_2.8.21_H2M3_H1.3M4	running
Next Patient Exam	140	-
Recon Status	Idle	-
Archive Status	Failed save exam. Paused 139	-
Network Status	Network Status	-
Filming Status	AutoFilm is paused	-

Web Applications Auto Update ☐ Update

Frequency (sec)



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How to Set Up InSite

1. Put a **Class-C+M** or **-DLS** CD in the drive.
 2. Open a UNIX shell on the system. Change to root user. Enter: **su -** and then the root password.
 3. To **install InSite software**, enter: **install.classC** (or **M**) or **install.DLS2003**, depending on the CD being used.
 4. Pay close attention for error messages during the load; they may indicate an unsuccessful load.
 5. If everything loads normally, continue with **InSite Configuration**. To start, enter: **ipadmin config**.
 6. Click **Accept** three times to enter.
 7. Click **Modem Tab** to configure the modem (if needed).
 8. Use the default settings for **Serial Port Name** and **Speed**; do not change unless troubleshooting.
 9. Select a **Dial-out prefix**, **Tone** or **Pulse**, **modem type**, and **Country** in which the site is located.
 10. Click **APPLY** to configure the serial port, set modem registers, and store the values in modem's NVRAM.
 11. Find the host list, the **/etc/hosts** file and the "loghost" entry to verify site **IP Address** is correct.
 12. Click the **InSite Checkout Tab**.
 13. Select either **CHECKOUT NOW**, **LATER** or **AUTO**. If **NOW**, call the local OnLine Center (OLC).
- Important! DO NOT press OK until told to by the OLC Engineer or you see the modem's OH (Off Hook) LED has gone out, meaning the OLC connection is idle, the Dial Out test from your console can be safely done.
- NOW** means the contacted OLC Engineer will start a two part script that establishes a PPP connection and downloads the **latest iLinq code**. If you press OK too soon at the console, the download will be interrupted, the process unsuccessful.
- LATER** disables the serial port for the modem. All previously saved information is retained. To check out later, restart **ipadmin config** at the **su level**, select the APPLY button on the Modem Tab, then request CHECKOUT NOW or AUTO.
- AUTO** can be used if a **sclink.cfg** file exists, meaning the site has completed a CHECKOUT or the file has been restored. AUTO will queue the task with the AutoSC to dial back. The Configuration Tool will wait ten minutes for the config file **sclink.cfg** to appear. The AutoSC will usually call back in 5-6 minutes.

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Key Software Commands

Software	Command
DASM (Laser Cam) status	showdasm
Disk Usage	df
OC/SBC active processes	ps -aef
OC/SBC hardware inventory	hinv
OC/SBC network configuration	ifconfig
OC/SBC network sockets	netstat -ian
OC/SBC route table	netstat -r
SCSI Bus errors	/var/adm/SYSLOG
SCSI Bus status	scsistat
System INFO	cat/usr/g/config/INFO
System software versions	showprods



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Acronyms

Gantry

Axial Power

Breathing Aid

Gantry Display

Gantry Keypad

Gantry Rotation

Axial Control Communication

Axial Control Interface Bus (ACIB)

Slip Ring Communication

DAS Data

Control Data

RCIB Faults

X-ray Generation

High Voltage Generation

Tube Filament

Tube Rotor

Slip Rings

X-ray Warning Light Control

DAS Detector

DAS Detector

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Z-Axis Cell Summation

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Image Generation Overview

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Post Processing

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Axial Drive Power

Rotating Gantry (OBC) Power

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Acronyms

Acronym	Translation
ACIB	Axial Control Interface Bus
AMD	Axial Motor Drive is a module
CAN	Controller Area Network
CCB	Collimator Control Board
DAS	Data Acquisition System
DCB	DAS Control Board in the S-DAS controls detector temperature
DDC	Diagnostic Data Collection tests DAS
DIP	DAS Interface Processor
ETC	Electronic Table Controller for patient setup
HCB	HEMRC Control Board
HEMRC	High Efficiency Motor Rotor Controller
HIF	HEMRC Interface Board
HSC	Helios Smart Collimator
ICE	VME Image Chain Engine
IG	Image Generator (recon software)
ISD	InterScan Delay is seconds between X-ray exposures

Acronym	Translation
OBC	On-Board Computer for scan control
OC	Operator Console
PDU	Power Distribution Unit
PMC	PCI Mezzanine Cards on SCSI and DIP
RCIB	Rotating Controller Interface Bus
RIP	Recon Interface Processor (Motorola PC)
SBC	Single Board Computer is not present in H2
SCIM	Scan Control Intercom Module
SDC	Scan Data Corrections (Mercury computer)
SRU	Scan Recon Unit for image generation
STC	Stationary Controller for gantry functions
TCB	Table Control Board
TNC	Table Network Controller
UPS	Uninterrupted Power Supply



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System Safety Guards and Resets

Emergency Stops (E-Stops) buttons are located on the console, gantry and PDU. Do not use the SCAN STOP buttons on the console or gantry keypads if you need to stop gantry rotation immediately. **Use an E-Stop.** The SCAN STOP switch only stops X-ray; it doesn't stop gantry rotation.

Gantry keypads located on four sides of the gantry allow manual control of patient positioning drives that are monitored by the **table ETC board** via the ETC I/F **CAN network**.

Microswitches installed throughout the table and gantry also work to assure safety. Magnetic reed switches located at the top and bottom of the **elevation** linear actuator assembly and firmware ensure that these extreme positions are never achieved. The limit positions can be achieved during the table characterization or firmware/hardware failure.

Tape switches located on the table where someone could get injured during **patient positioning** disable cradle feed, table lift and gantry tilt drives if any of these switches gets activated.

The **INTERFERENCE** indicator will light on the **Gantry Display**.

The gantry control panel RESET light will flash at a fast frequency.

Removing the **table side covers** will also activate this response. A jumper plug is provided to enable the drive circuits for service purposes.

Press the gantry control RESET switch to re-enable these drives. Gantry Tilt/Table Elevation Interference matrix switches are used as a safety back-up for a firmware controlled matrix that ensures the cradle and gantry never touch each other.

Tilt interference sensors are mounted on the front and rear gantry covers. When the sensor is not activated, the impedance should be 20 kOhms. When something activates the sensor, resistance approaches zero Ohms. A green LED means the sensor is not shorted. A yellow LED means more than 40 kOhms of resistance.

The **ETC I/F Board** interprets maximum impedance of the sensors, 40 kOhms as a **fault** condition and a short circuit as an **interference** condition. This is done for both front and rear sensors. If there is interference or a fault condition, both elevation and tilt are disabled. The ETC board can override this.

Internal gantry contact sensors inside the gantry detect loose hardware or cables. This protects you from possible projectiles while servicing the gantry. After you secure what's loose, do a **Scan HW Reset** at the console.



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Data Acquisition

Helical Scans

Images are created using data from all rows used during data acquisition dependent on slice thickness. **Row weighting**, or the amount of data used to reconstruct an image from a row, is dependent on table speed and pitch.

Axial Scans

In Axial mode, there are two ways relative to scanning and displaying images: $4 \times i$ or $2 \times i$, where i = thickness.

In a $4 \times i$ mode, images are created from the rows used as indicated by slice thickness. This mode should always be used to troubleshoot Detector/DAS related artifacts.

In a $2 \times i$ mode, Recon actually combines 2 rows of data to produce a thicker slice.

For Head First

Image	Data Compiled From Row:
1	2A
2	1A
3	1B
4	2B

For Feet First

Image	Data Compiled From Row:
1	2B
2	1B
3	1A
4	2A

For Slice Thickness

Thickness	Data Compiled From:
4×1.25	4 center rows (D2, D1, D1, D2)
4×2.50	8 center rows (D4+D3, D2+D1, D1+D2, D3+D4)
4×3.75	12 center rows (D6+D5+D4, D3+D2+D1, D1+D2+D3, D4+D5+D6)
4×5.00	16 center rows (D8+D7+D6+D5, D4+D3+D2+D1, D1+D2+D3+D4, D5+D6+D7+D8)

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Data Acquisition, *continued*

kV vs Slice Thickness

DAS Converter boards have 31 pre-amp gains based on the selected technique. These gains are not user selectable, but it's important to know if an artifact only appears at the same pre-amp gain. The pre-amp gain used can be found within the scan header by using Scan Analysis.

Calibration

The calibration technique is based on focal spot size, which is either small or large. A **small focal spot** is anything less than 24kW and a **large focal spot** is anything 24kW or bigger. The focal spot size must be known if it is suspected that the artifact may be caused by cal vectors data acquisition (the actual rows are dependent on slice thickness). **Row weighting**, or the amount of data used to reconstruct an image from a row, is dependent on table speed and pitch.

Refer to [Troubleshooting Ring Artifacts in a Helical Series](#).



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Data Management

Image Data

The system uses the **informix database** to store patient/exam/series information about the images and correlates it to the actual images (pixel data) stored in the image pool partitions.

The console host computer has two hard disks. One disk contains software and a portion of the system's image files. The **image disk** partitions are:

/usr/g/sdc_image_pool which is the primary hard disk for image storage (pixel data) and **/usr/g/sdc_image_pool2** which is on the second hard disk for image storage (pixel data).

The informix database is located at **/usr/g/informix** on the primary hard disk; it holds the patient/exam/series information database. This partition holds the system database, queue entries for Archive, Network, and Filming, and executable programs that handle job coordination.

Sometimes the system database and image partition lose synchronization. Scan data may have reconstructed with the pixel/image data going to the image partition but the entry did not get installed into the database. Or an image may have been removed from the image partition yet the entry did not get removed from the database.

The Disk Management process runs at startup, and periodically thereafter, to check for discrepancies between the database and image disk partition. It fixes the database if needed. This can take up to 30 minutes, running in the background.

Error messages appear in the error log when the Database Recovery process is running as well as when it makes changes to the database.

Scan Data

Raw scan data is stored in files on the **ICE Scan Data Disk**. These files can be accessed through **Common Service Desktop, continued** on the RIP as **/usr/g/data_management**.

Both the Scan Data Disk and System Disk directory trees contain recovery directories for analysis of a scan. On startup, recovery moves corrupted scan headers or scan data there. A software snap saves the files in the recovery directories so that engineering can analyze the data.

A map of the scan file database is maintained in RIP memory. This map defines the state of all the scan files in the scan file database, but contains none of the scan file data or header information.



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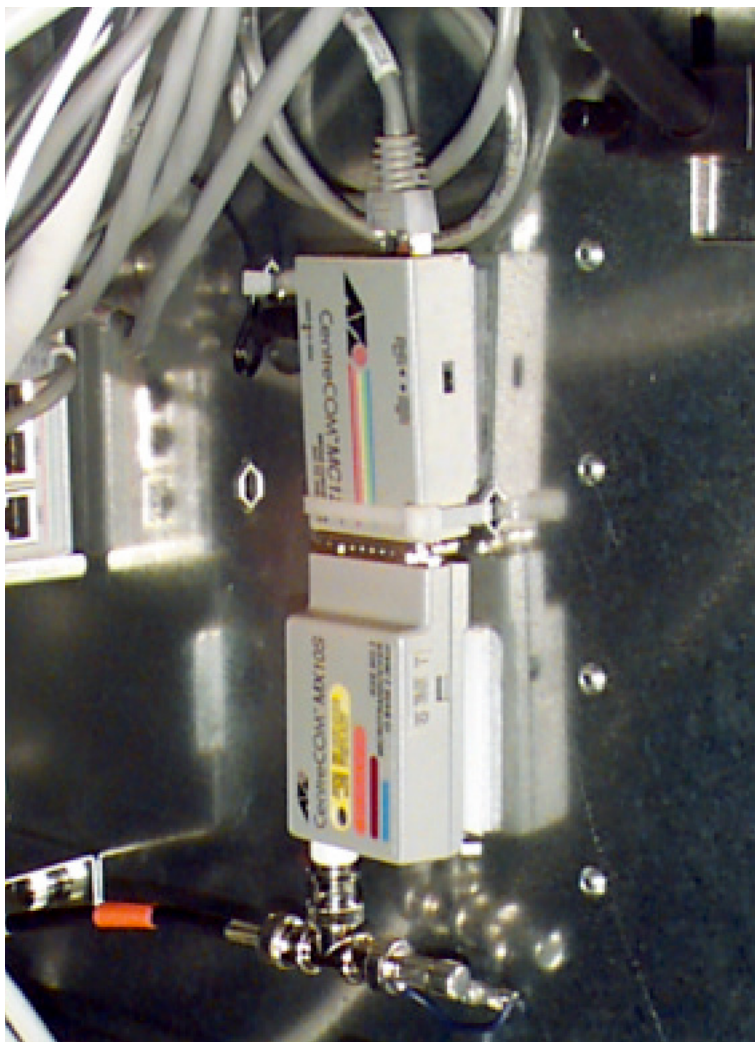


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System Control

Console LAN Connections



Console

The H2 replaces the O2 (SBC) with the **Power PC**.

Refer to **Console** drawing.

Patient Positioning

The table **CAN network** is the communications interface for the gantry display and four patient positioning control panels. On power-up, the **ETC I/F Board** tests communications to the gantry display and controllers. Faults are reported as **node failures**. A watchdog circuit disables pushbutton signals from leaving the ETC I/F if its microprocessor hangs.

Refer to the **Table Overview** drawing.

Gantry Communication

The **STC LSCOM** transfers instructions and status between host and gantry. The OBC transfers instructions and status between stationary and rotating gantry.

Refer to **Axial Control Communication Theory**.

Refer to **System** and **Slip Ring Communication Theory** for info on controls subsystems, and between rotating and stationary parts of the gantry.

Refer to **CAN Error Detection**.



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Slip Ring Communication

The system uses one High Speed Data Capacitive Device (**HSDCD**) Ring and 11 hard-wired **slip rings** to transfer signals between the rotating and stationary parts of the gantry. Five inner platters provide **HV and AC power**; three middle rings provide safety interlock signals; three outer rings provide **communication paths**.

Only the HSDCD ring for DAS data and the outer platters for control signals are discussed here.

The HSDCD slip ring consists of a circuit board material with two traces that run its entire circumference to carry the differential signal. It is called the emitter. There is also have a ground plane underneath it.

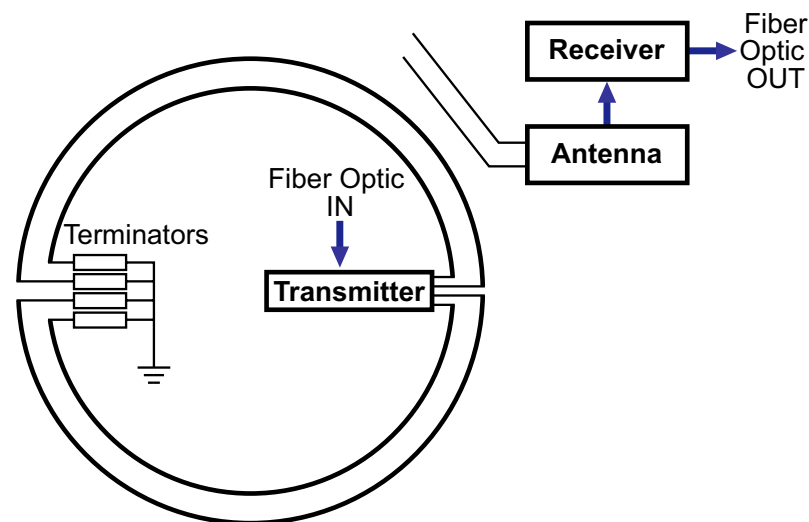
DAS Data

Data from the DAS crosses the HSDCD ring on its way to the console's DIP board. It transfers at up to 110 Mbaud. It travels from the rotating side to the stationary side through the HSDCD ring. This ring is a contact-less path consisting of the HSDCD transmitter on the rotating structure.

The stationary pickup is an antenna and receiver coupled together. The HSDCD ring and antenna form two plates of a capacitor. The transmitter input and receiver output are fiber optic connections. **Both have LED indicators**.

Forward Error Correction (FEC) is used in this path to ensure extremely high data reliability.

HSDCD Ring - DAS Data Transfer



DAS data received by the **transmitter** is converted into HSDCD differential signalling.

The **transmitter** converts its input to differential signals on the ring itself.

The **antenna** is capacitively coupled to the ring and passes the differential signal to the receiver.

The **receiver** converts and amplifies the differential signal to the original transmitter data input format.

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Slip Ring Communication, *continued*

DAS Data, *continued*

The HSDCD slip ring has two traces that run its entire circumference; they form the **emitter**. It also has a ground plane underneath. At the end of each trace, opposite the end fed by the transmitter, each trace is terminated with a surface mount resistor to the ground plane. Each board trace feeds one half of the ring with the HSDCD signal.

Control Data

A LAN connection and hard wired **slip rings enable communications** between the OC and Table/Gantry computers. The 10Base2 thin net 50 Ohm LAN cable provides the communication path between the OC computer in the console, the ETC in the table, and the STC in the stationary gantry. The hard slip rings provide a means to transfer data bidirectional between the STC and the rotating components including the OBC. The control rings transfer data at 2.5 Mbaud.

The hard slip ring communication connection acts as a sub-network between the STC and OBC but also carries the specific DAS trigger and Exposure Command signals multiplexed within the data stream. Control and status information travels in both directions across the rotating interface along two high speed serial links using three

rings, one inbound and one outbound. Both use an isolated ring as signal reference which provides excellent noise immunity. Brushes ride on the conductive ring material to provide signal connection to the stationary components and screws inserted into the back of each ring provide the connection for the rotating components.

Two LSCOM boards provide serial link control and signal multiplexing/de-multiplexing. **LEDs** located on these boards provide communication **status** of the data paths.

RCIB Faults

The Gantry Controller Area Network (GCAN) contains the Rotor Control Interface Bus (**RCIB**). This fault line is a 12 VDC signal originating on the HEMRC Control Board. It travels to the Collimator Control Board (CCB), DAS Control Board (DCB), and through the GCAN bus terminator at the DCB.

The RCIB creates a fault signal whenever a board on this bus detects a fault, or this bus or CAN bus terminator is open or missing. The FAULT line resides inside the CAN bus cabling.

Refer to **System Communication and Control Diagnostics**.



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DAS Detector

Detector Overview

The primary function of the Detector is to convert X-ray photons into electrical current, which is sent to the Data Acquisition System (DAS) for signal amplification and analog-to-digital conversion, before being sent to the Recon Unit for image reconstruction.

The x-rays pass through the patient (or object being scanned) and are attenuated by the density of material. The remaining energy of x-rays passes through to the detector. Wireless collimation improves the geometric dose efficiency.

Once the X-ray beam is collimated into cells/channels, the photons hit the **scintillator** pack, which causes it to emit light. The scintillator pack is made up of cast material and a GE exclusive material called Lumex. Lumex is an efficient material that converts X-ray energy to light, with low afterglow characteristics.

The light from the scintillator pack is picked up by a photodiode array. The photodiode array converts the emitted light into an electric current, which is passed through to the DAS. The current strength is dependent on the amount of X-ray energy absorbed into the Lumex, which corresponds to the light energy output. There is a photodiode output from each detector cell.

The detector assembly used on this system houses 57 Detector Modules. Each module has two sides—an A-side and a B-side—with 12 diodes or cells per channel per side. Cells are labeled D1 through D12, for a total of 24 cells across the detector face (both A and B sides).

Note There are sixteen 0.625 mm cells, eight per side, and eight 1.25 mm cells, four per side. The 1.25 mm cells comprise the four outer rows on both A and B sides. It must further be noted that only sixteen cells per channel are active during any given data collection.

Each module uses two flex connections to the Data Acquisition System (DAS). The flex connectors cannot be removed from the module. The flex end is connected to the DAS via the DAS Detector Interface (DDIF). Both the flex lead and the DAS backplane are outfitted with female pin connectors. There is a male to male pin interface called the “Interposer” that completes the connection. Each flex carries 128 signal lines, 6 FET control lines, 1 analog ground and 1 signal ground.

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DAS Detector, *continued*

Detector Overview, *continued*

Detector module temperature is regulated by the electrical resistance heater and the thermistor. This is a three-zone heater design. The heater and thermistor are incorporated into the detector assembly. The overall mass of the assembled detector system is approximately 25 Kg.

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DAS Detector, *continued*

Detector Module

Detector Module: A detector module consists of a 16 x 24 pixel array.

Detector Channel: A detector channel consists of 24 diodes arranged in the "Z" direction. In total, there are 912 channels on a detector. A single channel is 1 mm in length, in the azimuthal direction. A detector channel is sometimes referred to as a Detector Column.

Detector Row: A row of 16 cells across all detector channels designated by Diode Number AND Side. (Ex. Detector Row D2, Side "A").

Note A Detector Row is not the same as a Scan Slice. A detector row is 1.25 mm, 0.625 mm wide or a combination of the two (Z-Direction).

Cell: A cell is a single photodiode, and is 1/24th of a Detector Channel. In other words, there are 24 cells, or diodes, per Detector Channel.

Side A/B: There are 2 "sides" to a Detector, Side "A" and Side "B." The sides divide the Detector width in half, with 12 Rows per side. Side "A" is closer to the front of the Gantry (or Table side) and Side "B" is toward the back of the Gantry.

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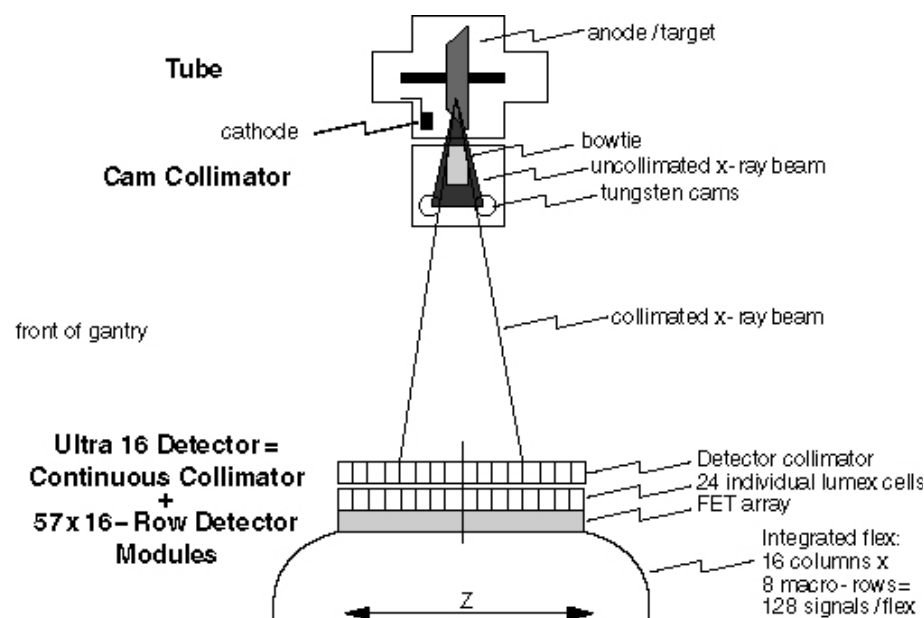


DAS Detector, *continued*

Z-Axis Cell Summation

The detector is segmented into cells in the Z dimension. Post-patient collimation is provided by the segmentation of the detector cells, not by a separate post-patient collimator. The post-patient collimation, along with summation of cells in the Z direction by the detector FET array, determines the Z-axis slice thickness of the scan data.

Detector Theory - X-Ray Collimation



The Z dimension extent of each cell is 0.625 and/or 1.25 mm at ISO center, as determined by the selected protocol. Cells are summed in Z to produce a macro cell. Macro cells are the combination or "ganging" of cells in a detector column to produce the desired slice thickness. Four 0.625 mm cells equal a 2.5 mm macro cell, or two 0.625 mm and one 1.25 mm cells equal a 2.5 mm macro cell. The selected protocol determines the combination of cells that are summed together.

All macro cells in the same Z plane form a macro row. A macro row is the detector row or combination of rows that generates a post-collimation slice thickness. A macro row consisting of a single cell in each column produces scan data with a thickness of 0.625 mm at ISO center. The largest macro row allowed produces scan data with a thickness of 5.0 mm at ISO center. There can be up to eight macro rows, labeled 8A, 7A, 6A, 5A, 4A, 3A, 2A, 1A, 1B, 2B, 3B, 4B, 5B, 6B, 7B, and 8B.

Each flex transmits eight macro cells to the DAS x 16 columns per detector module = 128 data channels per flex.

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DAS Detector, *continued*

Post Collimation: Z-Axis Beam Profile Considerations

The collimated beam has a Z-axis profile that consists of the umbra (essentially flat) and the penumbra. In order to avoid image artifacts, the system must always operate with the umbra region completely covering the detector cells contributing to the selected macro rows.

During gantry rotation, the position of the beam moves a small amount in the Z direction, due to various mechanical sags in the gantry, tube, collimator, etc. To ensure that the detector cells are completely covered by the umbra region, the Z dimension extent of the umbra is increased so that the detector is covered regardless of Z-axis beam motion. This is true for standard protocol profiles.

Low Dose protocol profiles differ in that an additional cell is summed to the outer macro row. The penumbra is allowed to cover the outer two cells of the two or three macro cells as determined by the selected protocol. This methodology reduces overall patient dose while maintaining signal levels for consistent image quality.

Detector reference cells estimate the actual position of the X-ray beam on the detector, and real-time feedback is provided to the collimator to compensate for beam motion.

All tracking schemes require a calibration of the x-ray beam optics. All tracking modes require prediction of the beam position based on the prescription. Static, predictive tracking estimates the beam position once per scan, based on prescription parameters such as gantry tilt, gantry speed, and selected focal spot. Based on these parameters, the beam width and position in Z is estimated such that the detector cells selected are always covered by umbra.

Closed-loop tracking provides more accurate control of the beam position in Z. This scheme uses detector reference cells to estimate the actual position of the X-ray beam on the detector, and then to compensate for beam motion by providing real-time feedback to the collimator. In this scheme, the collimator corrects for beam motion many times per gantry rotation.

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DAS Detector, *continued*

Detector FET Control

The Detector Field Effect Transistors (FETs) are used as switches to combine detector rows, to achieve the prescribed slice thickness. These FETs are physically located in the detector assembly.

Important! FETs are EXTREMELY ESD SENSITIVE.

ALWAYS use ESD precautions when handling the Detector or Detector flexes. A bad FET requires replacement of the entire Detector.

After entering a Scan prescription at the Host Computer, the Scan Rx parameters are sent to the appropriate controllers. For slice thickness, the parameters are sent to both the Collimator control board, to select the proper Collimator CAM positions, and the DAS Control Board (DCB), to select the macro row width.

There are three sets of six FET control lines driven by the DCB. These FET control lines form a binary value that gets decoded in the Detector and controls Detector Diode selection.

The DCB uses five quad SPST analog switches, which drive the FET_BUS.

The FET Control circuitry is the basis for all scanning techniques. FET modes are used for patient and DDC scanning protocols, and the DASTOOLS XRAY Verification test.

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DAS Detector, *continued*

Detector FET Control, *continued*

4-Slice FET Mode Assignment

Normal data modes with Z-chan same as H1.x								4-slice configuration										Detector Mode name	Protocol Mode name	FET 1 LED	FET 2 LED	FET 3 LED	FET 4 LED	FET 5 LED	FET 6 LED
A8 (D8)	A7 (D7)	A6 (D6)	A5 (D5)	A4 (D4)	A3 (D3)	A2 (D2)	A1 (D1)	B1 (D1)	B2 (D2)	B3 (D3)	B4 (D4)	B5 (D5)	B6 (D6)	B7 (D7)	B8 (D8)	Slice mode	FET mode								
2	1													1	2	Z-chan	45	4Ca16	Ca16	OFF	ON	OFF	OFF	ON	OFF
2	2	2	2	1	1	1	1	1	1	1	1	2	2	2	2	4x5.00	36	4x5.00	4x5.00	ON	ON	OFF	ON	ON	OFF
	2	1												1	2	Z-chan	44	4Ca15	Ca15	ON	ON	OFF	OFF	ON	OFF
		2	2	2	1	1	1	1	1	1	2	2	2			4x3.75	35	4x3.75	4x3.75	OFF	OFF	ON	ON	ON	OFF
			2	1								1	2			Z-chan	42	4Ca13	Ca13	ON	OFF	ON	OFF	ON	OFF
				2	2	1	1	1	1	2	2					4x2.50	34	4x2.50	4x2.50	ON	OFF	ON	ON	ON	OFF
				2	2	1	1	1	1	2	2					Z-chan	34	4x2.50	4x2.50	ON	OFF	ON	ON	ON	OFF
						2	1	1	2							4x1.25	33	4x1.25	4x1.25	OFF	ON	ON	ON	ON	OFF
						2	1	1	2							Z-chan	33	4x1.25	4x1.25	OFF	ON	ON	ON	ON	OFF
						2	1	1	2							1x1.25	33	4x1.25	4x1.25	OFF	ON	ON	ON	ON	OFF
						2	1	1	2							Z-chan	33	4x1.25	4x1.25	OFF	ON	ON	ON	ON	OFF
						2	1	1	2							Thin twin	33	4x1.25	4x1.25	OFF	ON	ON	ON	ON	OFF

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DAS Detector, *continued*

Detector FET Control, *continued*

8-Slice FET Mode Assignment

Normal data modes with Z-chan same as H1.x								8-slice configuration										Detector Mode name	Protocol Mode name	FET 1 LED	FET 2 LED	FET 3 LED	FET 4 LED	FET 5 LED	FET 6 LED
A8 (D8)	A7 (D7)	A6 (D6)	A5 (D5)	A4 (D4)	A3 (D3)	A2 (D2)	A1 (D1)	B1 (D1)	B2 (D2)	B3 (D3)	B4 (D4)	B5 (D5)	B6 (D6)	B7 (D7)	B8 (D8)	Slice mode	FET mode								
2	1					4	3	3	4					1	2	Z-chan	17	8Cal6	Cal6	OFF	ON	ON	ON	OFF	ON
2	2	2	2	1	1	1	1	1	1	1	1	2	2	2	2	4x5.00	8	4x5.00	4x5.00	ON	ON	ON	OFF	ON	ON
3	2	1					4	4					1	2	3	Z-chan	16	8Cal5	Cal5	ON	ON	ON	ON	OFF	ON
		2	2	2	1	1	1	1	1	1	2	2	2			4x3.75	7	4x3.75	4x3.75	OFF	OFF	OFF	ON	ON	ON
	4	3	2	1							1	2	3	4		Z-chan	14	8Cal3	Cal3	ON	OFF	OFF	OFF	ON	ON
				2	2	1	1	1	1	2	2					4x2.50	6	4x2.50	4x2.50	ON	OFF	OFF	ON	ON	ON
				2	2	1	1	1	1	2	2					Z-chan	6	4x2.50	4x2.50	ON	OFF	OFF	ON	ON	ON
						2	1	1	2							4x1.25	5	4x1.25	4x1.25	OFF	ON	OFF	ON	ON	ON
	4	3	2	1							1	2	3	4		Z-chan	14	8Cal3	Cal3	ON	OFF	OFF	OFF	ON	ON
				4	3	2	1	1	2	3	4					8x1.25	0	8x1.25	8x1.25	ON	ON	ON	ON	ON	ON
2	1					4	3	3	4					1	2	Z-chan	17	8Cal6	Cal6	OFF	ON	ON	ON	OFF	ON
4	4	3	3	2	2	1	1	1	1	2	2	3	3	4	4	8x2.50	1	8x2.50	8x2.50	OFF	ON	ON	ON	ON	ON
						2	1	1	2							Z-chan	5	4x1.25	4x1.25	OFF	ON	OFF	ON	ON	ON
						2	1	1	2							1x1.25	5	4x1.25	4x1.25	OFF	ON	OFF	ON	ON	ON
						2	1	1	2							Z-chan	5	4x1.25	4x1.25	OFF	ON	OFF	ON	ON	ON
						2	1	1	2							Thin twin	5	4x1.25	4x1.25	OFF	ON	OFF	ON	ON	ON



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DAS Detector, *continued*

Detector FET Control, *continued*

16-Slice FET Mode Assignment

ULTRA 16 APPLICATIONS FET MODE SELECTIONS - 16 SLICE MODE																															
A12	A11	A10	A9	A8	A7	A6	A5	A4	A3	A2	A1	B1	B2	B3	B4	B5	B6	B7	B8	B9	B10	B11	B12	Slice mode	FET mode	FET 1 LED	FET 2 LED	FET 3 LED	FET 4 LED	FET 5 LED	FET 6 LED
2	1									4	3	3	4									1	2	Z-Chan	49	OFF	ON	ON	ON	OFF	OFF
8	7	6	5	4	4	3	3	2	2	1	1	1	1	2	2	3	3	4	4	5	6	7	8	16x1.25	54	ON	OFF	OFF	ON	OFF	OFF
2	1									4	3	3	4									1	2	Z-Chan	49	OFF	ON	ON	ON	OFF	OFF
4	4	3	3	2	2	2	2	1	1	1	1	1	1	1	1	2	2	2	2	3	3	4	4	8x2.5	1	OFF	ON	ON	ON	ON	ON
3	2	1								4	3	3	4								1	2	3	Z-Chan	48	ON	ON	ON	ON	OFF	OFF
		2	2	2	2	1	1	1	1	1	1	1	1	1	1	1	1	2	2	2	2			4x3.75	7	OFF	OFF	OFF	ON	ON	ON
			2	1	1													1	1	2				Z-Chan	39	OFF	OFF	OFF	ON	ON	OFF
				8	7	6	5	4	3	2	1	1	2	3	4	5	6	7	8					16x0.625	53	ON	OFF	OFF	ON	ON	ON
			2	1	1													1	1	2				Z-Chan	39	OFF	OFF	OFF	ON	ON	OFF
				4	4	3	3	2	2	1	1	1	1	2	2	3	3	4	4					8x1.25	0	ON	ON	ON	ON	ON	ON
4	4	3	3	2	2	2	2	1	1	1	1	1	1	1	1	2	2	2	2	3	3	4	4	Z-Chan	1	OFF	ON	ON	ON	ON	ON
								4	3	2	1	1	2	3	4									8x0.625	3	OFF	OFF	ON	ON	ON	ON



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System Power

Axial Drive Power

The **PDU Axial Power Contactor** controls the 440 VAC input to the **Axial Motor Drive (AMD)**. The Axial Power Contactor is an AC Rated relay that connects or disconnects the 3 phase 440 VAC.

The **STC Axial Enable/Disable switch** can be used to disable axial power.

Refer to **Axial Power Theory**.

Rotating Gantry (OBC) Power

The **OBC PWR I/F board** provides a single, 120 VAC power distribution point on the rotating gantry for the X-ray Generation Subsystem. It provides a convenient location for fusing the various subsystem circuits. In this way the major pieces of the XRTGEN subsystem have been grouped logically while adequately protecting the harnessing from faults. It provides "Open Fuse Detection" for the tube fan and pump circuit. A small voltage sensing circuit monitors the level at the load end of F3. In the event fuse F3 should open, the optically coupled solid state relay, U1, will interrupt the XRT pressure switch signal to the Gentry I/O board. This causes an immediate scan abort and prevents additional energy from being dumped into the XRT.

Adhesive **insulating pads** on the back of the OBC PWR I/F board protect you from contacting live components.



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Axial Power

There is a safety interlock in the axial power contactor circuit. It is the **AX DR Disable switch on the STC backplane**. It works independent of firmware control to assure complete safety. A second hardware reset is not required.

Caution

Disable the Axial Drive Mechanism before servicing the gantry interior. The Axial Loop Contactor Status LED on the STC backplane indicates the status of the Axial Power control circuit only, not the state of the PDU Axial Power Contactor.

The PDU **Axial Power Contactor A3K4** supplies the 440 VAC input to the **Axial Motor Drive**. The Axial Power Contactor is an AC rated relay that connects or disconnects the three phase 440 VAC. The firmware function provides status monitoring and reporting of this power contactor.

The base requirements for axial power are:

- Power to the PDU
- DRIVE ON enabled
- Axial drive (AX DR) switch enabled on the STC backplane

At this point, the firmware is needed to close the Axial Drive Power Contactor. This means the scan hardware needs to be downloaded successfully. The Axial Control Board, Altera (FPGA), now contains the necessary firmware. When the FPGA receives a valid request to rotate the gantry the LOOPCONT signal is generated. This signal energizes the K3 relay on the Axial Control Board. The following sequence then occurs. The 24V-B from the PDU relay board goes through the Axial drive enable switch, S2, on the STC backplane. Then 24V-B becomes AX_DR_EN_SW/SW2. When Axial Control Board relay K3 closes, AX_DR_EN_SW/SW2 becomes CLOSELOOP as it leaves the Axial Control Board. The CLOSELOOP signal goes to the PDU Relay bd, K6. This in turn energizes K13, on the PDU relay bd, generating the LOOPHI command. LOOPHI energizes **A3K4**, which closes and provides 3 phase 440 VAC power to the Axial Motor Drive.

Note 440 VAC will be present at the Axial Motor Drive under normal conditions. Remember that this function is firmware controlled.

Note A Power Contactor failure could leave 440 VAC connected to the Axial Motor Drive at all times unless you use the AX DR disable switch.



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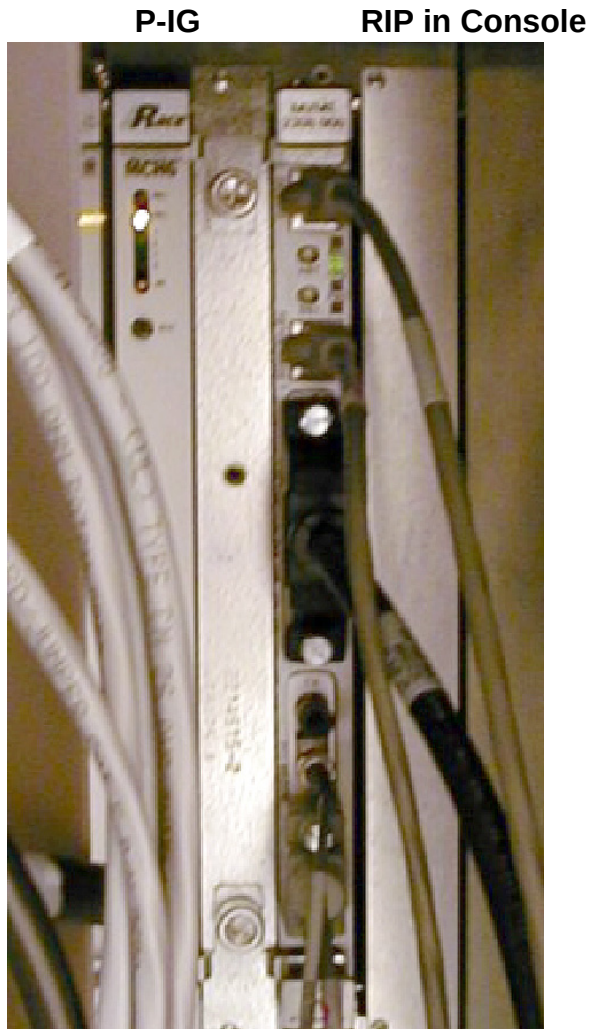


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Image Generation

Another name for reconstruction.



The **DAS Interface Processor (DIP)** acts as the main interface between the S-DAS and the Scan Reconstruction Unit (SRU) or Image Chain Engine (ICE). It receives TAXI data from the **DAS HSDCD slip ring**, buffering it, and sending an interrupt to Scan Data Save process to have it put on the **Scan Data Disk (SDD)**.

The interface between Scan Data Acquisition and the Scan Reconstruction Unit is the TAXI interface of the DIP. This board is a plug-in adapter card, with a PMC/PCI standard interface, to the Reconstruction Interface Processor (**RIP**) board.

The **RIP** Board from Motorola has two PCI Mezzanine Cards (**PMC's**), piggybacked on the RIP board. These two PMC's are the SBS SCSI Controller Card and the GE DIP Card.

The RIP CPU transfers data from the Scan Data Disk to the P-IG board to perform back projection.

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Image Generation Overview

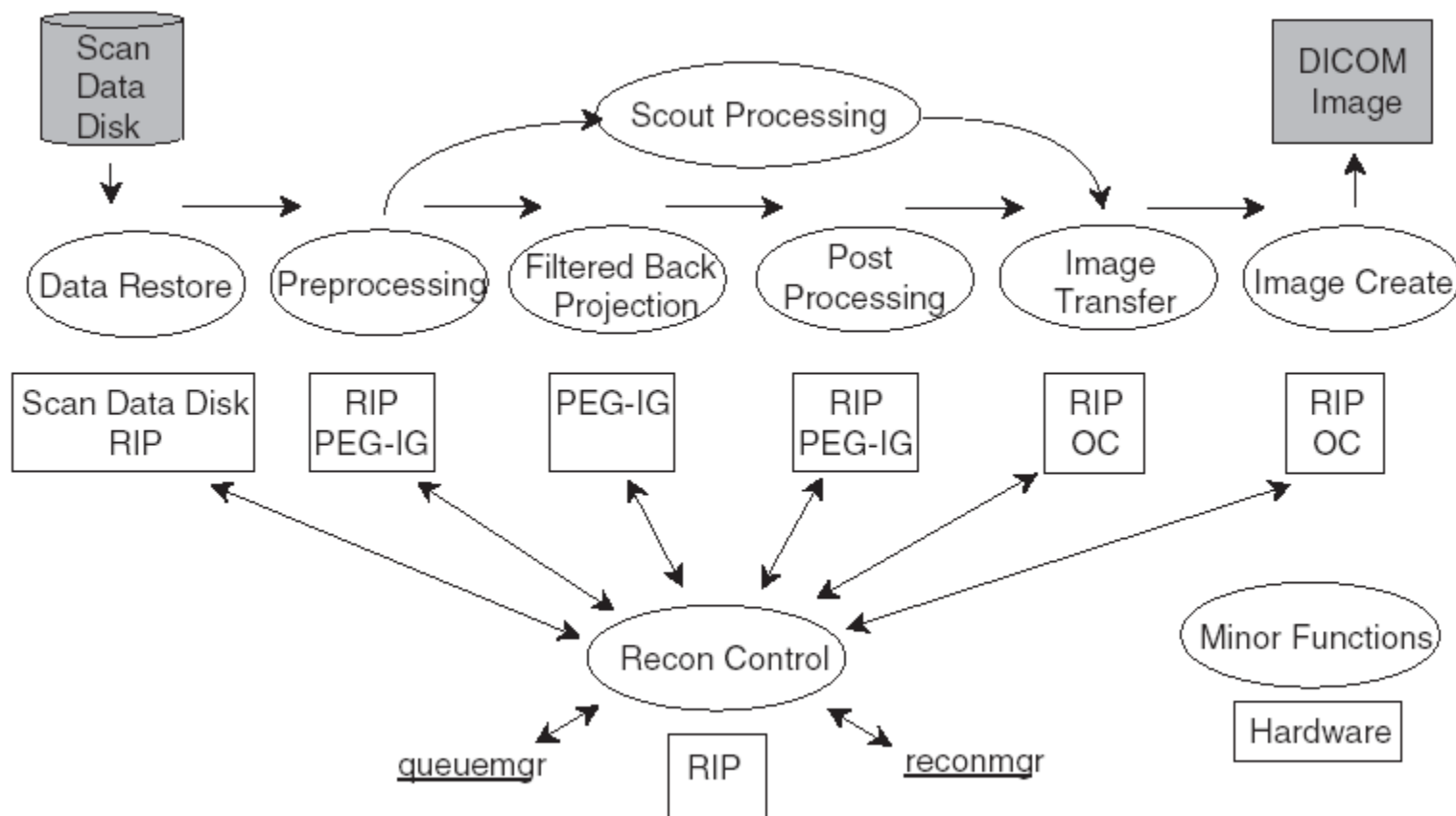




Image Generation, *continued*

Data Restore

The beginning of the reconstruction process occurs when enough scan data (from Scan Data Save) is on the Scan Data Disk to create projection data.

Applicable Diagnostic Coverage

1. Recon Data Path
2. **RIP** Diagnostics

Signal Flow #	Software Function	(FRUs) Involved
1		Scan Data Disk, RIP
2	Refer to Pre-Processing.	

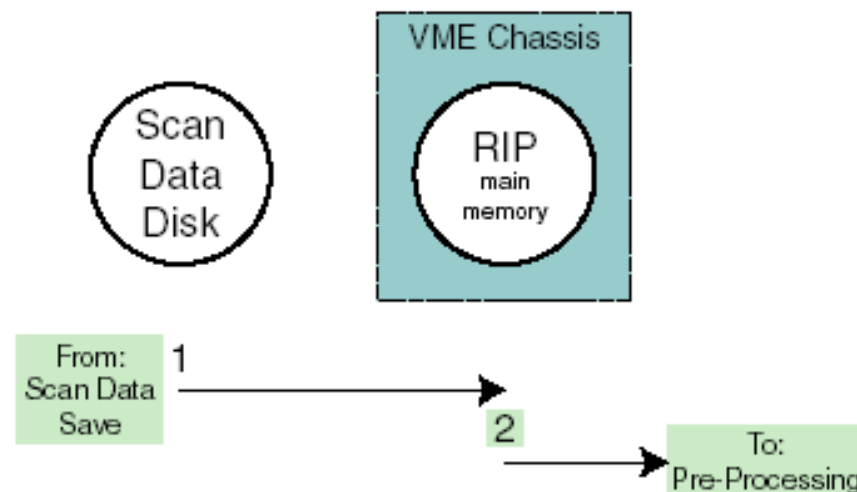




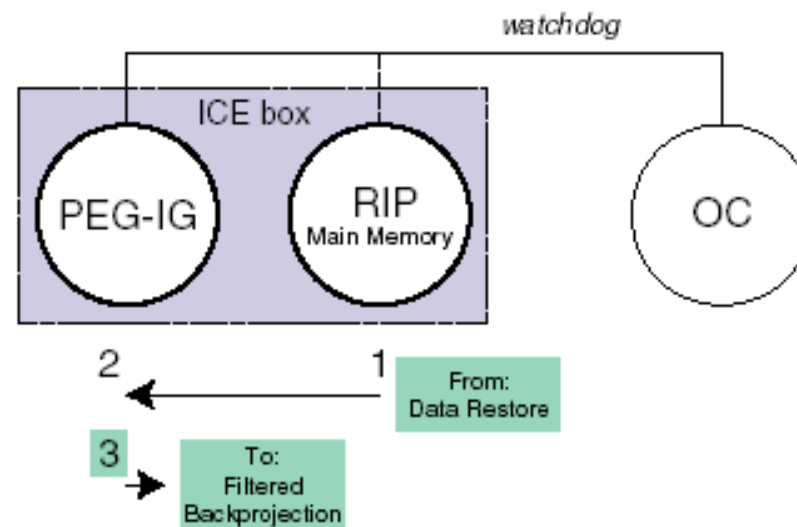
Image Generation, *continued*

Pre-Processing

The **PEG-IG** pulls blocks of scan data out of the RIP's main memory through the VME Bus. Before pre-processing, each view's checksum is checked against the checksum created during Data Restore.

Applicable Diagnostic Coverage

1. Recon Data Path
2. **RIP** Diagnostics
3. **PEG-IG** Diagnostics
4. **LAN** Diagnostics



Signal Flow #	Reporting Host	Software Process	Software Function	Associated Errors	(FRUs) Involved
watchdog		procICE	procICE sends ping messages attempting to talk to the RIP (every 15 seconds)	260005002	OC, PEG-IG , RIP and LAN
1	sdctrl	prep	PEG-IG pulls data from RIP using the function: dx_modify	230016024	
2	DSP	prep	Verify Checksum. Function: EM_PREP_VIEW_LOOP	230016024	PEG-IG , RIP
3	N/A	N/A	Refer to Filtered Back Projection.		



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Filtered Back Projection

The **PEG-IG** performs convolution (filtering) followed by back projection. After all views have been back projected, the **PEG-IG** generates a “back projection done” message to the **RIP**. This starts the postprocessing phase.

Applicable Diagnostic Coverage

1. Recon Data Path
2. **RIP** Diagnostics
3. **IG** Report Self Test

Software Function	Associated Errors	(FRUs) Involved
Receiving view blocks into input FIFO	242605	PEG-IG
All views consumed "back projection done"	230003020	PEG-IG
Refer to Post-Processing.		



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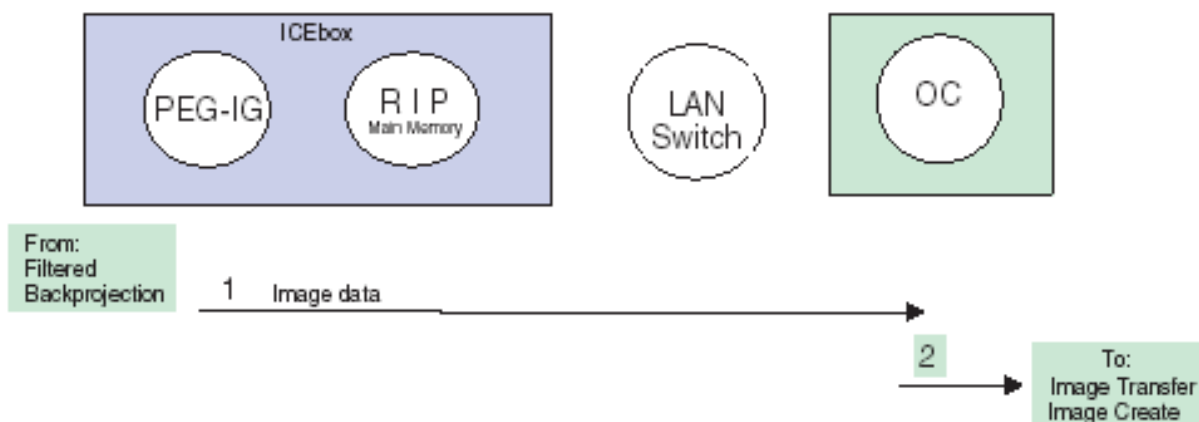


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Post Processing

Applicable Diagnostic Coverage

1. Recon Data Path
2. **RIP** Diagnostics
3. **LAN** Diagnostics



Signal Flow #	Reporting Host	Software Process	Software Function	Associated Errors	(FRUs) Involved
1	RIP	icProxy	Image out of the PEG-IG (checksum failure)	230007513	PEG-IG, RIP
2	N/A	N/A	Refer to Image Transfer & Image Create.		



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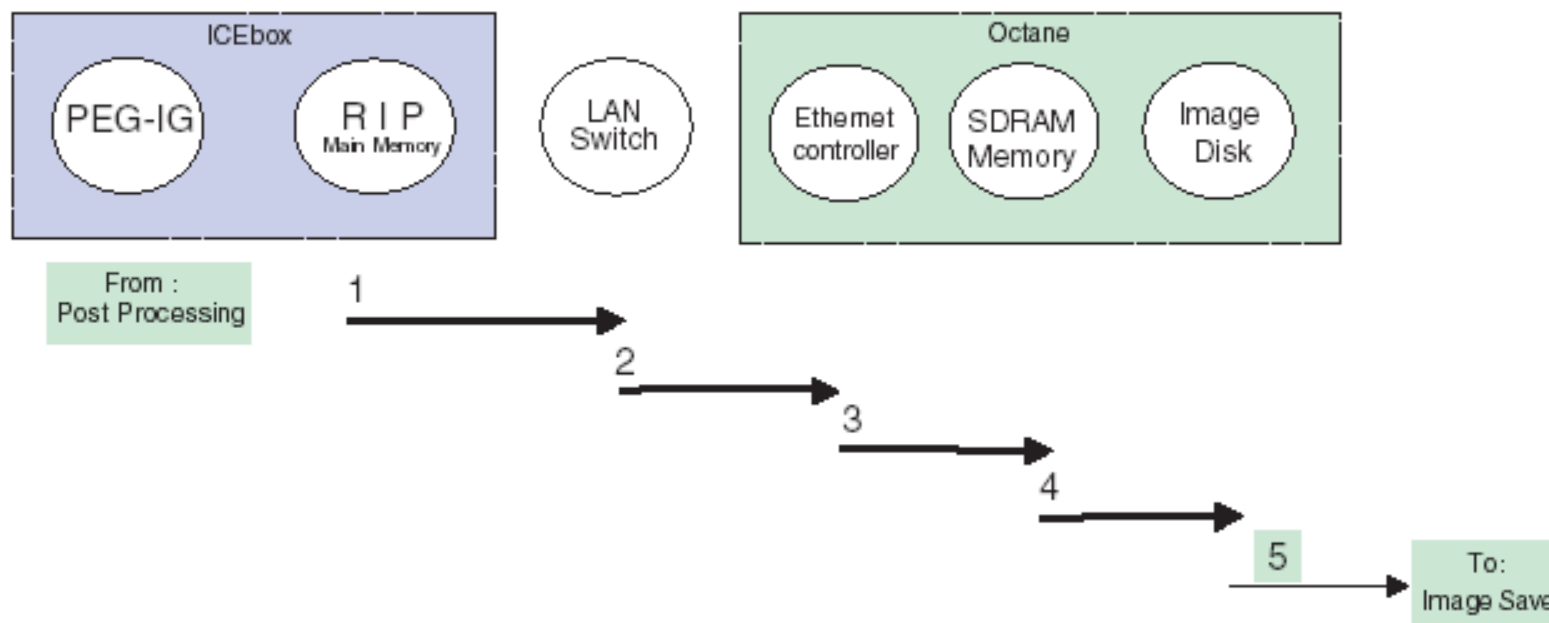
Image Generation, *continued*

Image Transfer

At completion of Post Processing, image pixel data is sent out of the **RIP**'s memory via Ethernet (100BaseTx) through the **LAN** Switch through the ethernet controller in the OC Computer, into memory, where image is checksum compared, then onto the OC Image Disk. The Image Create minor function describes how the OC controls the receiving of the image.

Applicable Diagnostic Coverage

1. Recon Data Path
2. **RIP** Diagnostics
3. **LAN** Diagnostics
4. OC **ide>** level Diagnostics



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Image Transfer, *continued*

Signal Flow #	Reporting Host	Software Process	Software Function	Associated Errors	(FRUs) Involved
1	RIP	icProxy	Ethernet communications	230007511	RIP, LAN switch, SBC
1	RIP	icProxy	Image out of RIP 's memory to LAN switch	230007510	
2	OC	imagecreate	LAN switch to ethernet control		LAN switch
3	OC	imagecreate	Ethernet control to SDRAM memory		Memory, Ethernet Control
4	OC	imagecreate	SDRAM memory to Image Disk	200002720	LAN Switch, RIP
5	N/A	N/A	Refer to Image Save.		

Note LAN switch is common to software processes involved. For LAN suspected failure, try swapping ports on LAN switch.



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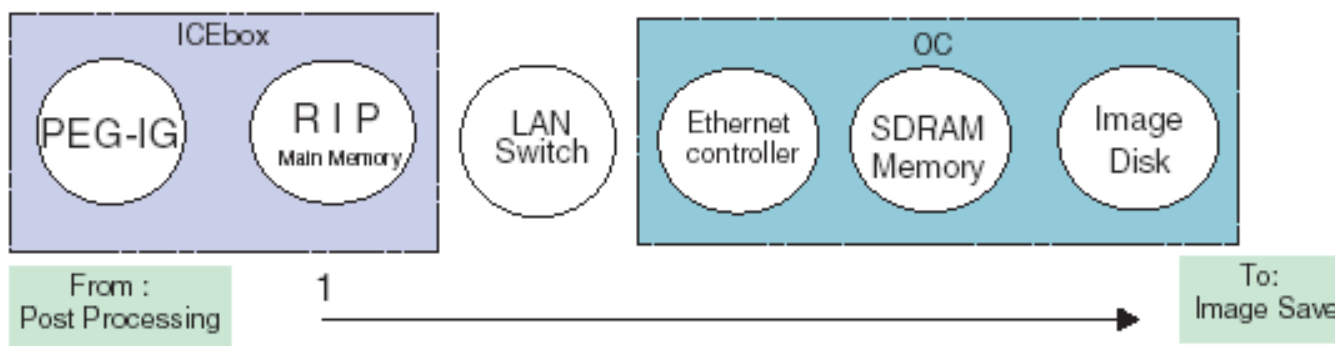


Image Generation, *continued*

Image Create

Applicable Diagnostic Coverage

1. Recon Data Path
2. **RIP** Diagnostics
3. **LAN** Diagnostics



Signal Flow #	Reporting Host	Software Process	Software Function	Associated Errors	(FRUs) Involved
1	RIP	icProxy	Ethernet communications	230007511	LAN switch, or cable, RIP



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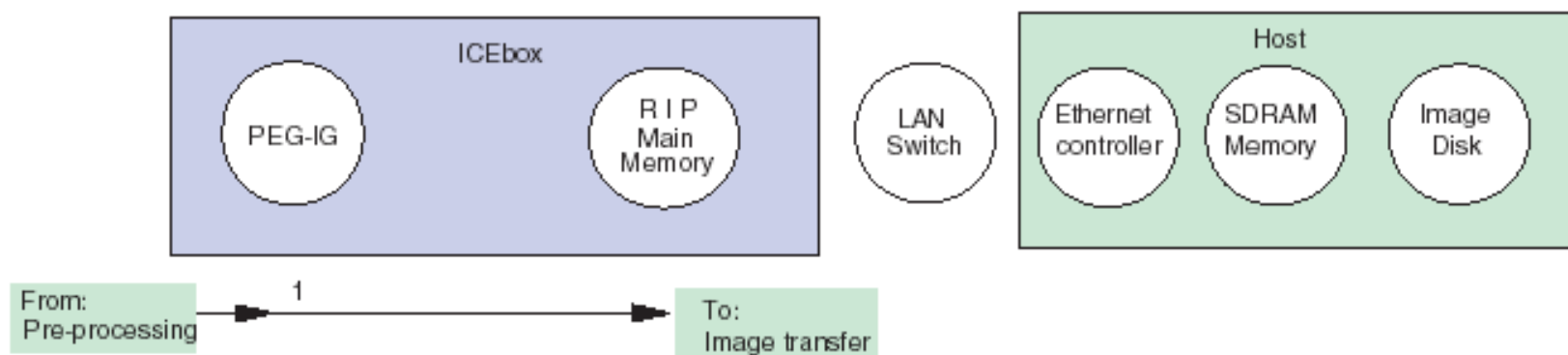
Scout Processing

Scout has exactly the same data restore path (see Image Generation/Data Restore) and image install path (see [Image Transfer](#) and [Image Create](#)) as all the other recon modes, except that the **IG** is not involved. That is, after the data is processed in **SDC**, it will be directly sent **DMA** (direct memory access) from **SDC** internal memory

to the **RIP** system memory, where IC proxy will forward data to Image Transfer.

Applicable Diagnostic Coverage

1. Scan Data Path
2. **RIP** Diagnostics



Signal Flow #	Reporting Host	Software Process	Software Function	Associated Errors	(FRUs) Involved
1	PEG-IG	sdc_post	Data transfer from PEG-IG to RIP	230016024	PEG-IG, RIP



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Image Generation, *continued*

Scan File Management

There are two main functions in Scan File Management, saving scan data and restoring scan data.

Saving Scan Data

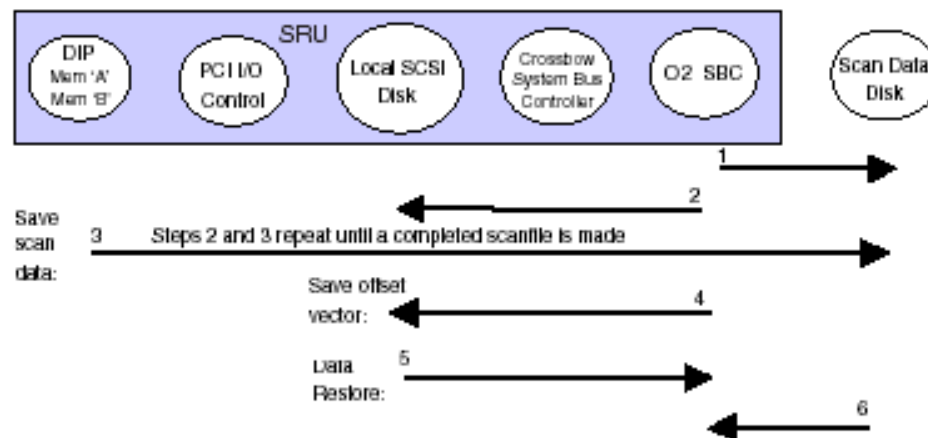
At the start of a scan, a scan file is allocated and a scan header is written to the Local SCSI Disk. As the raw data comes into the DIP memory, the DIP board verifies the integrity of the data and interrupts the SRU when one bank of the DIP memory is full. The SRU then initiates the data write from DIP memory to the Scan Data Disk. The scan header on the Local SCSI Disk (system disk) is periodically updated to indicate how much raw data has been written to the Scan Data Disk. The process continues until all the scan data is collected, then there is a final write to the Scan Header indicating that all the raw data is present on the Scan Data Disk making a complete scanfile.

During scanning, offset views are saved in the scan file as described above. For image reconstruction an offset vector is needed using the collected offset views. To facilitate faster retro reconstruction, during prospective reconstruction an offset vector is written back to the scanfile. The saving of the offset vector is done by

initiating a write from physical SBC memory to the Local SCSI Disk.

Restoring Scan Data

The retrieval of scan data starts with a read of the scan header from Local SCSI disk to SBC physical memory. Based on the data views requested, the software then initiates a read of the raw scan data from Scan Data Disk to SBC physical memory.



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Image Generation, *continued*

Scan File Management, *continued*

Signal Flow #	Reporting Host	Software Function	Associated Errors	(FRUs) Involved
1	SBC	SBC allocates scan file		SRU , Local Disk
2	SBC	Write scan header to Local Disk		SRU , Local Disk
3	SBC	Write scan data to Scan Data Disk		DIP , SRU , Scan Disk
4	SBC	Write offset vector to scan header		SRU , Local Disk
5	SBC	Read scan header to SBC memory		SRU , Local Disk
6	SBC	Read scan data to SBC memory		SRU , Local Disk

Applicable Diagnostic Coverage

1. Scan Data Path
2. **DIP** Diagnostics
3. Recon Data Path

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Image Generation, *continued*

Recon Control

The two main processes involved in recon control are **queuemgr** and **reconmgr**. The queuemgr process receives all recon requests and makes sure that reconmgr creates the images in the correct order. The queuemgr process is the main interface between recon and the rest of the system.

The reconmgr process pulls image requests from queuemgr sequentially and manages the reconstruction process in a pipeline fashion. The output of recon is a **DICOM** file (created on the OC) that can be installed in the image database.

The basic flow of recon is as follows:

- image request received by queuemgr
- queuemgr queues the request
- queuemgr notifies reconmgr
- reconmgr reads the next request in the queue
- reconmgr processes request
- reconmgr sends image header and image pixel data to image create on OC
- reconmgr tells queuemgr image is being created

- image create creates a **DICOM** file
- queuemgr is notified by OC when image is installed in database
- queuemgr removes image queue request

Since reconmgr is pipelined, it can process more than one image at time. Queuemgr makes no assumptions about the depth of the reconmgr pipeline and allows reconmgr to pull as many image requests as it can handle.

Image requests are generated only when there is enough scan data for the image on disk. During scanning, image requests are generated as soon as enough scan data is available to disk so that image generation can begin as soon as possible. A basic image request consists of the following:

- a scan key to indicate where the scan data is located on disk
- an image key for pre-allocated space in the image database
- scan-independent reconstruction parameters

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Image Generation, *continued*

Recon Control, *continued*

All image requests go through queuemgr and are queued into one of four queues according to their priority. The priority and contents of the queues are defined as:

Priority	Que #	Image Request
Highest	1	Scout
	2	Priority Recon
	3	Prospective Axial and Helical
Lowest	4	Retrospective Axial and Helical

Images are grouped by exam in each queue, and exams are queued in a last-in first-out (**LIFO**) basis. The images within each exam are queued in a first-in first-out basis (**FIFO**). The queuemgr process will notify reconmgr of new image requests in the queue as they come in.

An image is not removed from the queue until it has been installed in the image database. If an image fails to recon or install, queuemgr will allow recon to reprocess the image once more. If the image fails twice, it is put into a suspended state [error 230009013] and does not get removed from the queue. Suspended images will not be retried until they are un-suspended by the operator.

If five images are suspended in sequence, queuemgr will shut down and not allow recon to process any more images. When the queue is shut down, images can be still be queued, but new images will not be reconstructed until the operator selects **Resume** in the recon management window.

When each image is queued, a file is created on the SBC in the **/usr/g/queue** directory. This file is to ensure that an image is made. This file maintains the image request information in the event of an unexpected shutdown or power failure. The file also allows suspended images to be maintained across system restarts so that the images can be reconstructed (unsuspended) once the system is serviced. Image requests on the queue are handled by reconmgr whenever there is space in the recon pipeline.

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Image Generation, *continued*

Recon Control, *continued*

When reconmgr retrieves an image request, queuemgr starts a timer to wait for an acknowledge from reconmgr. The acknowledge tells queuemgr the results of reconstruction as follows:

- completed
- failed
- canceled

Completed means that the **DICOM** image data has been transferred to the OC, but not necessarily installed. In the image database. Queuemgr then starts a timer to wait for the an indication that the image has installed. Only after the DMS process says the image is installed is the image removed from the queue and its file deleted on **/usr/g/queue**.

Failed means that the image could not be reconstructed because one or more recon processes encountered an error of some kind while reconstructing the image. Failures can be flagged as "non-retrieable".

Canceled means that a high-priority image forced recon to stop processing the image in order to handle the higher priority image. For instance, this happens normally when a scout image is queued while the recon pipeline is full of images from a previous series.





Gantry Rotation

The hardware involved in the axial servo drive consists of the Axial Control Board, Axial Motor Drive Module, Axial Motor, Axial Motor Dynamic Brake Assembly, Axial Encoder, Axial Drive Belt and Axial Home Flag.

The **Axial Control Board (ACB)** interfaces the axial drive control firmware of the gantry Stationary Controller (**STC**) microprocessor to the **Axial Motor Drive (AMD)**. The ACB supplies the AMD with enable, speed, brake and direction information. The Axial Control Board, located in the STC backplane, uses the VME bus to communicate with the STC CPU. It uses the CAN (Controller Area Network) serial interface, called the ACAN (AXIAL CAN) and discrete signals to communicate with the AMD. It uses the encoder to coordinate scan events with azimuth location.

The ACB generates **DAS triggers** for all data collection modes. The DAS trigger signal uses inputs from the axial encoder, table, and X-ray command circuitry to coordinate data collection with gantry and table motion. This function also produces triggers for DAS offset characterization and view triggers for data acquisition.

The **Axial Motor Drive (AMD)** uses a customized version of an Allen-Bradley Model 1336 Plus II variable frequency AC motor drive. It has its own microprocessor, power supplies and three phase full bridge inverter.

The AMD communicates with the Axial Control Board through a CAN (Controller Area Network) serial bus. The AMD monitors axial speed and direction; it receives axial encoder feedback from the Axial Control Board for fine speed control. The combination of internal and external control allows the drive module to modulate the frequency of the AC output allowing the gantry to rotate at a maximum speed of 0.5 seconds per revolution in the clockwise direction. The AMD supplies isolated 12 V power (1.2 A capability) to the ACAN and Fault circuitry.

The AMD module also monitors the AXDC Bus Voltage. Errors are reported to the Axial Control Board.

The Axial Motor Drive (AMD) function provides three phase motor power to the Axial AC induction motor. The AMD is a two part integrated device: Axial Interface board (AIF) and AC Drive. The axial firmware controls the acceleration, run, and deceleration cycles of the axial motor. This firmware resides in the STC CPU and communicates drive instructions to the Axial Control Board (ACB). The ACB communicates with the AC Drive through the (AIF), located on the AMD assembly. The AMD interprets the instructions and configures itself to execute the commands.

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Gantry Rotation, *continued*

The AMD has its own firmware. The drive converts three phase 440 VAC input power to three phase Axial HEM (High Efficiency Motor Drive). This power drives the acceleration, run and braking functions.

The Axial Motor is an AC induction type. It is coupled to the rotating **bearing** with a **belt**, with a gear ratio of 13 motor rotations to 1 gantry bearing rotation. The axial motor requires the external Axial Dynamic Brake Module and the Axial Control Board.

An optical "slip-flag" detector senses a metal plate that rotates through it once every bearing revolution. The **Home Flag** information is used for axial coincidence as a position reference and over-speed sensing.

The axial encoder is directly coupled to the rotating bearing next to the Axial Drive Belt. It is interfaced with a direct contact gear as weighted assembly with a 13 to 1 gear ratio, the same as the axial motor to rotating bearing. This arrangement allows for simplified transfer functions for speed control and more accurate positioning of DAS trigger generation versus motor mounted encoder packaging. The encoder is incremental and provides 2048 counts per encoder rotation. Information from the encoder is decoded on the Axial Control Board and used for gantry position and speed control as well as DAS trigger generation. The STC

firmware verifies the encoder counts to be 106496 ± 3 counts per gantry revolution referenced to home coincidence.

The **Axial Dynamic Brake assembly** dissipates induction motor energy. Axial braking is accomplished by reducing the frequency applied to the motor leads to a level lower than the current operating point of the motor windings. This is called negative slip.

Negative slip generates electrical field resistance or negative torque within the windings. At the same time the motor windings generate voltage raising the motor lead voltage. This excess voltage is dissipated by the dynamic braking module via load resistors and chopper circuit control via the AXDC bus. This means that 440 VAC is always applied to the Axial Motor Drive module to both accelerate and decelerate the gantry. **Loss of this voltage will result in the gantry coasting to stop.** This can take more than a minute.

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Gantry Rotation, *continued*

This assembly provides logic and control power for the AMD when the Axial Power Contactor is open. During gantry deceleration, excess AMD AXDC bus power is dissipated through chopper load resistors.

The Axial Dynamic Brake Assembly is a modified HEMRC circuit. All components are identical to the Rotor Controller but configured differently.

The **axial brake** engages the motor shaft. It is a static brake meant to hold the gantry once it has been positioned by the axial drive. When the Axial Drive Enable Switch is in the "Disable" position, the axial brake is released to enable you to rotate the gantry by hand. This brake cannot hold the gantry when you are servicing the rotating base, so use the **rotational lock**.

Axial Control Communication

A hard-key processor on the keyboard assembly is hardwired to the Axial Control Board via a **UART serial link**. It monitors the status of the "Start Scan" and "Stop Scan" switches on the keyboard. Closure of the Start Scan Switch Contacts sends a start scan message to the STC via the ACB indicating that the scan should start.

This same communications link is also used to signal the start of **AutoVoice** messages.

The **ACAN** interface is the axial CAN communication located in the Stationary Gantry. It carries discrete control signals that enter the AIF for AC Drive control.

Fault detectors and fault status feedback from the ACAN monitor hardware operation. The AMD notifies the ACB when it detects a fault, or a monitored value falls out of tolerance. Refer to **Axial Drive Error Code** descriptions.

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Gantry Rotation, *continued*

Axial Control Interface Bus (ACIB)

The ACIB is the umbilical cord that links the Axial Motor Drive to the Axial Control Board. Each signal is a differential pair for noise immunity. This bus consists of these signals:

- **Isolated 12 VDC** is generated by the HEMRC Control Board which powers the Controller Area Network (CAN) Drivers.
- **Axial Enable** is available to the AMD to determine when gantry rotation begins and ends.
- **CAN Serial Line** is used for the transmission of control signals. It must be terminated by a 120 Ohm resistor at the beginning and end of the cable. This particular CAN line is referenced as the Axial CAN (ACAN) bus.
- **Fault Line** is the primary means to inform the Axial Control Board of a fault. It is asserted by the AMD whenever a reset or fault is detected by the controller.

- **Reset** is asserted by the STC when it becomes desirable to reset the AMD. The ACB resets the AMD via a command register or during an STC reset.

Note The drive module will not drive in a counter clockwise direction under normal control.

Note The **Axial Power Contactor** in the PDU controls the three phase, 440 VAC input of the Axial Motor Drive Module. There are interlocks in the power contactor control to the Axial Drive Enable/Disable switch located on the **STC backplane**.



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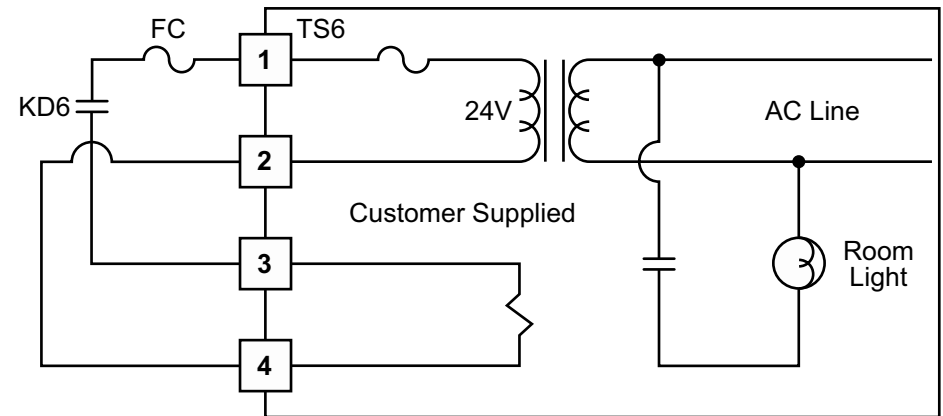


X-ray Warning Light Control

The **Axial Control Board (ACB)** provides a set of relay contacts, K1, to control the site X-ray ON Light relay in the PDU. The relay contacts close (X-ray light turned on) whenever the Axial Control Board commands X-ray to the OBC.

The ACB drives a 24 V 100 mA LED mounted in the rear of the gantry. This light, separate from the gantry display, indicates the presence of the ACB's X-ray command and will coincide with the PDU X-ray ON light signal.

The light is controlled by firmware rather than hardware so that it can be coordinated with actual X-ray state and not just the X-ray exposure command. STC firmware receives notification via the **communication slip rings** from the OBC when X-ray turns on and off.





Scan Control and Coordination

The **STC Axial Control Board (ACB)** trigger circuitry supports four scan modes: static, scout, axial, and helical. The modes and offset or view trigger output are selected by firmware. ACB firmware sets up the hardware by pre-programming the modes and parameters before the triggers are generated.

Scout and helical modes require a SYNC pulse from the **table** to coordinate the start of triggers. Scout scans use a fixed clock input reference to generate the triggers. Static mode also uses a fixed clock reference. Helical and axial use the axial ENCODER signal as a reference to generate triggers.

The **DAS trigger counter** controls the **duration** of a scan by counting the number of DAS triggers generated by the Axial Control Board (ACB). When the programmed number of triggers is received, the ZERO TRIGGER signal is asserted causing: EXPOSURE command to deactivate, trigger generation to deactivate and firmware to be interrupted by a maskable interrupt on VME IRQ4.

For variable length scans, such as the CT Smartview option, four counters make the DAS trigger circuit: pretrigger, minimum triggers, maximum triggers and cycle triggers.

Note Counter output is active when set to zero.

When the pre-trigger counter expires, the other counters begin counting. This allows for a fixed number of triggers to be generated outside the control of the other counters.

The minimum counter inhibits the zero trigger signal until at least a minimum number of triggers in addition to pretriggers have been generated.

The maximum counter asserts ZERO TRIG when the maximum number of triggers in addition to pre-triggers have been generated.

The cycle counter, with a bit programmable via the VME, can be used to force the system to collect an integer number of triggers in addition to pre-triggers. The cycle counter allows for the acquisition of a variable number of "sectors" during a scan.

Speed control is achieved by using the **Axial Encoder** input in two places: STC and AMD via the ACB. The AMD uses the ENCODER feedback to close its control loop and regulate speed. The STC uses the encoder feedback and compares it to the SPEED feedback from the AMD whose status is polled every 25 ms. A speed fault will occur if five consecutive samples are out of tolerance. The speed regulation is $\pm 3\%$ of commanded velocity.



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X-ray Generation

High Voltage Generation

KV is regulated/monitored by the OBC kV board which senses the feedback voltage dividers in each HV Tank. The kV board maintains kV consistency by adjusting the operating frequency and duty cycle of each Inverter. If the positive and negative KV's differ by 3% or more, an error appears. When they differ by 5% or more, the scan aborts.

The kV reported to the screen does not necessarily indicate what the actual kV is due to potential kV gain calibration. KV imbalance is usually due to the result of a HV component being "opened."

Tube Filament

The mA Filament Control Board in the OBC monitors the output voltage of the filament inverter. An improper voltage is reported as an open filament. Both large and small filament use the same board circuitry and relay **K78 on the OBC backplane**. K78 is switched by filament circuitry on the mA Control board.

Every time someone removes the cathode HV cable from the tube well, a **Filament Open Error** gets logged.

Tube Rotor

The HEMRC Rotor has nine phases of operation: three acceleration phases, two Run phases, three Brake phases, and idle. The Acceleration and first Run phase use HVDC; the rest use 120 VAC stepped up to 380 VAC by T1. When the HEMRC AC Drive internal DC bus voltage drops below 325 V, the tube rotor fails. The Internal bus is supplied by either HVDC through CR1 or 120 VAC through T1.

Determining which phase of rotor operation is producing an error is a clue to the **possible cause**.

Slip Rings

The gantry transfers High Voltage from the Generator to the tube, from the stationary to the rotating portion of the gantry, through the inner most **high voltage slip rings**.

The gantry transfers Line Voltage from the PDU to the rotating portions of the gantry through the **Power** slip rings #4 and #5.



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Table

If the system fails the [Interference Test](#), recharacterize table drives; **do not** adjust microswitches.

Table Elevation

Control of the **elevation** closed loop drive system is provided by the ETC computer, control and interface boards. Interlocks and enables are set by a table/gantry interference matrix and firmware.

The **drive amplifier** is supplied with 170 VDC and creates a three phase half wave rectified drive voltage that is pulse width modulated at a switching frequency of 17 kHz.

The resulting output is supplied through enable and motor select relays to the **table elevation drive motor**. The control circuit has no adjustments.

Elevation feedback is provided by a 6:1 geared **encoder** which the [ETC](#) board firmware converts to elevation information. The encoder turns one complete revolution over the entire table elevation range. Control signals are routed via the **ETC I/F board** and the signal enabling elevation is intercepted and another enable is created **so the Interface board can disable elevation if the interference sensor is in fault or interference is detected**.

Cradle Drive

Control of the **cradle drive** closed loop drive system is provided by a single chip motion controller located on the ETC Board. The controller sets velocity, direction, acceleration, and position.

The **drive amplifier** is supplied with 24 VDC and creates a three phase half wave rectified drive voltage that is pulse width modulated at a switching frequency of 17 kHz.

The resulting output is supplied through an enable relay to the **cradle drive motor**. The motor turns a drive roller at the front of the table that the cradle rests on thus causing the cradle to move.

Direction and speed feedback is supplied by an **encoder** and a 10 turn potentiometer driven by a cable and spool assembly attached to the cradle mounting hardware. The cradle encoder outputs approximately ten pulses per mm of cradle movement and makes eight full revolutions over full cradle range. The potentiometer determines which of the eight revolutions the encoder is in. A **tachometer** is used for additional stabilization of the control loop.



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Gantry Display



Refer to [Gantry Display Check](#).

- [1] **ECG Gating** lit means ECG gating working.
- [2] **Scan Location** is mm (S/I) distance from landmark.
- [3] **Elevation** distance in mm from isocenter.
- [4] **Tilt** indicates degree of gantry tilt and direction Superior (head) or Inferior (feet).
- [5] **Alignment Lasers** On when lit.
- [6] **Interference** between tilt and cradle limits.

[7] **Cradle Not Latched** indicates cradle released or collision table sensor strip activated. Slide the cradle by hand to its rearmost location to relatch it.

[8] **Respiratory** is a future application.

[9] **X-ray On** indicator is lit during a scan.

[10] **Breathing Lights** cue patient when to breathe and when to hold their breath.



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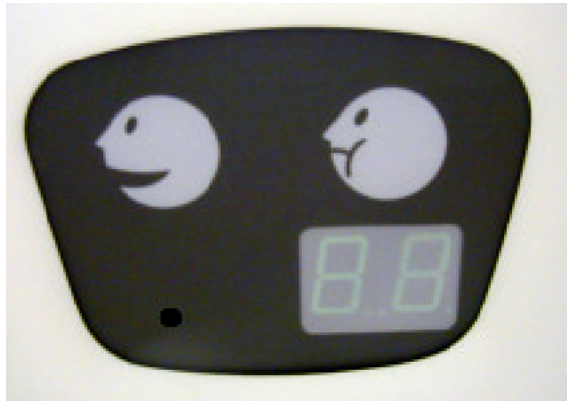


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Breathing Aid

Gantry Display Breathing Aid



The Breathing Lights get enabled on the AutoVoice menu.

The Selection field is called Voice/Lights/Timer. Breathing Lights must be prescribed to enable the Timer.

Meaning of the lights:

solid green:	breathe
blinking green:	get ready
solid yellow-orange:	hold breath

Count Down indicator: seconds left to hold breath

Refer to [Gantry Display](#).



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Gantry Keypad



Refer to [Gantry Controls](#) for additional key information.

[1] **Breathing Lights Demo** use to demonstrate breathing aid indicators. There are four steps, blinking green: get ready, solid yellow: hold breath, count down

timer, and solid green: breathe. Press the DEMO button to see or demonstrate each step.

[2] **Tilt and Table Travel limits** indicates available tilt range and table travel based on landmark. A table height of 118 allows maximum tilt.

[3] **Internal Landmark** sets landmark, the 0.0 scan location, using internal laser light location.

[4] **External Landmark** sets landmark using external laser lights location. The external landmark is ~ 240mm from scan plane.

[5] **Alignment Light** turns alignment light ON/OFF

[6] **Cradle Release** releases the cradle so it is free to move in or out of gantry.

[7] **Table collision sensor Indicator/RESET** - LED next to button indicates when a proximity switch on the table or gantry has been activated. Up/down table motion will be stopped until this button is pressed to reset the interlock. Used also to **Reset an Emergency Stop** activation.

[8] **Home** moves table down and back, untilts gantry.

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Gantry Keypad, *continued*



[9] **PET** moves the Patient Table to the PET acquisition point.

[10] **Table Down** lowers the table at slow speed of 5 mm per sec.

[11] **Cradle In** moves the cradle into the gantry at slow speed. LED indicates a prescribed table movement into the gantry.

[12] **Table Up** raises the table up at slow speed of 5mm per sec. LED indicates a prescribed table movement out of the gantry.

[13] **Fast** used with CRADLE IN/OUT or TABLE UP/DOWN buttons to move cradle and table at fast speed.

[14] **Cradle Out** moves the cradle out of the gantry at slow speed.

[15] **CT** moves the Patient Table to the CT acquisition point.

Note You can still press both tilt buttons at the same time to tilt to a prescribed tilt as done on the QX/i control panel.



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SCIM

The **Scan Control Intercom Module (Scan Control Intercom Module (SCIM))** and a commercial off the shelf **keyboard** replace the current custom user interface keyboard. The SCIM provides:

- **Scan Control** (Start Scan, Pause Scan, Stop Scan, Move To Scan, Stop Move),
- **Emergency Stop**
- **Intercom** (Patient and Operator Comm, Auto Voice)
- **X-ray On indicator** like the gantry one.

The SCIM is connected to the Operator Console Back Bulkhead at J19. The physical interface is a 1.8 meter 50 conductor cable with micro HD 50 male connectors at each end. This interface cable provides DC power (+5, +12 V and -12 V) along with communication (RS-422) and discrete signal paths.

On power up, the SCIM performs a self test of the processor and memory. Upon passing all self tests, all LEDs will be illuminated for two seconds then turned off. If any of the self tests fail, the LED On/Off sequence will not occur. The results of the self test are available for retrieval by the Operator Console processor. This self test can also be initiated by the Operator Console processor.

A microprocessor within the SCIM monitors the **communication path** from the Operator Console processor for requests and the state of the keys on the SCIM. Any time a key state change is detected, the state of all keys (pressed or not pressed) is sent to the Operator Console Processor.

The SCIM has been designed to be withstand accidental spills in the clinical environment such as coffee or soft drinks. An **elastomer** within the SCIM routes spilled liquids to drain holes on the underside of the unit. **Over current protection** has also been incorporated into the SCIM in the event liquid does reach internal electronic components.



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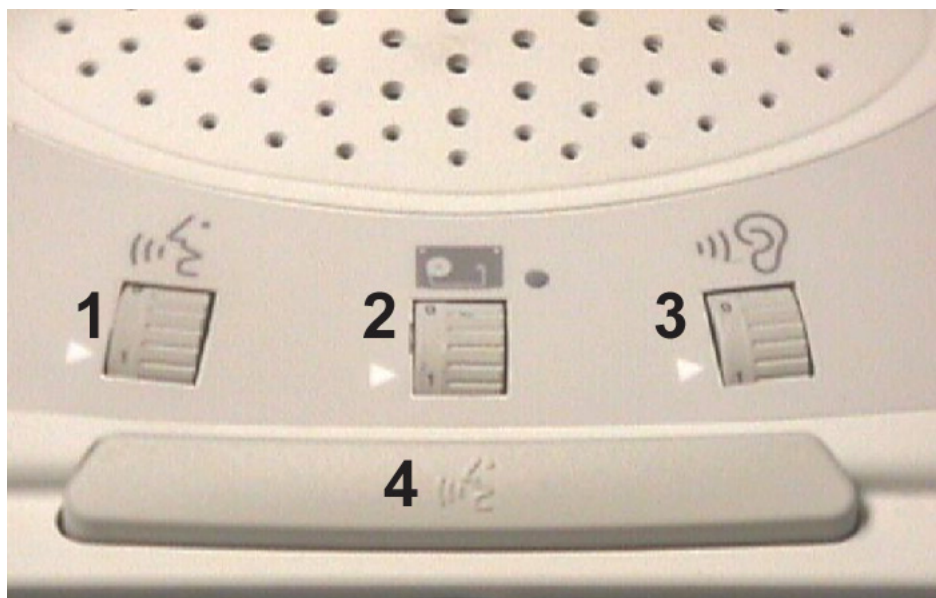
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Intercom



Speaker for intercom.

[1] **Volume Control to Patient** for instructions spoken using the TALK button. Volume is adjusted by rolling control wheel up and down. Control wheel is numbered. Higher number setting produces a louder sound. Triangle at bottom of wheel indicates volume level selected.

[2] **Volume Control for AutoVoice** messages. LED illuminates when AutoVoice is in use. Volume is adjusted by rolling the control wheel up or down.

[3] **Volume Control from Patient** for volume the operator hears when the patient speaks from inside the gantry room. Volume is adjusted by rolling control wheel up and down. Control wheel is numbered. Higher number setting produces a louder sound. Triangle at bottom of wheel indicates volume level selected.

[4] **Talk button** is held down to send the sound of your voice to gantry speaker.

Refer to [Intercom Functional Check](#).



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Safety

In order for the LightSpeed to be safely operated, all covers must be in place. Equipment covers are the primary means of protecting patients, operators and service personnel. Removal of any cover potentially exposes all to hazards.

If any of the covers should become damaged, you should contact your local GE Sales or Service representative immediately for replacement or repair.

Only qualified service personnel trained in the service and operation of this scanner should remove any cover or service this equipment.

Refer to [How to Safely Remove Covers](#).

Refer to [System Safety Guards and Resets Theory](#).

See [Emergency Stops \(E-Stops\) Button Locations](#).

See [Gantry Rotation Control Locations](#).

Refer to [Gantry Safety Warnings](#).

Refer to [Gantry Display Indicators](#).

Refer to [Gantry Controls](#).

Refer to [Console Scan Controls](#).

Refer to [X-ray Warning Light Control](#).

Potential Hazards

When the system covers are all in place, two potential hazards exist during the operation of this scanner unless proper safety precautions are followed. They are:

- **X-ray** - Radiation generated during a patient or service scan.
- **Laser Alignment Lights** - Eye damage from looking directly into the alignment light beam for an extended period of time could occur.

To prevent injury from these potential hazards, the following precautions must be taken:

- Proper radiation training and shielding for operators and service personnel.
- Check that the scan room is clear before you scan.
- Instruct patients and operators to refrain from looking directly into the patient alignment beams.



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- Additional References
- Gantry Safety
- Slip Ring Communication
- Control Slip Ring Bypass

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Data Acquisition

- DAS Pre-Check (Image Quality)
- Detector Temperature
- DAS Noise

DAS Data (Slip Ring)

- DAS Pre-Check (Image Quality)
- Symptoms
- Suggested Troubleshooting

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X-ray Generation

- General Checks
- kV Generation
- mA Generation
- Tube Current (mA Loop)

Tube Filament

- Tube Rotor
- X-ray Collimation

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Artifacts

- Hard
- Soft
- Troubleshooting DAS Artifacts
- Troubleshooting Ring Artifacts in a Helical Series

X-ray Generation Troubleshooting

X-ray Generation Troubleshooting

- X-ray Generation Troubleshooting, continued
- Ohm Filament
- mA Tolerance or Imbalance Errors
- Rotor Control and DC Bus Monitor
- RCIB Errors: System Comm./Control
- kV Loop Imbalance
- Filament Inverter
- Z-Axis Tracking

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[Scan Data Disk & SCSI Controller Troubleshooting](#)

[PMC DIP Board Troubleshooting](#)

[Pegasus Image Generator Board Troubleshooting](#)

Image Generation

[Scan Control Intercom Module \(SCIM\) Installation](#)

System

System Communication and Control

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[Gantry](#)

Refer to Software section for [Image Analysis](#) info.

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System Communication and Control

Refer to [System Control Theory](#).

Refer to [Slip Ring Communication Theory](#).

Refer to [Slip Ring Communication Maintenance](#).

Console

Refer to [Console LANVIEW LEDs](#).

Refer to [Common Service Desktop, continued](#).

Gantry

Refer to [Slip Ring Communication Checks](#).

- Verify 12 VDC at the HEMRC Control Board (HCB)
- Verify 12 VDC at the Collimator Control Board (CCB)
- Verify 12 VDC at the DAS Control Board (DCB)

Refer to [DAS Data \(Slip Ring\) Troubleshooting](#).

Refer to [Control Slip Ring Bypass Procedure](#).



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X-ray Generation

General Checks

- Ensure both jumpers are installed in the Artesyn boards for 20 MHz operation for Helios systems.
- New **Artesyn boards** have built in Ethernet connections; SBX cards are no longer needed.
- Run X-ray Interlock Functional Test
- Run X-ray Exposure Manual Test
- Check **Slip-ring Stats** when troubleshooting **LAN Watch Dog** errors.
- Ensure the 50 Ohm inline terminator is installed at the **ETC ethernet connection**.
- The DCB must have a terminator connected to the RCIB connection.

kV Generation

- Verify this is the primary cause error.
- Note which kV board is present.
- Run **kV LOOP test** and use result screens to determine FRU - If this diagnostic fails, the tube can be eliminated as the problem.

- IF kV LOOP does NOT fail, run **kV & mA test** and use result screens to determine FRU.
- Scope kVp at HV Tank Measurement Bd test points. 10,000/1 divider, 2 VDC Vertical and 50 msec/div Horizontal are good starting points.

mA Generation

If an Inverter or mA loop error occurs:

- Check whether the **mA Control Board** is generating a square wave with inductive spikes on the center tap. TP27 can be used to look at this square wave. Use 10X probes, time base of 20 usec/div, voltage 5 V/div. Use TP29 or 30 (FGND) for ground. System on standby.
- Check the 30 VDC Filament Power Supply, harness, and Filament Control Board. If mA Control Bd TP 21 has 30 VDC but fails to make square wave, this board is bad.

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X-ray Generation, *continued*

Tube Current (mA Loop)

- Verify mA Loop error in log; note all pertinent details
- Run kV and mA diags to determine whether mA out of tolerance is the primary error
- Verify mA meter
- Check Focal Spot Relay
- Troubleshoot Filament Drive error
- Check 36 VDC Filament Power Supply
- Check **OBC Power Supply**
- Redo software **.char** files

Tube Filament

- Check Generator 38 VDC Power Supply
- Ohm Filament and HV Cable
- Verify Filament Drive
- Verify Filament Command
- Verify filament select relay K78 on OBC backplane
- Test Transformer & Measurement Board
- Try shorting out the filament until open goes away

Tube Rotor

- Check connectors, fuses and T1 transformer to ensure there is power to the rotor
- Rotor errors can occur during the following phases:
 - **Acceleration**
 - **Run**
 - **Brake**

A	R	B	Possible Cause
X	X	X	HEMRC Interface Board (HIF)
X	X	X	HEMRC AC Drive
			Loose Connector
			HIF F1 F2 (when no HVDC)
X	first		HVDC Bus
X	first		CR1
	last	X	T1
		X	120 VDC
		X	X-ray Tube

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X-ray Generation, *continued*

X-ray Collimation

- Check tightness of Collimator Control Harness
- Check Filter Amp
- Check Filter Motor
- Check Filter Encoder
- Check CAM Motor(s)
- Check CAM Encoder
- Check CAM AMP / Driver
- Check CAM Home Switch
- Test Collimator Control Board (CCB)
- Run Characterization (software)

Use these X-ray Generation diagnostics:

- Final Exposure Command
- Exposure Enable (Interlocks)
- Rotating HVON Sense
- Table Gantry Synchronization



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Artifacts

Refer to [Image Analysis](#) in the Software and Diagnostics Section to further assist with image quality issues.

Hard

Hard artifacts are the ones that can be reproduced, such as Bad Cals, DAS, and Elastomer or Detector faults. These are the most likely causes for a hard image artifact.

1. Verify that the artifact is hard by re-scanning using same technique.
2. Do Fast Cal, then re-scan.
3. Run a full set of cals, then re-scan. If time is limited, cal the technique where artifact is most conspicuous.
4. Troubleshoot the DAS.
5. Troubleshoot the Detector.

Soft

Soft artifacts are subtle because they come and go, and typically disappear with a recalibration. A tube or collimator fault, or beam obstruction are the more likely causes for soft artifacts. Detector non-uniformity in gain or temperature can also cause **image drift**.

1. Do a full set of FAST CALS to re-establish a clean cal baseline. System should now be "Warm."
 2. Create base line images and check for the artifact. This step requires that the detector has been spinning and therefore detector temperature is uniform across the detector.
 3. Do "Warm" DDC air scans; save them for later use.
- Note** Artifact should be gone. If not, it is a hard artifact.
4. Check the **detector heater**. If there's a detector temperature error in the log, [check hardware](#).
 5. Wait two hours to create a "Cold" system with gantry parked. Then repeat baseline image scans.
 6. Normalize detector temperature by spinning the gantry for 20 minutes. Then repeat baseline image scans.
 7. Analyze MS DD files. Ratio Warm against Cold.

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Artifacts, continued

Troubleshooting DAS Artifacts

Reproducing the Artifact

1. Find the image with rings, bands, smudges or streaks.
2. Record the parameters that the operator used:
 - Slice thickness
 - MA
 - KV
 - Axial or helical
 - Number of slices
 - Rotation speed
 - Scan Field of View
 - Cradle speed (pitch)
3. Try to create the artifact on a phantom in axial mode.
4. Reproduce the operator's scan.
5. If necessary, vary the mA and kV values.
6. At this point, you have either recreated the artifacts on an image, or must use the patient image with the artifact. It is easier to identify the failing channel on an axial phantom image than on a patient image. In either case, record the:

- Exam number
- Series number
- Image number

Note Streaks can be caused by:

- Straps hanging down onto the Mylar window and interfering with the reference channels. You can spot this in the images by turning up the window/level settings.
- Low techniques, which usually show up as streaks coming off the clavicles or the top of the vertebrae or hips. This should be checked with the application specialist to determine whether this is normal or can be corrected.
- Metal objects inside the patient or within the scan plane usually show up as streaks or a star pattern pointing to a white spot. The white spot is the hard object causing the streaks.
- Any contrast on the head holder, pads or cradle can be reproduced by scanning them. The cradle and head holder can be cleaned. Sometimes the pads can be cleaned as well.

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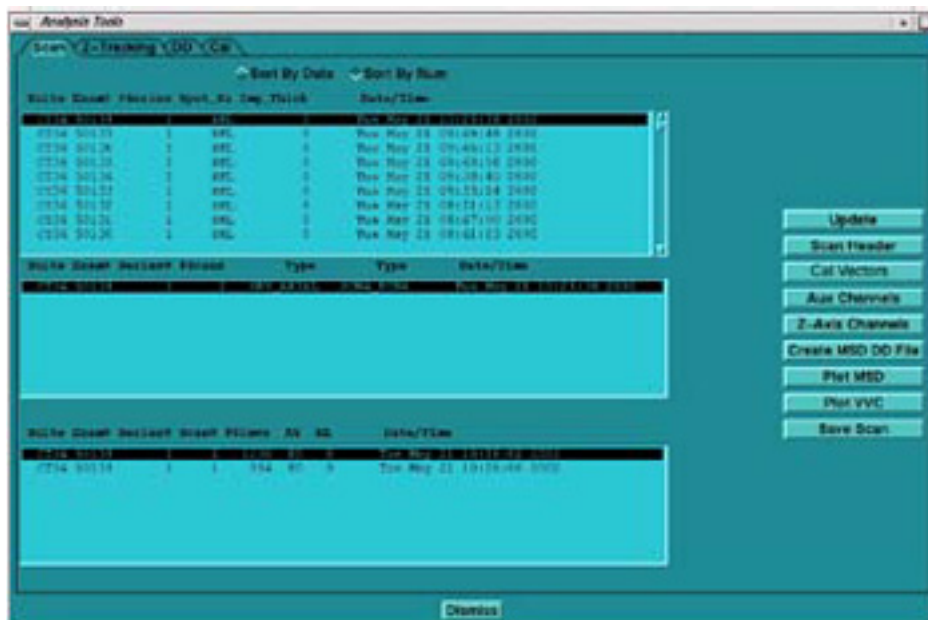
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Troubleshooting DAS Artifacts, continued

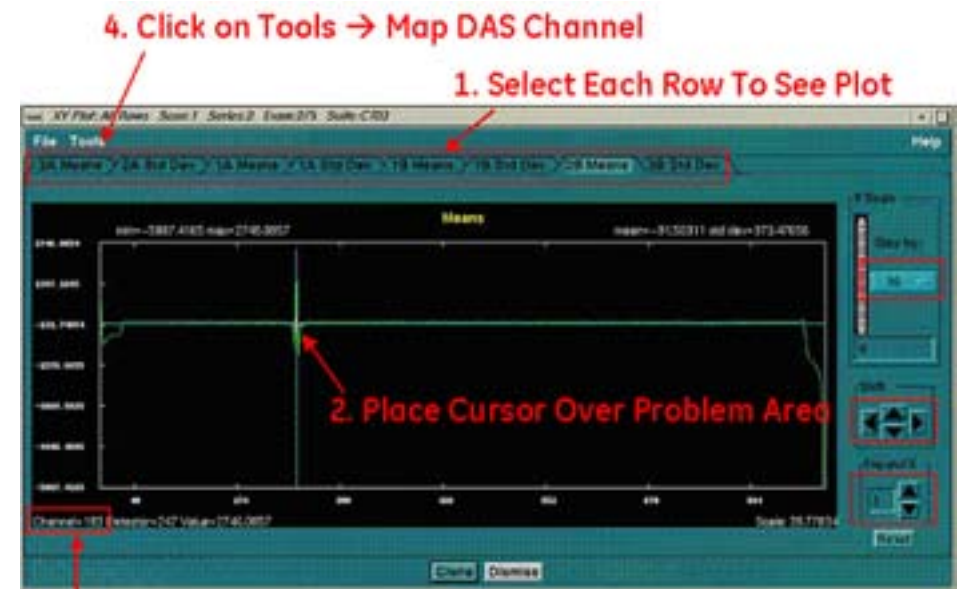
Using Scan Analysis to Identify Failed Channel Numbers

1. Select **Service Desktop > Image Quality > Scan Analysis** for the following Scan List Select screen:



2. On this screen, highlight **Exam > Series and Image**. Click on the **Plot MSD** button.

3. On the next screen, select **Offset Correction**, **Normalization**, and **Convolved Data**, and click on the **OK** button. This prompts the following **Means Plots** screen:



4. Select each tab (step 1); tabs 2A to 2B are for 4-Slice, 4A to 4B for 8-Slice, and 8A to 8B for 16-slice. Check each one for possible spikes, recording all that are found.

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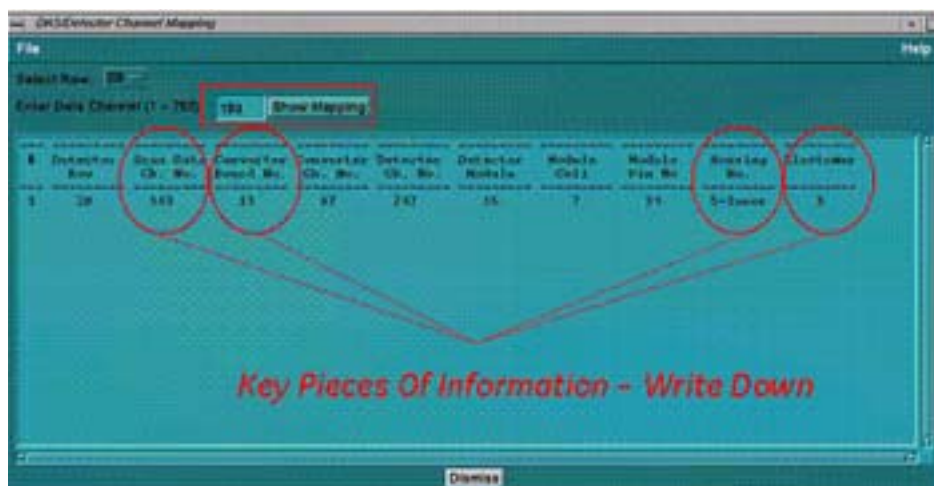


Artifacts, continued

Troubleshooting DAS Artifacts, continued

Identifying Failed Channel Numbers, continued

- When a spike is found, place the cursor over the spike (step 2) and record the Slice (1A, 1B, etc.) and DAS channel number (step 3). Do not record the detector number because the whole detector is a Field Replacement Unit (FRU).
- Select **Tools > Map DAS Channel** (step 4).



- A screen appears with information for the mapped DAS Channel number (183 in above example).
- Record the information from the columns shown above.

- Return to the previous screen by selecting the **Dismiss** button.
- Repeat steps 6 through 9 until each tab mentioned in step 4 has been checked. When finished, you should have the following information for each spike found:
 - Slice Number (2a to 2B/4A to 4B/8A to 8B)
 - DAS Channel Number
 - Converter Board Number
 - Housing Number
 - Elastomer
- Consider any of the following steps as possible causes of the spikes.

Step 1: Check for Dirty Parts

- Remove the Mylar window.
- Clean the surface of the detector and collimator with a **damp but not wet** cloth.
- Replace the Mylar window.
- Rescan and see if the problem is gone.

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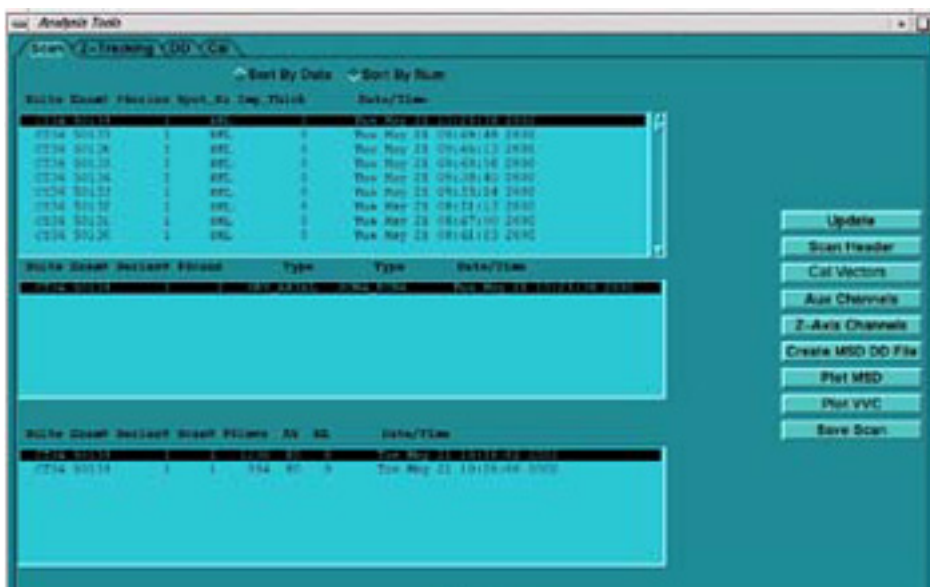


Artifacts, continued

Troubleshooting DAS Artifacts, continued

Step 2: Check Power Supplies

1. Select **Service Desktop > Image Quality > Scan Analysis** for the following Scan List Select screen:



2. On this screen, highlight **Exam > Series and Image**. Click on the **Aux Channels** button.
3. The next screen is a menu selection of auxiliary channels. Select all of them and click on the **OK** button.

4. Step through all the Power Supply voltages. If you see any spikes, you must replace that Power Supply.

Step 3: Check Converter Boards

1. Open the DAS section that contains the board identified by channel number.
2. Reseat ALL boards, and move the suspected bad board or boards.
3. Rescan.
 - **If the problem disappears**, then the loosened board has been secured. Run fast cals.
 - **If the problem remains**, then replace any identified board(s). Rescan to confirm that the problem is resolved.

Step 4: Check Elastomers or Interposers

1. Remove affected inner and outer flex leads.
2. Remove the plastic plate with the elastomers, or remove the interposers.

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Artifacts, *continued*

Troubleshooting DAS Artifacts, *continued*

Step 4: Check Elastomers or Interposers, *cont'd*

3. For elastomers:

- Clean the DAS backplane and flex leads that come into contact with the elastomers.
- Replace the elastomers.
- Reverse procedures in steps 1 and 2.

For interposers:

- Remove them and check for bent pins.
- Replace the elastomers.
- Reverse procedures in steps 1 and 2.

4. Rescan to confirm the problem(s) is resolved.

Step 5: Check FET Switching

- Cycle DAS power and make sure all LED's on the DAS backplane turn on and off.
- If not, then it is possible that the DSB or something else is shorting the DAS backplane.
- Rescan to confirm the problem is resolved.

Step 6: Check DAS Backplane Interconnections

- Remove the flex leads between the parts of the DAS backplane.
- Use contact cleaner to clean the pins and sockets.
- Replace the flex leads.
- Rescan to confirm the problem is resolved.

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Troubleshooting DAS Artifacts, continued

Step 7: Check the Collimator

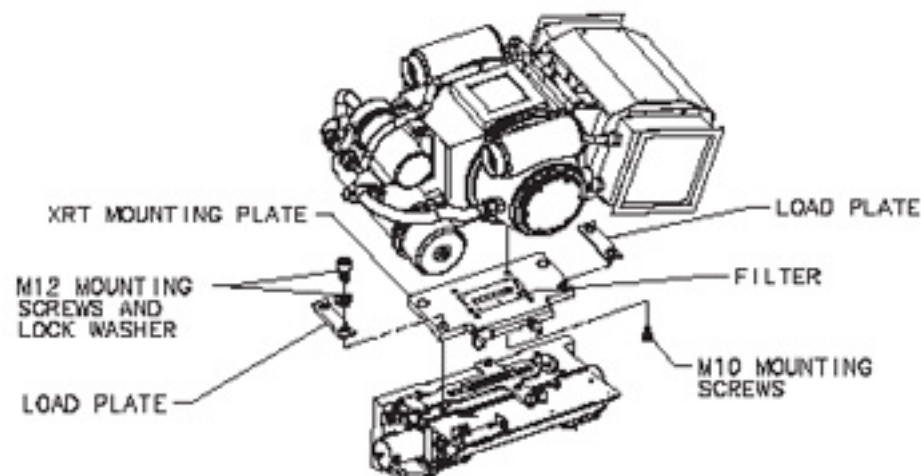
1. Open the collimator, and remove the filter.

Important! Use only GE approved alcohol pads (P/N 46-183039P1) for the next step. DO NOT use alcohol prep pads found in hospitals because they are often not PURE alcohol.

2. Clean the inside of the collimator with GE alcohol pads.
3. Use a magnifying glass to check for VERY faint cracks in the collimator filter.
4. If cracks are found, replace the filter. If not, reinstall the filter and close the collimator.
5. Rescan to confirm the problem is resolved.

Step 8: Check the Copper Filter

1. Depending on the unit, remove the copper filter from the X-ray tube OR remove the tube to access the filter.



2. Look for scratches in the copper filter. If something obvious is found, replace the filter. If nothing is found, rotate the filter; reinstall filter in tube.
3. Reinstall tube, if necessary.
4. Rescan to confirm the problem is resolved.

Step 9: Contact Support

If none of the previous steps resolve the problem, then contact OLC or ZSE Support.

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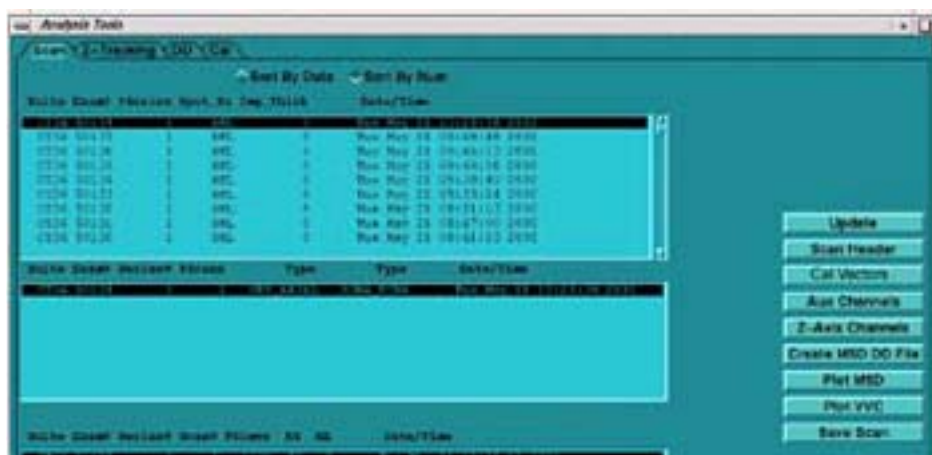
Troubleshooting Ring Artifacts in a Helical Series

It is difficult to troubleshoot using helical data because

- the bad data can be in any of the four data rows that produce a helical image and
- the number of views to produce a helical image are weighted differently based on table speed and pitch.

Therefore, the best way to troubleshoot ring artifacts in a helical series is to duplicate the problem in an Axial Mode.

1. Select **Service Desktop > Image Quality > Scan Analysis** for the following Scan List Select screen:



2. Select the **Exam Number, Series, and Scan Number**. Remember that 1 scan in a 4 x i mode is made up of 4 data sets (or rows) which produces 4 images. Images 1 through 4 are created from scan 1.
3. Once the scan is highlighted, select **Plot MSD**.
4. On the next screen, select **Convolved Data**, and click on the **OK** button. This prompts the following **Means Plots** screen:



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Troubleshooting Ring Artifacts in a Helical Series, continued

5. The Means Plot screen include tabs to view means and standard deviations for any of the four rows.

Note Refer to **Helical Scans** and **Axial Scans** Theory for the following examples.

6. Select the tab where the ring is expected. If necessary, adjust the level to find a spike in the data, or view other rows looking for any abnormal spikes.

For example:

If the data was acquired in a **4 x 5.00 mode** and the ring was on Row 2A, the Detector rows or diodes used were D8+D7+D6+D5 to produce row 2A.

If another scan was taken at **4 x 1.25 mode**, then the row 2A data would be acquired from detector row D2. If the ring was a hard problem (consistent every time) and is now gone after changing slice thickness, then the ring may have been caused by a suspect detector. Refer to further detector verification before replacing the detector. If the ring was still present, the problem could still be the detector, but may also be a DAS board or elastomer interface connection.

7. Select **File > DAS/Detector Architecture Tool**. A window appears and prompts for Detector Row and DAS channel. The program will display the associated DAS Converter board, Detector channel, module number, elastomer number, and other important information.
8. Find the associated DAS converter board number. If there are multiple rings, look for a pattern in each ring's information (Converter Board, Row, etc.).
9. **If a converter board is suspect:**
- Swap the suspect converter board with a board that is four boards away.
 - Repeat Scan Analysis.
 - If the bad channel follows the board**, then replace the board.
 - If the bad channel stays at same location**, then replace the elastomer associated with the failing channel. The elastomer number is found within the information given using the DAS/Detector Architecture Tool.

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Troubleshooting Ring Artifacts in a Helical Series, *continued*

10. If a **"PASS"** is received, then there is no more Ring and the problem is fixed.
11. If a **"FAILS"** is received and the same bad channel fails, run an X-ray verification test. If X-ray verification test fails, see X-ray Verification Troubleshooting in the Advanced Service Manual.



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Data Acquisition

DAS Pre-Check (Image Quality)

Note Always follow ESD procedures.

1. Record every step you performed on the system, including data and any moved or replaced components.
2. Mark all of the original parts.
3. Back up the existing calibrations before running any new calibrations. We recommend using a new disk.
4. Gather as much information as possible from the technician. Focus on the following data points:
 - On what protocols do the artifacts show up.
 - Ask the technician to duplicate the artifact, using a phantom while scanning in Axial mode.

Detector Temperature

- Check fuse in Alignment light switch box and fuse on 24 VDC Detector Heater Power Supply
- Check 120 to 24 VDC PS Voltages; if missing, check K2 Relay in Alignment light switch box
- Check Detector Heater Voltage is 20-32 VDC at caps
- Check Detector Heater Resistance is 10.4 Ohms
- Use Scan Analysis to test **DCB Board** Auxiliary Channels

Refer to [DAS Power Supplies Checks](#).

Refer to [Slip Ring Communication Checks](#).

Refer to [Data Acquisition Checks](#).

Refer to [DAS Data \(Slip Ring\) Troubleshooting](#).

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DAS Noise

Important! Always troubleshoot noise problems with all DAS covers and Air Fan plenum installed.
BEFORE troubleshooting DAS/Detector issues, make sure the proper ESD and Detector handling materials are available.

- Use DAS Tools diagnostic to check DAS Noise
- Reseat all DAS cards if noise is RANDOM
- Investigate failing channels if noise is NON-RANDOM
- Use Scan Analysis to plot scan file
- Check for fan noise; adjust fan position or swap polarity if necessary.
- Check for Logic **Power Supply** noise
- Check elastomer
- Check detector flexes
- Check DAS backplane



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Image Generation

Refer to Image Generation **diagnostics** for tests that help find problems with the reconstruction hardware.

If you suspect a LAN failure, try swapping ports on the LAN Switch.

To evaluate image quality, refer to **Image Analysis**.

The **SDC (Mercury) Board** (1x only) and the **RIP (Motorola) Board** are easily unseated from their slots. It is difficult to tell this, except that the board(s) do not work properly. They must always be held in place with the provided retaining bracket(s). It may be necessary, when seating these boards to either tap on them to securely seat them in their slots, or to remove the front panels and push directly on the circuit boards to effectively seat them in their slots.

Note Anytime you work on either the SDC (Mercury) Board or the RIP (Motorola) Board, always be sure to check and re-seat the other board before re-installing the retaining bracket(s).



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Gantry Rotation

The opto-isolated Enable, Start, and Stop signals from the **Axial Control Board** (ACB) to the **Axial Motor Drive** (AMD) provide a contact closure as an input to the AMD. The **Enable** contacts close electrically to enable the AMD, the **Start** contacts close electrically to start the AMD, and the **Stop** contacts close electrically to enable the AMD to run and open electrically to stop the AMD. The Enable, Start, and Stop opto-isolators carry 10 mA with less than a 3 V drop when closed, and withstand 5 V when the contacts open.

The fault signal from the AMD to the ACB consists of the AX_FLT and AX_FLT_RTN signal wires. The circuit uses drives with a normally-open fault contact. If either the fault signal wires open electrically, the ACB generates a fault condition. If the AX_SPD_FRQ_RTN signal wire opens during operation the system can report either AX_AT_SPD or AX_AT_FREQ errors.

The AX_AT_SPD*, AX_AT_FREQ* AND AX_AT_POS* are active low or NO FAULT conditions during normal axial operations. These signals should be high with the gantry rotation idle. The AX_FLT is a normally low signal. If this signal goes high then the AX_FAULT_CONTACTS in the AMD have opened and you have a fault.

Additional References

Refer to [ACB Error Message Descriptions](#).

Refer to [Axial Drive Logic Diagram](#).

Refer to [Axial Drive Theory](#).

Refer to [Axial Power Theory](#).

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Gantry Rotation, *continued*

Gantry Safety

Review each of the following safety topics before proceeding:

DANGER Extreme caution is necessary when working in the **slip ring area**, as both HVDC primary and 120 VAC are present and exposed.

DANGER Disable the Axial Drive Mechanism before servicing the gantry interior so the rotating gantry cannot hurt anyone.

DANGER Check the Axial Drive ENABLE LED, DS4, on the **STC backplane**. It indicates the actual Axial Power Control circuit status.

DANGER A failure in the Axial Power Controller could enable the Axial Motor Drive and cause un-commanded motion.

DANGER

The Chopper Control Circuit on the Interface Board is referenced to the AXDC rail at all times. This is a potentially lethal voltage source. Do not connect to ground. Chassis mounted dropping resistors R4 and R5 limit the Chopper power supply current derived from the AMD AXDC bus to <50 mA. The Chopper Control supply is referenced to the AXDC bus return, NOT to ground. NEVER reference this voltage to ground.



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DAS Data (Slip Ring)

DAS Pre-Check (Image Quality)

Note Always use the proper ESD equipment and follow the proper ESD procedures.

1. Record every step you performed on the system, including data and any moved or replaced components.
2. Mark all of the original parts.
3. Back up existing calibrations before running any new calibrations. We recommend using a new disk.
4. Gather as much information as possible from the technician. Focus on the following data points:
 - On what protocols do the artifacts show up.
 - Ask the technician to duplicate the artifact, using a phantom while scanning in Axial mode.

Symptoms

- **Data Collection Errors** resulting in Scan Aborts
- Some cases fail in all scan modes; others scout scans work fine while axial scans fail
- Error 2300000064: **TAXI data violation detected by Forward Error Correction** error
- An increase in **DIP statistics**

- Stationary errors related to specific gantry position
- Rotating errors only
- Both stationary and rotating failures
- Partial functionality when the antenna/receiver is purposely maladjusted, either skewed in the Z-axis or with maximum gap adjustment

Suggested Troubleshooting

Caution

Ensure proper ESD procedures are followed, especially when removing the transmitter.

1. Carefully remove the transmitter.
2. Review **DIP Stats** to establish baseline.
3. Quantify failure. Stationary, rotating, positional. Use **ScanDataPath diagnostic**, make 50 passes.
4. Scope power supplies, verify **antenna alignment**, fiber optic connections, Power and Signal LED's lit on both TX and RX units (LED's always lit).

Note Use care with fiber connections. These are keyed connectors with BNC style locking posts and can be easily damaged with excessive twisting force.

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DAS Data (Slip Ring), *continued*

Suggested Troubleshooting, *continued*

5. **Bypass DAS slip ring** by connecting Receiver and DCB J1 connectors and verify no error generation. Do not rotate the gantry.
6. **Turn OFF** gantry 120 VAC. Do not connect/disconnect power connector from TX or RX units with power on (**Power LED lit**). This can damage these units.
7. Take resistance reading on the **HSDCD ring** itself, near the surface mount terminating resistors. Track-to-track ~18 Ohms, same track opposite sides of terminator resistors ~2.5 Ohms. If you read 9 Ohms track-to-track, then the terminator resistors are open or the track trace is open. If the track is open, then you should read 48 Ohms instead of 2.5 Ohms. These failures require ring replacement.
8. Take a resistance reading between the center pins of the 10 pin, 2 row male connector located on the inside ring diameter (to what the transmitter mounts). This resistance should be ~18 Ohms. If the resistance is significantly different, there is damage between this point and the HSDCD ring (Emitter PCB), requiring ring replacement.

9. Re-install transmitter, reconnect power connectors, **turn ON** gantry 120 VAC.
10. Start **ScanDataPath diags** and run multiple groups. 500 rotating passes without error is a high confidence result. If errors return after 10 minutes, the **receiver** may have a thermal breakdown.

Warning

Never rotate the gantry with a misadjusted antenna.

11. Position HSDCD terminator resistors directly under the antenna and run stationary diags. Manually rotate HSDCD terminator about 5 inch increments. Verify 10 passes per location.

When a failing position is found, you can misadjust the antenna height by varying the gap to improve performance. If this improves performance, it is likely the receiver has gain or noise immunity problems and should be replaced.

Refer to **Data Acquisition Troubleshooting**.

Another troubleshooting tool is to **bypass** the slip rings that transfer **control signals** between the STC and OBC to see if system communication improves.



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Control Slip Ring Bypass

Note To perform this operation, remove all gantry covers and have the system ready for operation.

1. **Turn OFF** all three power switches on the **STC backplane**.
2. At the rear of the gantry, look around the inside (rear) of the slip ring until you find the gray signal cable. It is the only cable with a deconnector.
3. Rotate the gantry until the gray signal cable aligns with the center of the slip ring brush block assembly. Use a screwdriver to unscrew and then unplug the connector.
4. On the left center of the slip ring brush block assembly (behind the rear cover mounting bracket), you will see another gray signal cable. Remove this cable, push it behind the slip rings, and connect it to the first gray signal cable.
5. Engage the **rotational lock**. Check that the gantry is securely locked by trying to rotate it by hand.
6. **Turn ON** the 120 VAC and HVDC switches but NOT the Axial Drive ENABLE switch on the **STC backplane**.

7. To reset the scanning hardware, select **Service Desktop > System Resets > Scan Hardware > Run**.
8. Once the scanning hardware reset is completed, press the RESET switch on the STC backplane.
9. From one of the gantry keypads, press the CRADLE RELEASE button (1), and then the TABLE UP (2) and FAST (3) buttons simultaneously to reference the table.



Note Make certain the gantry is disabled before performing the next step.

10. Select **Service Desktop > System Control > RCIB Diagnostics > Test Selection > Ping All > Run**.

This test should complete quickly without any errors.

11. Disconnect the two signal cables and reinstall them to their original locations.
12. Unlock the rotational lock so the gantry can rotate.
13. Restore power, reset the cradle and reinstall the gantry covers.



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Slip Ring Communication

If communication failures (**TAXI Link Failures**) still occur after vacuuming, you may have to clean the **comm rings** with alcohol or perhaps even sand mechanical damage with Cratex® abrasive sticks.

Alcohol cleaning should only be done if necessary and only AFTER vacuuming the slip rings and gantry has not corrected the problem.

1. Check baseline, then remove the **Slip Ring Brushes Block Assembly** to avoid contaminating the brushes.

Important! Use only GE approved alcohol (P/N 46-183039P1) for the next step. DO NOT use alcohol prep pads found in hospitals; they can contaminate the slip ring and brushes because they are often not PURE alcohol.

2. Using only GE approved alcohol, clean in 4" (10.2 cm) sections by applying the alcohol and then immediately wiping off the wet residue with a lint-free towel.
3. Once the alcohol cleaning is completed, allow the rings to AIR DRY for 15 minutes.
4. Reinstall the Brushes Block Assembly. Follow the proper replacement procedure; this is critical to the life of the slip ring components.

5. Replace all gantry covers.

Cratex abrasive stick sanding is for mechanical damage to the ring ONLY (i.e., pits and arc marks). Use Cratex to fix pit and arc marks only. Smoothing the surfaces is a time consuming task and if not done properly and completely, will result in either permanent damage or more arcs. Remove the "clogged" end of the abrasive stick with a coarse file to help speed up the process of smoothing out pitted areas.

1. Remove the **Brushes Block Assembly** because the filings will contaminate the brushes.
2. Use a Cratex **fine** abrasive stick (P/N 46-297961P2) to smooth the pitted area or deposits on the slip ring. Do not attempt to clean areas more than 2 cm (3/4 inch) at a time. ONLY use Cratex on the ring needing repair.
3. If the area(s) is still not smooth, use the Cratex **medium** abrasive stick (46-297961P1). Do not attempt to clean areas more than 2 cm (3/4 inch) at a time. When done, repeat step 2 with the fine stick.
4. When done sanding, remove ALL traces of abrasive with a thorough alcohol cleaning.
5. Reinstall the Brushes Block Assembly.



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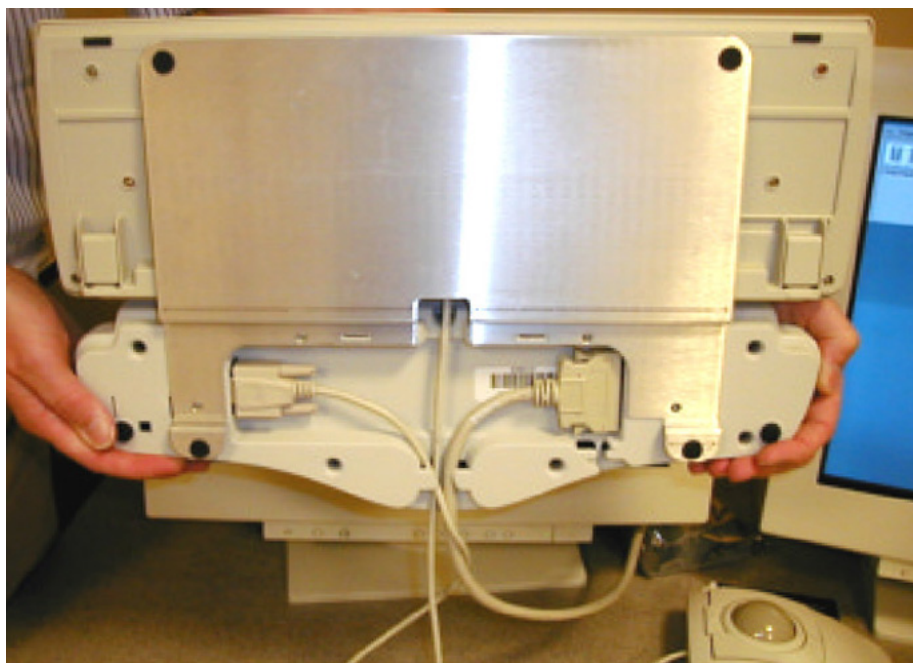
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Scan Control Intercom Module (SCIM) Installation



Supplied Velcro Strip



Note The trackball and mouse cables should be routed through the cable raceway so that neither will be able to touch the floor when you are finished.

If the SCIM is not functioning properly, verify that:

- The keyboard cable is routed under the SCIM.
- The mouse and keyboard cables (Run 5) are routed through the console desktop raceway and the 1" (25.4 mm) hole on top of the console.
- The mouse and keyboard cables are connected to the back of the Octane computer.
- The angled end of the keyboard cable is connected to the keyboard.
- The SCIM **50 pin cable** (P/N 2262158-9) is run under the monitor table top and connected to the console back bulk head **J19**.

The proper overlay for your system is selected and installed (1 with Tilt or 2 without Tilt).

- The keyboard to the SCIM fits snugly against the SCIM with the supplied Velcro strip.
- No cables are pinched and a service loop is formed when the computer tray is pushed in.



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Console

This section covers general console troubleshooting information. Refer to your Console Service Methods for detailed troubleshooting procedures.

RIP Boot Sequence (MVME-24xx)

Step 1: Motorola Firmware ("PPC4-Bug>" prompt)

1. Entered after power-on reset.
2. Can execute board level diagnostics.
3. RIP does not respond to ping in this state.
4. RIP can ping other hosts in this state.
5. RIP cannot reset VME bus in this state.
6. Transfers control to VxWorks bootrom via romboot ("rb") command.
7. Useful commands:

rb;v - transfer control to VxWorks bootrom

onping 0 0 192.9.220.11 192.9.220.1 1 - ping the OC one time

he - help

Step 2: VxWorks bootrom ([VxWorks Boot]: prompt)

1. RIP does not respond to ping in this state.
2. RIP can reset the VME bus in this state.

3. Two commands can assist with status menu:

b - does a local reset of the board and returns control to Motorola Firmware

v - does a VME reset of the ICE chassis and returns control to the Vxworks Bootrom

4. Boot string is sent by procICE to boot the kernel and transfer control to VxWorks.

Step 3: VxWorks shell (-> prompt)

1. Functions like a Unix shell, but it is not the same.

Step 4: Useful VxWorks Shell Commands:

"ipstatShow - similar to netstat on OC, but shows only IP statistics

"tcpstatShow - similar to netstat on Unix, but shows only TCP statistics

"ifShow - Shows information about the Ethernet interface including the MAC address

"arpShow - Shows the mapping of IP addresses to MAC addresses

"i - Shows the state of all running tasks (like Unix **"ps"**)

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RIP Boot Sequence (MVME-24xx), continued

Step 5: Boot Sequence

Note Numbers in the left column of the sample iceConsole.log output (Table 1) correspond to the numbers in the sequence below.

1. After power is turned on, RIP goes into Motorola Firmware and waits for input from the serial port.
2. proclCE executes **sbcutil -resetvme** script, which does the following:
 - a. Transfers control to the VxWorks bootrom.
 - b. Resets the VME bus.
 - c. Transfers control back to the VxWorks bootrom.
3. proclCE sends the boot string to the VxWorks bootrom to bring up the system and waits for the RIP applications to reply.
4. RIP downloads the VxWorks kernel by doing an **rsh** to the OC as indicated by the "Loading . . . " message. When it is complete, the number of bytes downloaded is printed on the console.
 - a. Note that this normally takes about one second. Network or installation problems can cause the RIP to hang at this point.
 - b. When it succeeds you can see the **rsh** command printed in the Unix syslog on the OC.
5. Bootrom transfers control to VxWorks operating system.
6. RIP probes the SCSI bus looking for SCSI devices. All devices present are printed on the console. Note that normally this takes only a couple of seconds. Problems on the SCSI bus can cause the RIP to hang at this point.
7. RIP creates a proxy network, which allows the IG to boot. The IG cannot download its VxWorks kernel until this happens.
8. The RIP mounts all exported NFS file systems from the OC. These are printed on the console.
 - a. Should take only a few seconds. Network or installation problems can cause this to hang or fail.
 - b. The only failure indicator is that no file systems are printed.

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RIP Boot Sequence (MVME-24xx), continued

9. VxWorks downloads the symbol table.
 - a. This normally takes less than a second. Network or installation problems can cause this to hang or fail.
 - b. When it succeeds you can see the **rsh** command printed in the Unix syslog on the OC.
 - c. Applications cannot start without the symbol table.
10. RIP monitors IG board waiting for it to boot.
 - a. If the IG fails to boot, it will reset the IG and try again.
 - b. The IG fails to boot on occasion due to glitches in the VxWorks operating system.
 - c. If the IG requires more than one retry to boot, or the **iceConsole.log** file shows many retries, it could indicate a problem with the IG or with the VME backplane.
 - d. Successful boot of IG is indicated by "Image Generator is UP" message
- e. When IG boot succeeds you can see the **rsh** commands printed in the Unix **syslog** on the OC.
11. RIP prints out the Pegasus hardware configuration.
12. RIP downloads CT applications.
13. RIP applications read the ECOMM database.
14. RIP applications read the ERMES database. No messages go the **gesyslog** from the ICE until this succeeds.
15. Application is started.
16. Raw data disk is initialized by **sdiomgr**. Output goes to console.

This normally takes only a few seconds. A corrupted file system or SCSI hardware problems can cause this to hang which will result in a message from Vx monitor indicating **sdiomgr** took too long to start up.
17. This message appears when recon has successfully initialized. It means that the recon subsystem is functioning and ready to make images.

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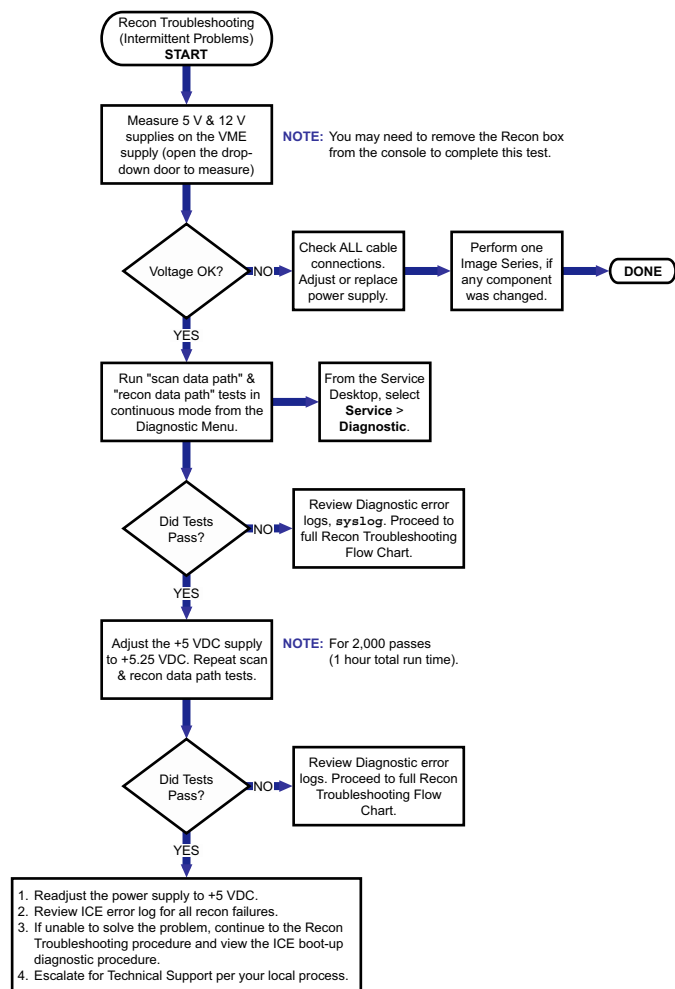
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RIP Boot Sequence (MVME-24xx), continued

Intermittent Recon Troubleshooting Flowchart



Intermittent Recon Troubleshooting

To troubleshoot intermittent recon problems:

1. Adjust 5 V supply to 5.25 VDC to increase the noise generated by the on board regulators.
2. Run recon data path in the continuous mode for at least 2000 passes.
3. If any failures occur, replace the Pegasus board.

Image Artifacts: Try a retrospective reconstruction of the image. If the artifacts are exactly the same it is unlikely that there is a problem with reconstruction; check the Scan Data Path.

Note You can operate the recon engine with the service doors open for short periods without further damaging the recon engine. Max airflow through the recon engine is dependant on having all service doors closed and the recon engine placed in the console over the air intake vents.

Use this procedure for:

- First pass recon failure screening
- Intermittent Recon failures
- Recon Power Supply intermittent failure screening

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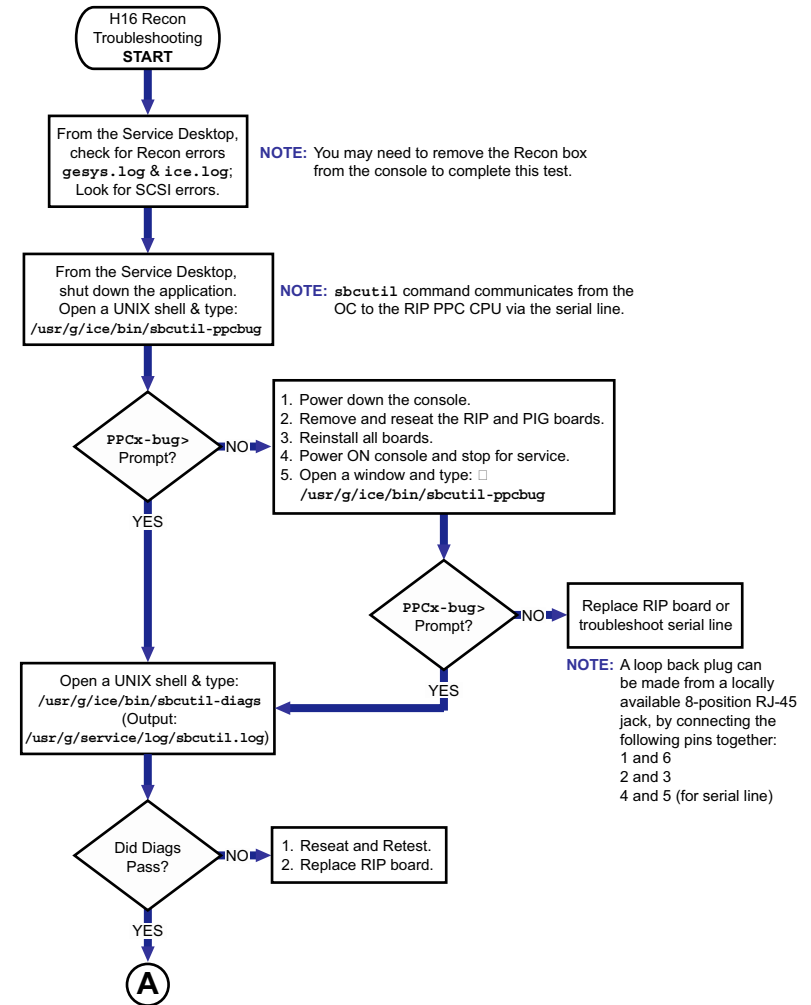
RIP Boot Sequence (MVME-24xx), continued

Intermittent Recon Troubleshooting, continued

When troubleshooting recon problems, confirm that:

- All connecting cables are tight.
- All fans are operational.
- The recon engine is placed over the air intake vents.

"Hard Down" Recon Troubleshooting Flowchart (1 of 4)



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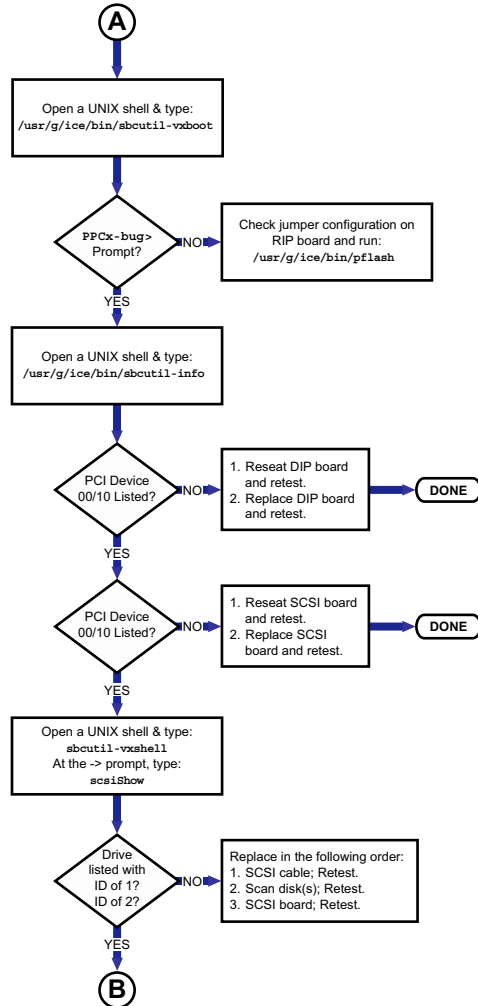
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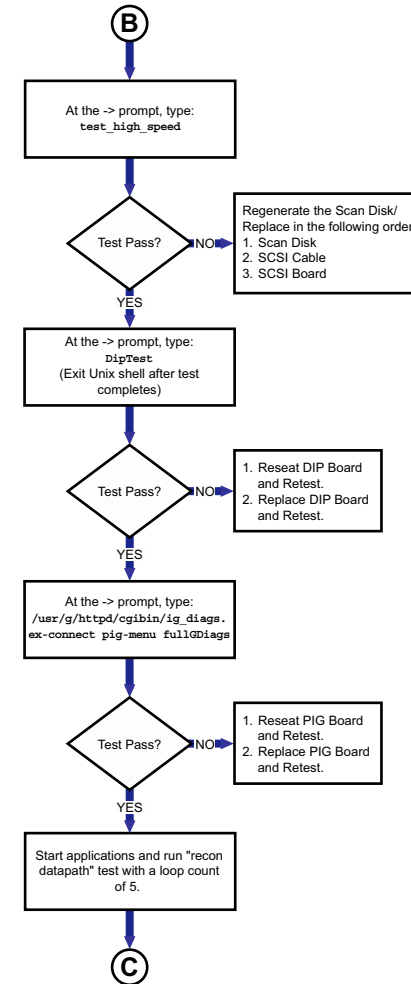
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RIP Boot Sequence (MVME-24xx), continued

"Hard Down" Recon Troubleshooting Flowchart (2 of 4)



"Hard Down" Recon Troubleshooting Flowchart (3 of 4)



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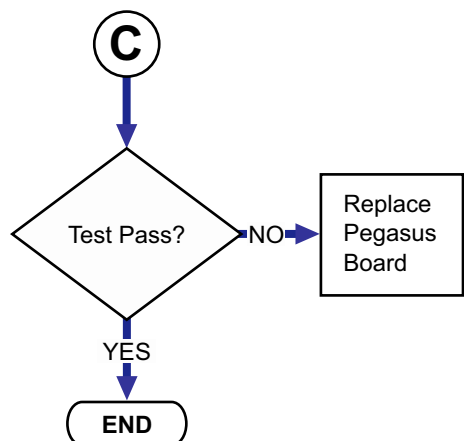
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RIP Boot Sequence (MVME-24xx), continued

"Hard Down" Recon Troubleshooting Flowchart (4 of 4)



"Hard Down" Recon Troubleshooting

1. A loop back plug can be made from a locally available 8-position RJ-45 jack, by connecting the following pins together:
 - 1 and 6
 - 2 and 3
 - 4 and 5 (for serial line)

2. Output of `/usr/g/ice/bin/sbcutil` info on 16-slice scanner (dual-channel SCSI card is implemented as two PCI "functions", so it is listed twice):

PCI Function 00/10/0 (00008000) ID/
Revision = 435417B6/00 (DIP)

PCI Function 00/11/0 (00008800) ID/
Revision = 00211000/01 (SCSI)

PCI Function 00/11/1 (00008900) ID/
Revision = 00211000/01 (SCSI)

3. Output of `/usr/g/ice/bin/sbcutil` info on MDAS16 scanner:

PCI Function 00/10/0 (00008000) ID/
Revision = 43540000/41 (DIP)

PCI Function 00/11/0 (00008800) ID/
Revision = 000F1000/04 (SCSI)

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RIP Boot Sequence (MVME-24xx), continued

"Hard Down" Recon Troubleshooting, continued

4. Output of [/usr/g/httpd/cgi-bin/ig_diags.ex - connect pig -menu full IGDiagnostics](#):

8-Slice Pegasus [2216467], 64 MByte SPAMs,
Version is: E

Found Pegasus Daughter Card, 64 MByte SPAMs,
Version is: A

IG Temperature Sensors:

40.50C(UF) 44.00C(UB) 43.00C(LF) 41.50C(LB)

SPAM 0 SDRAM Quick Bus Test Pass: 1 / Fail: 0

SPAM 0 SHARC Quick Bus Test Pass: 1 / Fail: 0

SPAM 0 SHARC Host Bus Memory Test Pass: 1 /
Fail: 0

SPAM 0 SDRAM Memory Test Pass: 1 / Fail: 0

SPAM 0 PowerPC Collision Tests Pass: 1 /
Fail: 0

SPAM 0 ALL SHARCs Collision Tests Pass: 1 /
Fail: 0

SPAM 1 SDRAM Quick Bus Test Pass: 1 / Fail: 0

SPAM 1 SHARC Quick Bus Test Pass: 1 / Fail: 0

SPAM 1 SHARC Host Bus Memory Test Pass: 1 /
Fail: 0

SPAM 1 SDRAM Memory Test Pass: 1 / Fail: 0

SPAM 1 PowerPC Collision Tests Pass: 1 /
Fail: 0

SPAM 1 ALL SHARCs Collision Tests Pass: 1 /
Fail: 0

SPAM 2 SDRAM Quick Bus Test Pass: 1 / Fail: 0

SPAM 2 SHARC Quick Bus Test Pass: 1 / Fail: 0

SPAM 2 SHARC Host Bus Memory Test Pass: 1 /
Fail: 0

SPAM 2 SDRAM Memory Test Pass: 1 / Fail: 0

SPAM 2 PowerPC Collision Tests Pass: 1 /
Fail: 0

SPAM 2 ALL SHARCs Collision Tests Pass: 1 /
Fail: 0

SPAM 3 SDRAM Quick Bus Test Pass: 1 / Fail: 0

SPAM 3 SHARC Quick Bus Test Pass: 1 / Fail: 0

SPAM 3 SHARC Host Bus Memory Test Pass: 1 /
Fail: 0

SPAM 3 SDRAM Memory Test Pass: 1 / Fail: 0

SPAM 3 PowerPC Collision Tests Pass: 1 /
Fail: 0

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"Hard Down" Recon Troubleshooting, continued

SPAM 3 ALL SHARCs Collision Tests Pass: 1
Fail: 0

IMAX Configuration Register Test Pass: 1
Fail: 0

IMAX Data Bus Test Pass: 1 Fail: 0

IMAX Address Bus Test Pass: 1 Fail: 0

IMAX Memory Test (IM0 & IM1) Pass: 1 Fail: 0

IMAX Pixel Localization Test Pass: 1 Fail: 0

IMAX Mirror Image Test Pass: 1 Fail: 0

IMAX Image Attenuation Test Pass: 1 Fail: 0

IMAX Image Accumulation Test Pass: 1 Fail: 0

C67/HPI Bus: 512kB DP Data Bus Test Pass: 1
Fail: 0

C67/HPI Bus: 512kB DP Addr Bus Test Pass: 1
Fail: 0

C67/HPI Bus: 512kB DP Memory Test Pass: 1
Fail: 0

C67/HPI Bus: 256kB DP Data Bus Test Pass: 1
Fail: 0

C67/HPI Bus: 256kB DP Addr Bus Test Pass: 1
Fail: 0

C67/HPI Bus: 256kB DP Memory Test Pass: 1
Fail: 0

C67/HPI Bus: 16MB SDRAM Data Bus Test Pass: 1
Fail: 0

C67/HPI Bus: 16MB SDRAM Addr Bus Test Pass: 1
Fail: 0

C67/HPI Bus: 16MB SDRAM Memory Test Pass: 1
Fail: 0

APU/BPC BPC Fifo Depth Test Pass: 1 Fail: 0

APU/BPC ATR Test Pass: 1 Fail: 0

APU/BPC S&T Test Pass: 1 Fail: 0

APU/BPC DR/R Test Pass: 1 Fail: 0

APU/BPC APU Projection Mem Test Pass: 1
Fail: 0

APU/BPC Circular Dual Port Test Pass: 1
Fail: 0

APU/BPC Apu Pipeline Test Pass: 1 Fail: 0

APU/BPC Homogeneity Test Pass: 1 Fail: 0

APU/BPC Individual Apu Test Pass: 1 Fail: 0

Full IG Diags Completed.



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RIP Diagnostic Tests Overview

The Reconstruction Interface Processor (RIP), or Motorola Board, has built-in diagnostic routines that are executed by running either of two executable diagnostic scripts. These scripts are **diags** and **fulldiags**, and are called from a UNIX Shell window.

The diags test is a short board level diagnostic test (approximately 1 minute), and the fulldiags test is a longer test that performs more extensive, lower level testing of the board's RAM and VME3 channels (approximately 6 minutes). These executable scripts are located in the **/usr/glice/bin** directory of the OC-Octane.

Starting the diagnostic will reboot the Motorola Board into a diagnostics mode. When the tests are completed, the script will automatically return to the Octane ctuser prompt at the **/usr/glice/bin** directory level.

Note Because the RIP Board is rebooted when these tests are launched, you must shut down Applications before running either of these tests.

RIP Diagnostic Tests Execution

Perform the following steps to launch either the short or long Motorola diagnostic tests:

1. Select **Service Desktop > Utilities > Application Shutdown**.
2. Open a UNIX Shell.
3. Change directory to **/usr/glice/bin**. At the ctuser prompt, type: **cd /usr/glice/bin <ENTER>**.
4. Launch either the short or long RIP diagnostic test. At the ctuser prompt, type: **ice diags <ENTER>** OR **ice fulldiags <ENTER>**.

Each test starts RIP diagnostic testing at the VME **PPC1-Diag>** prompt level. Verify that all tests result in either a PASSED or BYPASS indication. The **diags** test takes about 1 minute to complete, and the **fulldiags** test takes approximately 6 minutes to complete. When the test routines complete, the routines automatically return the system to the **ctuser** prompt at the **/usr/glice/bin** directory.

5. Close the Shell. At the ctuser prompt, type: **exit <ENTER>**.
6. Restart Applications. At the ctuser prompt, type: **st <ENTER>**.



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Scan Data Disk & SCSI Controller Troubleshooting

Diagnostic Testing Overview

The disk subsystem should work correctly, if it passes the tests described in the sections that follow. Three questions are answered:

1. Is the disk subsystem hardware recognized by the RIP "CPU"?
2. Does the RIP's operating system ("Vx Works") recognize the hardware as a disk subsystem?
3. Can data be reliably written and read from the disk subsystem?

Because of the tight linkage between a SCSI controller and a single disk drive, the tools used to identify and resolve problems are similar.

Note Before executing any of the VxWorks software commands, it's highly recommended that CT applications be shutdown. There is a potential for CT Applications to become inoperable as a result of the system commands you execute, thus requiring a restart anyway.

Recognition of SCSI Controller Hardware by RIP Board

Verify the RIP board is able to identify the existence of the SCSI card. Perform the following steps to verify that the SCSI Board is being recognized by the Operating System.

1. Select **Service Desktop > Utilities > Application Shutdown**.
2. Connect to ICE Box (RIP Board), using one of the following methods.

Serial Connection:

- a. Open a Unix shell (as ctuser)
- b. Type the following at the command prompt:
ice stop start connect <ENTER>

Network Connection:

- a. Open a Unix shell and become root:
su - <ENTER>
- b. Enter password: **#bigguy <ENTER>**
- c. Remote login into ICE box: **rlogin ice <ENTER>**

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Recognition of SCSI Controller Hardware by RIP Board, continued

- Execute the VxWorks command to view the PCI Devices information: **pciDeviceShow <ENTER>**

The system will provide a text output page similar to the following: **pciDeviceShow**

Scanning function 0 of each PCI device on bus 0
Using configuration mechanism 1

bus	device	function	vendorID	deviceID	class
00000000	00000000	00000000	00001057	00004801	00060000
00000000	0000000b	00000000	000010ad	00000565	00060100
00000000	0000000d	00000000	000010e3	00000000	00068000
00000000	0000000e	00000000	00001011	00000009	00020000
00000000	00000010	00000000	00001000	0000000f	00010000
00000000	00000011	00000000	00000001	00000001	00ff0000

<-- SCSI bd
value = 0 = 0x0

In the above example, the line of information in the printout (as marked) identifies the SCSI Board, and shows that it is properly recognized by VxWorks.

- Exit VxWorks and return to the Octane ctuser prompt. At the **VxWorks** prompt, type the following to end a serial connection: **~. <ENTER>**

- The SCSI controller must be listed above in order to be recognized by the RIP board. If not, you most likely have a defective SCSI controller or PCI interface (RIP board). If recognized, proceed to next section.

Recognition of SCSI Disk Subsystem by Operating System

The operating system must see the existence of a SCSI disk drive before software support is loaded. Perform the following steps to verify that the SCSI Board and disk are properly recognized.

- Select **Service Desktop > Utilities > Application Shutdown**.
- Connect to ICE Box (RIP Board), using one of the following methods.

Serial Connection:

- Open a Unix shell (as ctuser)
- Type the following at the **command** prompt:
ice stop start connect <ENTER>

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Scan Data Disk & SCSI Controller Troubleshooting, continued

Recognition of SCSI Disk Subsystem by Operating System, continued

Network Connection:

- a. Open a Unix shell and become root:
su - <ENTER>
 - b. Enter password: **#bigguy <ENTER>**
 - c. Remote login into ICE box: **rlogin ice <ENTER>**
3. Execute the following VxWorks command to list SCSI devices recognized by the operating system:
scsiShow <ENTER>

The system will provide a text output page SIMILAR to the (non-LightSpeed 4.X) examples shown below, depending on the make and model of disk drive recognized. A disk drive must be listed, regardless of make and model.

SEAGATE ST318404LW MODELS:

ID	LUN	VendorID	ProductID	Rev.	Type	Blocks	Size	pScsi	PhysDev
1	0	SEAGATE	ST318404LW	0002	0	35843670	512	0x017dd	a88

value = 0 = 0x0

SEAGATE ST318451LW MODELS:

ID	LUN	VendorID	ProductID	Rev.	Type	Blocks	Size	pScsi	PhysDev
1	0	SEAGATE	ST318451LW	0003	0	35843671	512	0x01bdd	b18

value = 0 = 0x0

4. Inspect the output of the **scsiShow** command. Is a disk drive listed in the output? If no, the SCSI controller, cable or disk drive may be defective. You must correct this problem before proceeding. If listed, proceed to **Recognition of the Disk File System by the Operating System**.
5. Exit VxWorks and return to the Octane ctuser prompt. At the **VxWorks** prompt, type the following to end a serial connection: **~. <ENTER>**

Recognition of Disk File System by Operating System

There must be a file system on the disk drive. The file system on the disk is created during a software load. A file system must exist for CT applications to operate correctly. In the following check, you will verify the existence of that file system.

1. Select **Service Desktop > Utilities > Application Shutdown**.

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Scan Data Disk & SCSI Controller Troubleshooting, continued

Recognition of the Disk File system by the Operating System, continued

2. Connect to ICE Box (RIP Board), using one of the following methods.

Serial Connection:

- a. Open a Unix shell (as ctuser)
- b. Type the following at the **command** prompt:
ice stop start connect <ENTER>

Network Connection:

- a. Open a Unix shell and become root:
su - <ENTER>
 - b. Enter password: **#bigguy <ENTER>**
 - c. Remote login into ICE box: **rlogin ice <ENTER>**
3. Execute the following **VxWorks** command to perform a listing of the directory raw_data:
ls "/raw_data" <ENTER>
(remember to include the quotes)

4. You should get a listing of directory raw_data if it exists:

```
/raw_data/info_file  
/raw_data/recovery  
/raw_data/ex163.CT18.1003803242.952008  
/raw_data/ex171.CT18.1003832725.458539
```

5. Inspect the screen output.
 - If the "ls" command lists the contents of the raw_data directory, you're done and everything is okay. Proceed to **Testing for Reliable Disk Subsystem Operation**.
 - If you receive file not found or some other error message, the directory is defective.

Testing for Reliable Disk Subsystem Operation

To test the disk subsystem reliability (RIP Board, SCSI Controller, and Scan Data Disk), perform the following test:

1. Select **Service Desktop > Utilities > Application Shutdown**.
2. Open a UNIX Shell.

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Scan Data Disk & SCSI Controller Troubleshooting, *continued*

Testing for Reliable Disk Subsystem Operation, *continued*

3. Serial connect to the **VxWorks** prompt. At the Octane ctuser prompt, type:
ice stop start connect <ENTER>

4. Launch the Scan Data Disk test:
test_high_speed <ENTER>

The system will proceed to transfer data to and from the Scan Data Disk. If any errors are detected, they will be displayed on the terminal screen. If the test runs to completion without error, you can be assured that the Scan Data Disk, SCSI Controller, and RIP Board are capable of performing acceptably well to handle the scan data save and restore operations that are used during normal scanning.

5. Exit VxWorks and return to the Octane ctuser prompt. At the **VxWorks** prompt, type: **~. <ENTER>**





Console, continued

PMC DIP Board Troubleshooting

System Operating Errors Commonly Associated with the DIP Board

The RIP board is a common place for scan data acquisition system errors to be logged and reported. Scan data acquisition errors are reported to the RIP, using interrupts, real-time. This section attempts to sort out those failures so that the failing subsystem can be correctly identified.

Failures can occur almost anywhere along the scan data acquisition chain, including the DIP and RIP board. Error detection and correction schemes built into slip ring communication (SRC) are the first level of defense. To prevent lost/missing and/or corrupt data from being processed.

The first step is to identify the failing subsystem. Failures common to the SRC subsystem are FEC errors, serial data modem violations and view length errors. If you receive any of these errors, the SRC subsystem should be thoroughly checked out first. If you receive data checksum or parity errors error messages, then the DIP and RIP board (ICE box subsystem) should be thoroughly tested.

Common SRC Errors

Forward Error Correction (FEC): Protects the integrity of the link between the DAS DCB and the DIP. Without any form of error correction, the system bit error rate is 10-12. This translates to a data error every one week of scanning. With FEC, the not correctable failure rate is reduced to 10-15 or a data error every 6 years. Therefore FEC virtually eliminates the possibility of a data error between the DCB and the DIP. In addition, the FEC on board the DIP will maintain a running count in the board status register, per scan, of the number of corrected bit errors. This register is readable from the RIP and can be used to maintain statistics and indicate potential problems.

Serial Data Modem Violation: A very basic failure of the communication link, indicating that the link has broken.

View Length Error: Another very basic failure of the communication link, indicating lost data and that the link has been broken.

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PMC DIP Board Troubleshooting, *continued*

System Operating Errors Commonly Associated with the DIP Board, *continued*

Common PCI Interface Errors

SRC errors indicate problems with the DAS to DIP communication link, not the DIP or the DIP to RIP interface. Since the PCI bus link in the ICE box is orders of magnitude more reliable than the SRC path, error correction is not used. However, error detection is used with the RIP's PCI interface to the DIP board. The following errors indicate a detection problem in either the DIP or the RIP board.

Data Checksum Error: A checksum for each view is generated by the DCB (in the DAS) and stored in the view data. View data is stored on the scan data disk by the DIP and RIP. The checksum is tested when view data is restored from disk. Any failure that generates a checksum error should have also generated one of the above mentioned SRC errors. If not, the error could have only occurred on the DIP board or during the transfer of data from the DIP to the RIP.

Parity Error: The PCI interface uses parity error detection. A parity bit is generated for each scan data word transmitted across the PCI interface between the

DIP and the RIP. Hardware on board the RIP checks parity and produces an abort condition when an error is detected. A parity error can only occur during the transfer of data from the DIP to the RIP.

Common DIP Board Errors

To isolate the DIP from the RIP board as the failing, the following should be done:

1. **View the system error log.** FEC, serial data modem, or view length errors indicate a DAS to DIP error, not a DIP failure. PCI parity errors indicate a DIP to RIP error. Data checksum errors indicate a DIP, RIP or possibly a scan data disk error.
2. **Run DIP BLDs.** These tests will run known data patterns through the DIP at full data rates, indicating problems on board the DIP or the DIP to RIP interface.
3. **Run the system data path tests.** These tests will run known data patterns from the DAS to the RIP, through the DIP at full data rates. By monitoring the aforementioned system error log, more information can be gathered to determine the failing FRU.

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PMC DIP Board Troubleshooting, continued

System Operating Errors Commonly Associated with the DIP Board, continued

Diagnostic Tests Overview

If the DIP and RIP boards are to work correctly, they must be able to communicate reliably. Three simple questions must be answered first:

1. Is the DIP board recognized by the RIP "CPU"?
2. Can data be transferred from the DIP to the RIP board reliably?
3. Is data received reliably from the slip ring?

Recognition of DIP Board by RIP

Performing the following steps to determine whether the RIP Board (VxWorks) sees the existence of the DIP Board through the PCI interface.

1. Select **Service Desktop > Utilities > Application Shutdown**.
2. Open a UNIX Shell.

3. Serial connect to the **VxWorks** prompt. At the Octane **ctuser** prompt, type:
ice stop start connect <ENTER>
4. Execute the VxWorks command to view the PCI Devices information: **pciDeviceShow <ENTER>**
5. Inspect the screen output. Verify the DIP board is recognized by the RIP through its PCI interface.

Scanning function 0 of each PCI device on bus 0
Using configuration mechanism 1

bus	device	function	vendorID	deviceID	class
00000000	00000000	00000000	00001057	00004801	00060000
00000000	0000000b	00000000	000010ad	00000565	00060100
00000000	0000000d	00000000	000010e3	00000000	00068000
00000000	0000000e	00000000	00001011	00000009	00020000
00000000	00000010	00000000	00001000	0000000f	00010000
<-- DIP bd					
00000000	00000011	00000000	00000001	00000001	00ff0000
value = 0 = 0x0					

In the above example, the last line of information in the printout (as marked) identifies the DIP Board, and shows that it is properly recognized by VxWorks.

6. Exit VxWorks and return to the Octane **ctuser** prompt. At the **VxWorks** prompt, type: **~. <ENTER>**

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PMC DIP Board Troubleshooting, *continued*

Testing DIP Board Functionality (DIP Diagnostics)

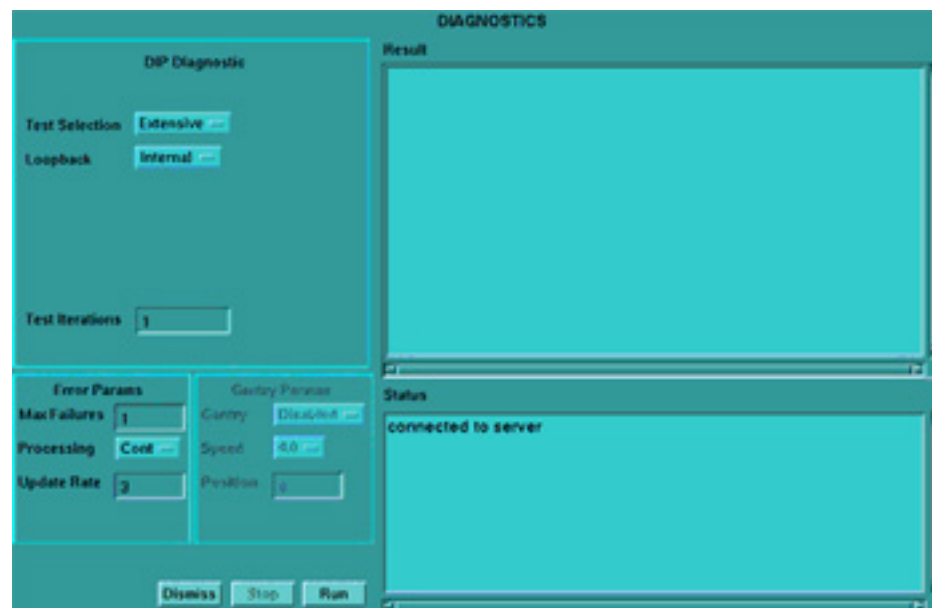
Diagnostics for the DIP board are located on the Service Desktop, under the Diagnostics Menu. There are two types of test selections available (quick and extensive), and two sources of data (internal and external loopbacks).

If you wish to perform the external loopback tests, you will need to loop back data between the fiber-optic jacks (J2 and J3) on the DIP board. A suggested cable for this purpose is the fiber-optic cable used to connect the DCB to the slip ring transmitter (fiber-optic cable P/N 2293408.)

Select **Service Desktop > Diagnostics > DIP Diagnostics**.

A diagnostic GUI will appear on the console that allows you to select the testing mode and the loopback mode; see DIP Diagnostics GUI example on the next page. If you select "external" for the loopback mode, a separate window will pop up, reminding you of the need to install an external fiber-optic jumper.

DIP Diagnostics GUI



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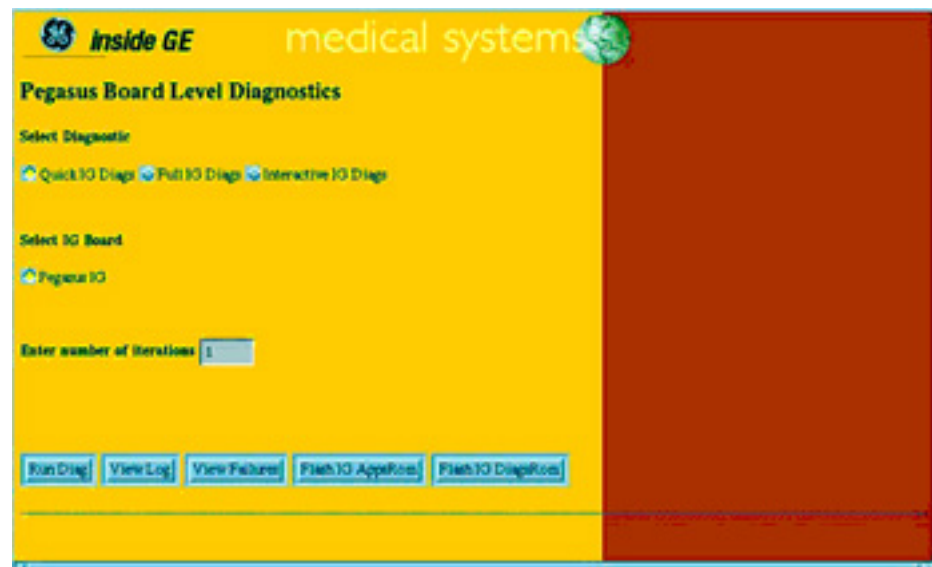
There are two (2) levels of Pegasus diagnostics:

- **Low Level Board Diagnostics:** Run from the service browser with applications shutdown. Used to test the functionality of the Pegasus board specifically.
- **High Level Recon Data Path Tests:** Run from the service desktop at applications level. Tests the ability of the Pegasus board to communicate and operate with the rest of the system.

Low Level Board Diagnostics Procedures

1. Select **Service Desktop > Utilities > Application Shutdown**.
2. Wait for the "Attention" window to disappear and then open a shell. Within that shell, execute the following command to start the service browser:
service_browser <ENTER>.
3. The Service Desktop screen appears again. Select **Diagnostics > IG Diagnostic ICON**.

4. The following IG Board level diagnostic screen should appear:



5. Select either **Quick IG Diags** or **Full IG Diags** and enter number of iterations.
6. Select **Run Diag**. A shell window will open and the results of either test is displayed (see next two examples). When testing has completed, close the shell window before attempting to run any other tests.

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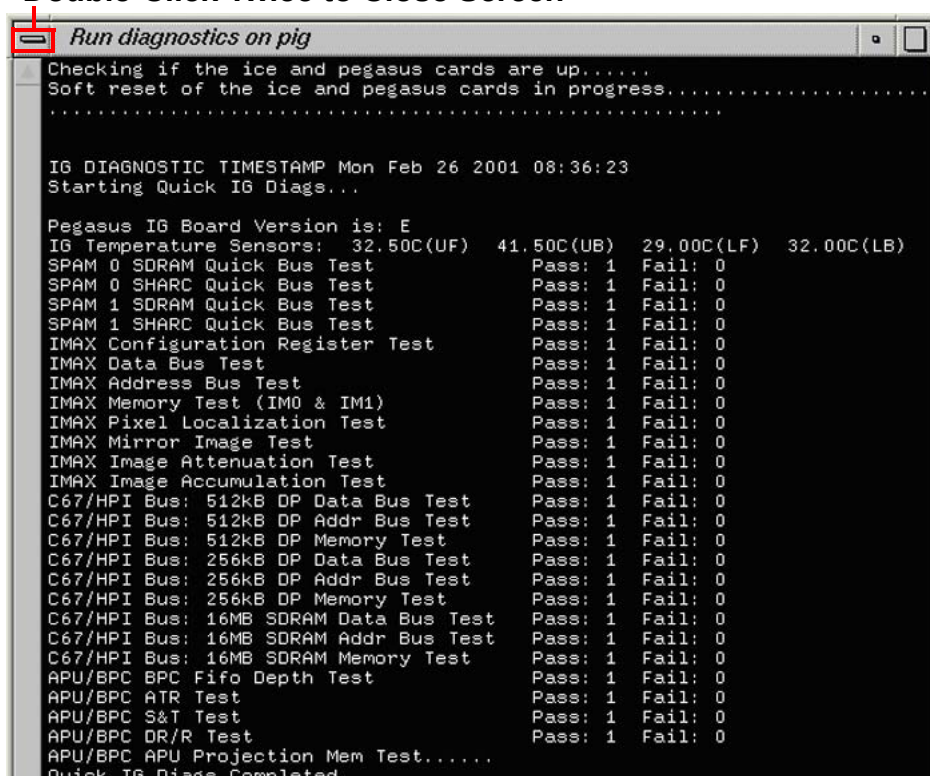
Pegasus Image Generator Board Troubleshooting, continued

Low Level Board Diagnostics Procedures, cont'd

- To exit testing, close all diagnostic windows by double-clicking the bar in the upper left corner of the screen.

Quick IG Diags Screen Example

Double-Click Twice to Close Screen

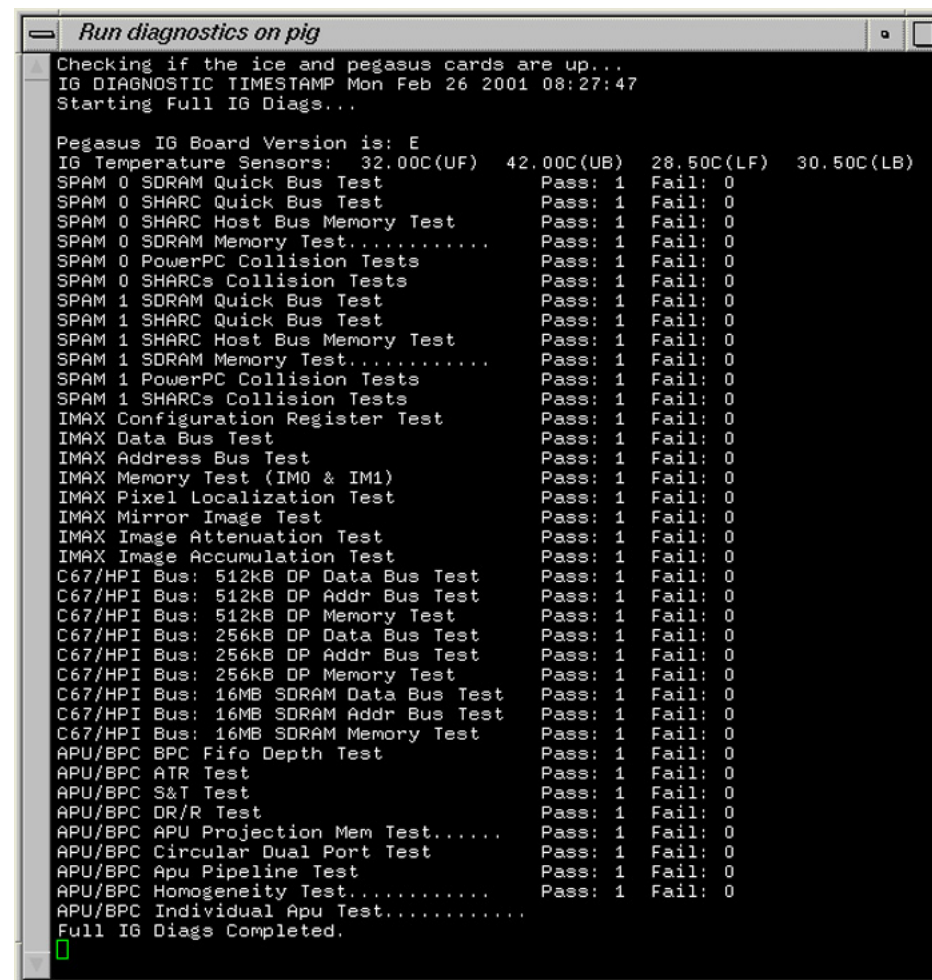


```

Run diagnostics on pig
Checking if the ice and pegasus cards are up.....
Soft reset of the ice and pegasus cards in progress.....
.....
IG DIAGNOSTIC TIMESTAMP Mon Feb 26 2001 08:36:23
Starting Quick IG Diags...

Pegasus IG Board Version is: E
IG Temperature Sensors: 32.50C(UF) 41.50C(UB) 29.00C(LF) 32.00C(LB)
SPAM 0 SDRAM Quick Bus Test Pass: 1 Fail: 0
SPAM 0 SHARC Quick Bus Test Pass: 1 Fail: 0
SPAM 1 SDRAM Quick Bus Test Pass: 1 Fail: 0
SPAM 1 SHARC Quick Bus Test Pass: 1 Fail: 0
IMAX Configuration Register Test Pass: 1 Fail: 0
IMAX Data Bus Test Pass: 1 Fail: 0
IMAX Address Bus Test Pass: 1 Fail: 0
IMAX Memory Test (IM0 & IM1) Pass: 1 Fail: 0
IMAX Pixel Localization Test Pass: 1 Fail: 0
IMAX Mirror Image Test Pass: 1 Fail: 0
IMAX Image Attenuation Test Pass: 1 Fail: 0
IMAX Image Accumulation Test Pass: 1 Fail: 0
C67/HPI Bus: 512kB DP Data Bus Test Pass: 1 Fail: 0
C67/HPI Bus: 512kB DP Addr Bus Test Pass: 1 Fail: 0
C67/HPI Bus: 512kB DP Memory Test Pass: 1 Fail: 0
C67/HPI Bus: 256kB DP Data Bus Test Pass: 1 Fail: 0
C67/HPI Bus: 256kB DP Addr Bus Test Pass: 1 Fail: 0
C67/HPI Bus: 256kB DP Memory Test Pass: 1 Fail: 0
C67/HPI Bus: 16MB SDRAM Data Bus Test Pass: 1 Fail: 0
C67/HPI Bus: 16MB SDRAM Addr Bus Test Pass: 1 Fail: 0
C67/HPI Bus: 16MB SDRAM Memory Test Pass: 1 Fail: 0
APU/BPC BPC Fifo Depth Test Pass: 1 Fail: 0
APU/BPC ATR Test Pass: 1 Fail: 0
APU/BPC S&T Test Pass: 1 Fail: 0
APU/BPC DR/R Test Pass: 1 Fail: 0
APU/BPC APU Projection Mem Test..... Pass: 1 Fail: 0
Quick IG Diags Completed.
  
```

Full IG Diags Screen Example



```

Run diagnostics on pig
Checking if the ice and pegasus cards are up...
IG DIAGNOSTIC TIMESTAMP Mon Feb 26 2001 08:27:47
Starting Full IG Diags...

Pegasus IG Board Version is: E
IG Temperature Sensors: 32.00C(UF) 42.00C(UB) 28.50C(LF) 30.50C(LB)
SPAM 0 SDRAM Quick Bus Test Pass: 1 Fail: 0
SPAM 0 SHARC Quick Bus Test Pass: 1 Fail: 0
SPAM 0 SHARC Host Bus Memory Test Pass: 1 Fail: 0
SPAM 0 SDRAM Memory Test..... Pass: 1 Fail: 0
SPAM 0 PowerPC Collision Tests Pass: 1 Fail: 0
SPAM 0 SHARCs Collision Tests Pass: 1 Fail: 0
SPAM 1 SDRAM Quick Bus Test Pass: 1 Fail: 0
SPAM 1 SHARC Quick Bus Test Pass: 1 Fail: 0
SPAM 1 SHARC Host Bus Memory Test Pass: 1 Fail: 0
SPAM 1 SDRAM Memory Test..... Pass: 1 Fail: 0
SPAM 1 PowerPC Collision Tests Pass: 1 Fail: 0
SPAM 1 SHARCs Collision Tests Pass: 1 Fail: 0
IMAX Configuration Register Test Pass: 1 Fail: 0
IMAX Data Bus Test Pass: 1 Fail: 0
IMAX Address Bus Test Pass: 1 Fail: 0
IMAX Memory Test (IM0 & IM1) Pass: 1 Fail: 0
IMAX Pixel Localization Test Pass: 1 Fail: 0
IMAX Mirror Image Test Pass: 1 Fail: 0
IMAX Image Attenuation Test Pass: 1 Fail: 0
IMAX Image Accumulation Test Pass: 1 Fail: 0
C67/HPI Bus: 512kB DP Data Bus Test Pass: 1 Fail: 0
C67/HPI Bus: 512kB DP Addr Bus Test Pass: 1 Fail: 0
C67/HPI Bus: 512kB DP Memory Test Pass: 1 Fail: 0
C67/HPI Bus: 256kB DP Data Bus Test Pass: 1 Fail: 0
C67/HPI Bus: 256kB DP Addr Bus Test Pass: 1 Fail: 0
C67/HPI Bus: 256kB DP Memory Test Pass: 1 Fail: 0
C67/HPI Bus: 16MB SDRAM Data Bus Test Pass: 1 Fail: 0
C67/HPI Bus: 16MB SDRAM Addr Bus Test Pass: 1 Fail: 0
C67/HPI Bus: 16MB SDRAM Memory Test Pass: 1 Fail: 0
APU/BPC BPC Fifo Depth Test Pass: 1 Fail: 0
APU/BPC ATR Test Pass: 1 Fail: 0
APU/BPC S&T Test Pass: 1 Fail: 0
APU/BPC DR/R Test Pass: 1 Fail: 0
APU/BPC APU Projection Mem Test..... Pass: 1 Fail: 0
APU/BPC Circular Dual Port Test Pass: 1 Fail: 0
APU/BPC APU Pipeline Test Pass: 1 Fail: 0
APU/BPC Homogeneity Test..... Pass: 1 Fail: 0
APU/BPC Individual APU Test..... Pass: 1 Fail: 0
Full IG Diags Completed.
  
```

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Pegasus Image Generator Board Troubleshooting, *continued*

Tool Errors

On rare occasions, diagnostic tests may fail to execute or an error message will be displayed. If this happens, perform the following:

1. Close all open shell windows created by executing **Run Diag**. Diagnostics will not run with multiple windows opened by **Run Diag**.
2. Shut down console, restart and then re-run IG Diagnostics. On rare occasions, run diags cannot allocate the OS system resources necessary to execute diagnostics. When this occurs, you will see multiple PTY (pseudo TTY) errors reported.

None of the items identified above are related to the operation of the PEG-IG board. They're only related to the operation of the diagnostic tool itself.



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X-ray Generation Troubleshooting

The Lightspeed system provides valuable errors to assist with troubleshooting. The errors are indicated by **major** and **minor functions**. This section covers X-ray generation errors, which include:

- **X-ray Generation Troubleshooting, continued**
- **Ohm Filament**
- **mA Tolerance or Imbalance Errors**
- **Rotor Control and DC Bus Monitor**
 - **Error 260003116**
 - **Error 260003120**
 - **Error 260003121**
 - **Error 260003124**
 - **Error 260003125**
 - **Error 260003140**
 - **Error 260003142**
 - **Error 260003143**
 - **Error 260003147**
 - **Error 260004020**
- **Errors 219800 & 183327**
- **Errors 183316 & 185548**
- **Errors 183124 & 214335**
- **RCIB Errors: System Comm./Control**
- **kV Loop Imbalance**
- **Filament Inverter**
- **Z-Axis Tracking**

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X-ray Generation Troubleshooting, *continued*

Open Filament Error 183121

Error

Error # 183121 - Open Filament

Wed Apr 1 17:02:23 1998

Suite: HSA1 Host: OBC Proc: Generator Error:
183121

File: gen_abort.c 1.30 Method: No Method Line: 340

Function: XRAY GENERATION : Filament Selection

Scan: 0/0/0 Type: None/Unknown

Exception Level: Pri/Most Severe Time:

17:02:22:885 Log Series: 26

FIL/MA fault detected: Open filament circuit.

Desired focal spot = 0. (0 = large, 1 = small)

Fault latch address = ffd43 hex, Bit = D7

Field Replacement Units (FRUs) Involved

- Tube
- Cathode **HV** cable
- Cathode Tank (contains the filament transformer)
- Wiring from tank to **OBC** backplane
- Filament select relay
- mA Control Board
- Harness
- 38 VDC Power Supply

Troubleshooting Summary

1. 38 VDC Power Supply.
2. Ohm Filament and **HV** Cable.
3. Filament Drive Verification.
4. Filament Command Verification.
5. Verify filament select relay K78 on the backplane of the **OBC**.
6. Transformer & Measurement Board verification
7. Escalate to Level 2 Support.

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X-ray Generation Troubleshooting, *continued*

Ohm Filament

Procedure Objective

The objective of this procedure is to check the continuity of the HVDC cables and the X-ray tube small and large filaments. The HVDC cables can fail generally in an open condition. There are three (3) conductors, labeled “L” for large, “S” for small, and “C” for common. The X-ray tube filaments can fail in either open or shorted conditions.

You will take resistance measurements through the cables to measure the resistance of the filaments.

Procedure Summary

1. Measure resistance of all HVDC conductors to frame ground.
2. Measure resistance of the small and large filament through the cathode HVDC cable.
3. Measure resistance of the of the cathode HVDC cable.

Procedure Details

1. Remove power to the gantry using proper Lockout/Tagout procedures.
2. Remove the cathode HVDC cable from the cathode HV tank.
3. Connect the common lead of the DVM to frame ground.
4. Connect the positive lead of the DVM to “S”, “L”, “C” pins of the HVDC cable each in turn.

The readings from this step should all be OPEN.

5. Connect the common lead of the DVM to the “C” pin of the HVDC cable candlestick.
6. Connect the positive lead of the DVM to the “S” and “L” pins of the HVDC cable candlestick each in turn.
7. Reverse polarity of the test leads and repeat steps “3e and 3f”.
8. Compare readings to the table below.

If a reading of OPEN or high resistance was measured in steps 5 through 7, proceed with the following steps.

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Ohm Filament, *continued*

Procedure Details, *continued*

9. Remove the cathode HVDC cable at the X-ray tube.
10. Connect the DVM leads across the HVDC cable, "C" to "C", "S" to "S", "L" to "L" each in turn.

If a reading of OPEN or high resistance is measured in step 9, replace the Cathode HVDC cable.

11. Inspect the X-ray tube cathode well and verify:
 - No arcing or cracks.
 - Contact springs present in each pin receptacle.

12. HVDC cable pins are tight, if applicable, or the pin gap is present.

Tighten HVDC pin to 1.5 ft-lbs (2 Nm) or spread the pin gap using tool, part number 46-208572P44.

13. Insert HVDC cable back into X-ray tube. Repeat steps 5 through 8.

If readings are still abnormal, replace the X-ray tube, as the filaments are damaged.

Note Failures in steps 1 through 4 are a result of a damaged HVDC cable or filament to case short. Replace appropriate part as determined in this procedure.

Resistance Reading Through the HVDC Cable (Performix Tube Cathode)

Path	Ohms (Approx.)	Path	Ohms (Approx.)
L to C	0.5	C to L	0.9
S to C	2.1	C to S	0.5
L to S	0.7	S to L	2.3

Note The resistance measurements of the filament are polarity sensitive. The first test point is always referenced as the positive lead of the DVM. There is a diode-like effect due to the construction of the filament and focal cup interaction.

At no time should you read any continuity to ground. Readings are approximates, variability in meter leads and measurement methods should be taken into consideration.

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mA Tolerance or Imbalance Errors

Error

Application Errors: #183101, #183296, #183186

Diagnostic Errors: #214321, #214343, #214343

Wed Apr 1 22:00:56 1998

Suite: HSA1 Host: OBC Proc: Generator Error:
183186

File: gen_tech.c 1.32 Method: No Method Line: 923

Function: XRAY GENERATION : MA Loop

Scan: 50006/1/2 Type: Static

Exception Level: Pri/Soft Time: 22:00:55:785 Log
Series: 39

mA imbalance over 10 % tolerance.

Selected technic: 100 kV, 50 mA

Imbalance threshold = 5.0 mA.

Average anode mA = 49.7 mA.

Average cathode mA = 0.2 mA.

Wed Apr 1 22:00:56 1998

Suite: HSA1 Host: OBC Proc: Generator Error:
183296

File: gen_tech.c 1.32 Method: No Method Line: 932

Function: No System Function Reported

Scan: 50006/1/2 Type: Static

Exception Level: Pri/Soft Time: 22:00:55:790 Log
Series: 41

Selected Technic: 100 kV, 50 mA, 0 % Mod, Backup
Timer 1080.

VarmA: (Index: Start 0, Skip 0, Last 0) Last Cmd
50 mA.

Monitoring: (Measured 49.7 mA vs Ref 50 mA).

Total Samples, Tube Spit Ctr 0.

Field Replacement Units (FRUs) Involved

- X-ray Tube
- mA Board
- Cathode tank (contains the filament transformer)
- Focal Spot Relay (on the back of the OBC Backplane)
- 38 VDC Filament Power Supply

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mA Tolerance or Imbalance Errors, *continued*

FRUs Involved, *continued*

- OBC Power Supply
- Software **.char** files
- Wiring from tank to OBC backplane.

Troubleshooting Summary

1. Verify error in log and note all pertinent details.
2. Run kV & mA Diags to verify mA out of tolerance as the primary error
3. Verify mA meter verification
4. Verify Focal Spot Relay
5. Troubleshoot Filament Drive error
6. Verify 36 VDC Filament Power Supply
7. Verify OBC Power Supply
8. Call the OLC (tentative)
9. Software **.char** files
10. Escalate:
 - Tips and tricks
 - Zone Support

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Rotor Control and DC Bus Monitor

Error 260002008

Error

901784298 3 1 Thu Jul 30 07:38:18 1998 200352008 2

baya_sbc collimator_func

collimator_func.c 1062

Aperture positioned incorrectly.

Width : 268604568

Offset : 7919

Note This is not a primary error. Look into Sys Log for other application, CAM errors.

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Rotor Control and DC Bus Monitor, *continued*

Error 260003116

Error

01812389 1 1 Thu Jul 30 15:26:29 1998 260003116 2

HSC CCB Firmware

ccb_drive_timers.c 503

Function: XRAY GENERATION : Xray Collimation and Filtration

Scan: 60375/1/0 Type: None/Unknown

Exception Level: Sec/Soft Time: 15:26:29:260 Log Series: 118

Filter drive tracking detected no motion when motion was requested.

Verify operation of stepper motor and encoder feedback.

Start Position = 1

Desired Position = 3780

Current Motor Position = 1440

Current Encoder Position= 7

Field Replacement Units (FRUs) Involved

- Collimator Control Board
- Harness
- Motor
- CAM Driver / Amp
- Encoder

Troubleshooting Summary

1. Verify Harness
2. Check Filter Amp
3. Check Filter Motor
4. Check Filter Encoder
5. Check CCB
6. Check software
7. Escalate

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Rotor Control and DC Bus Monitor, *continued*

Error 260003120

Error

901641431 4 1 Tue Jul 28 15:57:11 1998 260003120 2

HSC CCB Firmware

aperture_home.c 192

Function: XRAY GENERATION : Xray Collimation and Filtration

Scan: 60316/2/0 Type: None/Unknown

Exception Level: Pri/Soft Time: 15:57:11:810 Log Series: 91

An error occurred attempting to home CAM B.(or CAM A)

Field Replacement Units (FRUs) Involved

- Collimator Control Board (CCB)
- Harness
- CAM B Home Switch (or CAM A Home Switch) (Collimator if the FRU)
- CAM B Motor (or CAM A MOTOR) (Collimator if the FRU)
- CAM B Encoder (or CAM A Encoder) (Collimator if the FRU)

Troubleshooting Summary

1. Check Collimator Control Board
2. Verify Harness
3. Check CAM Home Switch
4. Check CAM Motor
5. Verify CAM Encoder
6. Escalate

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Rotor Control and DC Bus Monitor, *continued*

Error 260003121

Error

901641432 0 1 Tue Jul 28 15:57:12 1998 260003121 2

HSC CCB Firmware

aperture_move.c 225

Function: XRAY GENERATION : Xray Collimation and Filtration

Scan: 60316/2/0 Type: None/Unknown

Exception Level: Pri/Soft Time: 15:57:12:350 Log Series: 93

The Aperture failed to move to the requested position.

Note This is not a primary error. Look into Sys Log for other application, CAM errors.

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Rotor Control and DC Bus Monitor, *continued*

Error 260003124

Error

901725195 1 1 Wed Jul 29 15:13:15 1998 260003124 2

HSC CCB Firmware

aperture_home.c 143

Function: XRAY GENERATION : Xray Collimation and Filtration

Scan: 60362/2/0 Type: None/Unknown

Exception Level: Pri/Soft Time: 15:13:16:065 Log Series: 78

Aperture Homing Problem.

Field Replacement Units (FRUs) Involved

- Collimator Control Board
- Harness
- Motor
- CAM Driver / Amp
- Encoder
- Characterization (software)

Troubleshooting Summary

1. Verify Harness
2. Check CAM Motors
3. Check CAM Encoders
4. Check CAM AMP / Driver
5. Check Collimator Control Board
6. Check Characterization (software)
7. Escalate

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Rotor Control and DC Bus Monitor, *continued*

Error 260003125

Error

901641431 1 1 Tue Jul 28 15:57:11 1998 260003125 2

HSC CCB Firmware

aperture_home.c 404

Function: XRAY GENERATION : Xray Collimation and Filtration

Scan: 60316/2/1 Type: Static

Exception Level: Pri/Soft Time: 15:57:11:485 Log Series: 90

Did not detect a encoder index pulse when homing CAM B. (or CAM A)

EN 330

Field Replacement Units (FRUs) Involved

- Collimator Control Board (CCB)
- Harness
- CAM B Home Switch (or CAM A Home Switch) (Collimator if the FRU)
- CAM B MOTOR (or CAM A MOTOR) (Collimator if the FRU)
- CAM B Encoder (or CAM A Encoder) (Collimator if the FRU)

Troubleshooting Summary

1. Check Collimator Control Board
2. Verify Harness
3. Check CAM Home Switch
4. Check CAM Motor
5. Verify CAM Encoder
6. Escalate

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Rotor Control and DC Bus Monitor, *continued*

Error 260003140

Error

901785079 0 1 Thu Jul 30 07:51:19 1998 260003140 2

HSC CCB Firmware

filter.c 78

Function: XRAY GENERATION : Xray Collimation and Filtration

Scan: 60371/2/0 Type: None/Unknown

Exception Level: Pri/Soft Time: 07:50:59:615 Log Series: 163

Filter Switch Error. NO/NC Poles are not opposite.

Register value: 0

Field Replacement Units (FRUs) Involved

- Collimator Control Board
- Harness
- Filter Home Switch

Troubleshooting Summary

1. Verify Harness
2. Check Filter Home Switch
3. Check Collimator Control Board (CCB)
4. Escalate

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Error 260003142

Error

901812388 3 1 Thu Jul 30 15:26:28 1998 260003142 2

HSC CCB Firmware

filter_home.c 189

Function: XRAY GENERATION : Xray Collimation and Filtration

Scan: 60375/1/0 Type: None/Unknown

Exception Level: Pri/Soft Time: 15:26:28:470 Log Series: 117

Filter homing problem.

EN 302

Field Replacement Units (FRUs) Involved

- Harness
- Filter Drive / Motor
- Filter Home Switch
- Filter Encoder
- Characterization (software)

Troubleshooting Summary

1. Verify Harness
2. Check Collimator Control Board
3. Check Filter Drive / Motor
4. Check Filter Encoder
5. Check Filter Home Switch
6. Check Characterization (software)
7. Escalate

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Rotor Control and DC Bus Monitor, *continued*

Error 260003143

Error

901783557 3 1 Thu Jul 30 07:25:57 1998 260003143 2

HSC CCB Firmware

filter_move.c 99

Function: XRAY GENERATION : Xray Collimation and Filtration

Scan: 60370/1/0 Type: None/Unknown

Exception Level: Pri/Soft Time: 07:25:58:880 Log Series: 96

The Filter failed to move to the requested position.

Note This is not a primary error. Look into Sys Log for other application, CAM errors.

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Rotor Control and DC Bus Monitor, *continued*

Error 260003147

Error

901812388 1 1 Thu Jul 30 15:26:28 1998 260003147 2

HSC CCB Firmware

filter_hisr.c 125

Function: XRAY GENERATION : Xray Collimation and Filtration

Scan: 60375/1/0 Type: None/Unknown

Exception Level: Sec/Soft Time: 15:26:28:110 Log Series: 116

The filter amplifier short circuit detection interrupt cannot be cleared

Collimator Register = 0x30

Field Replacement Units (FRUs) Involved

- Collimator Control Board (CCB)
- Harness
- Filter Amplifier
- Filter

Troubleshooting Summary

1. Verify Harness
2. Check Filter Amplifier
3. Check Collimator Control Board (CCB)
4. Escalate

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Rotor Control and DC Bus Monitor, *continued*

Error 260004020

Error

901785079 3 1 Thu Jul 30 07:51:19 1998 260004020 5

OBC Collimator

col_cc_in_mail.c 297

Function: No System Function Reported

Scan: 60371/2/0 Type: None/Unknown

Exception Level: Pri/Most Severe Time:
07:51:09:600 Log Series: 166

Collimator firmware object: Command Timed Out.

Time out waiting for Message Code = 19.

Watchdog Timer was set to 10000 msec.

Field Replacement Units (FRUs) Involved

- Harness
- Filter Drive / Motor
- Filter Home Switch
- Filter Encoder
- Characterization (software)

Troubleshooting Summary

1. Verify Harness
2. Check Collimator Control Board
3. Check Filter Drive / Motor
4. Check Filter Encoder
5. Check Filter Home Switch
6. Check Characterization (software)
7. Escalate

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Rotor Control and DC Bus Monitor, *continued*

Errors 219800 & 183327

Errors

Mon Apr 20 07:38:04 1998 Error: 219800

Suite: KC03 Host: OBC Process: Generator

File: HEMRClogFault.c 1.2 Method: No Method Line: 196

Function: XRAY GENERATION : Tube Rotor Control

Scan: 526/1/1 Type: Scout

Exception Level: Secondary Time: 07:38:04:300 Log Series: 357

The HEMRC AC Drive reported fault code: F4 - Undervoltage Fault

The Drive internal DC bus voltage dropped below 325V. Possible low voltage condition on 120 VAC in gantry or power interruption. Also, may indicate excessive run or braking power required due to sluggish tube.

Mon Apr 20 07:38:04 1998 Error: 183327

Suite: KC03 Host: OBC Process: Generator

File: gen_abort.c 1.31 Method: No Method Line: 598

Function: XRAY GENERATION : Tube Rotor Control

Scan: 526/1/1 Type: Scout

Exception Level: Pri/Most Severe Time: 07:38:04:200 Log Series: 357

HEMRC fault detected.

Fault latch address = ffb82b hex, Bit = D0

Field Replacement Units (FRUs) Involved

- HEMRC Interface Board - HIF (All Phases)
- HEMRC AC Drive (All Phases)
- Fuses F1, F2 on Interface Board (would produce a Primary Error of "NO HVDC")
- HVDC Bus (1st 4 Phases)
- 120 VAC from slip ring assembly (last 4 Phases)
- T1 transformer (last 4 phases)
- Bridge Rectifier - CR1 (1st 4 Phases)

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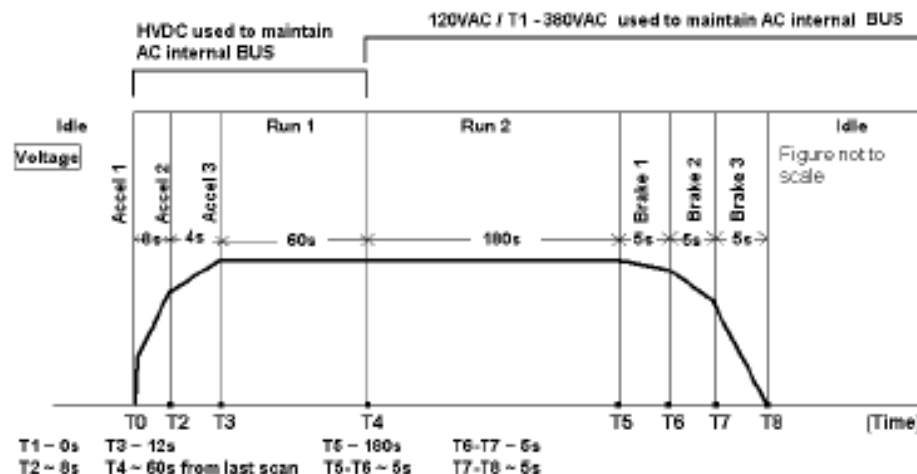
Rotor Control and DC Bus Monitor, *continued*

Errors 219800 & 183327, *continued*

Troubleshooting Summary

1. Verify error in log is the primary error and note all pertinent details. Troubleshoot HVDC errors.

Important! The HEMRC Rotor has nine (9) phases of operation: 3 accel phases, 2 Run phases, 3 Brake phases, and idle. The 3 accel phases and the 1st run phase use HVDC. The rest of the phases use 120 VAC stepped up to 380 VAC by T1. A graphical interpretation of the 9 Phases is shown at right.



2. Determine the mode of rotor operation when the error occurs:
 - Acceleration mode
 - Run mode
 - Brake mode
 - HVDC-related
 - 120/380 VAC-related
3. Escalate:
 - Tips and Tricks
 - Zone Support

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Rotor Control and DC Bus Monitor, *continued*

Errors 183316 & 185548

Errors

Thu Apr 9 01:19:07 1998 Error: 183316

Suite: CT01 Host: OBC Process: Generator

File: hemrc_manager.c 1.14 Method: No Method Line: 968

Function: XRAY GENERATION : Tube Rotor Control

Scan: 0/0/1 Type: Static

Exception Level: Pri/Most Severe Time:

02:37:06:930 Log Series: 30

The HEMRC failed to reach the Run speed in the allotted time.

Thu Apr 9 01:19:21 1998 Error: 185548

Suite: CT01 Host: OBC Process: Diag Exec

File: hemrc_manager.c 1.5 Method: gen_rotor Line: 859

Function: Diagnostics XRAY GENERATION : Tube Rotor Control

Test Name: HEMRC Manual Unique ID: 11553

Iteration: 1

Exc. Level: Hard Time: 294048 Error Type: Hardware

The HEMRC operating frequency is below the minimum value.

Actual frequency: 88.394 Minimum frequency limit: 249.995

At Frequency signal: Inactive Expected: Inactive
Rotor State = Run

Field Replacement Units (FRUs) Involved

- HEMRC Interface Board
- HEMRC AC Drive
- HVDC BUS
- HEMRC Control Board (HCB)
- Fuses F1, F2 on interface Bd (would produce a Primary Error of "NO HVDC")
- Check all the J3A series connections

Troubleshooting Summary

1. Verify error in log is the primary error and note all pertinent details.
2. Troubleshoot HVDC errors.

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RCIB Errors: System Comm./Control

Errors

DIAG MESSAGE

01525711 1 1 Mon Jul 27 07:48:31 1998 260001013 1

OBC HOP

hop_fault_test.c 236

Function: No System Function Reported

Scan: 0/0/0 Type: None/Unknown

Exception Level: Pri/Most Severe Time:
07:48:29:535 Log Series: 25

Fault line test failed:

The fault line was still set on one or more nodes
after being disabled.

See log for additional information.

DIAG MESSAGE

901525711 2 1 Mon Jul 27 07:48:31 1998 260001057 1

OBC HOP

hop_fault_status.c 190

Function: No System Function Reported

Scan: 0/0/0 Type: None/Unknown

Exception Level: Sec/Soft Time: 07:48:29:570 Log
Series: 25

RCIB Fault Line failure detected.

The status of the fault bus is as follows:

Parallel or Serial Fault line: [1=Fault 0=OK -
1=Unknown]

Command: [1=Fault Asserted 0=OK -1=Unknown]

(Note: An unknown status is due to message failure.)

HEMRC fault status: 1 [1=serial or parallel fault
set. 0=OK]

Node Parallel Serial Command

HEMRC - 1 0

CCB 0 1 0

DCB 0 1 0

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RCIB Errors: System Comm./Control, (*continued*)

Field Replacement Units (FRUs) Involved

- HEMRC Control Board (HCB)
- Collimator Control Board
- Rotating Controller Interface Bus (RCIB)
- CAN Harnessing

Troubleshooting Summary

1. Verify 12 VDC at the HEMRC Control Board (HCB)
2. Verify 12 VDC at the Collimator Control Board (CCB)
3. Verify 12 VDC at the DAS Control Board (DCB)
4. Escalate:
 - Tips and tricks
 - Zone Support

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kV Loop Imbalance

Errors

Wed Jun 21 16:59:42 1995

Suite: CT02 Host: OBC Proc:

Generator Error: 183183

File: Method: No

950621 1713 03294 Method Line: 571

Function: XRAY GENERATION : KV Loop

Scan: 16922 1 1 Type: Scout

Exception Level: Sec Soft Time: 16:59:39:810

Log Series: 38

kV out of 3 % tol

Selected technic: 80 kV, 40 mA

Tot. samples = 2

Tot. kV samples above tol. = 0, below tol. = 2

950621 1712 03294 Avg. of kV samples above tol. =
0.000 kV, below tol. =

40.000 kV

Avg. of tot. kV = 40.000 kV

Wed Jun 21 16:59:42 1995

Suite: CT02 Host: OBC Proc:

Generator Error: 183185

File: Method: No

Method Line: 661

950621 1712 03294 Function: XRAY GENERATION : KV
Loop

Scan: 16922 1 1 Type: Scout

Exception Level: Sec Soft Time: 16:59:39:810

Log Series: 38

kV imbalance over 10 % tolerance.

Selected technic: 80 kV, 40 mA

Imbalance threshold = 8.000 kV

Average anode kV = 0.000 kV.

Average cathode kV = 40.049 kV

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KV Loop Imbalance, *(continued)*

Field Replacement Units (FRUs) Involved

- KV Board
- Both Inverters
- Both Tanks
- OBC Quad Power Supply
- Fiber Optics Sequencing
- OBC Backplane Wiring
- X-ray Tube
- DCRGS or HVDC

Troubleshooting Summary

1. Verify that this is the primary error.
2. Determine which KV board is installed in system.
3. Run KV LOOP test and use result screens to determine FRU; this is the diagnostics you want to fail because the tube can be verified/eliminated as the problem.
4. IF KV LOOP does NOT fail, run KV & mA test and use result screens to determine FRU.
5. Scope KVp.

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Filament Inverter

Errors 183124 & 214335

Error

Wed Mar 5 10:42:53 1997 Error: 183124
 Suite: TMC2 Host: OBC Process: Generator
 File: gen_abort.c 1.30 Method: No Method Line: 368
 Function: XRAY GENERATION : Filament Selection
 Scan: 0/0/1 Type: Static
 Exception Level: Pri/Most Severe Time:
 10:42:52:865 Log Series: 32
 FIL/MA fault detected: Inoperative filament
 inverter.
 Desired focal spot = 0. (0 = large, 1 = small)
 Fault latch address = ffd4d hex, Bit = D7
 Wed Mar 5 10:53:55 1997 Error: 214335
 Suite: TMC2 Host: OBC Process: Diag Exec
 File: isr1_handler.c 1.33 Method: obc_isr1_handler
 Line: 516
 Function: Diagnostics XRAY GENERATION : MA Loop
 Test Name: XRay - Manual Unique ID: 52105
 Iteration: 1

Exc. Level: Hard Time: 85769 Error Type: Hardware
 Filament error.
 A filament inverter error has been detected.
 MA address: FFDF4DH Bit: D7

Field Replacement Units (FRUs) Involved

- mA/Filament Control Board
- 30 VDC Filament Power Supply
- Harness between Power Supply/mA boards

Troubleshooting Summary

1. Observe the signal at TP27 with an oscilloscope. Look for a wave shape similar to [Figure 1](#).
2. Observe the signal at TP27 with an oscilloscope. Look for a wave shape similar to [Figure 2](#).
3. Observe the signal at TP27 with an oscilloscope. Look for a wave shape similar to [Figure 3](#).
4. Check TP21 for 30 VDC. Use TP29 or TP30 for FGND.

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Filament Inverter, *(continued)*

Troubleshooting Summary, *continued*

5. Check the Filament Power Supply for proper voltages.
6. Gather all previous data, then escalate:
 - OLC
 - Zone Support

Figure 1: TP27 Wave Form



Figure 2: TP27 Square Wave

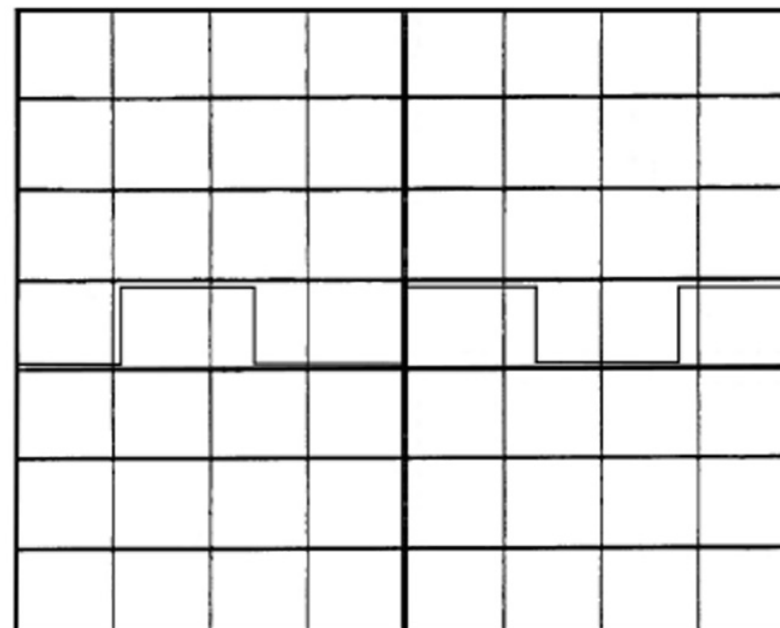


Figure 3: TP27 Flat-Line



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X-ray Generation Troubleshooting, *continued*

Z-Axis Tracking

Theory

The purpose of Z-axis tracking is to follow the focal spot so that the system can keep the uniform portion (umbra) of the X-ray beam for the narrowest possible beam on the detector to reduce dose and still avoid artifacts. The focal spot moves in the Z-axis due to thermal changes in the tube, mechanical forces during gantry rotation and due to gantry tilt angle.

Z-axis tracking takes advantage of the Collimator's ability to independently control the cams that set the Z-axis beam thickness and Detector's ability to sense the Z-axis edges of the X-ray beam.

A logically separate detection and control feedback loop is used for each cam's position. The tracking design treats the A and B Cams as independent mechanisms used to control the 'z' thickness of the X-ray beam.

The applications or 'run time' system operation sets an initial location for each of the cam's based upon the

selected acquisition mode similar to QX/i's original 1.0 Software implementation. The initial cam positions are set wide enough to ensure that the X-ray beam will fully flood the selected area of the detector.

Troubleshooting Summary

Troubleshooting is the same for either 8-slice or 4-slice system configuration. In 4-slice mode, any **even** converter cards reported are the fault of the DCB since these slots are empty.

Troubleshooting Strategy

Service actions for repair can be broken down into four activities defined as:

- **Detect:** Error Messages or Image Artifacts
- **Diagnose:** Diagnostics, Tools
- **Correct:** Adjustment, Replacement
- **Verify:** Scan Error Free

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Z-Axis Tracking, *continued*

Troubleshooting Details

1. Tracking Fault Detection

Detection of hardware faults for Z-Axis tracking is expected to take place in the run-time or application software. The Collimator and DAS hardware incorporate features that provide feedback as to normal and out-of-bounds operation. This includes subsystem communications link integrity checking, command feedback, and operating status. This functionality should provide good runtime monitoring for out of specification conditions in the tracking loop.

Within the collimator, rotary optical encoders mechanically attached to each of the cams provide positional feedback. This feedback coupled with a factory created Cam Encoder position database assures accurate positioning of the Cams. Since this mechanism relies on the accuracy of the collimator calibration database an independent means to test the cam position accuracy was developed and is used.

2. Collimator Aperture Test

The collimator aperture test uses the detector to 'see' the edge of the collimated X-ray beam in the 'z' axis and relate that to where the collimator mechanism has positioned the cams.

The detector cells used for this test are not the 'z-channel' cells used for the run-time Z-axis tracking. This allows confirmation that the collimator control loop can control the collimator cams independent of the detector Z-channels and that the detector DAS combination can correctly detect and communicate the edge of the X-ray beam.

This test method depends upon the fixed mechanical dimensions of the detector and the detectors fixed mechanical relationship to the X-ray tube. The other factors are the relationship between detector cells signal output and the amount of X-rays striking the detector cells. The test starts a X-ray exposure with the cams positioned in such a way that the detector cells should be fully flooded with X-rays. The cam position is then changed until the monitored detector cells output drops to 50% of the detector row's fully flooded output level. Because of the fixed mechani-

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Z-Axis Tracking, *continued*

2. Collimator Aperture Test, *(continued)*

cal relationships for the tube detector and their distances from one another, we can determine how many encoder counts should be required to get to the 50% cell output position. This number is compared with the actual number of encoder counts just moved in order to determine if the positioning accuracy is within specification. The collimator aperture test provides Pass/Fail results.

Therefore, the collimator aperture test verifies that the collimator control circuitry can correctly control the collimator cams independently of the detector Z-channels.

3. Collimation and Filtration Functional Test

The collimation and filtration functional test is used to verify that the collimator electro-mechanical elements can operate in a repeatable manner. This test commands the collimator electronics to move the cams and beam shaping filter to specific locations in their operating range checking the actual encoder counts to the expected position counts. Discrepancies result in error reports detailing the discrepancy.

4. DAS Tools Interconnect Test

5. Tracking Analysis Tool

a. Tracking Analysis General Information

The Tracking Analysis Tool provides detailed information regarding closed loop performance that obtained from analysis of patient scan data.

This is necessary to assess the health of the tracking loop and to associate artifacts with tracking abnormalities.

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Z-Axis Tracking, *continued*

5. Tracking Analysis Tool, (*continued*)

b. What does Tracking Analysis Provide?

For each scan data set available in the scan file database, the analysis package can plot various data versus the scan.

For each scan data set available in the scan file database, the analysis package can various data views in a default filtered (20 pt boxcar) or unfiltered format.

The data plotted and its presentation versus scans and views is given in the **Figure 4** Table.

When data is plotted versus views, the numerical information (plotted versus scans) is displayed. Pass/ fail criteria is indicated if possible.

6. Loop Error Plot

Loop error is the difference between the calculated position of the beam minus the desired target position (operating point). The calculation requires data from the Z-channels and from the DAS Gain Cal results.

7. Loop Error Relative to Fast Cal Beam Position Plot

Same as Loop Error Plot except that the display represents the loop error relative to the mean beam position during Fast Cal. The fast cal beam position is stored in calibration database.

8. Time to Initialize Plot

Compute the time required for the absolute value of the filtered Loop Error to be < 0.1 mmd. Time depends on magnitude of initial error but in all cases should be tracking to position within 150 ms.

9. Z-Ratio Plot

Compute the Z ratio 'R' as described for the loop error computation in step 6, Loop Error Plot.

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Z-Axis Tracking, *continued*

10. Blocked Z Module Plot

Compute the views where a blocked Z-channel condition occurs. The calculation requires: mA DAS count value from the auxiliary channel input, DAS trigger frequency, DAS gain factor for channel 762 from DAS Gain Cal, and Blocked Channel scale factor determined during fast cal.

The percent blocked value per scan is the number of blocked views divided by the total views x 100.

11. Cam Positions Plot

Obtained the cam positions from the scan file view header.

12. Focal Spot Position Plot

The calculated focal spot position.

13. Normalized Aperture

The calculated normalized aperture.

14. Cam Ringing

Cam ringing is computed to provide a plot of the high frequency variations that are 180 degrees out of phase such as typical cam ringing. A spec is not available but typical expected values are less than 0.1 mmf.

15. Rotor Run

Rotor run is computed to provide a plot of the high frequency variations that are in phase such as the small periodic movement of the anode at the rotor run frequency. A spec is not available but typical expected values are less than 0.1 mmf.

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Z-Axis Tracking, *continued*

Figure 4: Tracking Information Plots

Function Plotted	How Data is Obtained or Other Info	Output Type			Displayed (Sustained) Specification
		vs. scan	filtered* vs. views	raw vs. views	
Loop Error A and B	Computed at isocenter from Z data and collimator positions-target value	mean not blocked Peak to Peak (PP) error filtered	X	X	0.3 notify 0.6 abort
Loop Error A and B relative to mean beam position at fast cal	Same Loop error as above except subtract beam post at fast cal	mean	X	X	typical <0.3
Time to initialize	How long until error <.1 mmd	time in mS			150 mS typical <40 mS
Z ratio A and B	Computed	PPfiltered	X	X	Info tool
Blocked	Overlay on any plot - shade blocked region	% views blocked	X	X	Info tool
Cam positions A and B	From view header	Data mean, PP	X	X	Info tool
Focal spot position A and B	Computed from Z data and collimar positions	Mean, PP	X	X	Info tool
Normalized aperture	Computed	Mean, PP	X	X	1 ±0.05 typical 1 ±0.15 notify 1 ±0.30 abort
Focal spot position	Computed	Mean, PP	X	X	+0.30 typical (cold) -0.20 typical (hot)
Cam ringing	Computed	Stdev, PP	X	X	<0.1 pp typical
Rotor run	Computed	Stdev, PP	X	X	<0.1 pp typical



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Acrobat Reader 4.0 Controls

Here are some shortcut keystrokes for Acrobat Reader 4.0:

[HOME] takes you to the first page of the EPST.

[PAGE UP], [PAGE DOWN], and [END] work as titled.

[CTRL][←] takes you back to the previous display.

[CTRL][N] allows you to select a specific page number.

[F5] toggles the Bookmark display. Click on a Bookmark to go there.

[F6] toggles the Thumbnail display.



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Linking and Navigation

On the bottom of every page, you will find a linking tool bar. Click on the section's icon to take you to that section. In addition to the linking toolbar on each page, you will find other links throughout the EPST.

Convention	Link
Blue text	Takes you to supplemental information about that topic.
See logic diagram XX	Takes you to the specified logic/block diagram.
	Links to another area on the same block/logic diagram. Note Not all of these have an active link; they are linked only when you need information you cannot see on a particular part of a diagram.

For additional navigation options, you may use the Acrobat Reader bookmark feature. To display bookmarks:

1. Press [F5] or click on the PageWithLeftMargin button on the Acrobat tool bar to open the Bookmark window.
2. Click on the section or page listed in the Bookmark window that you wish to view.
3. Press [F5] or click on the PageWithLeftMargin button to close the Bookmark window.



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Adding Your Personal Notes

To add your personal notes:

1. Click on the Notes icon.
A Word document opens.
2. Use the [TAB] key to navigate through the tables.
3. Type in the corresponding EPST page number and your notes.
4. When you finish making your notes, save the file and close Word.
5. To access your Notes, simply return to the EPST and click on the Notes icon at the bottom of any page.

Important! Personal Notes cannot work if you move the file; the path is hard-coded into the EPST. Personal Notes may not work when the computer is networked.



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Printing

To print, perform the following actions.

1. On the Acrobat toolbar, select **File, Print**.
2. Select whether to print all pages, only the current page, or a range of pages. (Default is Print All Pages.)
3. Enable the "Shrink to Fit" option when printing the 11" x 17" block diagrams on 8.5" x 11" paper.
4. Click on **OK** to begin printing.



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About This EPST

- The EPST is not intended to replace OEM system documentation. It also does not cover every problem or situation you may encounter while working on the system. There are probably many situations in which it may be helpful to refer to OEM system documentation.
- The link icons located at the bottom of every page usually take you to the first page of that section. When a link icon takes you to a specific reference, the link icon on the destination page will generally take you to the first page of that section.
- The setup program will create a Word document for your notes. The file is located at:
`c:\mvgdi\epst_notes\fcxxxx\persnote`
 - Personal Notes cannot work if you move the file; the path is hard-coded into the EPST.



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Chemicals and Materials

Always wear personal protection equipment to prevent inhalation, digestion and absorption of any substance through the skin, eyes, nose or mouth.

Slip Ring Brush Dust and Debris

Avoid inhalation and digestion of slip-ring brush dust. The slip ring brushes are made of carbon (refer to the MSDS for these materials). Carbon is the conductive material used to pass power and signals to the slip ring platters.

Cleaning Equipment

Use a HEPA (High Efficiency Particulate Air) vacuum cleaner to remove residual brush debris. Do not use a standard vacuum cleaner, because it cannot filter out all of the debris and will exhaust it back into the air. If a HEPA vacuum cleaner is not available, use alcohol soaked paper toweling to wipe surfaces clean. This method must be followed up by vacuuming with a HEPA vacuum, when available.

Personal Protection Equipment (PPE)

- Safety Glasses
- Neoprene or Nitrile Gloves

Recommended Cleaning Procedure



Possible electric shock

Servicing hardware with Power ON

Apply Tag and Lockout system power before working around or performing any slip-ring maintenance.

To clean the slip rings, proceed as follows: Remove all existing brush debris from the brush blocks, brackets, and surrounding areas, using a HEPA (High Efficiency Particulate) vacuum cleaner. Use the HEPA vacuum cleaner to remove all brush debris that may have been deposited on the gantry base and floor during servicing. Never use rags or cloths on the slip ring and brushes.

When cleaning the slip ring and brush area, do not touch the slip ring or the brushes with your bare fingers. Skin oil can damage the rings and brushes.



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Chemicals and Materials, *continued*

Slip Ring Brush Dust and Debris, *continued*

Clean-up and Personal Hygiene

After servicing slip ring component and the area, wash your hands thoroughly with soap and water. Dispose of gloves and rags properly.

Place all materials that contain brush debris (such as the HEPA vacuum cleaner filter, paper toweling, gloves, and old brush assemblies) in a sealed plastic bag and return to the GE Recycling Center:

Oils (Tube, Tank and Hydraulic)

Oil resistant gloves should be worn when working with oil or performing clean-up of it. Dispose of chemicals, rags and materials in accordance with hospital and local government environmental and safety requirements.

Lead

Wear gloves and wash hands after handling lead. The CT detector uses a small amount of lead tape and a trace amount of Chromium (Cr) in its construction. The lead tape is located on the corners of the detector. Detectors can be returned to GEMS for proper disposal.

Heat Sources

The X-ray tube casing, plumbing and radiator can become hot enough to cause minor burns, if not allowed to cool sufficiently. Please allow sufficient time for this equipment to cool adequately before performing any service.



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Power Distribution Unit

DANGER Note the following before working on the Power Distribution Unit (PDU).

- Do not perform any work within the PDU, unless it is de-energized. More than 100 Kilowatts of power exists in the PDU at various periods of time. Therefore, consider all points in the PDU as hazardous.
- Connect voltage measuring equipment only when power is removed and the wall power box is locked and tagged.
- Always wear safety glasses because of the high voltages that exist in the PDU. Components can literally EXPLODE when power is applied.
- Be sure that all secondary protective covers on the PDUs are in place before the PDU is energized.

Electrical

Potential Hazards

- Axial drive power for gantry rotation (AC)
- High voltage DC for X-ray generation (floating DC)
- Distributed console, table and gantry power (AC)

Hazard Awareness Indicators

With the PDU's top cover hinged open, a small power lamp is visible. When illuminated, this lamp indicates power is present within the **PDU**.

Protected Service Outlets

The service outlet is protected by a circuit breaker. The outlet is located on the A4 panel.



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Power Distribution Unit, *continued*

Electrical, *continued*

Circuit Breakers and Switches

There are three (3) groups of circuit breakers in the PDU used to protect various parts of the system.

Group	Circuit Breaker
1	CB6 - Main Axial drive power
2	CB1 - Console AC power CB3 - Table and Gantry AC service outlets CB4 - Table and Gantry AC to stationary electronics CB5 - Gantry rotating power, tilt power and communications
3	CB7 - Master 120/208VAC power

The Auxiliary gantry power switch should be left “ON” at all times. It is used to disable 120VAC.

Mechanical

The PDU’s top cover employs latches on both sides to hold the cover in the open position.



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Gantry

Overview

With the gantry's primary covers removed, secondary covers are used to help prevent accidental contact with electrical contacts. The most electrically dangerous area in the gantry is the exposed slip ring platter. The system should be tagged and locked out whenever the gantry covers are removed.

When the gantry is rotating, the left and right sides of the gantry are where objects are most likely to be ejected, if not properly fastened.

Caution It is important that all hardware be properly fastened (Torqued) to its proper specification.

Take the following precautions when working on, near or around the gantry:

- Never wear loose clothing or jewelry. Clothing might become entangled in the rotating assembly and jewelry can short to high voltages.
- Avoid standing near the rotating assembly when it is operational, to avoid being struck by the assembly or ejected objects. Always torque fasteners to their proper specification.

- Avoid standing or kneeling near the slip ring platter. High voltages exist on the exposed rings. Always disable power to the rings by using the service switches before performing service.
- Never put any part of your body into the gantry, unless the gantry is locked. Axial drive power must be disabled. If working on the tilt assembly, the tilt bracket should be installed.
- Wear and use personal protection equipment.
- Tag and lockout power at the main disconnect.
- Always use and follow procedures described in your service documentation, when servicing this equipment.



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Electrical

Potential Hazards

All un-insulated electrical contacts, including the slipring, have secondary covers in place to protect service personnel from accidental contact. Removal of any **secondary cover** exposes service personnel to potentially deadly voltages.

All secondary covers must be in place before primary covers are installed and during routine service.

Un-insulated high voltage areas in the brush-block area include:

- High voltage DC for X-ray generation. Only measurement equipment isolated from ground can be used to measure HVDC on this system. Use of grounded measurement equipment can result in serious personal injury and/or equipment damage.
- 120VAC for power supplies.

Safety Awareness Indicators

If a secondary cover can be removed and it potentially exposes a service person to an uninsulated electrical hazard, a warning label is applied to or near the secondary cover. In the gantry, voltage hazards in excess of 120VAC have been labeled. However, the 120VAC present in the gantry also is capable of causing electrocution.



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Gantry, continued

Lamps and LEDs

There are a number of lamps/LEDs on the STC chassis backplane that indicate the functional state of the gantry. The descriptions apply when the associated LED is lit.

LED	Color	Label	Description
DS1	Green	C Pulse	C Pulse indicator from Axial Encoder.
DS2	Green	RST	Indicates status of the HVDC & gantry drives circuit in PDU. On steady = HVDC & Drives Enabled. Slow Flash = E-Stop activated. HVDC & Drives Disabled. Fast Flash = Table Tape Switch activated. Cradle, Tilt & Elevation Disabled.
DS3	Yellow	AX DR ON	Indicates the Axial Drive Contactor in the PDU is energized.
DS4	Green	ENBL	Indicates the Axial Drive Contactor in the PDU is enabled.

LED	Color	Label	Description
DS5	Yellow	HVDC ON	Indicates the HVDC Contactor in the PDU is energized.
DS6	Green	ENBL	Indicates the HVDC Contactor in the PDU is enabled.
DS7	Yellow	120VAC ON	Indicates the Gantry 120Vac Contactor in the PDU is energized.
DS8	Green	ENBL	Indicates the Gantry 120Vac Contactor in the PDU is enabled.



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Gantry, continued

Lamps and LEDs

Service Outlets and E-Stops

A number of features have been provided as means of disabling hazards at particular points in the gantry, for ease of service. This and the next few sections describe these features.

The **E-Stop** and **Service outlets** are located on both sides of the gantry on the stationary frame. The E-Stop is identified as a large red push button. The Service outlets are standard duplex 120VAC outlets. These are provided for ease of using AC powered test equipment near the gantry without requiring extension cords.

Service Switches

The Service switches are located at the top of the STC backplane. UP (enabled) is the normal operational position for these switches.

The service switches and circuit breakers described hereafter are not to be relied on as personal protection devices. They do not replace tag and lockout of main power to ensure personal safety. Switches and breakers are intended to only inhibit particular system functions and equipment operation. They do not eliminate or remove the electrical or mechanical hazards that exist. Because hardware can fail and defeat the functionality of these devices, only Lockout/Tagout ensures protection from unattended gantry rotation and electrocution. Be aware that not all power is removed on the gantry by use of these switches.



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Lamps and LEDs, continued

Label	Description
S1	Momentary Push button - Resets gantry drives enable circuit in PDU.
S2	Switch enables or disables the Axial Drive function. Default position up (enable)
S3	Switch enables or disables the HVDC function. Default position up (enable).
S4	Switch enables or disables Gantry 120VAC function - Default position up (enable). With S4 OFF, 120VAC to the gantry and table service outlets is controlled by CB3 in the PDU only.

Power Pan Circuit Breaker

The **circuit breaker** in the power pan, located at the rear of the gantry base, protects both the 170VDC and tilt drives (table and gantry respectively).

Gantry E-Stop

The gantry's internal E-Stop performs the same function as the E-Stops mounted to the console and the gantry covers.

Mechanical

Hazards

Inside the Gantry are several hazards that can cause personal injury from:

- moving assemblies (rotational and tilt)
- assembly weights (tube and covers)
- chemicals (slip ring brush dust and oils {Tube, HV Tank and Tilt Drive Hydraulic Oil})
- heat sources (tube)

Fastener Torque Specifications

To prevent assembly and part separations from the rotating assembly, all fasteners must be torqued to their proper specification, using a calibrated torque wrench. The torque specification for a fastener is specified in its associated replacement procedure. For further information on torque, including a conversion factor chart, refer to **Torque Wrench Information**.



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Rotational Locking Pin

To prevent un-commanded motion of the rotating assembly, the rotational locking pin must be engaged anytime the rotating assembly is serviced.

The rotational lock is located on the rear side of the gantry, near the top. It is positioned directly across from the teeth in the rotational assembly. To operate the lock:

1. Turn the handle clockwise until the teeth on the lock fully engage the teeth on the rotating assembly. If necessary, rock the rotating assembly slightly to align the teeth. Hand tighten until snug. Do not over-tighten. Visually verify that the teeth are engaged.
2. Turn the handle counter-clockwise until the teeth on the lock and the rotating assembly are fully disengaged and the teeth clear each other sufficiently.

Tilt “Locking” Brackets

Anytime the tilt assembly is serviced, the tilt locking brackets must be installed.

1. Position the gantry at zero degrees. Start on one side.
2. While holding the bracket in place, secure the bracket to the stationary frame at locations 1 and 2. The two tilt brackets are identical.]
3. Next, secure the bracket to the pivoting frame at location 3.
4. Repeat these steps on the other side.

When the brackets and associated hardware are not in use, store them in the top compartment of the PDU.



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X-Ray Tube Hoist

Whenever the X-ray tube is removed or installed, a tube hoist must be used. With the help of the tube hoist, one (1) person can change an X-ray tube.

Front and Rear Covers Dollies

The front and rear covers are designed to be safely removed by one (1) person, using the cover dollies supplied with your system. Because the weight of these covers could cause injury, these cover dollies must always be used. Both the installation and system manuals describe how to assemble and use cover dollies.



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Table

Electrical

Potential Hazards

The table base contains several lethal voltages. There are a number of points in the table where the voltages are dangerous (120VAC and 170VDC present). These points pose a potential electrical hazard to anyone that accidentally comes in contact with them.

Safety Awareness Indicators

A power lamp is located on the table power assembly. It is illuminated whenever power is present at the table service outlets. See Illustration 1.

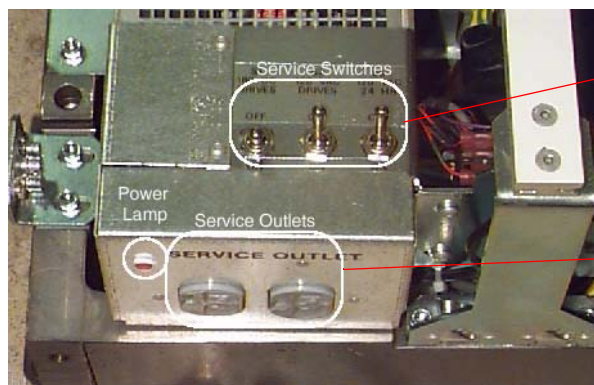
Service Outlet

A service outlet is located on the table's power assembly. It is protected by CB3 in the PDU.

Switches

The Table Safety Service switches are located on top of the power assembly. These switches are a subset of the 120VAC switch on the gantry.

S1	180 DC Drives	Enables/disables 170VDC Power Supplies for table elevation.
S2	120VAC Drives	Enables/disables 24V Power Supply for table cradle and elevation.
S3	120VAC 24 HR	Enables/disables 120VAC Table Power.



Service switches

Service outlet



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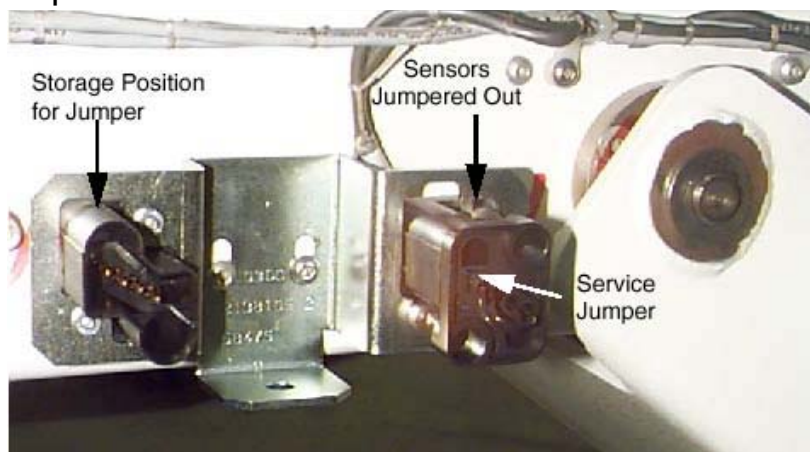
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Table, continued

Mechanical

During service, the table touch sensors must remain operable for the table to fully function. To operate the table with the covers removed, the sensors must be jumpered.



Chemical

Always wear personal protection equipment that prevents inhalation, digestion and absorption through skin of chemicals.



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Torque Wrench Information

This procedure contains the following information:

- Recommended Torque Wrench Practices
- General Torque Cross Reference
- Torque Formula
- Torque Wrench Accuracy

Recommended Torque Wrench Practices

- ALL FASTENERS HAVE A TORQUE REQUIREMENT. Default tables should be used only if the service documentation does not specify a torque value.
- Never use a torque wrench to loosen a tightened fastener. Permanent damage of the internal mechanism can occur due to excessive strain.
- Always approach the specified torque slowly. This is not a speed wrench.
- Hand location is important. Position one hand at the axis of rotation and one hand on the tool handle. This gives the user stability and accurate torque repeatability.
- Always approach the desired torque evenly and slowly. If the desired torque is 66 N-m on 4 bolts, then tighten each bolt 50 to 70% of desired value. Then set the wrench to the required torque and tighten slowly until the wrench “Just Clicks”.
- Always release the tension on the torque wrench to prevent “spring set” on adjustable or “clicker” type torque wrenches. This will ensure correct torque settings throughout the range of the tool.
- Always allow the tool to reach room temperature.
- Spring tension is the basis of “Clicker” type torque wrenches.
- A spring’s tension changes with temperature.
- Calibrate the tool on a regular schedule. Follow established local calibration processes.
- Do not drop or shock the tool. Internal damage can occur. Calibration should be performed to ensure accuracy.
- Do not attempt to straighten a bent “Beam” or non adjustable wrench. Replace it.



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Torque Wrench Information

Recommended Torque Wrench Practices, *continued*

- Never use a “Universal Joint” with a torque wrench. The angle of the universal joint can change the torque value by more than 50%.
- Always use the torque wrench with a 90 degree angle
- The 25 degree tilt is the physical limit of a Bondhus Ball End Hex key.
- Use the specified torque value for the HV tank mounting fasteners. Do not attempt to calculate the sin angle correction.
- There is less than 2% error for up to 10 degrees of tilt from the desired angle.
- Minimize the angle as much as possible.
- Always clean fastener threads to reduce friction. Fasteners should thread easily using finger pressure.
- Replace fasteners or clean threads using a tap or die, compressed air, brass brush.
- Never use a tap to clean thread inserts. It will damage them requiring replacement.

- Never lubricate a fastener unless specifically instructed. Loctite is considered in the design. It must be used when specified.
- Replace Nylon nuts if they are finger loose.

All cable connectors should be finger tightened by alternating between the two thumbscrews, until snug avoiding excessive over torque. The thumbscrew design includes a screwdriver slot designed to ensure access. All cable thumbscrews are required to be finally tightened by using a torque wrench type screwdriver in accordance to the defined torque specification of 45N-cm, (4 in-lbf) for all #4-40 Thumbscrews and receiver hex standoffs.

When secured properly, cable connectors will still have slight side to side movement. This is a normal condition.



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Torque Wrench Information, *continued*

General Torque Cross Reference

if the Service documentation does not contain specific torque values use these default Torque Values as specified by GEMS CT. Use the appropriate replacement procedure to verify the correct torque requirement for each specific fastener.

Note The Illustrated Parts List contains Engineering drawings that should also be used as a reference. These drawings call out specific instructions as notations, where needed.

It should also be noted, that some of the cable connector thumbscrews found on and around the DAS assembly on some gantries have a plastic slotted head. The plastic slot is designed to fail, (Strip out) if excessive torque is used to tighten these connectors. The torque specification for all cable connectors (#4-40 thumbscrew and receiving hex standoffs) is 45N-cm, (4 in-lbf). Engineering recommends that a torque screwdriver should be used on all cable connectors found in the system.

Fastener Size	Tool Size HEX Key	Tool Size Pocket	Torque in N-M	Torque in LBF-FT	Torque in LBF-IN	Torque Kg-cm
M3	2.5 mm	5.5 mm	--	--	--	--
M4	3 mm	7 mm	2.3	1.7	20.4	23.5
M6	5 mm	10 mm	7.9	5.8	70	80.8
M8	6 mm	13 mm	19	14	168	193.8
M10	8 mm	17 mm	38.4	28.3	339.6	391.8
M12	10 mm	18 mm	66.4	48.9	586.8	677.1
M14	12mm	--	105.9	78.1	937.2	1081.4
M16	14 mm	24 mm	160	117.8	--	--



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Torque Wrench Information, *continued*

General Torque Cross Reference, *continued*

Many service operations on this CT scanner require a torque wrench. The use of a torque wrench may appear complicated because there are several standards and metrics. Using conversion factors can simplify that task.

First, only use a calibrated torque wrench. Use a torque wrench that is on a Calibration schedule and is approved by GEMS-AM Service. The kit that can be used that is on a regular Calibration schedule is kit number 46-268445G1. This torque wrench kit has wrenches that measure inch pounds and foot pounds.

Second, make any necessary conversions for the torque wrench you are using. The units of measure are typically marked on most torque wrenches. To make conversions to Kg-cm and Newton meters, use the following conversion factors.

Torque Conversion Factors

To convert Kgcm to foot-lbs, multiply Kgcm by 0.07233

To convert Kgcm to inch-lbs, multiply Kgcm by 0.8679

To convert Kg-cm to N-m, multiply Kgcm by 0.0981

To convert N-m to inch-lbs, multiply N-m by 8.8508

To convert N-m to foot-lbs, multiply N-m by 0.73756

To convert foot-lbs to N-m, multiply lbf-ft by 1.3558

To convert foot-lbs to kg-m, multiply lbf-ft by 0.1383

To convert inch-lbs to N-m, multiply lbf-in by 0.11298

To convert inch-lbs to Kg-cm, multiply lbf-in by 1.1539

Torque Formula

$$T = R \times F \times \sin(\text{angle})$$

Where:

T = Torque in N-m

R = Distance from axis of rotation

F = Force Applied

$\sin(90) = 1$



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Torque Wrench Information, *continued*

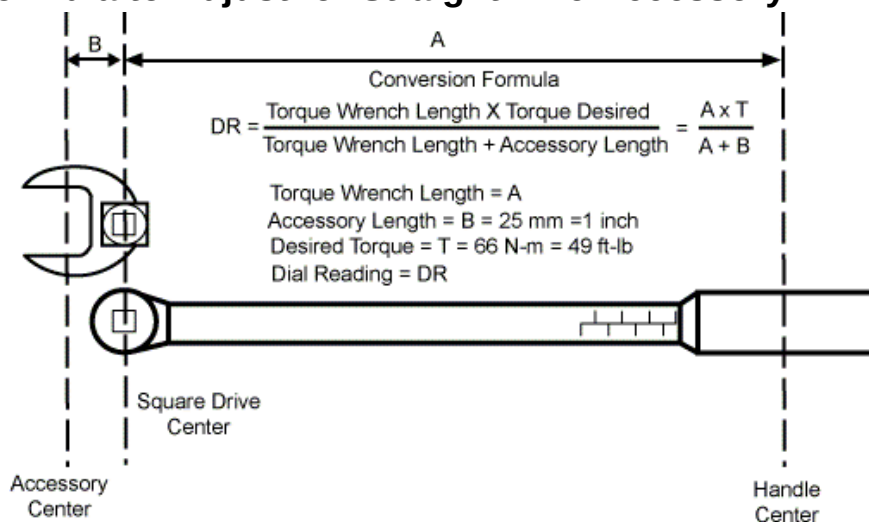
General Torque Cross Reference, *continued*

Torque Conversion Factors, *continued*

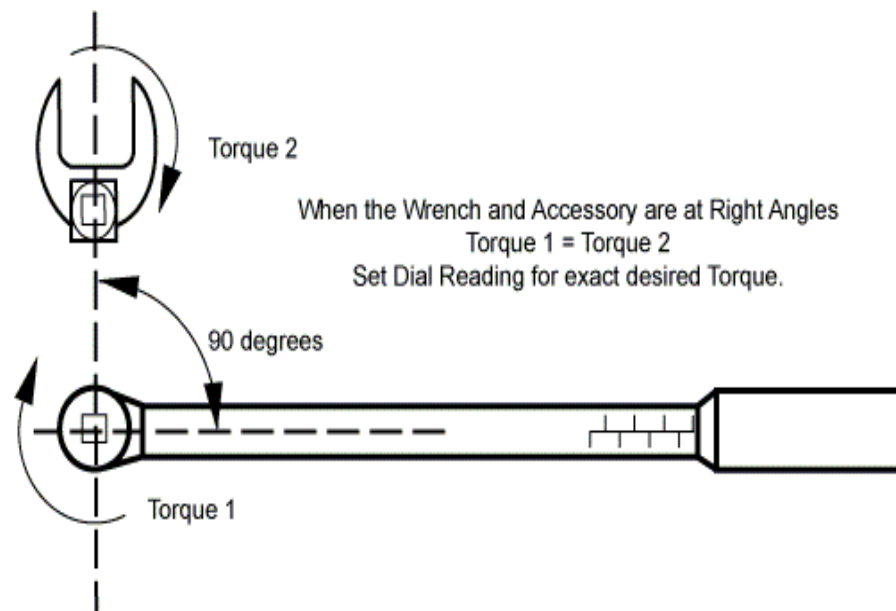
From this formula we can see that it is necessary to apply the force at a 90 degree angle to the axis of rotation to achieve accurate fastener torque. This same principle can be applied when using accessories with the torque wrench.

Note The length of a standard square drive extension has no effect on torque since it is along the axis of rotation. See Illustration 3.

Formula to Adjust for Straight Line Accessory



Formula for 90 Degree Accessory Usage



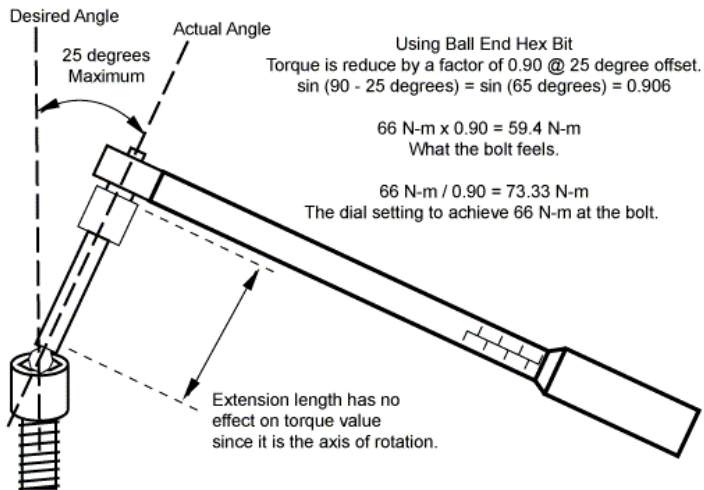


Torque Wrench Information, continued

General Torque Cross Reference, *continued*

Torque Conversion Factors, continued

Formula when not at 90 Degree to Axis of Rotation



Torque Wrench Accuracy

It needs to be clearly understood that “torque” is an indirect measure of tension or “preload force”. The components of a bolted joint can be defined as:

- Preload force (F_p), bolt stretch.
- Tension force (F_t), resistance of bolted materials.
- Clamping force (F_c), difference of preload and tension forces.
- Shear force (F_s), sideways or sliding force of bolted materials.

$$\text{Therefore, } F_c = F_p - F_t$$

With shear force, a properly designed and tightened joint, the friction between the bolted materials absorbs the stress and the bolt itself feels little to no load.

There are other factors that need to be considered as well. Fastener material has a large effect on torque versus preload force. Lubricants can also significantly change the effects of torque versus preload force. Anti-seize compounds can reduce the needed torque up to 20%.



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Torque Wrench Information, continued

Torque Wrench Accuracy, *continued*

In short, torque measurement is an economical method of achieving a properly tensioned joint. Other methods are available, but training needs and tool expense increase.

Note CT Engineering has taken into account the variability of using torque wrenches. The design standard applied is a safety factor of 8 on all fasteners, after the “G Force” load is calculated for each component. This is to ensure clamping force is maintained without exceeding the strength of the fastener.

Various studies have been performed on the effectiveness of torque wrench accuracy. The following conclusions have been made.

PRELOAD MEASURING METHOD ACTUAL PRELOAD
FORCE ERROR

“Feeling” > 35%

Torque Wrench +/-25%

Angle Torquing +/-15%

Indicating Washer +/-10%

Fastener Elongation +/-5%

Strain Gauge +/-1%

Not using a torque wrench is the worst case event. The “feeling” method also changes with the tool. A 1/4” drive “feels” different than a 1/2” drive.



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Lockout-Tagout and PPE



POSSIBLE ELECTRIC SHOCK

SERVICING HARDWARE WITH POWER ON.

EQUIPMENT SERVICE CAN ONLY BE PERFORMED SAFELY WITH THE MAIN POWER "DISCONNECT" TAGGED AND LOCKED OUT.



CRUSH HAZARD.

UNEXPECTED GANTRY ROTATION CAN CAUSE SERIOUS INJURY.

NEVER SERVICE THE GANTRY WITH THE AXIAL DRIVE ENABLED. USE THE GANTRY ROTATION LOCK TO LOCK OUT GANTRY ROTATION.

OF SERVICE DOCUMENT GUI FOR APPROPRIATE PROCEDURES.

Follow these general rules:

- Only qualified service personnel trained in the service and operation of this scanner should perform any service on this equipment.

- Equipment fuses, switches and circuit breakers are for fire and equipment protection only. Do not rely on them to protect you against electrical shock or un-commanded equipment motion.

- Personal Protection Equipment (PPE) is required and must be worn.

The service switches and circuit breakers described hereafter are not to be relied on as personal protection devices. They do not replace tag and lockout of main power to ensure personal safety. Switches and breakers are intended to only inhibit particular system functions and equipment operation. They do not eliminate or remove the electrical or mechanical hazards that exist. Because hardware can fail and defeat the functionality of these devices, only Lockout/Tagout ensures protection from unattended gantry rotation and electrocution.

Personal protection equipment must always be used when performing service on this equipment. Always use PPE when working with hazardous chemicals or materials.



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Lockout-Tagout and PPE, *continued*

LOTO Overview

The Lightspeed System requires the following LOTO rules to be followed:

- Apply 1 lock per person working on the system
- Only Authorized employees who are trained in LOTO can perform LOTO procedures
- Affected employees include customers, guests/visitors, field engineers, and contractors
- Verbally notify use of LOTO

System Energy Sources

Energy Source	Yes/No	Location of Energy Isolation Means	Magnitude of Energy
Electrical	Yes	A1 Panel or Main Disconnect Panel	Nominal 480VAC or 380VAC
Pneumatic	No		
Hydraulic	Yes	Inside Gantry (Tilt locking devices)	
Gas/Water/Steam	No		

Energy Source	Yes/No	Location of Energy Isolation Means	Magnitude of Energy
Chemical	No		
Mechanical Motion	Yes		Inside Gantry (Rotation lock)
Gravity	Yes		Inside Gantry (Rotation lock)
Stored Energy	No		
Springs	No		
Thermal	Yes		
Air Under Pressure	No		
Oil Under Pressure	No		
Water Under Pressure	No		
Gas Under Pressure	No		
Steam	No		
Other	No		



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Lockout-Tagout and PPE, continued

Following are the Lock Out Tag Out/Safety procedures relating to Thermal, **Electrical**, **Mechanical**, **Gravity**, and **Hydraulic energy** in the system. Be familiar with these procedures as they are referenced in the service procedures when appropriate.

Thermal Energy

When working on the CT LightSpeed gantry after the system has been scanning, you may have to wait for up to 60 minutes for the tube and heat exchanger to cool down if you will be touching the tube. A more specific cooling period for performing service on the tube will be found in the X-ray tube service procedures.



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Lockout-Tagout and PPE, *continued*

Electrical Energy

Remove Power from the System

1. Prepare for shutdown of equipment. Notify affected personnel working in the area, via verbal communication, that LOTO is being performed.
2. Bring the system software down at the operator console.
3. After software shutdown is complete, place console power switch in the OFF position.
4. Turn off power at the system A1 Panel or Main Disconnect Panel. All personnel servicing the system must place their personal LOTO devices and tags on the A1 Panel or Main Disconnect Panel. See Illustration 1 for example.
5. Verify that all energy has been dissipated as follows. Take the top and front cover off the PDU. Measure incoming system power at TS1 (for NGPDU) or Input Power Panel (for Compact PDU). Measure each phase to ground and verify zero potential.
6. Wait 5 minutes for any stored energy in capacitors to dissipate.

7. Service/repair the CT system per service instructions.

Restore Power to the System

1. Notify affected personnel that equipment will be re-energized and LOTO devices are being removed.
2. All service personnel are to remove their personal LOTO lock and tag from A1 Panel or Main Disconnect Panel.
3. Re-apply power to the PDU Cabinet.
4. Reset the Gantry drives by pushing the reset button on any of the control panels.
5. Bring the system software back up.



Electrical Panel Locked out and tagged.

Your electrical panel may look different but the process is the same. The handle of the electrical panel will have a means of locking the handle in place after the handle has been placed in the OFF position.



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Lockout-Tagout and PPE, *continued*

Gantry Rotation (Mechanical and Gravity Energy)

Disable Gantry Rotation

1. Remove the right side gantry cover.
2. Disable Axial Drive from the service switch panel or STC backplane depending on your system version.
3. Engage the Gantry rotational lock by turning the handle in a clockwise direction until the gantry will no longer move. See Illustration 2.
4. Pull safety lid over top of Gantry lock knob and lock with personal lock and tag.
5. **Remove power** from the system.
6. Service/repair the Gantry per service procedures.

Illustration 2: Gantry Rotational Lock

Illustration 3: Gantry Rotation Locked Out

Restore Gantry Rotation

1. Verify all test equipment has been removed from the Gantry and all components are installed and torqued per service procedures.
2. Notify affected personnel that Gantry LOTO devices are being removed.
3. Remove Gantry Lock-Out Tag-Out equipment and disengage the Gantry rotational lock.
4. **Restore power** to the system.
5. Enable Axial Drive.



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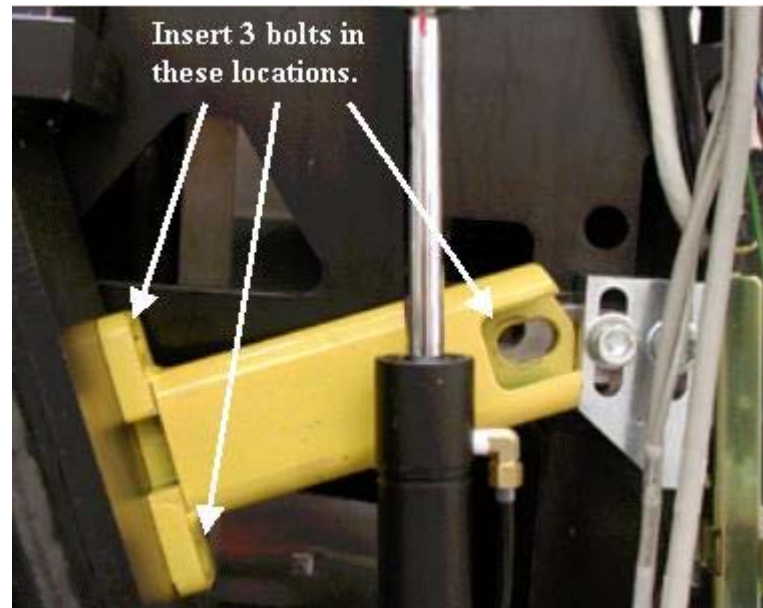
Lockout-Tagout and PPE, *continued*

Gantry Tilt (Hydraulic Energy)

Note This procedure is to lock the gantry from tilting. This is not for use when replacing the tilt hydraulic cylinders. The appropriate method is documented in that replacement procedure.

Disable Gantry Tilt

1. Position gantry at zero degrees tilt.
2. Take shipping brackets from storage compartment in top of the PDU (if compact PDU) or Gantry base (NGPDU).
3. Bolt them in place on both sides of the gantry using 3 bolts with each bracket.
4. **Remove power** from the system.
5. Service the gantry per service procedures.



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Lockout-Tagout and PPE, *continued*

Gantry Tilt (Hydraulic Energy), *continued*

1.4.2 Restore Gantry Tilt

1. **Restore power** to the system.
2. Notify affected personnel that gantry locking devices are being removed.
3. Loosen the bolts of the gantry tilt shipping brackets 1-2 turns and make sure the shipping bracket feels loose. If it is not loose, there may be a problem.

Caution If tilt bracket is not loose, stop and put the bolts back in and tighten tilt bracket back in place. If there is a load on the tilt bracket, removal may cause the gantry to suddenly tilt all the way back due to a possible lack of hydraulic pressure.

4. Check the hydraulic connections for leaks or lack of fluid. Use the tilt controls to relieve the load on the tilt bracket prior to removal. Do not use force to remove the bracket.
5. Remove the gantry tilt shipping brackets and place them back in their storage location



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http://bronze-web-prod.health.ge.com/itsurvey/docs/secure/survey_form.jsp?ID=16805&Key=WI1P6MM70C6ZC4
X



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